Bundles, Sheaves and Cohomology — Units 1–30

Thirty lectures and worksheets, independent English edition

Holger Brenner (source work)

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Contents

About this edition

This is an independent English edition of Holger Brenner's course *Bündel, Garben und Kohomologie (Osnabrück 2019–2020)*. Its scope is exactly 30 lectures and 30 worksheets, Units 1–30, in source order. All 495 exercises are retained. The translation scope includes the twenty-five publicly available solutions. For the other 470 exercises, the edition preserves documented negative search results; it does not invent solutions absent from the source.

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Frozen source authority

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Each of Units 1–30 has an authority manifest, official unit PDFs and component-rights records. The edition's scope covers the lecture, worksheet and available source solutions of every unit. Source closure records file identities, exercise order, public solutions and negative candidate results. English reader checks, reproducible-build reports and release-file identities are separate evidence from these frozen source records; Indonesian build evidence is not presented as proof of an English build.

Structure and study routes

The following order is that of the source lectures and worksheets; these routes do not authorise renumbering. Each range forms a coherent reading route.

Route	Units and topics	Exercises	Public solutions	Negative results
The language of bundles and sheaves	1 **Vector Bundles and Tangent Bundles**; 2 **Sections and Gluing Data**; 3 **Linear Constructions, Presheaves and Stalks**; 4 **Sheaves and Sheaf Morphisms**	71	2	69
Sheaf operations and spectra	5 **Sheafification and Quotient Sheaves**; 6 **Covering Spaces, Exactness, and Inverse and Direct Images of Sheaves**; 7 **Ringed Spaces and Local Rings**; 8 **Spectra and Maps of Spectra**	73	5	68
Affine and projective schemes	9 **Affine Schemes**; 10 **Schemes and Scheme Morphisms**; 11 **Irreducible Spaces and Noetherian Schemes**; 12 **The Projective Spectrum of a Graded Ring**	57	5	52

Route	Units and topics	Exercises	Public solutions	Negative results
Modules, invertible sheaves and differentials	13 **Cone Maps, Sheaves of Modules and Invertible Sheaves**; 14 **Quasi-Coherent Modules**; 15 **Modules on Projective Schemes**; 16 **Locally Free Sheaves**; 17 **Geometric Vector Bundles**; 18 **Kähler Differentials**	141	4	137
Tangent bundles and divisor classes	19 **Tangent Bundles**; 20 **The Picard Group**; 21 **Discrete Valuation Rings**; 22 **The Divisor Class Group**	58	5	53
Homological and cohomological tools	23 **Injective Modules**; 24 **Right-Derived Functors**; 25 **Sheaf Cohomology**; 26 **Čech Cohomology**; 27 **Cohomology on Projective Schemes**	61	1	60
Projective curves and Riemann–Roch	28 **Morphisms to Projective Space**; 29 **The Genus of a Curve**; 30 **The Riemann–Roch Theorem**	34	3	31
Total	**30 lectures and 30 worksheets**	**495**	**25**	**470**

Exercises and solutions

Public solutions occur in Unit 2 (Exercise 2.4), Unit 3 (3.1), Unit 5 (5.5), Unit 7 (7.14), Unit 8 (8.3, 8.4 and 8.11), Unit 11 (11.9, 11.13 and 11.14), Unit 12 (12.5 and 12.10), Unit 16 (16.12), Unit 18 (18.6, 18.17 and 18.18), Unit 19 (19.10), Unit 21 (21.3, 21.9 and 21.10), Unit 22 (22.19), Unit 25 (25.1), Unit 28 (28.6), and Unit 29 (29.5 and 29.12).

No public solutions were found at the source boundary for Units 1, 4, 6, 9, 10, 13, 14, 15, 17, 20, 23, 24, 26, 27 or 30.

In units with some public solutions, every other exercise also has its own negative result. Thus 470 is a count of documented negative search results, not blank solution pages and not proof that solutions have never existed elsewhere.

Media and accessibility

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Units 3, 5–7, 9–28 and 30 have no reader-media positions. The official unit and course PDFs are authority witnesses, not reader illustrations, so they are not counted as illustration assets. The absence of media positions in those units is not missing content.

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How to use this reader

Read each lecture before attempting the worksheet with the same number. Follow the routes in order: the language of bundles and sheaves; sheaf operations and spectra; affine and projective schemes; modules and differentials; tangent bundles, Picard groups and divisor classes; homological and cohomological tools; then projective-curve geometry through Riemann–Roch. Read edition notes where they occur to distinguish source content from verifiable corrections, translation decisions, notation notes and media provenance. Do not interpret a negative solution search as missing translated content. The terminology guide that follows helps identify corresponding terms across units without obscuring mathematical distinctions.

Terminology guide

This guide connects German terms with their use in Units 1–30. It adapts the independent Indonesian edition's connective glossary for English readers. Synonymous translations do not denote new mathematical objects: read them together with the definitions, hypotheses and notation in the relevant unit. The preferred forms here support cross-unit searching without changing formulae or wording inside source quotations.

Objects that must be distinguished

Source term	English equivalent	Reading guidance
Bündel; *Garbe*; *Prägarbe*	bundle; sheaf; presheaf	Bundle and sheaf are different source terms. The sheaf of sections of a bundle connects the two viewpoints.
Halm; *Faser*	stalk; fibre	For a sheaf of modules, the stalk at a point differs from the fibre obtained by passing to the residue field.

Source term	English equivalent	Reading guidance
Schnitt; *Durchschnitt*	section; intersection	A section of a sheaf or bundle is not a set-theoretic intersection. When splitting a surjection, a section means a right inverse.
Strukturgarbe; *Modulgarbe*	structure sheaf; sheaf of modules	The scalar structure matters: an $\mathcal{O}_X$-module is a module over the structure sheaf.
Homomorphismenmodul; *Homomorphismengarbe*	module of homomorphisms; sheaf of homomorphisms	Unit 13 also uses **global homomorphism module** to distinguish Hom from the sheaf $\mathcal{H}om$.
Garbenmorphismus; *Garbenhomomorphismus*	sheaf morphism; sheaf homomorphism	The second term emphasises the group or module structure being preserved; do not drop that structure from a statement.
quasikohärent; *kohärent*	quasi-coherent; coherent	Coherence includes additional finiteness conditions under the relevant definition; the two properties are not interchangeable.
Hyperebene; *Hyperfläche*	hyperplane; hypersurface	A hyperplane is linear, whereas a hypersurface may have degree greater than one.

Corresponding terms across units

Source term	Preferred form and variants	Limits of meaning
Vergarbung	**sheafification**; forming the associated sheaf	The construction of the sheaf associated to a presheaf, not an arbitrary sheaf construction.
Einheit; *Einheitengarbe*	**unit**; **sheaf of units**	A unit is an element with a multiplicative inverse, not just the identity element 1. "Unit 20" instead names a numbered part of the course.
Rang	**rank**	For a locally free sheaf or bundle, this means its rank, not its degree.
Einschränkung; *Restriktion*	**restriction**	Restricting an object or map to the specified subspace. In $\mathcal{F}|_U$, restriction is not a boundedness property.
Restriktionsabbildung	**restriction map**	Where the source emphasises the algebraic structure, use **restriction homomorphism**.
Integritätsbereich	**integral domain**	A nonzero commutative ring without zero divisors; "integral" here does not refer to integration.
Restklassenring	**quotient ring**; factor ring; residue-class ring	A ring modulo an ideal. Do not confuse this with *Quotientenkörper*, meaning **field of fractions**.
Restklassenmodul; *Restklassengruppe*	**quotient module**; **quotient group**	"Factor" and "residue-class" terminology may describe the same quotient construction. Modules, groups and rings must still be distinguished.

Source term	Preferred form and variants	Limits of meaning
Funktionenkörper	**function field**	The Indonesian variants *lapangan fungsi* and *medan fungsi* denote the same field; a **vector field** is a different notion.
Twist; *getwistete Strukturgarbe*	**twist**; **twisted structure sheaf**	A degree twist or a twist by an invertible sheaf, not a geometric rotation or dualisation.
glatt	**smooth**	Geometric smoothness is not automatically the same as *regulär* (**regular**) without suitable hypotheses on the base.
feine Monomgraduierung	**fine monomial grading**	"Fine" means a more detailed grading, not geometric smoothness.
welk; *welke Garbe*	**flasque** or **flabby**; **flasque sheaf**	Every restriction map is surjective. *Azyklisch* means **acyclic**; acyclic and flasque are not synonyms.
Überdeckung	**cover**; covering	A family of subsets covering a space. "Open", "affine" and "finite" remain mathematical qualifications.
affine Standardüberdeckung	**standard affine cover**	The cover used in the discussion of projective space; a standard cover is not an arbitrary cover.
Überlagerung	**covering map**	A covering-space map differs from a covering family of open sets.
projektiv	**projective**	The Indonesian spellings *projektif* and *proyektif* do not denote different objects; English uses projective in both courses.
Einbettung	**embedding**	Check the category and the properties of the map. An embedding of groups or modules does not automatically carry the properties of a scheme embedding.
beringter Raum	**ringed space**	A topological space equipped with a sheaf of rings. The earlier Indonesian wording *ruang berdering* has this defined meaning; *ruang bergelanggang* is that edition's preferred term.

Terms in the later units

Source term	Equivalent and guidance
Kähler-Differentiale; *Tangentialgarbe*; *Kotangentialgarbe*	**Kähler differentials**; **tangent sheaf**; **cotangent sheaf**. Preserve the distinctions among a sheaf, a module and a bundle.
kanonische Garbe; *antikanonische Garbe*	**canonical sheaf**; **anticanonical sheaf**. Do not omit the prefix anti-.
Syzygiengarbe	**syzygy sheaf**. The Indonesian spellings *syzygy* and *sizigi* refer to the same relation construction.
Weildivisor; *Hauptdivisor*	**Weil divisor**; **principal divisor**. A divisor here is a geometric object, not a divisor of an integer.
Nullstellendivisor; *Polstellendivisor*	The **divisor of zeros** and **divisor of poles** of a function. The former records zeros; it need not itself be the zero divisor.

Source term	Equivalent and guidance
injektive Auflösung; *rechtsabgeleiteter Funktor*	**injective resolution**; **right-derived functor**. A resolution is the whole complex, not a single injective map.
Čech-Kohomologie; *Čech-Kozykel*; *Čech-Koränder*	**Čech cohomology**; **Čech cocycles**; **Čech coboundaries**. Cohomology classes are cocycles modulo coboundaries.
lange exakte Kohomologiesequenz	**long exact cohomology sequence**; long exact sequence in cohomology. Both word orders retain the requirement of **exactness**.
kurze exakte Sequenz	**short exact sequence**. A sequence is not an *Ordnung*, an order in order theory.
lineares System; *volles lineares System*; *basispunktfrei*	**linear system**; **complete linear system**; **base-point-free**. Completeness and base-point-freeness are different properties.
Verzweigungsindex; *Verzweigungsordnung*	**ramification index**; **ramification order**. Unit 29 explains the naming difference between witnesses; both refer there to the same local ramification exponent.
Serre-Dualität; *Euler-Charakteristik*; *Satz von Riemann-Roch*	**Serre duality**; **Euler characteristic**; **Riemann–Roch theorem**. The names do not replace the hypotheses or the scope of the source statements.

Notation and origin of this guide

Prose terminology does not change source notation. For example, an **image** may still be written with the source operator `bild`, and the word **spectrum** does not require replacing `Spek` by `Spec` inside a formula. Differences in indices, equality signs, isomorphisms and hypotheses are not vocabulary variants.

This connective guide is based on Holger Brenner's course and the terminology decisions of the Indonesian edition, adapted to English. AI-assisted preparation: **OpenAI Codex gpt-5.6-sol, Ultra.**

Lecture 1: Parameter-Dependent Systems of Linear Equations and Vector Bundles

Parameter-dependent systems of linear equations

Consider the real linear equation

$$7u - 5v + 2w = 0.$$

Its solution set

$$L = \{(u, v, w) \in \mathbb{R}^3 \mid 7u - 5v + 2w = 0\} \subset \mathbb{R}^3$$

is a two-dimensional real vector subspace of $\mathbb{R}^3$. Solving such a linear equation means, among other things, finding a basis for L. In this case, for example,

$$L = \left\langle \begin{pmatrix} 5 \\ 7 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ -7 \end{pmatrix} \right\rangle .$$

The methods of solution are largely independent of the particular coefficients of the linear equation, although we shall see below the limitations of this statement. If we replace specific numbers by coefficients depending functionally on parameters, we may ask how the solution space varies with those parameters. For example, consider the linear equation depending on a parameter s,

$$7u - 5v + (s^2 - 3s - 10)w = 0.$$

For each s, the solution space L_s depends on s, but remains a two-dimensional subspace,

$$L_s \subset \mathbb{R}^3.$$

In other words, the solution space is a plane moving through space as s varies. We may ask for which values of s the vector

$$\begin{pmatrix} 5 \\ -3 \\ 8 \end{pmatrix}$$

is a solution, that is, belongs to L_s. We may also ask whether there are distinct parameters s, t for which

$$L_s = L_t$$

as subspaces of $\mathbb{R}^3$; whether the solution space always has a basis of the form

$$\left\langle \begin{pmatrix} a \\ b \\ 0 \end{pmatrix}, \begin{pmatrix} c \\ 0 \\ d \end{pmatrix} \right\rangle;$$

or whether there is always a solution vector of the form

$$\begin{pmatrix} 1 \\ 0 \\ e \end{pmatrix}.$$

Recall that the algorithm for solving systems of linear equations, Gaussian elimination, branches when certain coefficients are 0 or become 0 during the algorithm. The equation

$$7u - 5v + 0w = 0$$

has solution space

$$\left\langle \begin{pmatrix} 5 \\ 7 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\rangle$$

and contains no vector of the form $\begin{pmatrix} 1 & 0 & e \end{pmatrix}^\intercal$. Since $s = -2$ and $s = 5$ are the roots of the quadratic polynomial $s^2 - 3s - 10$, at these two parameter values the parametrised equation above becomes

$$7u - 5v + 0w = 0.$$

Thus, for these two values, L_s contains no vector of the form $\begin{pmatrix} 1 & 0 & e \end{pmatrix}^\intercal$. For all other parameter values, the solution space contains the vector

$$\begin{pmatrix} 1 \\ 0 \\ -\dfrac{7}{s^2 - 3s - 10} \end{pmatrix}.$$

A certain aspect of the solution space therefore itself depends functionally on the parameter.

It is natural to study the dependence of a linear equation or a system of linear equations on parameters in two stages. In the first stage, the coefficients of the equations themselves are treated as variables, or *universal parameters*, and we study how the solution spaces vary with them. In particular, we want to understand qualitative jumps in the behaviour of the solution spaces. In the second stage, we impose additional, more or less restrictive conditions on the universal parameters, or allow them to depend functionally on other parameters.

Example 1.1: one equation in two variables

Consider the general real linear equation

$$su + tv = 0$$

in the variables u, v and parameters s, t, which serve as indeterminate coefficients. We want to understand the solution space

$$L_{(s,t)} = \{(u,v) \mid su + tv = 0\} \subseteq \mathbb{R}^2$$

as a function of the parameters (s,t). An extreme case occurs at $(s,t) = (0,0)$: every (u,v) satisfies the equation, so the solution space is the whole two-dimensional space $\mathbb{R}^2$. If $(s,t) \neq (0,0)$, the solution space is one-dimensional, and a basis vector for this solution line is

$$\begin{pmatrix} t \\ -s \end{pmatrix}.$$

Thus, over the parameter space $\mathbb{R}^2 \setminus \{(0,0)\}$, the solution space has the uniform description

$$L_{(s,t)} = \left\{ c \begin{pmatrix} t \\ -s \end{pmatrix} \;\Big|\; c \in \mathbb{R} \right\}.$$

A more compact interpretation is obtained by considering the total solution space

$$L = \{(s,t,u,v) \mid su + tv = 0\} \subseteq \mathbb{R}^4.$$

Note that L is not a linear subspace of $\mathbb{R}^4$. The solution space for a particular parameter value (s,t) is obtained by intersecting L with the affine plane $(s,t) \times \mathbb{R}^2$. Under the total projection

$$L \longrightarrow \mathbb{R}^2 \times \mathbb{R}^2 \xrightarrow{p_{s,t}} \mathbb{R}^2, \qquad (s,t,u,v) \longmapsto (s,t),$$

$L_{(s,t)}$ is the fibre over (s,t). The total solution space displays both the variation of the solution lines with the parameter and their degeneration into a solution plane over the origin. The behaviour away from the parameter origin is described by the restriction

$$L' = L \setminus (\{(0,0)\} \times \mathbb{R}^2) = p^{-1}(\mathbb{R}^2 \setminus \{(0,0)\}) \longrightarrow \mathbb{R}^2 \setminus \{(0,0)\}.$$

Each fibre of this restricted projection is a one-dimensional solution space. Moreover, there is a bijection

$$\begin{aligned} (\mathbb{R}^2 \setminus \{(0,0)\}) \times \mathbb{R} &\longrightarrow L', \\ (s,t;c) &\longmapsto (s,t,ct,-cs), \end{aligned}$$

which is linear for each parameter (s,t). On the left is the direct product of the base space $\mathbb{R}^2 \setminus \{(0,0)\}$ and the fibre $\mathbb{R}$, which is independent of the base point. On the right is a family of varying lines in $\mathbb{R}^2$, but the bijection translates one description into the other.

Editorial note - order of factors and base space. In the definition of L', the source prints $\mathbb{R}^2 \times (0,0)$, whereas equality with $p^{-1}(\mathbb{R}^2 \setminus \{(0,0)\})$ requires removing the fibre over the origin, namely $\{(0,0)\} \times \mathbb{R}^2$. The source prose subsequently prints $\mathbb{R}^2 \times (0,0)$ again as the base space, whereas the domain of the immediately preceding bijection is $\mathbb{R}^2 \setminus \{(0,0)\}$. This edition follows the two displayed maps and explicitly records the discrepancy.

Example 1.2: one equation in three variables

Consider the general real linear equation

$$ru + sv + tw = 0$$

in the variables u, v, w and parameters r, s, t, which serve as indeterminate coefficients. We want to understand the solution space

$$L_{(r,s,t)} = \{(u,v,w) \mid ru + sv + tw = 0\} \subseteq \mathbb{R}^3$$

as a function of the parameters (r,s,t). At $(r,s,t) = (0,0,0)$, the solution space is the whole of $\mathbb{R}^3$. If $(r,s,t) \neq (0,0,0)$, it is two-dimensional. We exclude the origin from the parameter space and consider the total solution space

$$\begin{aligned} L = &\left\{(r,s,t,u,v,w) \,\middle|\, ru + sv + tw = 0, (r,s,t) \neq (0,0,0)\right\} \\ &\subseteq (\mathbb{R}^3 \setminus \{(0,0,0)\}) \times \mathbb{R}^3, \end{aligned}$$

together with the projection p to $\mathbb{R}^3 \setminus \{(0,0,0)\}$. The fibre of p over a particular parameter (r,s,t) is the solution space $L_{(r,s,t)}$ of the equation determined by that parameter tuple.

Editorial note - notation for the origin and scope of the product. In this inclusion the source prints $\mathbb{R}^3 \setminus \{0,0,0\} \times \mathbb{R}^3$, without writing the origin as a tuple or using parentheses to separate the base space from the fibre factor. The projection in the next sentence uniquely determines the intended meaning. This edition writes $(\mathbb{R}^3 \setminus \{(0,0,0)\}) \times \mathbb{R}^3$.

Can we give a basis for each solution space that depends on the parameters in an explicit computational and algebraic way? Since we have removed the origin,

$$\mathbb{R}^3 \setminus \{(0,0,0)\} = \{(r,s,t) \mid r \neq 0\} \cup \{(r,s,t) \mid s \neq 0\} \cup \{(r,s,t) \mid t \neq 0\}.$$

The base space can therefore be written as a union of three open sets. Over the open set $r \neq 0$, for example, a basis is given by

$$(s, -r, 0) \quad \text{and} \quad (t, 0, -r).$$

The condition $r \neq 0$ ensures that the two vectors are linearly independent. Indeed, the two vectors are well-defined solutions everywhere, but when $r = 0$ they lose their linear independence and hence do not form a basis everywhere. In any case, the map

$$\begin{aligned} \{(r,s,t) \mid r \neq 0\} \times \mathbb{R}^2 &\longrightarrow L|_{\{(r,s,t) \mid r \neq 0\}}, \\ (r,s,t;c,d) &\longmapsto (r,s,t;\, c(s,-r,0) + d(t,0,-r)) \end{aligned}$$

is a computationally simple bijection between the product of the base space with $\mathbb{R}^2$ and the solution space over $\{(r,s,t) \mid r \neq 0\}$.

Editorial note - base coordinates. The source writes only the fibre vector on the right-hand side of this map. Since its codomain is the total space L, the unchanged base coordinates (r,s,t) are displayed here as well.

We now ask whether it is possible to give, globally on all of $\mathbb{R}^3 \setminus \{(0,0,0)\}$, a basis of the solution space that varies with the base point. The question is whether there exist two functions $u(r,s,t)$ and $v(r,s,t)$ with values in $\mathbb{R}^3$ that always form a basis of the corresponding fibre, and in particular belong to it. With no further conditions on u and v, this is possible by a case-by-case definition. However, it is no longer possible if both functions are required to be continuous. By continuity, the global functions u and v are already determined by their values on the dense open set

$$U = \{(r,s,t) \mid r \neq 0\} \subseteq \mathbb{R}^3 \setminus \{(0,0,0)\}.$$

Using the basis over U given above, we can write

$$u = \alpha(r,s,t)\begin{pmatrix} s \\ -r \\ 0 \end{pmatrix} + \beta(r,s,t)\begin{pmatrix} t \\ 0 \\ -r \end{pmatrix}$$

and

$$v = \gamma(r,s,t)\begin{pmatrix} s \\ -r \\ 0 \end{pmatrix} + \delta(r,s,t)\begin{pmatrix} t \\ 0 \\ -r \end{pmatrix},$$

where $\alpha, \beta, \gamma, \delta$ are continuous real-valued functions on U. We cannot expect these coefficient functions to be defined on all of $\mathbb{R}^3$, so the argument in the continuous case becomes more complicated. The result will follow from Theorem 2.3; see Remark 2.4.

For now, we therefore restrict attention to rational functions whose denominators may contain a power of r, that is, rational functions on U. Consider

$$\begin{aligned} u &= \alpha\begin{pmatrix} s \\ -r \\ 0 \end{pmatrix} + \beta\begin{pmatrix} t \\ 0 \\ -r \end{pmatrix} \\ &= \frac{P}{r^m}\begin{pmatrix} s \\ -r \\ 0 \end{pmatrix} + \frac{Q}{r^n}\begin{pmatrix} t \\ 0 \\ -r \end{pmatrix}, \end{aligned}$$

where P, Q are polynomials and factors of r have been cancelled wherever possible. Since u as a whole is defined on all of $\mathbb{R}^3$, the exponent m, and likewise n, is at most 1; otherwise u would have a pole. For $m = n = 1$, the first component gives a polynomial equation of the form

$$rN + sP + tQ = 0, \qquad N, P, Q \in \mathbb{R}[r,s,t].$$

In this case, the relevant idea being the Koszul resolution,

$$(N, P, Q) = A(-s, r, 0) + B(t, 0, -r) + C(0, t, -s)$$

for polynomials $A, B, C \in \mathbb{R}[r,s,t]$. Similarly, v has a representation in terms of (N', P', Q') and (A', B', C'). Write

$$X = \mathbb{R}^3 \setminus \{(0,0,0)\}$$

and consider the map

$$\begin{aligned} \varphi : X \times \mathbb{R}^3 &\longrightarrow L \subseteq X \times \mathbb{R}^3, \\ (r,s,t; a,b,c) &\longmapsto (r,s,t;\, a(-s,r,0) + b(t,0,-r) + c(0,t,-s)). \end{aligned}$$

Under this map, the polynomial tuples $(-A, -B, -C)$ and $(-A', -B', -C')$, viewed as maps $X \to X \times \mathbb{R}^3$, are sent to u and v. By assumption, u and v form a basis of every fibre of L, so (A, B, C) and (A', B', C') are linearly independent at every point. The tuple $(t, s, -r)$ is sent by φ to 0 in every fibre. Therefore,

$$(A, B, C), \qquad (A', B', C'), \qquad (t, s, -r)$$

form a basis of $\mathbb{R}^3$ at every point: $(t, s, -r)$ cannot be a linear combination of the first two tuples, since applying φ would then give a nontrivial relation between u and v. However, the determinant of the matrix

$$\begin{pmatrix} A & B & C \\ A' & B' & C' \\ t & s & -r \end{pmatrix}$$

is a polynomial combination of the variables r, s, t, and so is not a unit in the polynomial ring. In the real case, we cannot yet conclude that this determinant has a real zero in X; for example, it might have the form $r^2 + s^2 + t^2$. However, if we replace $\mathbb{R}$ by $\mathbb{C}$, the algebraic argument is unchanged, and we can conclude that the determinant has a zero in

$$X_{\mathbb{C}} = \mathbb{C}^3 \setminus \{(0,0,0)\}.$$

Thus such a global basis cannot exist at every point.

> **Editorial note - sign of the lifted tuples.** The source says that (A, B, C) and (A', B', C') map to u and v. With its displayed convention $rN + sP + tQ = 0$, however, $u = -(N, P, Q)$, and similarly for v. Negating the two coefficient tuples gives the stated lifts. The linear-independence and determinant argument is unchanged.

Example 1.3: two equations in three variables

Consider the general real system of linear equations

$$au + bv + cw = 0$$

and

$$du + ev + fw = 0$$

in the variables u, v, w and parameters a, b, c, d, e, f, which serve as indeterminate coefficients of the system. If the parameters are sufficiently general, or more precisely, if there is no linear relation between the two equations, then the solution space

$$L_{(a,b,c,d,e,f)} = \left\{ (u, v, w) \;\middle|\; au + bv + cw = 0 \text{ and } du + ev + fw = 0 \right\} \subseteq \mathbb{R}^3$$

is a line. Under this condition, the parameters therefore determine a family of varying lines in $\mathbb{R}^3$. The relevant parameter space for this family of lines is

$$P = \left\{ (a, b, c, d, e, f) \;\middle|\; (a, b, c) \text{ and } (d, e, f) \text{ linearly independent} \right\}.$$

Altogether, we obtain the total solution space

$$\begin{aligned} L = \{&(a, b, c, d, e, f, u, v, w) \mid au + bv + cw = 0 \text{ and } du + ev + fw = 0\} \\ &\subseteq P \times \mathbb{R}^3, \end{aligned}$$

together with its projection to P.

Can this line, or a basis element of it, be specified globally as a function of the parameters? Viewing the two equations as orthogonality relations, we seek a nonzero vector perpendicular to both constraint vectors

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} d \\ e \\ f \end{pmatrix}.$$

Their cross product has this property, namely

$$\begin{pmatrix} bf - ce \\ -af + cd \\ ae - bd \end{pmatrix}.$$

For the properties of the cross product used here, the source refers to Lemma 33.3 in *Lineare Algebra (Osnabrück 2024-2025)*.

Thus there is a bijection

$$\begin{aligned} P \times \mathbb{R} &\longrightarrow L, \\ (a,b,c,d,e,f;s) &\longmapsto (a,b,c,d,e,f; s(bf-ce), s(-af+cd), s(ae-bd)). \end{aligned}$$

Editorial note - two coordinate names in the source. The source displays the second constraint vector as (e, f, g), although the system and parameter space define it as (d, e, f). In the final map, the source also prints the middle component as $-af + ce$, whereas the cross product printed immediately before it gives $-af + cd$. This edition uses (d, e, f) and $-af + cd$, which can be verified directly from the two equations, while recording both source typographical errors.

In Examples 1.1 and 1.3 there are *global polynomial trivialisations*: polynomial functions translate the complicated geometric object into the simple object $P \times \mathbb{R}$, where P is the base space. By contrast, such a global trivialisation is impossible in Example 1.2, although local trivialisations exist over the three specified open sets. Geometric objects of this kind are called vector bundles.

Real vector bundles

Definition 1.4: real vector bundle

Let X be a topological space and $r \in \mathbb{N}$. A *real vector bundle of rank* r is a topological space V together with a continuous map

$$p : V \longrightarrow X$$

such that every fibre $p^{-1}(x)$ is an r-dimensional real vector space, and there is an open cover

$$X = \bigcup_{i \in I} U_i$$

together with homeomorphisms over U_i,

$$\varphi_i : p^{-1}(U_i) \longrightarrow U_i \times \mathbb{R}^r,$$

which induce a linear isomorphism on each fibre,

$$(\varphi_i)_x : p^{-1}(x) \longrightarrow \mathbb{R}^r.$$

The space V is also called the *total space*, and X the *base space* of the vector bundle. The fibre over x is often denoted by

$$V_x = p^{-1}(x).$$

In the examples above, X is the relevant parameter space, namely the locus of parameters for which the solution spaces have minimal dimension. This dimension is the rank r in the definition above: respectively, $1, 2, 1$. In the first and third examples, the open cover consists only of the base space itself; these two bundles have a global trivialisation. In the second example, there is a cover by three open sets over which trivialisations have been given.

In the homeomorphism

$$p^{-1}(U) \longrightarrow U \times \mathbb{R}^r,$$

the right-hand side carries the product topology, $\mathbb{R}^r$ its natural Euclidean topology, and $p^{-1}(U)$ the topology induced from V. Thus every fibre V_x carries the natural topology of a finite-dimensional real vector space. A homeomorphism *over* U means that the diagram

$$\begin{array}{ccc} p^{-1}(U) & \xrightarrow{\varphi} & U \times \mathbb{R}^r \\ & \searrow p & \downarrow \mathrm{pr}_1 \\ & & U \end{array}$$

commutes.

> **Editorial note - rank symbol in the diagram.** The source diagram prints $U \times \mathbb{R}^n$, whereas the definition and all surrounding formulae specify the rank as r. This edition displays $\mathbb{R}^r$.

The product $X \times \mathbb{R}^r$ is a vector bundle called the *trivial vector bundle*.

Lemma 1.5: restriction of a vector bundle

Let

$$p : V \longrightarrow X$$

be a real vector bundle over a topological space X. For every open set $W \subseteq X$, the restriction

$$p^{-1}(W) \longrightarrow W$$

is also a vector bundle.

Proof Simply restrict the fibrewise linear homeomorphisms

$$\varphi_i : p^{-1}(U_i) \longrightarrow U_i \times \mathbb{R}^r$$

to

$$\varphi_i|_{W \cap U_i} : p^{-1}(W \cap U_i) \longrightarrow (W \cap U_i) \times \mathbb{R}^r.$$

□

The restriction of a vector bundle to each U_i is trivial. Thus every vector bundle is locally trivial.

Definition 1.6: homomorphism of vector bundles

Let E and F be real vector bundles over a topological space X. A *homomorphism of vector bundles*

$$\varphi : E \longrightarrow F$$

is a continuous map over X such that, for every $x \in X$, the induced map

$$\varphi_x : E_x \longrightarrow F_x$$

is $\mathbb{R}$-linear.

Definition 1.7: isomorphism of vector bundles

Let E and F be real vector bundles over a topological space X. A homomorphism of vector bundles

$$\varphi : E \longrightarrow F$$

is called an *isomorphism* if there is a homomorphism

$$\psi : F \longrightarrow E$$

whose composition with φ, in either order, is the identity map.

The tangent bundle of a manifold

We now discuss another particularly important vector bundle, present on every manifold: the tangent bundle.

Every point $P \in M$ of a manifold has a tangent space $T_P M$. The tangent space is an n-dimensional vector space, where n is the dimension of the manifold. Its elements are tangent vectors, or "infinitesimal directions" at that point. Initially, tangent directions at two distinct points have nothing to do with one another: their definitions depend only on arbitrarily small open neighbourhoods of the respective points, and the Hausdorff property allows these neighbourhoods to be chosen disjoint.

The picture for an open set $V \subseteq \mathbb{R}^n$ is quite different. For every $Q \in V$, the tangent space $T_Q V$ can be identified naturally with the ambient vector space $\mathbb{R}^n$. A vector $v \in \mathbb{R}^n$ is assigned the tangent vector determined by the linear curve $t \mapsto Q + tv$. Since this identification applies at every point, there is a direct parallelism between the tangent spaces for

$$Q \in V \subseteq \mathbb{R}^n.$$

A manifold is covered by open sets diffeomorphic to open subsets of Euclidean space. It is therefore natural to expect that its various tangent spaces are not completely isolated. The concept of the tangent bundle brings all the tangent spaces together and reflects their local interconnection.

> **Source illustration - `Tangent_bundle.svg`.** Two visualisations of the tangent bundle of a circle. In the upper picture, the tangent space at each point P of the circle is placed tangentially to the circle and realised as a one-dimensional affine subspace of $\mathbb{R}^2$. This embedding creates intersections that do not exist in the tangent bundle itself, since the base point P must also be taken into account. In the lower picture, the tangent spaces are arranged in parallel over the points of the circle, producing a cylinder.

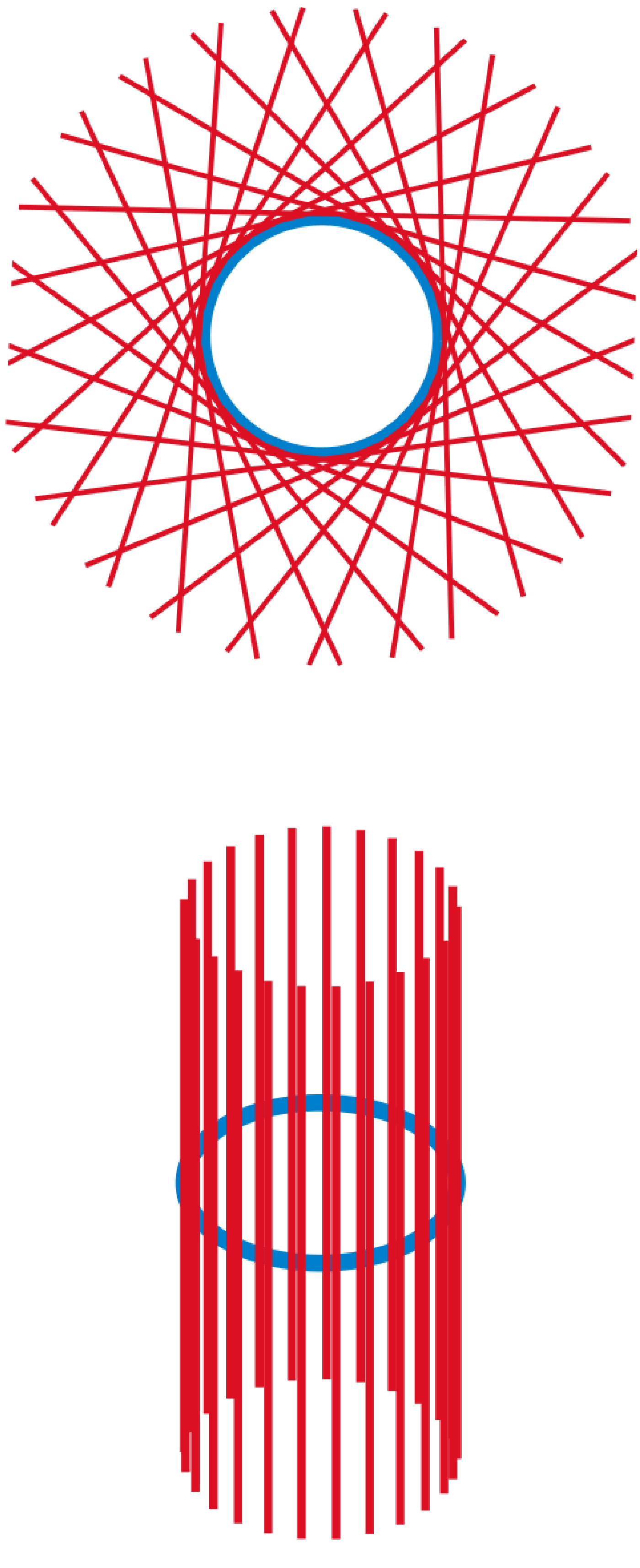

Figure 1: Diagram of the tangent bundle, with tangent spaces as fibres over the points of a manifold

Definition 1.8: the tangent bundle as a disjoint union

Let M be a differentiable manifold. The set

$$TM = \biguplus_{P \in M} T_P M,$$

together with the projection map

$$\begin{aligned} \pi : TM &\longrightarrow M, \\ (P, v) &\longmapsto P, \end{aligned}$$

is called the *tangent bundle* of M.

A point $u \in TM$ always has a base point $P \in M$ and is an element of the tangent space $T_P M$. It is usually written (P, v) with $P \in M$ and $v \in T_P M$. For an open set $V \subseteq \mathbb{R}^n$,

$$TV = V \times \mathbb{R}^n,$$

so it is a product space. This does not hold for an arbitrary manifold. Initially, the tangent bundle merely takes the disjoint union of the various tangent spaces, without identifying different tangent spaces with one another. However, the topology we shall shortly put on the tangent bundle adds a "neighbourhood structure" between the tangent spaces.

Definition 1.9: tangent map

Let M and N be differentiable manifolds and

$$\varphi : M \longrightarrow N$$

a differentiable map. Let TM and TN be the corresponding tangent bundles. The *tangent map*

$$T(\varphi) : TM \longrightarrow TN$$

is the disjoint union of the tangent maps at the individual points, namely

$$T(\varphi) = \biguplus_{P \in M} T_P(\varphi).$$

Example 1.10: local trivialisation from a chart

Let M be a differentiable manifold and

$$\alpha : U \longrightarrow V$$

a chart, where $V \subseteq \mathbb{R}^n$ is open. The chart induces a natural bijection

$$\begin{aligned} T(\alpha^{-1}) : TV = V \times \mathbb{R}^n &\longrightarrow TU, \\ (Q, v) &\longmapsto \left(\alpha^{-1}(Q), [s \mapsto \alpha^{-1}(Q + sv)]\right). \end{aligned}$$

Here s ranges over a real interval I chosen so that $Q + sv \in V$ (compare Lemma 77.5 in *Analysis (Osnabrück 2014-2016)*). Since $V \times \mathbb{R}^n$ is a product of topological spaces,

$$TV = V \times \mathbb{R}^n$$

is itself a topological space. It is natural to transfer this topology to TU, and then construct a topology on the whole tangent bundle TM.

Definition 1.11: topology of the tangent bundle

Let M be a differentiable manifold of dimension n and

$$TM = \biguplus_{P \in M} T_P M$$

its tangent bundle, with projection

$$\pi : TM \longrightarrow M,$$
$$(P, v) \longmapsto P.$$

Equip the tangent bundle with the following topology: a subset $W \subseteq TM$ is open if and only if, for every chart

$$\alpha : U \longrightarrow V,$$

the set

$$T(\alpha)\left(W \cap \pi^{-1}(U)\right)$$

is open in $V \times \mathbb{R}^n$.

In particular, for every open set $U \subseteq M$, the inverse image

$$\pi^{-1}(U) = TU \subseteq TM$$

is open; in other words, the projection π is continuous. With these conventions, the tangent bundle of a differentiable manifold is a real vector bundle. If

$$M = \bigcup_{i \in I} U_i$$

is an open cover by sets U_i homeomorphic to open sets $V_i \subseteq \mathbb{R}^n$, then the charts

$$\alpha_i : U_i \longrightarrow V_i$$

directly provide trivialisations

$$TM|_{U_i} = TU_i \overset{T(\alpha_i)}{\longrightarrow} TV_i = V_i \times \mathbb{R}^n.$$

A remarkable number of properties of a manifold are reflected in properties of its tangent bundle. The tangent bundle may be trivial even when M is not homeomorphic to an open subset of $\mathbb{R}^n$.

Worksheet 1: Vector Bundles and Tangent Bundles

In the following exercises, we use the notation

$$D(r) = \left\{(r, s, t) \in \mathbb{R}^3 \mid r \neq 0\right\}.$$

Exercise 1.1

For the vector bundle

$$L = \Big\{(r,s,t,u,v,w) \;\Big|\; ru+sv+tw=0,\ (r,s,t)\neq(0,0,0)\Big\}$$
$$\subseteq (\mathbb{R}^3\setminus\{(0,0,0)\})\times\mathbb{R}^3 \longrightarrow \mathbb{R}^3\setminus\{(0,0,0)\},$$

determine linear trivialisations over $D(r)$, $D(s)$ and $D(t)$, that is, bases depending on r,s,t over $D(r)$ and so on. Determine the change-of-basis maps on

$$D(rs) = D(r)\cap D(s).$$

Exercise 1.2

For the vector bundle

$$L = \Big\{(r,s,t,u,v,w) \;\Big|\; ru+sv+tw=0,\ (r,s,t)\neq(0,0,0)\Big\}$$
$$\subseteq (\mathbb{R}^3\setminus\{(0,0,0)\})\times\mathbb{R}^3 \longrightarrow \mathbb{R}^3\setminus\{(0,0,0)\},$$

determine all parameters (r,s,t) for which the vector $(3,7,4)$ belongs to the fibre $L_{(r,s,t)}$.

Exercise 1.3

Show that, in Example 1.2 and over $D(r)$, the formula

$$u(r,s,t) = \frac{t}{r}\begin{pmatrix} s \\ -r \\ 0\end{pmatrix} - \frac{s}{r}\begin{pmatrix} t \\ 0 \\ -r\end{pmatrix}$$

defines a parameter-dependent vector in the solution space that extends polynomially to all of $\mathbb{R}^3$, even though the coefficient functions t/r and $-s/r$ are defined only on $D(r)$ and do not extend. Is $u(r,s,t)$ part of a basis at every point?

Exercise 1.4

In Example 1.3, determine the parameters for which the solution space $L_{(a,b,c,d,e,f)}$ is one-, two- or three-dimensional. Are these parameter sets open or closed?

Exercise 1.5

Show that, over an arbitrary field K, for two linearly independent vectors

$$u = \begin{pmatrix} a \\ b \\ c\end{pmatrix} \quad\text{and}\quad v = \begin{pmatrix} d \\ e \\ f\end{pmatrix},$$

the family consisting of u, v and their cross product

$$\begin{pmatrix} a \\ b \\ c\end{pmatrix} \times \begin{pmatrix} d \\ e \\ f\end{pmatrix}$$

need not form a basis of K^3.

Exercise 1.6

Consider the topological space

$$Y := \{(s,t,u,v) \in \mathbb{R}^4 \mid su + tv = 1\}$$

with projection

$$\begin{aligned} p : Y &\longrightarrow \mathbb{R}^2 \setminus \{(0,0)\} = X, \\ (s,t,u,v) &\longmapsto (s,t). \end{aligned}$$

1. Show that every fibre of p is homeomorphic to a real line.
2. Show that

   $$\varphi(s,t) = (s,t,u(s,t),v(s,t)) = \left(s,t,\frac{s}{s^2+t^2},\frac{t}{s^2+t^2}\right)$$

   defines a continuous map $\varphi : X \to Y$ with

   $$p \circ \varphi = \mathrm{Id}_X\,.$$

3. Define a homeomorphism between Y and $X \times \mathbb{R}$.
4. Show that there is no polynomial map $\psi : X \to Y$ with

   $$p \circ \psi = \mathrm{Id}_X\,.$$

Exercise 1.7

Show that a real vector bundle over a point, that is, over a one-point topological space, is the same as a finite-dimensional real vector space.

Exercise 1.8

Let $p : V \to X$ be a real vector bundle over a topological space X. Show that V is a Hausdorff space if and only if X is a Hausdorff space.

Exercise 1.9

Let X be a topological space. Show that the identity map

$$\mathrm{Id}_X : X \longrightarrow X$$

can be regarded as a real vector bundle of rank 0.

Exercise 1.10

Let X be a topological space. Show that a homomorphism of trivial vector bundles

$$\varphi : X \times \mathbb{R}^n \longrightarrow X \times \mathbb{R}^m$$

is the same as an $m \times n$ matrix whose entries are continuous functions from X to $\mathbb{R}$.

Exercise 1.11

Let $U \subseteq \mathbb{R}^n$ be open and $f : U \to \mathbb{R}^m$ a continuously differentiable map. Show that the total differential, in the form

$$\begin{aligned} U \times \mathbb{R}^n &\longrightarrow U \times \mathbb{R}^m, \\ (x, v) &\longmapsto (x, (Df)_x(v)), \end{aligned}$$

defines a homomorphism from the vector bundle $U \times \mathbb{R}^n$ to the vector bundle $U \times \mathbb{R}^m$.

Exercise 1.12

Show that the tangent bundle TS^1 of the 1-sphere S^1 is homeomorphic to the product $S^1 \times \mathbb{R}$.

How is this exercise related to Example 1.1?

Exercise 1.13

Give an example of a differentiable curve

$$\gamma : [0, 1) \longrightarrow S^1$$

such that the limit

$$\lim_{t \to 1} \gamma(t)$$

exists, but the limit

$$\lim_{t \to 1} (\gamma(t), T_t(\gamma)(1))$$

does not exist in TS^1.

Exercise 1.14

Show that the map

$$\begin{aligned} TS^1 &\longrightarrow \mathbb{R}^2, \\ ((a, b), t(-b, a)) &\longmapsto (a, b) + t(-b, a) \end{aligned}$$

has two preimages for every point $(x, y) \in \mathbb{R}^2$ outside the unit disc, one preimage for every point on the unit circle, and no preimage for any point inside the open unit disc. Interpret this geometrically.

Exercise 1.15

Give an example of an injective differentiable map

$$\varphi : M \longrightarrow N$$

between two differentiable manifolds M and N such that the associated tangent map

$$T(\varphi) : TM \longrightarrow TN$$

is not injective.

Exercise 1.16

Give an example of a surjective differentiable map

$$\varphi : M \longrightarrow N$$

between two differentiable manifolds M and N such that the associated tangent map

$$T(\varphi) : TM \longrightarrow TN$$

is not surjective.

Exercise 1.17

Let M and N be differentiable manifolds and $\varphi : M \to N$ a differentiable map. Show that the associated tangent map

$$T(\varphi) : TM \longrightarrow TN$$

is continuous.

Public Solution Coverage for Worksheet 1

At the frozen revision boundary, the source provides no public solution for any of the 17 exercises on Worksheet 1. The frozen solution-candidate query and exercise map record negative results for Exercises 1.1-1.17.

No new solutions have been created for this edition. This section merely documents the scope of the source so that the absence of solutions is not mistaken for content lost during translation.

Lecture 2: Sections, the Hairy Ball Theorem, and Gluing Data

Sections

Definition 2.1: continuous section

Let X and Y be topological spaces, and let

$$p : Y \longrightarrow X$$

be a continuous map. A *continuous section* of p is a continuous map

$$s : X \longrightarrow Y$$

such that

$$p \circ s = \mathrm{Id}_X \,.$$

For example, we may think of Y as a vector bundle over X. A section can exist only if p is surjective, as is always the case for a vector bundle. A section is sometimes identified with its image; this causes no difficulty, since every section is injective. The *zero section* plays a special role: to each base point P, it assigns the zero vector in the vector space V_P. Sections of tangent bundles have a name of their own.

Definition 2.2: vector field

Let M be a differentiable manifold. A map

$$F : M \longrightarrow TM$$

satisfying

$$F(P) \in T_P M$$

for every point $P \in M$ is called a (time-independent) *vector field*.

The hairy ball theorem

> **Source illustration - `Hairy_ball_one_pole.jpg`.** A continuous vector field on the 2-sphere must have at least one zero.

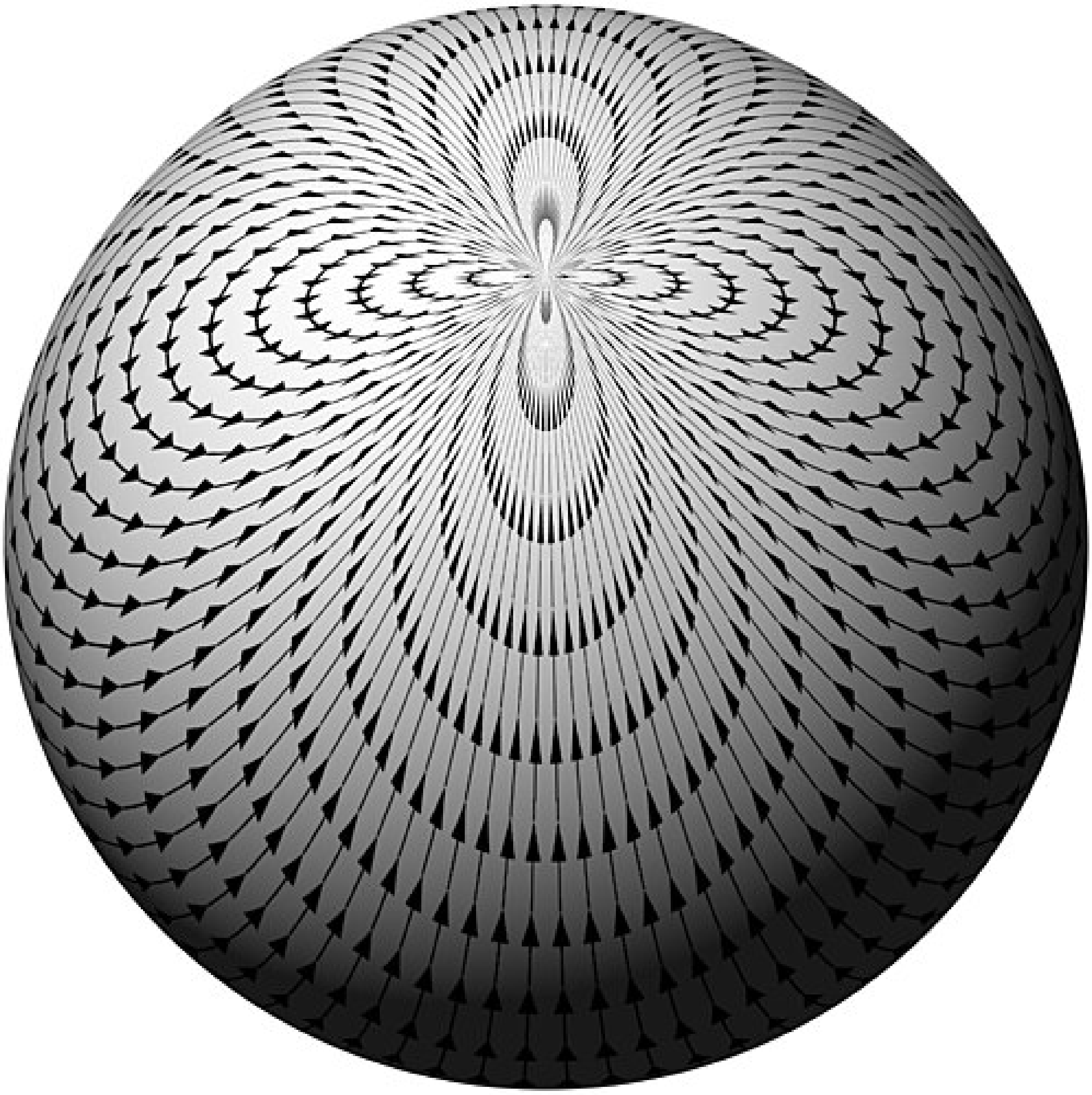

Figure 2: A hairy ball with a single whorl at a pole, illustrating that a continuous tangent vector field on a sphere must have a zero

Theorem 2.3: the hairy ball theorem

On the 2-sphere, every continuous vector field

$$f : S^2 \longrightarrow TS^2$$

has at least one zero.

In particular, the tangent bundle of the 2-sphere is not trivial. There are various interpretations of this theorem. For example, it says that there is always a point on the Earth's surface where there is no wind, when the instantaneous horizontal wind is regarded as a continuous vector field. Similarly, it is impossible to lay all the spines of a hedgehog flat against its body.

Remark 2.4: application to Example 1.2

The hairy ball theorem explains why the vector bundle L from Example 1.2, over

$$\mathbb{R}^3 \setminus \{(0,0,0)\},$$

has no continuous trivialisation. First,

$$S^2 \subset \mathbb{R}^3 \setminus \{(0,0,0)\},$$

so we can restrict L to S^2. If L itself were trivial, this restriction would also be trivial. But the restriction of L to the unit sphere is the tangent bundle of the unit sphere. Indeed, the condition

$$ru + sv + tw = 0$$

can be interpreted as an orthogonality relation, and the extrinsic tangent space at a point (r, s, t) of the sphere is determined by this relation. If the tangent bundle were trivial, there would be two continuous vector fields u and v forming a basis of the tangent space at every point of the sphere. The hairy ball theorem, however, says that even a single vector field must have a zero, and 0 cannot belong to a basis.

Gluing data for topological spaces

A vector bundle $V \to X$ is "assembled" from the trivial vector bundles $V|_{U_i} \to U_i$ for an open cover of X. The precise way these pieces are assembled determines the vector bundle, and can be described conveniently by gluing data. We first need gluing data for topological spaces in general.

The underlying question is: what must we know about an open cover

$$X = \bigcup_{i \in I} U_i$$

in order to reconstruct the space X? The short answer is that we need to know the U_i, the pairwise intersections $U_i \cap U_j$ as subsets both of U_i and of U_j, how these two copies are identified, and a compatibility condition on the identifications involving each triple of sets.

> **Source illustration** - `Inclusion-exclusion.svg`**.** Three overlapping sets illustrate the need for a compatibility condition on triple intersections.

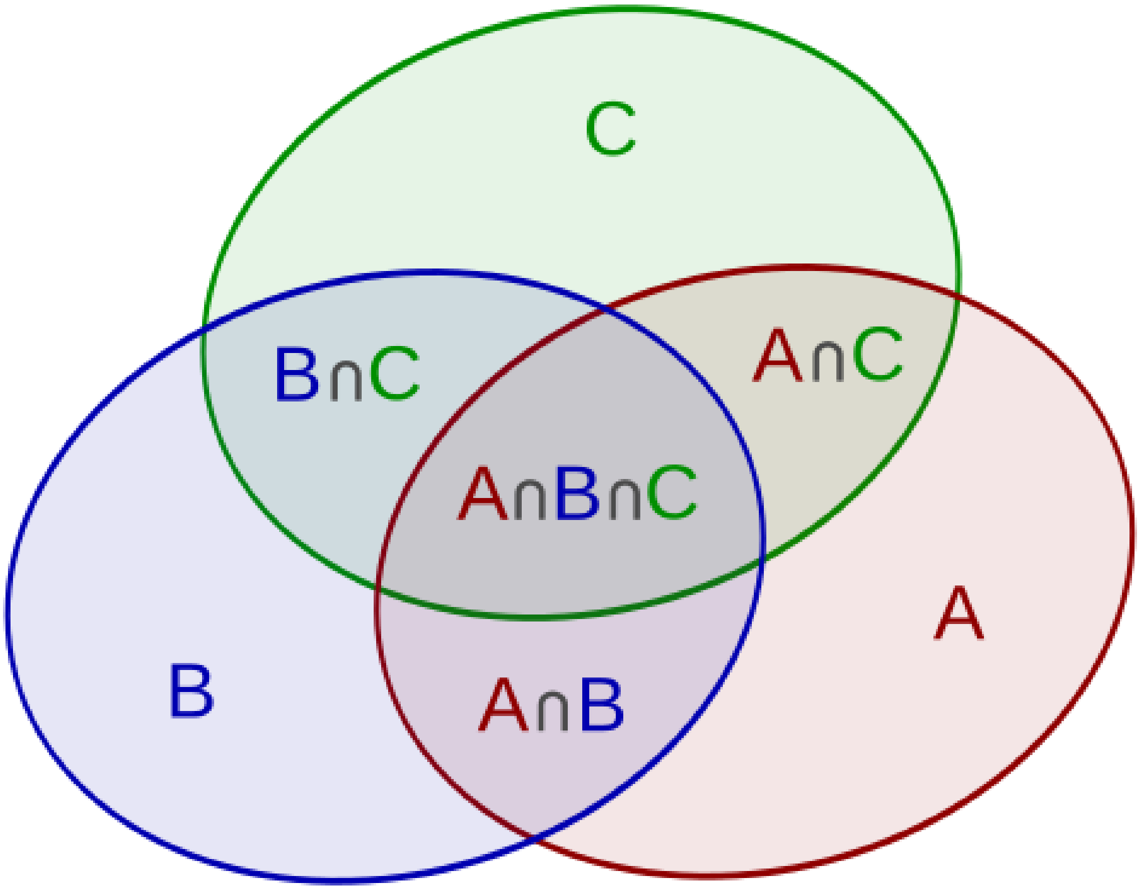

Figure 3: Diagram of three overlapping circular sets, with the regions of intersection distinguished by colour

Definition 2.5: gluing data for topological spaces

Gluing data for topological spaces consist of:

1. a family of topological spaces $(U_i)_{i \in I}$;
2. for each pair (i, j), an open subset
$$U_{ij} \subseteq U_i,$$
with $U_{ii} = U_i$;
3. for each pair (i, j), a homeomorphism
$$\varphi_{ji} : U_{ij} \longrightarrow U_{ji},$$
with $\varphi_{ii} = \mathrm{Id}_{U_i}$;
4. for all $i, j, k \in I$, the *cocycle condition*
$$\varphi_{kj} \circ \varphi_{ji} = \varphi_{ki}$$
holds as an equality of maps from $U_{ik} \cap U_{ij}$ to U_k.

Lemma 2.6: reconstructing a space from gluing data

Suppose gluing data $(U_i)_{i \in I}$ for topological spaces are given. Then there exist a uniquely determined topological space X, an open cover

$$X = \bigcup_{i \in I} V_i,$$

and homeomorphisms

$$\psi_i : U_i \longrightarrow V_i$$

such that

$$\psi_i(U_{ij}) = V_i \cap V_j$$

and

$$\psi_i|_{U_{ij}} = \psi_j|_{U_{ji}} \circ \varphi_{ji}.$$

Proof Let Y be the disjoint union of the U_i. Define an equivalence relation $\sim$ on Y by declaring $x_i \in U_i$ and $x_j \in U_j$ equivalent when

$$x_i \in U_{ij}, \qquad x_j \in U_{ji}, \qquad \varphi_{ji}(x_i) = x_j.$$

The properties of an equivalence relation are ensured by the cocycle condition; see Exercise 2.14. Set

$$X := Y/\sim$$

and equip X with the quotient topology. The composites

$$U_i \longrightarrow Y \longrightarrow X$$

are the maps ψ_i, and the V_i are their images. Thus $\psi_i : U_i \to V_i$ are homeomorphisms. For $x \in U_i$,

$$\psi_i(x) \in V_j$$

if and only if $x \in U_{ij}$, since precisely in this case x is identified with $\varphi_{ji}(x)$. Hence

$$\psi_i(U_{ij}) = V_i \cap V_j.$$

> **Editorial note - order of indices in the proof.** Under the convention in Definition 2.5, $\varphi_{ji} : U_{ij} \to U_{ji}$. In the preceding sentence, the source prints $\varphi_{ij}(x)$ for $x \in U_{ij}$, although the correctly typed map is $\varphi_{ji}(x)$. This edition displays the indices consistent with the domain and codomain in the proof and preserves the source form in this note.

Commutativity of the diagram

$$\begin{array}{ccc} U_{ij} & \xrightarrow{\varphi_{ji}} & U_{ji} \\ & \searrow \psi_i & \downarrow \psi_j \\ & & V_i \cap V_j \end{array}$$

follows in the same way. □

Lemma 2.7: gluing continuous maps

Suppose gluing data $(U_i)_{i\in I}$ for topological spaces are given. Let Z be another topological space, and suppose continuous maps

$$\theta_i : U_i \longrightarrow Z$$

are given satisfying

$$\theta_i|_{U_{ij}} = (\theta_j|_{U_{ji}}) \circ \varphi_{ji}.$$

Then there is a unique continuous map

$$\theta : X \longrightarrow Z$$

such that

$$\theta|_{V_i} \circ \psi_i = \theta_i,$$

where X is the topological space determined by the gluing data as in Lemma 2.6, whose notation we also use.

> **Editorial note - ill-typed composition identity.** The source prints $(\psi_i)^{-1} \circ \theta|_{V_i} = \theta_i$, but this composition is not defined: $\theta|_{V_i}$ takes values in Z, whereas $(\psi_i)^{-1}$ has domain V_i. This edition displays the correctly typed identity $\theta|_{V_i} \circ \psi_i = \theta_i$ in the lemma and preserves the source form in this note.

Proof See Exercise 2.18.

Gluing data for vector bundles

Definition 2.8: gluing data for real vector bundles

Gluing data for a real vector bundle of rank r over a topological space X consist of:

1. an open cover

$$X = \bigcup_{i\in I} U_i;$$

2. a family of real vector bundles of rank r,

$$(E_i \longrightarrow U_i)_{i\in I};$$

3. for each pair (i, j), an isomorphism of vector bundles

$$\varphi_{ji} : E_i|_{U_i\cap U_j} \longrightarrow E_j|_{U_i\cap U_j}$$

   over $U_i \cap U_j$;

4. for all $i, j, k \in I$, the cocycle condition

$$\varphi_{kj} \circ \varphi_{ji} = \varphi_{ki}$$

   holds as an equality of maps from $E_i|_{U_i\cap U_j\cap U_k}$ to $E_k|_{U_i\cap U_j\cap U_k}$.

Remark 2.9: matrix description

Typically, the vector bundles in item (2) of Definition 2.8 are trivial bundles over U_i, namely

$$E_i = \mathbb{R}^r \times U_i.$$

The isomorphisms in item (3) are then simply bijective linear maps

$$\varphi_{ji} : \mathbb{R}^r \longrightarrow \mathbb{R}^r$$

depending continuously on the base point in $U_i \cap U_j$. They can be described compactly as continuous maps

$$\varphi_{ji} : U_i \cap U_j \longrightarrow \mathrm{GL}_r(\mathbb{R})$$

into the general linear group. Thus an invertible $r \times r$ matrix is assigned continuously to each base point; continuity means that every matrix entry is a continuous function. This is called a *matrix description* of the bundle. The cocycle condition still applies.

Lemma 2.10: gluing vector bundles

Suppose gluing data $(E_i)_{i\in I}$ over a topological space

$$X = \bigcup_{i\in I} U_i.$$

are given. Then there exist a uniquely determined real vector bundle $E \to X$ and isomorphisms

$$\psi_i : E_i \longrightarrow E|_{U_i}$$

such that

$$\psi_i|_{E_i|_{U_i\cap U_j}} = \psi_j|_{E_j|_{U_i\cap U_j}} \circ \varphi_{ji}.$$

Proof The existence of a topological space E with these properties follows from Lemma 2.6. The open sets to be glued are

$$W_{ij} := E_i|_{U_i\cap U_j},$$

and the existence of a continuous map to X follows from Lemma 2.7. Every fibre E_x has a well-defined vector space structure inherited from E_i for any open neighbourhood $x \in U_i$. Independence of the choice of i follows because, for $x \in U_i \cap U_j$, the hypothesis supplies an isomorphism of vector bundles

$$\varphi_{ji} : E_i|_{U_i\cap U_j} \longrightarrow E_j|_{U_i\cap U_j},$$

inducing a vector space isomorphism

$$(E_i)_x \longrightarrow (E_j)_x.$$

Editorial note - order of indices in the fibre isomorphism. Definition 2.8 specifies $\varphi_{ji} : E_i|_{U_i\cap U_j} \to E_j|_{U_i\cap U_j}$. The source prints φ_{ij} in the sentence of the proof whose map goes from E_i to E_j. This edition displays φ_{ji}, consistent with the domain and codomain, in the proof and preserves the source form in this note.

□

Source illustration - `Fiddler_crab_mobius_strip.gif`. A Möbius strip, arising by reversing the fibre on one overlap component when two local trivialisations are glued.

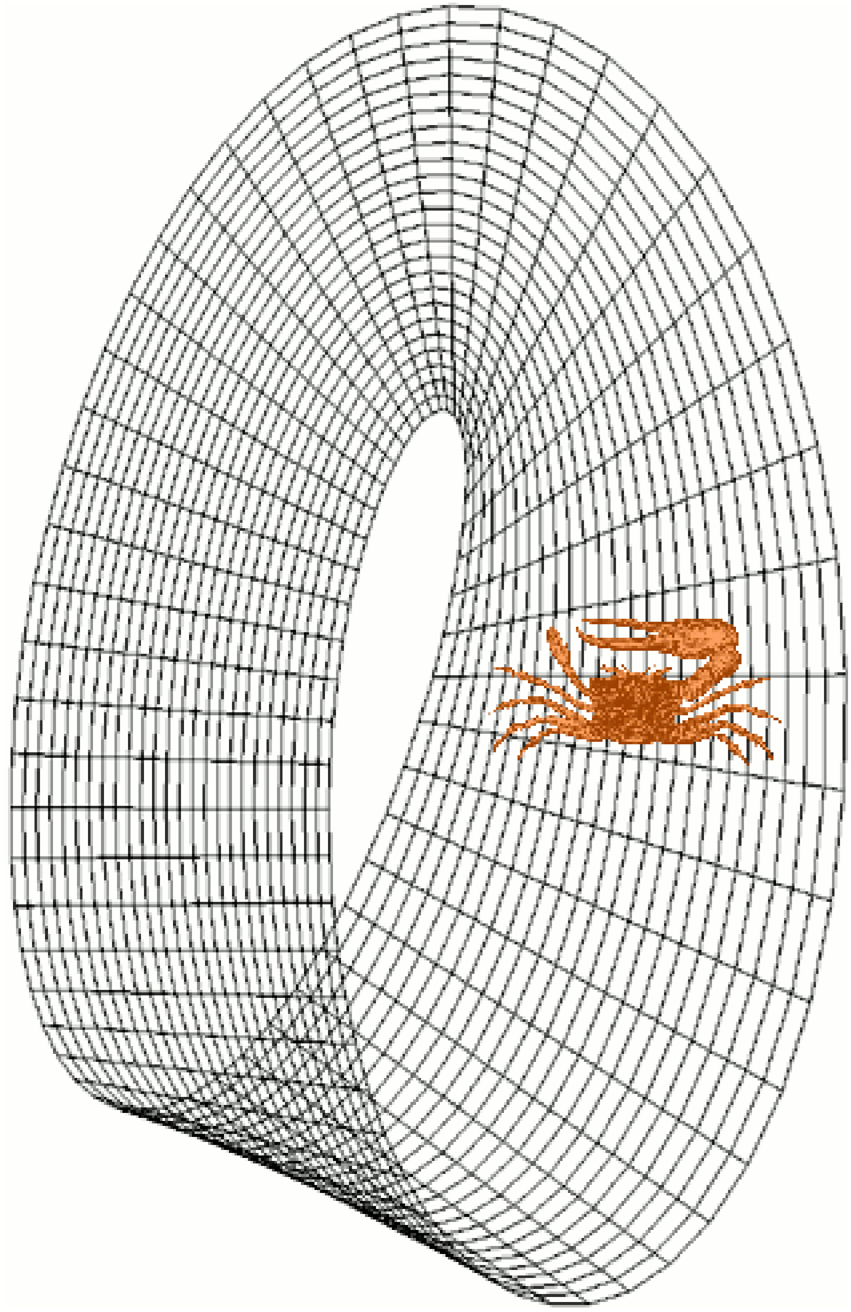

Figure 4: Animation of a crab making one circuit of a Möbius strip and returning with reversed orientation

Example 2.11: the Möbius strip from gluing data

On the one-dimensional sphere

$$S^1 = \left\{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\right\},$$

consider the open cover

$$S^1 = U \cup V,$$

with

$$U = S^1 \setminus \{(0,1)\}, \qquad V = S^1 \setminus \{(0,-1)\}.$$

We shall describe gluing data for a real vector bundle of rank 1. Both open sets are homeomorphic to the real line. Their intersection is

$$\begin{aligned} U \cap V &= S^1 \setminus \{(0,1),(0,-1)\} \\ &= \left\{(x,y) \in S^1 \mid x \neq 0\right\}. \end{aligned}$$

This set is not connected, but is homeomorphic to two disjoint open real half-lines (or, equivalently, two real lines). Set

$$L = U \times \mathbb{R}, \qquad M = V \times \mathbb{R}.$$

Define an isomorphism

$$\varphi : L|_{U\cap V} \longrightarrow M|_{U\cap V}$$

by

$$\varphi(x,y,t) := \begin{cases} (x,y,t), & x > 0, \\ (x,y,-t), & x < 0. \end{cases}$$

The map φ is continuous because the two formulae apply on disjoint open sets. On one half, the fibre is mapped identically; on the other, it is reversed. In the sense of Remark 2.9, the continuous matrix description, constant on each component,

$$\psi(x,y) := \begin{cases} (1), & x > 0, \\ (-1), & x < 0 \end{cases}$$

holds on $U \cap V$. Since there are only two open sets, the cocycle condition is automatically satisfied. By Lemma 2.10, these gluing data determine a real vector bundle of rank 1 on the sphere, called the *Möbius strip*.

Example 2.12: an algebraic realisation of the Möbius strip

We give a direct algebraic realisation of the Möbius strip in $\mathbb{R}^4$. Consider

$$Y := \left\{(x,y,z,w) \in \mathbb{R}^4 \;\middle|\; x^2+y^2 = 1,\ (1-y)z = xw,\ xz = (1+y)w\right\}$$

together with its natural projection to the one-dimensional sphere

$$S^1 = \left\{(x,y) \in \mathbb{R}^2 \mid x^2+y^2 = 1\right\} = U \cup V,$$

with

$$U = S^1 \setminus \{(0,1)\}, \qquad V = S^1 \setminus \{(0,-1)\}.$$

We claim that Y is a vector bundle of rank 1 isomorphic to the Möbius strip. On U, we have $y \neq 1$, so the second equation can be solved for z:

$$z = \frac{x}{1-y} w.$$

The third equation is then automatically satisfied, since

$$xz = \frac{x}{1-y}xw = \frac{x^2}{1-y}w = \frac{1-y^2}{1-y}w = (1+y)w.$$

Similarly, on V we have

$$w = \frac{x}{1+y}z,$$

and the other equation is automatically satisfied. Thus, over U and V, Y is a trivial vector bundle of rank 1, with fibre variables w and z, respectively. Its transition map on $U \cap V$ is given by

$$\frac{x}{1-y} = \frac{1+y}{x},$$

so one matrix description of this bundle is

$$\left(\frac{x}{1-y}\right).$$

Unlike the constant matrix in Example 2.11, this matrix depends explicitly on $(x, y) \in U \cap V$. Nevertheless, the two bundles are isomorphic. Using Exercise 2.21, take the nowhere-zero continuous functions $\sqrt{1-y}$ on U and $\sqrt{1+y}$ on V. We obtain

> **Editorial note - undefined variable in the source.** The source prints $\sqrt{1-t}$ on U and $\sqrt{1+t}$ on V, but there is no variable t in this example. The calculation immediately following uses y and uniquely determines the intended corrections as $\sqrt{1-y}$ and $\sqrt{1+y}$. This edition displays these correctly typed expressions in the text and discloses the normalisation here.

$$\begin{aligned}
\frac{1}{\sqrt{1+y}} \cdot \frac{x}{1-y} \cdot \sqrt{1-y} &= \frac{1}{\sqrt{1+y}} \cdot \frac{x}{\sqrt{1-y}} \\
&= \frac{x}{\sqrt{1-y^2}} \\
&= \frac{x}{\sqrt{x^2}} \\
&= \frac{x}{|x|} \\
&= \pm 1,
\end{aligned}$$

depending on the sign of x. Hence the two bundles are isomorphic.

Worksheet 2: Sections and Gluing Data

Exercise 2.1

Show that a real line bundle $L \to X$ over a topological space X is trivial if and only if it has a nowhere-zero continuous section.

Exercise 2.2

Let $s : X \to V$ be a continuous section of a real vector bundle $p : V \to X$ over a topological space X. Show that the image

$$s(X) \subseteq V$$

is a closed subset homeomorphic to X.

Exercise 2.3

Let $p : V \to U$ be a vector bundle over an open set $U \subseteq \mathbb{R}^m$. Suppose that it is given by a system of linear equations in n variables with m universal parameters, and that U is characterised by the condition that the fibre dimension is constant. Show that a continuous section of V is the same as a continuous universal solution of the system of linear equations.

Exercise 2.4

Give a continuous vector field on S^2 with exactly one zero.

Exercise 2.5

Show that the restriction of the vector bundle from Example 1.2 to the open set

$$D(r) \cup D(s) = \{(r, s, t) \mid r \neq 0 \text{ or } s \neq 0\} \subset \mathbb{R}^3$$

has a trivialisation.

Exercise 2.6

Let

$$X = \bigcup_{i \in I} U_i$$

be an open cover of a topological space X. In what sense does this cover provide topological gluing data? Can the topological space X be reconstructed from these gluing data?

Exercise 2.7

Clarify precisely what constitutes gluing data with two open sets.

Exercise 2.8

Clarify precisely what constitutes gluing data with three open sets.

Exercise 2.9

Let U and V each be a real line. Glue them along the open half-lines

$$\mathbb{R}_+ \subseteq U \qquad \text{and} \qquad \mathbb{R}_+ \subseteq V$$

using the identity map. Is the resulting space Hausdorff?

Exercise 2.10

Let U and V each be a real line. Glue them to one another along the punctured lines

$$\mathbb{R} \setminus \{0\} \subseteq U \qquad \text{and} \qquad \mathbb{R} \setminus \{0\} \subseteq V$$

using the identity map. Is the resulting space Hausdorff?

Exercise 2.11

Consider the topological space

$$X = \mathbb{R} \cup \{0'\},$$

obtained from the real numbers by adjoining a new element $0'$. Declare the following two types of sets to be open:

- open sets $U \subseteq \mathbb{R}$ with $0 \notin U$;
- sets $V \cup \{0'\}$ for open $V \subseteq \mathbb{R}$ with $0 \in V$.

Show that these specifications give X a topology. Is the point 0 closed in this space? Is the point $0'$ closed? How is this space related to the one constructed in Exercise 2.10?

Exercise 2.12

Let U and V each be a real line. Glue them along the punctured lines

$$\mathbb{R} \setminus \{0\} \subseteq U \qquad \text{and} \qquad \mathbb{R} \setminus \{0\} \subseteq V$$

using the inversion map $t \mapsto t^{-1}$. Which topological space results?

Exercise 2.13

Let U and V each be a complex line $\mathbb{C}$ (that is, the complex plane). Glue them along the punctured planes

$$\mathbb{C} \setminus \{0\} \subseteq U \qquad \text{and} \qquad \mathbb{C} \setminus \{0\} \subseteq V$$

using the inversion map $z \mapsto z^{-1}$. Which topological space results?

Exercise 2.14

Show that the relation defined in the proof of Lemma 2.6 is indeed an equivalence relation.

Exercise 2.15

Suppose gluing data $(U_i)_{i \in I}$ are given with

$$U_{ij} = \emptyset$$

for all $i \neq j$. Which topological space do these gluing data determine?

Exercise 2.16

Consider two cylinders

$$U = V = S^1 \times (0, 3).$$

On the open subsets $S^1 \times (0, 1)$ and $S^1 \times (2, 3)$, in U and V respectively, consider the homeomorphisms formed from the identity or central reflection (the antipodal map) on the circle, together with the identity or reflection about the midpoint of the interval. Which geometric objects result from these various gluing data?

> **Editorial note - interval coordinates.** The source speaks of the identity between the intervals $(0, 1)$ and $(2, 3)$, although the literal identity does not map one onto the other. Interpret this using their translated interval coordinates: the two interval maps are $t \mapsto t + 2$ and $t \mapsto 3 - t$.

Exercise 2.17

Let M be a topological manifold and

$$\alpha_i : U_i \longrightarrow V_i$$

a family of charts with transition maps

$$\varphi_{ij} = \alpha_j \circ (\alpha_i)^{-1} : V_i \cap \alpha_i(U_i \cap U_j) \longrightarrow V_j \cap \alpha_j(U_i \cap U_j).$$

Show that the manifold M can be reconstructed from the family $(V_i)_{i \in I}$, the subsets $V_{ij} \subseteq V_i$, and the transition maps

$$\varphi_{ij} : V_{ij} \longrightarrow V_{ji}.$$

1. On

   $$N := \biguplus_{i \in I} V_i,$$

   consider the equivalence relation declaring $P \in V_i$ and $Q \in V_j$ equivalent when they are mapped to one another by φ_{ij}.
2. Equip the quotient set $N/\sim$ with a suitable topology.
3. Define charts on $N/\sim$.
4. Show that M and $N/\sim$ are homeomorphic.

Exercise 2.18

Suppose gluing data $(U_i)_{i \in I}$ for topological spaces are given. Let Z be another topological space, and suppose continuous maps

$$\theta_i : U_i \longrightarrow Z$$

are given satisfying

$$\theta_i|_{U_{ij}} = (\theta_j|_{U_{ji}}) \circ \varphi_{ji}.$$

Show that there is a unique continuous map

$$\theta : X \longrightarrow Z$$

with

$$\theta|_{V_i} \circ \psi_i = \theta_i,$$

where X is the topological space determined by the gluing data as in Lemma 2.6, whose notation we also use.

> **Editorial note - ill-typed composition identity.** The source prints $(\psi_i)^{-1} \circ \theta|_{V_i} = \theta_i$, but the composition is not defined. This edition displays the correctly typed identity $\theta|_{V_i} \circ \psi_i = \theta_i$ in the exercise and preserves the source form in this note.

Exercise 2.19

Show that gluing data for a line bundle relative to an open cover

$$X = \bigcup_{i \in I} U_i,$$

are the same as a family of nowhere-zero continuous functions

$$f_{ij} : U_i \cap U_j \longrightarrow \mathbb{R}$$

satisfying, on every $U_i \cap U_j \cap U_k$,

$$f_{ki} = f_{kj}\, f_{ji}.$$

Exercise 2.20

Determine gluing data for the vector bundle from Example 1.2.

Exercise 2.21

Let

$$X = \bigcup_{i \in I} U_i$$

be an open cover of a topological space X. Let

$$\varphi_{ij} : U_i \cap U_j \longrightarrow \mathrm{GL}_r(\mathbb{R})$$

and

$$\psi_{ij} : U_i \cap U_j \longrightarrow \mathrm{GL}_r(\mathbb{R})$$

be matrix descriptions giving rise to vector bundles E and F, respectively. Show that these bundles are isomorphic if and only if there are continuous maps

$$\alpha_i : U_i \longrightarrow \mathrm{GL}_r(\mathbb{R})$$

such that, after restriction to the appropriate domains,

$$\alpha_i \circ \varphi_{ij} \circ \alpha_j^{-1} = \psi_{ij}$$

for all i, j.

Exercise 2.22

Suppose a vector bundle $V \to X$ of rank m over a topological space X is given, relative to an open cover

$$X = \bigcup_{i \in I} U_i,$$

by continuous matrix-valued maps

$$\varphi_{ij} : U_i \cap U_j \longrightarrow \mathrm{GL}_m(\mathbb{R}).$$

Show that a continuous section $s : X \to V$ is the same as a family of continuous maps

$$t_i : U_i \longrightarrow \mathbb{R}^m$$

satisfying

$$t_j = \varphi_{ij} t_i$$

for all i, j.

> **Editorial note - local index convention.** The source uses φ_{ij} in this exercise for the transition from the i-coordinates to the j-coordinates, as the displayed equation states. In Definition 2.8 the same direction is denoted by φ_{ji}. The exercise retains its own index convention; the identity is understood on $U_i \cap U_j$.

Exercise 2.23

Consider the algebraic realisation of the Möbius strip from Example 2.12,

$$Y = \left\{ (x, y, z, w) \in \mathbb{R}^4 \;\middle|\; x^2 + y^2 = 1,\ (1-y)z = xw,\ xz = (1+y)w \right\} \longrightarrow S^1.$$

Show that the image of the continuous map

$$\begin{aligned} \psi : [0, 2\pi] &\longrightarrow \mathbb{R}^4, \\ t &\longmapsto \left(\sin t, \cos t, \cos \frac{t}{2}, \sin \frac{t}{2} \right) \end{aligned}$$

lies in Y, that $p \circ \psi$ is the trigonometric parametrisation of the unit circle, and that the image of ψ never meets the zero section.

> **Editorial note - order of the first two coordinates.** The source statement prints $(\cos t, \sin t, \cos(t/2), \sin(t/2))$. Substitution of $t = 0$ into the two equations defining Y gives $(1, 0, 1, 0) \notin Y$, so the claim that the image “lies in Y” is false as printed. If the first two coordinates are interchanged to $(\sin t, \cos t)$, the half-angle identities ensure that both equations hold. This edition displays the corrected order in the exercise and preserves the source formula in this note.

The following statement can easily be checked with scissors on the closed version of the Möbius strip.

Exercise 2.24

Deduce from Exercise 2.23 that the complement of the zero section in the Möbius strip is path-connected.

Exercise 2.25

Show that the complement of the zero section in a trivial line bundle over a nonempty base is not connected.

> **Editorial note - nonempty base.** The source leaves this hypothesis implicit. Over the empty base the complement is empty and hence connected, so the stated nonempty-base condition is needed.

Exercise 2.26

Take a narrow rectangular strip, twist it through one full turn about its long axis, and glue the two short edges together. Now cut the strip lengthwise along its centre line with scissors. Is the resulting object connected? What happens if the strip is given n half-turns?

Exercise 2.27

Determine the limit of the function

$$\frac{1-y}{x} = \frac{x}{1+y}$$

on the unit circle as $(x, y) \to (0, 1)$. Also consider the trigonometric parametrisation of the unit circle.

Public Solutions for Worksheet 2

At the frozen authority boundary, the source provides exactly one public solution among the 27 exercises on Worksheet 2, namely the solution to Exercise 2.4. The frozen exercise map records negative results for Exercises 2.1-2.3 and 2.5-2.27. No new solutions have been created for this edition.

Source solution to Exercise 2.4

On $\mathbb{R}^2$, consider the continuous vector field F given by

$$F(x,y) = \frac{1}{1+x^2+y^2} e_1.$$

This field is nowhere zero and continuous. We transport it by stereographic projection to

$$\mathbb{R}^2 \cong S^2 \setminus \{N\}$$

and extend it at the north pole by the value

$$0 \in T_N S^2.$$

We claim that this vector field is continuous. Let (P_n) be a sequence on S^2 converging to N. We may immediately assume that $P_n \neq N$ for all n. The image of this sequence in the chart is

$$P_n' = (x_n, y_n).$$

Since P_n converges to the north pole, $\|P_n'\|$ diverges to ∞. Hence the sequence

$$F(P_n') = F(x_n, y_n) = \frac{1}{1+x_n^2+y_n^2} e_1$$

converges to 0.

> **Editorial note - the source formula is undefined at the origin.** The source defines F on all of $\mathbb{R}^2$ by $F(x,y) = (x^2+y^2)^{-1} e_1$, but this expression is undefined at $(0,0)$. Thus the assertion that F is continuous and nowhere zero on the entire plane is false as printed. This edition displays the minimal correction $F(x,y) = (1+x^2+y^2)^{-1} e_1$: this function is continuous and nowhere zero on all of $\mathbb{R}^2$, tends to 0 as $\|(x,y)\| \to \infty$, and, after being pushed forward by inverse stereographic projection, also tends to the zero vector at the north pole (the differential of that inverse projection remains bounded and even decays at infinity). The change therefore repairs the local defect without altering the structure of the source argument; the source form is preserved in this note. In the limiting expression, the source also prints x, y without subscripts; this edition displays x_n, y_n to match the sequence just defined. The source also duplicates the verb *sei* in the sentence introducing (P_n); that typographical repetition is omitted in the translation and recorded here.

Lecture 3: Linear Constructions of Vector Bundles and Presheaves

Linear constructions of vector bundles

There are many constructions for vector spaces, such as the direct sum, tensor product and dual space. We want to introduce corresponding constructions for vector bundles. Fibrewise, they should agree with the constructions of linear algebra, while also taking into account how the fibres depend on the base space. We shall work with gluing data for vector bundles and use the fact that, given two vector bundles on a topological space X, there is always a sufficiently fine open cover of X relative to which both bundles admit trivialisations. In particular, we can reduce to the case where both bundles are given by matrix descriptions. The constructions then take place at the level of matrix operations.

Definition 3.1: direct sum

Let E and F be real vector bundles over a topological space X, with trivialisations

$$\alpha_i : E|_{U_i} \longrightarrow U_i \times \mathbb{R}^m$$

and

$$\beta_i : F|_{U_i} \longrightarrow U_i \times \mathbb{R}^n.$$

The vector bundle obtained from the gluing data

$$G_i = U_i \times \mathbb{R}^m \times \mathbb{R}^n$$

and

$$\varphi_{ij} : G_i|_{U_i \cap U_j} \longrightarrow G_j|_{U_i \cap U_j},$$

with

$$\varphi_{ij}(x, v, w) = (x, \alpha_j(\alpha_i^{-1}(x, v)), \beta_j(\beta_i^{-1}(x, w))),$$

is called the *direct sum* of E and F, denoted by $E \oplus F$.

> **Editorial note - fibre coordinates.** In the source formula above, $\alpha_j(\alpha_i^{-1}(x, v))$ and $\beta_j(\beta_i^{-1}(x, w))$ are themselves pairs containing the base point. In the second and third slots of $\varphi_{ij}(x, v, w)$, only their fibre coordinates are intended; formally, apply pr_2 to each pair. The base coordinate remains x. The tensor, exterior-power and homomorphism constructions below likewise use the induced linear maps on each fibre.

If E is given by the matrix description

$$\varphi_{ij} : U_i \cap U_j \longrightarrow \mathrm{GL}_m(\mathbb{R})$$

and F by

$$\psi_{ij} : U_i \cap U_j \longrightarrow \mathrm{GL}_n(\mathbb{R}),$$

then a matrix description of $E \oplus F$ is obtained by placing the two matrices in the diagonal blocks of an $(m + n) \times (m + n)$ matrix and filling the other blocks with zeros.

Definition 3.2: tensor product

Let E and F be real vector bundles over X, with trivialisations α_i and β_i as above. The vector bundle obtained from the gluing data

$$G_i = U_i \times (\mathbb{R}^m \otimes \mathbb{R}^n)$$

and

$$\varphi_{ij} : G_i|_{U_i \cap U_j} \longrightarrow G_j|_{U_i \cap U_j}, \qquad \varphi_{ij} = (\alpha_j \circ \alpha_i^{-1}) \otimes (\beta_j \circ \beta_i^{-1}),$$

is called the *tensor product* of E and F, denoted by $E \otimes F$. Here the tensor product of the linear maps is taken at each base point.

Given matrix descriptions of the two bundles, a matrix description of their tensor product is obtained by the *Kronecker product*: every entry of one matrix is multiplied by every entry of the other.

Definition 3.3: exterior power

Let E be a real vector bundle of rank m over a topological space X, with trivialisations

$$\alpha_i : E|_{U_i} \longrightarrow U_i \times \mathbb{R}^m,$$

and let $r \in \mathbb{N}$. The vector bundle obtained from the gluing data

$$G_i = U_i \times \bigwedge^r \mathbb{R}^m$$

and

$$\varphi_{ij} : G_i|_{U_i \cap U_j} \longrightarrow G_j|_{U_i \cap U_j}, \qquad \varphi_{ij} = \bigwedge^r (\alpha_j \circ \alpha_i^{-1}),$$

is called the *rth exterior power* of E, denoted by $\bigwedge^r E$. At each base point, the rth exterior power of the corresponding linear map is taken.

Given a matrix description of E, a matrix description of $\bigwedge^r E$ is obtained by assembling all determinants of $r \times r$ submatrices into a matrix.

Definition 3.4: determinant bundle

Let E be a real vector bundle of rank m over a topological space X. The mth exterior power

$$\bigwedge^m E$$

is called the *determinant bundle* of E, denoted by $\det E$.

The determinant bundle is a line bundle. Its matrix description is given by the determinant.

Definition 3.5: homomorphism bundle

Let E and F be real vector bundles over a topological space X, with trivialisations

$$\alpha_i : E|_{U_i} \longrightarrow U_i \times \mathbb{R}^m, \qquad \beta_i : F|_{U_i} \longrightarrow U_i \times \mathbb{R}^n.$$

The vector bundle obtained from the gluing data

$$G_i = U_i \times \mathrm{Hom}_{\mathbb{R}}(\mathbb{R}^m, \mathbb{R}^n)$$

and

$$\varphi_{ij} : G_i|_{U_i \cap U_j} \longrightarrow G_j|_{U_i \cap U_j},$$

with

$$\varphi_{ij}(\theta) = (\beta_j \circ \beta_i^{-1}) \circ \theta \circ (\alpha_i \circ \alpha_j^{-1}),$$

is called the *homomorphism bundle* from E to F, denoted by $\mathrm{Hom}(E, F)$.

Definition 3.6: dual bundle

For a real vector bundle E over a topological space X, the homomorphism bundle

$$\operatorname{Hom}(E, X \times \mathbb{R})$$

is called the *dual bundle* of E, denoted by E^*.

On a manifold, the dual of the tangent bundle is called the *cotangent bundle*.

Presheaves

Definition 3.7: presheaf

Let X be a topological space. A *presheaf* $\mathcal{F}$ on X is an assignment associating a set $\mathcal{F}(U)$ to each open set $U \subseteq X$, and a map

$$\rho_{V,U} : \mathcal{F}(V) \longrightarrow \mathcal{F}(U),$$

to each pair of open sets $U \subseteq V$, subject to the following two conditions.

1. For $U = V$,

$$\rho_{U,U} = \mathrm{Id}_{\mathcal{F}(U)}.$$

2. For open sets $U \subseteq V \subseteq W$,

$$\rho_{W,U} = \rho_{V,U} \circ \rho_{W,V}.$$

The maps $\rho_{V,U}$ are called *restriction maps*. The set $\mathcal{F}(U)$ is also called the value of the presheaf on the open set U.

The following constructions are basic examples of presheaves, and later of sheaves.

Example 3.8: continuous maps

Let X and Z be topological spaces. To each open set $U \subseteq X$, associate the set of continuous maps from U to Z, namely

$$C^0(U, Z) = \{\varphi : U \to Z \mid \varphi \text{ continuous}\}.$$

Every continuous map $\varphi : U \to Z$ can be restricted to an open subset $V \subseteq U$. Moreover, for $U \subseteq V \subseteq W$, restriction from W to U can be performed either in one step or in two. Thus this construction is a presheaf.

The following special case has additional structure, namely that of a ringed space.

Example 3.9: continuous real-valued functions

Let X be a topological space. To each open set $U \subseteq X$, associate the set of continuous real-valued functions on U,

$$\mathcal{C}(U) = C^0(U, \mathbb{R}) = \{f : U \to \mathbb{R} \mid f \text{ continuous}\}.$$

Since every continuous function on U can be restricted to any open subset $V \subseteq U$, this construction is a presheaf.

Example 3.10: differentiable functions

Let X be a differentiable manifold. To each open set $U \subseteq X$, associate the set of differentiable real-valued functions on U,

$$\mathcal{C}(U) = C^1(U, \mathbb{R}) = \{f : U \to \mathbb{R} \mid f \text{ continuously differentiable}\}.$$

Since every differentiable function on U can be restricted to any open subset $V \subseteq U$, this construction is a presheaf.

Example 3.11: constant presheaf

On a topological space X, for a fixed set M, the assignment associating M to every open set $U \subseteq X$ and the identity on M to every inclusion is a presheaf. It is called the *constant presheaf.*

For the next example, think of a vector bundle over the base X.

Example 3.12: the presheaf of continuous sections

Let X and Y be topological spaces, and let

$$p : Y \longrightarrow X$$

be a fixed continuous map. For each open set $U \subseteq X$, this induces a continuous map

$$Y|_U = p^{-1}(U) \longrightarrow U.$$

To U, associate the set of continuous sections of this map over U,

$$S(U, Y) = \{s : U \to p^{-1}(U) \mid s \text{ a continuous section of } p\}.$$

A continuous section can be restricted to any open subset $V \subseteq U$, with its codomain restricted accordingly to $p^{-1}(V)$. Thus this construction is a presheaf.

Because of this important example, an element $s \in \mathcal{F}(U)$ is also called a *section* of the presheaf $\mathcal{F}$ over U. For the restriction of a section to a smaller open set $V \subseteq U$, we also write

$$s|_V = \rho_{U,V}(s).$$

Definition 3.13: subpresheaf

Let $\mathcal{F}$ be a presheaf on a topological space X. A presheaf $\mathcal{G}$ is called a *subpresheaf* of $\mathcal{F}$ if, for every open set $U \subseteq X$,

$$\mathcal{G}(U) \subseteq \mathcal{F}(U),$$

and, for every $U \subseteq V$, the restriction maps are compatible:

$$\rho^{\mathcal{G}}_{V,U} = \rho^{\mathcal{F}}_{V,U}|_{\mathcal{G}(V)}.$$

Editorial note - restriction condition missing from the source. The source definition states only $\mathcal{G}(U) \subseteq \mathcal{F}(U)$ for each U and does not state compatibility of the restriction maps. This edition includes the restriction condition above so that the object defined is indeed a subpresheaf; the shorter source form is preserved in this note.

Since differentiable functions on a manifold are in particular continuous, the presheaf of differentiable functions forms a subsheaf of the presheaf of continuous real-valued functions.

Presheaves with additional structure

Definition 3.14: presheaf of groups

A presheaf $\mathcal{F}$ on a topological space X is called a *presheaf of groups* if $\mathcal{F}(U)$ is a group for every open set $U \subseteq X$ and, for every inclusion $U \subseteq V$, the restriction map

$$\rho_{V,U} : \mathcal{F}(V) \longrightarrow \mathcal{F}(U)$$

is a group homomorphism.

Definition 3.15: presheaf of commutative rings

A presheaf $\mathcal{F}$ on a topological space X is called a *presheaf of commutative rings* if $\mathcal{F}(U)$ is a commutative ring for every open set $U \subseteq X$ and, for every inclusion $U \subseteq V$, the restriction map

$$\rho_{V,U} : \mathcal{F}(V) \longrightarrow \mathcal{F}(U)$$

is a ring homomorphism.

Remark 3.16: presheaves as functors

A presheaf $\mathcal{F}$ on a topological space $(X, \mathcal{T})$ can be viewed as a contravariant functor

$$\mathcal{F} : \mathcal{T} \longrightarrow \text{MEN},$$

where $\mathcal{T}$ is regarded as a category as in Appendix Example 1.11, and MEN denotes the category of sets. Likewise, a presheaf of commutative groups is a contravariant functor to the category of commutative groups, and a presheaf of commutative rings is a contravariant functor to the category of commutative rings, and so on.

Definition 3.17: topological group

A *topological group* is a group G which is also a topological space, such that the group operation

$$G \times G \longrightarrow G, \qquad (g, h) \longmapsto g \circ h,$$

and inversion

$$G \longrightarrow G, \qquad g \longmapsto g^{-1},$$

are continuous maps.

Examples of topological groups are

$$(\mathbb{R}, +), \quad (\mathbb{R} \setminus \{0\}, \cdot), \quad (\mathbb{C}, +), \quad (\mathbb{C} \setminus \{0\}, \cdot), \quad (\mathbb{R}^n, +),$$

the circle S^1 with addition of angles, the general linear groups $\mathrm{GL}_n(\mathbb{R})$ and $\mathrm{GL}_n(\mathbb{C})$, and a complex torus $\mathbb{C}/\Gamma$ for a lattice $\Gamma \subseteq \mathbb{C}$. Every group becomes a topological group when equipped with the discrete topology.

For a topological space X, the set of continuous maps from X to a topological group G is itself a group under the natural operation. Restriction to an open subset is a group homomorphism. Therefore the assignment

$$U \longmapsto C^0(U, G)$$

is a presheaf of groups on X.

Stalks of presheaves

A fundamental idea behind vector bundles and presheaves is to distinguish meaningfully between local and global properties of geometric objects and to understand their interplay. A local property, for example, is one that holds on "small" open sets. We often want to replace small open sets by still smaller ones, particularly to understand behaviour in an arbitrarily small neighbourhood of a point. We introduce the following concepts for this purpose.

Definition 3.18: topological filter

Let X be a topological space. A collection F of open subsets of X is called a *filter* if the following hold for open sets U and V:

1. $X \in F$;
2. if $U \in F$ and $U \subseteq V$, then $V \in F$;
3. if $U \in F$ and $V \in F$, then $U \cap V \in F$.

The most important examples here are neighbourhood filters of points: such a filter consists of all open neighbourhoods of a fixed point.

> **Editorial note - corrupted source sentence.** The source prints "Die wichtigsten Filter sind für und die Umgebungsfilter zu einer Punkt, der aus allen offenen Mengen eines fixierten Punktes besteht." This sentence is grammatically corrupted. Based on the definition just given and the use of filters in the definition of a stalk below, this edition gives a complete contextual reading about neighbourhood filters, without attributing this reconstruction to the source author.

Definition 3.19: directed set

A nonempty ordered set $(I, \preccurlyeq)$ is called *directed* if for every $i, j \in I$ there is a $k \in I$ such that

$$i, j \preccurlyeq k.$$

> **Editorial note - nonempty directed systems.** The source does not explicitly require $I \neq \emptyset$. This standard hypothesis is included here because the later assertion that a directed colimit of groups has a group structure would otherwise fail for the empty system. All neighbourhood-filter systems used here are nonempty.

We regard a topological filter as a set ordered by inclusion. The intersection property of a filter makes it a directed set; the direction convention is $\preccurlyeq = \supseteq$.

Definition 3.20: ordered and directed systems

Let $(I, \preccurlyeq)$ be an ordered index set. A family

$$M_i, \qquad i \in I,$$

is called an *ordered system of sets* if:

1. for $i \preccurlyeq j$ there is a map $\varphi_{ij} : M_i \to M_j$;
2. for $i \preccurlyeq j \preccurlyeq k$ we have $\varphi_{ik} = \varphi_{jk} \circ \varphi_{ij}$.

If the index set is also directed, the family is called a *directed system of sets.*

If all the M_i are groups, respectively rings, and all maps between them are group homomorphisms, respectively ring homomorphisms, we speak of an ordered or directed system of groups, respectively rings.

Definition 3.21: colimit

Let $(M_i)_{i\in I}$ be a directed system of sets. The *colimit*, also called the direct or inductive limit, of the system is

$$\operatorname{colim}_{i\in I} M_i = \left(\biguplus_{i\in I} M_i \right) / \sim .$$

Here $\sim$ is the equivalence relation declaring two elements $m \in M_i$ and $n \in M_j$ equivalent if there is a $k \in I$ with $i, j \preccurlyeq k$ and

$$\varphi_{ik}(m) = \varphi_{jk}(n).$$

In particular, $s_i \in M_i$ is equivalent to its image $\varphi_{ik}(s_i) \in M_k$ for all $i \preccurlyeq k$. If the system is a directed system of groups or rings, the colimit of sets can also be given a group or ring structure. This is because two colimit elements represented by $s_i \in M_i$ and $s_j \in M_j$ can be identified with their images in some M_k with $i, j \preccurlyeq k$, where the operation is then performed. See Exercise 3.13.

Our main example is the directed system determined by a topological filter for a presheaf $\mathcal{F}$ on X, namely

$$\mathcal{F}(U), \qquad U \in F.$$

Definition 3.22: stalk at a point

For a presheaf $\mathcal{F}$ on a topological space X and a point $P \in X$,

$$\mathcal{F}_P := \operatorname{colim}_{P \in U} \Gamma(U, \mathcal{F})$$

is called the *stalk* of the presheaf at P.

In particular, every section $s \in \mathcal{F}(U)$ and every point $P \in U$ determine a unique element

$$s_P \in \mathcal{F}_P,$$

called the *germ* of s at P. The map

$$\mathcal{F}(U) \longrightarrow \mathcal{F}_P, \qquad s \longmapsto s_P,$$

is called a restriction map and is denoted by $\rho_{U,P}$. For $P \in V \subseteq U$, the following diagram commutes:

$$\begin{tikzcd} \mathcal{F}(U) \arrow[r, "\rho_{U,V}"] \arrow[dr, "\rho_{U,P}"'] & \mathcal{F}(V) \arrow[d, "\rho_{V,P}"] \\ & \mathcal{F}_P. \end{tikzcd}$$

The following definition is slightly more general.

Definition 3.23: stalk at a filter

For a presheaf $\mathcal{G}$ on a topological space X and a topological filter F,

$$\mathcal{G}_F := \operatorname{colim}_{U \in F} \Gamma(U, \mathcal{G})$$

is called the *stalk* of the presheaf at the filter F.

Morphisms of presheaves

Definition 3.24: morphism of presheaves

Let $\mathcal{F}$ and $\mathcal{G}$ be presheaves on a topological space X. A *morphism of presheaves*

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

is a family of maps

$$\varphi_U : \mathcal{F}(U) \longrightarrow \mathcal{G}(U)$$

for every open set $U \subseteq X$, such that for every open inclusion $U \subseteq V$ the following diagram commutes:

$$\begin{array}{ccc} \mathcal{F}(V) & \xrightarrow{\varphi_V} & \mathcal{G}(V) \\ \downarrow\rho^{\mathcal{F}}_{V,U} & & \downarrow\rho^{\mathcal{G}}_{V,U} \\ \mathcal{F}(U) & \xrightarrow{\varphi_U} & \mathcal{G}(U). \end{array}$$

Editorial note - reversed source diagram. For $U \subseteq V$, the source diagram places $\mathcal{F}(U)$ and $\mathcal{G}(U)$ in the top row, $\mathcal{F}(V)$ and $\mathcal{G}(V)$ in the bottom row, and labels the downward vertical arrows $\rho_{U,V}$. Presheaves are contravariant, so restriction maps instead go from the value on V to the value on U. This edition displays the correctly typed diagram above, with superscripts distinguishing the two presheaves; the source layout is preserved in this note.

Definition 3.25: isomorphism of presheaves

A morphism of presheaves $\varphi : \mathcal{F} \to \mathcal{G}$ on X is called an *isomorphism* if, for every open subset $U \subseteq X$, the map

$$\varphi_U : \mathcal{F}(U) \longrightarrow \mathcal{G}(U)$$

is a bijection.

Lemma 3.26: identity, composition and inclusion

Let X be a topological space and $\mathcal{F}, \mathcal{G}, \mathcal{H}$ presheaves on X. The following statements hold.

1. The identity $\mathcal{F} \to \mathcal{F}$ is a morphism of presheaves.
2. If $\varphi : \mathcal{F} \to \mathcal{G}$ and $\psi : \mathcal{G} \to \mathcal{H}$ are morphisms of presheaves, then $\psi \circ \varphi$ is also a morphism of presheaves.
3. For a subpresheaf $\mathcal{F} \subseteq \mathcal{G}$, the natural inclusion is a morphism of presheaves.

Editorial note - source typographical error. In the third item, the source and Exercise 3.17 print *Prägraben*, an evident typographical error for *Prägarben* (presheaves). This edition uses the mathematically correct term.

Proof See Exercise 3.17.

Lemma 3.27: induced maps on stalks

A morphism of presheaves

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

on a topological space X defines, for every point $P \in X$, a map between stalks

$$\varphi_P : \mathcal{F}_P \longrightarrow \mathcal{G}_P$$

compatible with the restriction maps. That is, for $P \in U$, the diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\varphi_U} & \mathcal{G}(U) \\ \downarrow\rho_{U,P} & & \downarrow\rho_{U,P} \\ \mathcal{F}_P & \xrightarrow{\varphi_P} & \mathcal{G}_P \end{array}$$

commutes.

Proof Let $s_P \in \mathcal{F}_P$. This means that there are an open neighbourhood $P \in U \subseteq X$ and an $s \in \mathcal{F}(U)$ with $\rho_{U,P}(s) = s_P$. Set

$$\varphi_P(s_P) := \rho_{U,P}(\varphi_U(s)).$$

We must show that this definition is independent of the representative s and of U. Let $t \in \mathcal{F}(V)$ be another representative. Since $s_P = t_P$, there is an open neighbourhood

$$P \in W \subseteq U \cap V$$

such that $s|_W = t|_W$. Then

$$\varphi_U(s)|_W = \varphi_W(s|_W) = \varphi_W(t|_W) = \varphi_V(t)|_W,$$

and hence

$$\rho_{U,P}(\varphi_U(s)) = \rho_{V,P}(\varphi_V(t)).$$

Thus the map φ_P is well-defined and the diagram above commutes.

Worksheet 3: Linear Constructions, Presheaves and Stalks

The Kronecker product of the matrices

$$A = (a_{ij})_{1 \le i \le m,\, 1 \le j \le n}$$

and

$$B = (b_{k\ell})_{1 \le k \le p,\, 1 \le \ell \le r}$$

is the matrix

$$(a_{ij} b_{k\ell})_{1 \le i \le m,\, 1 \le k \le p;\, 1 \le j \le n,\, 1 \le \ell \le r}.$$

Exercise 3.1

Compute the Kronecker product of the two matrices

$$\begin{pmatrix} 3 & -4 \\ 5 & -2 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} -2 & 7 \\ 6 & 3 \end{pmatrix}.$$

Exercise 3.2

Let K be a field, and let

$$A = (a_{ij})_{1 \le i \le m,\, 1 \le j \le n}, \qquad B = (b_{k\ell})_{1 \le k \le p,\, 1 \le \ell \le r}$$

be matrices with associated linear maps

$$A : K^n \longrightarrow K^m, \qquad B : K^r \longrightarrow K^p.$$

Show that the tensor product of these linear maps, relative to the basis

$$(e_j \otimes e_\ell)_{1\leq j\leq n,\, 1\leq \ell\leq r}$$

of $K^n \otimes K^r$ and the basis

$$(e_i \otimes e_k)_{1\leq i\leq m,\, 1\leq k\leq p}$$

of $K^m \otimes K^p$, is described by the Kronecker product of A and B.

Exercise 3.3

Show that the tensor product of the Möbius strip with itself is a trivial line bundle.

Exercise 3.4

Let $\mathcal{F}$ and $\mathcal{G}$ be presheaves on a topological space X. Show that the assignment

$$U \longmapsto \mathcal{F}(U) \times \mathcal{G}(U),$$

together with the natural product maps as restriction maps, defines a presheaf on X.

Exercise 3.5

Let I be an index set and $(\mathcal{F}_i)_{i\in I}$ a family of presheaves on a topological space X. Show that the assignment

$$U \longmapsto \prod_{i\in I} \mathcal{F}_i(U),$$

together with the natural product maps as restriction maps, defines a presheaf on X.

Exercise 3.6

Interpret Example 3.8 in the framework of Example 3.12.

Exercise 3.7

Let

$$p : V \longrightarrow X$$

be a real vector bundle of rank m on a topological space X. Show that for every open set $U \subseteq X$ on which V is trivial, the corresponding presheaf of continuous sections is isomorphic to

$$C^0(U, \mathbb{R})^m.$$

Explain the sense in which this isomorphism is meant.

Exercise 3.8

Show that the groups

$$(\mathbb{R}, +), \quad (\mathbb{R}\setminus\{0\}, \cdot), \quad (\mathbb{C}, +), \quad (\mathbb{C}\setminus\{0\}, \cdot), \quad (\mathbb{R}^n, +),$$

the circle S^1 with addition of angles, and the general linear groups $\mathrm{GL}_n(\mathbb{R})$ and $\mathrm{GL}_n(\mathbb{C})$ are topological groups.

Exercise 3.9

Let G be a topological group and $H \subseteq G$ a subgroup. Show that, on every topological space X, the presheaf $C^0(-,H)$ is a subpresheaf of $C^0(-,G)$.

A differentiable manifold G which is also a group, and for which inversion and the group operation are differentiable maps, is called a *real Lie group*.

Exercise 3.10

Show that the groups

$$(\mathbb{R},+), \quad (\mathbb{R}\setminus\{0\},\cdot), \quad (\mathbb{C},+), \quad (\mathbb{C}\setminus\{0\},\cdot), \quad (\mathbb{R}^n,+),$$

the circle S^1 with addition of angles, and $\mathrm{GL}_n(\mathbb{R})$ and $\mathrm{GL}_n(\mathbb{C})$ are Lie groups.

Exercise 3.11

Show that the tangent bundle of a Lie group is trivial.

> **Hint.** Show that the tangent space at the identity element can be transported naturally to the other tangent spaces.

Exercise 3.12

Let I be a directed index set and $(M_i)_{i\in I}$ a directed system of sets, with system maps $\varphi_{ij} : M_i \to M_j$. Let N be another set, and suppose that for every $i \in I$ a map

$$\psi_i : M_i \longrightarrow N$$

is given such that

$$\psi_i = \psi_j \circ \varphi_{ij}$$

for all $i \preccurlyeq j$. Prove the universal property of the colimit: there is a unique map

$$\psi : \mathrm{colim}_{i\in I} M_i \longrightarrow N$$

such that

$$\psi_i = \psi \circ j_i,$$

where $j_i : M_i \to \mathrm{colim}_{i\in I} M_i$ are the natural maps.

Show also that if (M_i) is a directed system of groups, N is a group, and all the ψ_i are group homomorphisms, then ψ is a group homomorphism.

Exercise 3.13

Let I be a directed index set and $(G_i)_{i\in I}$ a directed system of commutative groups. Show that its colimit is a commutative group.

Exercise 3.14

Let R be a commutative ring and $S \subseteq R$ a multiplicative system. Consider the following partial order on S: write $f \preccurlyeq g$ if f divides a power of g, identifying two elements if this relation holds in both directions.

Show that the commutative rings

$$R_f, \qquad f \in S,$$

form a directed system, and that

$$\operatorname{colim}_{f \in S} R_f = R_S.$$

Exercise 3.15

Let M be a differentiable manifold and $P \in M$. Show that the stalk at P of the presheaf of continuous sections of the tangent bundle $TM \to M$ depends only on the dimension of the manifold at P.

> **Editorial note - clarification of the object whose stalk is taken.** The source asks for a statement about the "stalk of the tangent bundle". Stalks belong to presheaves, not directly to bundles. The context of Lecture 3 and Example 3.12 identifies the intended object as the presheaf of continuous sections of the tangent bundle. This edition states that object explicitly and preserves the source's shorthand in this note.

Exercise 3.16

Let $\mathcal{F}$ and $\mathcal{G}$ be presheaves on a topological space X, and let $\mathcal{F} \times \mathcal{G}$ be their product presheaf. Show that for every point $P \in X$,

$$(\mathcal{F} \times \mathcal{G})_P = \mathcal{F}_P \times \mathcal{G}_P.$$

Exercise 3.17

Let X be a topological space and $\mathcal{F}, \mathcal{G}, \mathcal{H}$ presheaves on X. Prove the following statements.

1. The identity $\mathcal{F} \to \mathcal{F}$ is a morphism of presheaves.
2. If $\varphi : \mathcal{F} \to \mathcal{G}$ and $\psi : \mathcal{G} \to \mathcal{H}$ are morphisms of presheaves, then $\psi \circ \varphi$ is also a morphism of presheaves.
3. If $\mathcal{F} \subseteq \mathcal{G}$ is a subpresheaf, the natural inclusion is a morphism of presheaves.

> **Editorial note - source typographical error.** In the third item, the source prints *Prägraben*, an evident typographical error for *Prägarben* (presheaves). The correct form is used above; the same defect is also recorded in Lemma 3.26.

Exercise 3.18

Let I be an index set, $(\mathcal{F}_i)_{i \in I}$ a family of presheaves on a topological space X, and $\prod_{i \in I} \mathcal{F}_i$ their product presheaf. Let $\mathcal{G}$ be another presheaf on X. Show that a morphism of presheaves

$$\psi : \mathcal{G} \longrightarrow \prod_{i \in I} \mathcal{F}_i$$

is the same as a family of morphisms of presheaves

$$\psi_i : \mathcal{G} \longrightarrow \mathcal{F}_i, \qquad i \in I.$$

Public Solutions for Worksheet 3

At the frozen authority boundary, the source provides exactly one public solution among the 18 exercises on Worksheet 3, namely the solution to Exercise 3.1. The frozen exercise map and candidate evidence record negative results for Exercises 3.2-3.18. No new solutions have been created for this edition.

Source solution to Exercise 3.1

The Kronecker product of the two matrices is

$$\begin{pmatrix} 3 & -4 \\ 5 & -2 \end{pmatrix} \otimes \begin{pmatrix} -2 & 7 \\ 6 & 3 \end{pmatrix} = \begin{pmatrix} 3\begin{pmatrix} -2 & 7 \\ 6 & 3 \end{pmatrix} & -4\begin{pmatrix} -2 & 7 \\ 6 & 3 \end{pmatrix} \\ 5\begin{pmatrix} -2 & 7 \\ 6 & 3 \end{pmatrix} & -2\begin{pmatrix} -2 & 7 \\ 6 & 3 \end{pmatrix} \end{pmatrix}$$

$$= \begin{pmatrix} -6 & 21 & 8 & -28 \\ 18 & 9 & -24 & -12 \\ -10 & 35 & 4 & -14 \\ 30 & 15 & -12 & -6 \end{pmatrix}.$$

Lecture 4: Sheaves and Sheaf Morphisms

Figure 5: Sheaves of spelt wheat standing upright in a field

Sheaves of spelt wheat. André Karwath aka Aka, CC BY-SA 2.5; see the Unit 4 media credits.

Sheaves

Definition 4.1: sheaf

Let X be a topological space. A *sheaf* $\mathcal{F}$ on X is a presheaf $\mathcal{F}$ on X satisfying the following two properties.

1. For every open cover

   $$U = \bigcup_{i \in I} U_i$$

   and every $s, t \in \mathcal{F}(U)$ with

   $$\rho_{U,U_i}(s) = \rho_{U,U_i}(t)$$

   for all $i \in I$, we have $s = t$.

2. For every open cover

   $$U = \bigcup_{i \in I} U_i$$

   and every compatible family $s_i \in \mathcal{F}(U_i)$, meaning that

   $$\rho_{U_i,U_i \cap U_j}(s_i) = \rho_{U_j,U_i \cap U_j}(s_j)$$

   for all $i, j \in I$, there exists an $s \in \mathcal{F}(U)$ with

   $$s_i = \rho_{U,U_i}(s)$$

   for all $i \in I$.

These two properties are called the *Serre conditions.* The first says that equality of sections can be checked locally on an open cover. The second says that compatible local sections come from a global section. That global section is unique by the first condition.

The set $\mathcal{F}(\emptyset)$ has exactly one element. Set-theoretically, this follows by applying the two conditions to the cover of the empty set indexed by the empty set.

As a representative of many similar examples, we show that the presheaf of sections of a continuous map is a sheaf.

Example 4.2: the sheaf of continuous sections

We continue Example 3.12. Let X and Y be topological spaces and

$$p : Y \longrightarrow X$$

a fixed continuous map. The presheaf of continuous sections in Y is given by

$$U \longmapsto S(U, Y) = \{s : U \to p^{-1}(U) \mid s \text{ a continuous section of } p\}.$$

This presheaf is a sheaf. The first Serre condition holds because two sections are equal when their values agree at every point $P \in U$, and this equality can be checked locally on an open cover. For the second condition, a compatible family of continuous sections

$$s_i : U_i \longrightarrow Y|_{U_i}$$

directly defines a section

$$s : U \longrightarrow Y|_U$$

extending all the s_i simultaneously. The map s is continuous because continuity can be checked locally.

Example 4.3: the sheaf of continuous group-valued maps

Let G be a topological group and X a topological space. The assignment

$$U \longmapsto C^0(U, G)$$

is a sheaf: the sheaf of groups of continuous maps with values in G. The sheaf properties follow from two facts: equality of continuous maps can be checked pointwise, and continuous maps on open sets which agree on every intersection can be glued to a global continuous map.

Lemma 4.4: a local test for equality of sections

Let $\mathcal{F}$ be a sheaf on a topological space X, and let

$$s, t \in \mathcal{F}(X).$$

If

$$s_P = t_P$$

in the stalk $\mathcal{F}_P$ for every $P \in X$, then $s = t$.

Proof By hypothesis, for every $P \in X$ there is an open neighbourhood

$$P \in U_P \subseteq X$$

such that

$$\rho_{X,U_P}(s) = \rho_{X,U_P}(t).$$

Since

$$X = \bigcup_{P \in X} U_P,$$

the first sheaf property gives $s = t$.

Sheaf morphisms

A sheaf morphism is simply a presheaf morphism between two sheaves. Nevertheless, there are important special features concerning injectivity, surjectivity, images and local tests for isomorphisms.

> **Editorial note - typographical error in the source heading.** The source prints *Garbenmorpismen*; the intended German word is *Garbenmorphismen*. This edition uses the correct mathematical term, “sheaf morphisms”.

Lemma 4.5: injectivity can be tested on stalks

Let X be a topological space and

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

a sheaf morphism. The following statements are equivalent.

1. The map

$$\varphi_U : \mathcal{F}(U) \longrightarrow \mathcal{G}(U)$$

   is injective for every open set $U \subseteq X$.
2. The stalk map

$$\varphi_P : \mathcal{F}_P \longrightarrow \mathcal{G}_P$$

   is injective for every $P \in X$.

Proof First suppose that all maps on sections over open sets are injective. Let $s_P, t_P \in \mathcal{F}_P$ with

$$\varphi_P(s_P) = \varphi_P(t_P).$$

We may represent both germs by sections $s, t \in \mathcal{F}(U)$ on an open neighbourhood U of P. Equality in the stalk $\mathcal{G}_P$ gives a smaller open neighbourhood

$$P \in U' \subseteq U$$

with

$$\varphi_{U'}(s|_{U'}) = \varphi_{U'}(t|_{U'}).$$

Injectivity of $\varphi_{U'}$ gives $s|_{U'} = t|_{U'}$, hence $s_P = t_P$.

Conversely, suppose that all stalk maps are injective. Let $s, t \in \mathcal{F}(U)$ with $\varphi_U(s) = \varphi_U(t)$. For every $P \in U$, we obtain

$$\varphi_P(s_P) = \varphi_P(t_P),$$

so $s_P = t_P$. Applying Lemma 4.4 to the restriction of the sheaf to U, we obtain $s = t$.

Lemma 4.6: testing isomorphisms on stalks

Let X be a topological space and

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

a sheaf morphism. The morphism φ is a sheaf isomorphism if and only if, for every $P \in X$, the stalk map

$$\varphi_P : \mathcal{F}_P \longrightarrow \mathcal{G}_P$$

is an isomorphism.

Proof The forward direction is immediate. For the converse, we must show that

$$\varphi_U : \mathcal{F}(U) \longrightarrow \mathcal{G}(U)$$

is bijective for every open set $U \subseteq X$. By restricting both sheaves, it suffices to consider $U = X$. Injectivity follows from Lemma 4.5.

For surjectivity, take $t \in \mathcal{G}(X)$. For every $P \in X$, there is a unique $s_P \in \mathcal{F}_P$ with

$$\varphi_P(s_P) = t_P.$$

Choose a representative $r_P \in \mathcal{F}(U_P)$ on an open neighbourhood U_P of P. Since $\varphi(r_P)$ and t have the same germ at P, after shrinking U_P if necessary we obtain

$$\varphi_{U_P}(r_P) = t|_{U_P}.$$

The sets U_P cover X. On $U_P \cap U_Q$, for every $Z \in U_P \cap U_Q$, both germs $(r_P)_Z$ and $(r_Q)_Z$ are sent by the isomorphism φ_Z to t_Z. Thus

$$(r_P)_Z = (r_Q)_Z.$$

By Lemma 4.4,

$$r_P|_{U_P \cap U_Q} = r_Q|_{U_P \cap U_Q}.$$

The second sheaf property then glues all the r_P to an $r \in \mathcal{F}(X)$. On each U_P, we have $\varphi(r)|_{U_P} = t|_{U_P}$, so the first sheaf property gives $\varphi(r) = t$.

This statement holds neither for presheaves—for example, consider the sheafification of a presheaf—nor without the existence of a morphism between the two sheaves. Two sheaves whose stalks are isomorphic at every point need not be isomorphic as sheaves. Important examples are locally free sheaves: they are locally isomorphic to free sheaves, but in general are not globally free.

At first sight, it may be surprising, perhaps even disappointing, that for a sheaf morphism surjectivity on sections over open sets differs from surjectivity on stalks. What initially appears to be a shortcoming is actually a strength of sheaf theory: the failure of global surjectivity for a stalkwise-surjective morphism can reflect topological properties of the underlying space.

Definition 4.7: surjective sheaf morphism

A sheaf morphism

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

on a topological space X is called *surjective* if, for every point $P \in X$, the stalk map

$$\varphi_P : \mathcal{F}_P \longrightarrow \mathcal{G}_P$$

is surjective. This is substantially weaker than surjectivity of the map on sections over every open set.

Example 4.8: surjective on stalks, but not always on sections

Consider the continuous group homomorphism

$$\varphi : \mathbb{R} \longrightarrow S^1, \qquad t \longmapsto (\cos t, \sin t),$$

that is, the periodic trigonometric parametrisation of the unit circle. On every topological space X, this map induces a sheaf morphism

$$C^0(-, \mathbb{R}) \longrightarrow C^0(-, S^1),$$

sending a continuous function $f : U \to \mathbb{R}$ to the composite

$$\varphi \circ f : U \longrightarrow S^1.$$

This morphism is surjective because φ is locally invertible. However, the map on sections is not always surjective. For example, if $X = S^1$, the identity on S^1 has no continuous lift to $\mathbb{R}$.

Lemma 4.9: the sheaf of morphisms

For two sheaves $\mathcal{F}$ and $\mathcal{G}$ on a topological space X, the assignment

$$U \longmapsto \mathrm{Mor}(\mathcal{F}|_U, \mathcal{G}|_U)$$

is a sheaf.

Proof Restricting a sheaf morphism

$$\varphi : \mathcal{F}|_U \longrightarrow \mathcal{G}|_U$$

to any open set $V \subseteq U$ gives a morphism

$$\varphi|_V : \mathcal{F}|_V \longrightarrow \mathcal{G}|_V.$$

Thus the assignment above is, to begin with, a presheaf.

Let $U = \bigcup_{i \in I} U_i$. For the equality condition, let φ and ψ be two morphisms on U whose restrictions agree on every U_i. For every open set $V \subseteq U$ and every $s \in \mathcal{F}(V)$, the sections $\varphi_V(s)$ and $\psi_V(s)$ of $\mathcal{G}(V)$ agree after restriction to each $V \cap U_i$. The first sheaf property of $\mathcal{G}$ gives $\varphi_V(s) = \psi_V(s)$. Since this holds for all V and s, we obtain $\varphi = \psi$.

For the gluing condition, suppose morphisms

$$\varphi_i : \mathcal{F}|_{U_i} \longrightarrow \mathcal{G}|_{U_i}$$

are given satisfying

$$\varphi_i|_{U_i \cap U_j} = \varphi_j|_{U_i \cap U_j}.$$

For every open set $V \subseteq U$ and every $s \in \mathcal{F}(V)$, set, on $V \cap U_i$,

$$t_i = (\varphi_i)_{V \cap U_i}(s|_{V \cap U_i}).$$

The family (t_i) is compatible on all intersections, so there is a unique $t \in \mathcal{G}(V)$ with $t|_{V \cap U_i} = t_i$. Set

$$\varphi_V(s) := t.$$

If $W \subseteq V$, then $\varphi_V(s)|_W$ and $\varphi_W(s|_W)$ have the same local restrictions on every $W \cap U_i$; uniqueness of gluing shows that they are equal. Thus the family φ_V is compatible with all restriction maps and genuinely defines a sheaf morphism

$$\varphi : \mathcal{F}|_U \longrightarrow \mathcal{G}|_U.$$

Its restriction to each U_i is φ_i, again by uniqueness of gluing.

> **Editorial note - completion of the source proof.** The source proof checks equality and constructs the gluing only for sections over U. A sheaf morphism must have a component on every open set $V \subseteq U$ and must commute with restrictions. The proof above supplies the missing standard step, using the cover $(V \cap U_i)_i$ and uniqueness of gluing. The abbreviated source form remains preserved within the Unit 4 authority boundary.

Corollary 4.10: gluing sheaf morphisms

Let

$$X = \bigcup_{i \in I} U_i$$

be an open cover of a topological space X, and let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on X. For each $i \in I$, suppose a sheaf morphism

$$\alpha_i : \mathcal{F}|_{U_i} \longrightarrow \mathcal{G}|_{U_i}$$

is given with

$$\alpha_i|_{U_i \cap U_j} = \alpha_j|_{U_i \cap U_j}$$

for all i, j. Then there is a unique sheaf morphism

$$\alpha : \mathcal{F} \longrightarrow \mathcal{G}$$

satisfying $\alpha|_{U_i} = \alpha_i$ for every i.

Proof This follows directly from Lemma 4.9.

Corollary 4.11: testing equality on stalks

Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on a topological space X, and let

$$\alpha, \beta : \mathcal{F} \longrightarrow \mathcal{G}$$

be sheaf morphisms. Then

$$\alpha = \beta$$

if and only if

$$\alpha_P = \beta_P$$

for every $P \in X$.

Editorial note - inconsistent source index. The source writes $\alpha_p = \beta_P$ after quantifying over the point P. This edition uses the consistent indices $\alpha_P = \beta_P$.

Proof This follows directly from Lemma 4.9 and Lemma 4.4.

Worksheet 4: Sheaves and Sheaf Morphisms

Exercise 4.1

Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on a topological space X. Show that the assignment

$$U \longmapsto \mathcal{F}(U) \times \mathcal{G}(U),$$

together with the natural product maps as restriction maps, defines a sheaf on X.

Exercise 4.2

Let $\mathcal{G}$ be a sheaf on a disconnected space X with a decomposition

$$X = U \uplus V$$

into two disjoint nonempty open sets. Show that

$$\mathcal{G}(X) = \mathcal{G}(U) \times \mathcal{G}(V).$$

Exercise 4.3

Let X be a topological space with a decomposition

$$X = Y \uplus Z$$

into two disjoint nonempty open subsets. Let $\mathcal{G}$ be a sheaf on Y and $\mathcal{H}$ a sheaf on Z. Show that, for each open set $U \subseteq X$, the assignment

$$\mathcal{F}(U) = \mathcal{G}(U \cap Y) \times \mathcal{H}(U \cap Z)$$

defines a sheaf $\mathcal{F}$ on X.

Exercise 4.4

Let X be a Hausdorff space with at least two points and let M be a set with at least two elements. Show that the constant presheaf with value M is not a sheaf.

Editorial note - source hypothesis too weak. The source assumes only $M \neq \emptyset$. The conclusion is false if M is a singleton, since the constant singleton-valued presheaf satisfies both sheaf conditions. This edition states the intended hypothesis that M has at least two elements.

Exercise 4.5

Show that the restriction of a sheaf to an open subset

$$U \subseteq X$$

is a sheaf.

Exercise 4.6

Show that the stalk at $0 \in \mathbb{C}$ of the sheaf of holomorphic functions is isomorphic to the ring of convergent power series in one variable.

Exercise 4.7

Let

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

be a sheaf morphism on a topological space X. Suppose

$$\varphi_U : \mathcal{F}(U) \longrightarrow \mathcal{G}(U)$$

is surjective for every open set $U \subseteq X$. Show that every stalk map

$$\varphi_P : \mathcal{F}_P \longrightarrow \mathcal{G}_P$$

is also surjective.

Exercise 4.8

Let $\mathcal{G}$ be a sheaf of commutative groups on a topological space X. Show that

$$\mathcal{G}(\emptyset) = 0,$$

that is, the value of the sheaf on the empty set is the trivial group.

Exercise 4.9

Let X be a topological space, $P \in X$ a point, and G a commutative group. Consider the assignment

$$U \longmapsto \mathcal{G}(U) := \begin{cases} G, & \text{if } P \in U, \\ 0, & \text{if } P \notin U, \end{cases}$$

with the natural restriction maps for each inclusion of open sets $V \subseteq U$.

1. Show that $\mathcal{G}$ is a sheaf of commutative groups.
2. Determine the stalk $\mathcal{G}_P$.
3. Now suppose that P is a closed point. Determine the stalk $\mathcal{G}_Q$ at every point $Q \neq P$.

   Editorial note - two source typographical errors. The last instruction in the source reads *Besitmme die Halm.* The intended wording is *Bestimme die Halme*, meaning to determine the stalks at all points $Q \neq P$.

The sheaf constructed in the preceding exercise is called the *skyscraper sheaf* with value G at P.

Public Solution Coverage for Worksheet 4

At the frozen revision boundary, the source provides no public solution for any of the nine exercises on Worksheet 4. The frozen solution-candidate query and exercise map record negative results for Exercises 4.1-4.9.

No new solutions have been created for this edition. This section merely documents the scope of the source so that the absence of solutions is not mistaken for content lost during translation.

Lecture 5: Sheafification, Homomorphisms and Quotient Sheaves

Sheafification

A presheaf can be assigned a sheaf in a canonical way. This construction is called *sheafification.*

Definition 5.1: sheafification

Let $\mathcal{F}$ be a presheaf on a topological space X. The presheaf given by

$$\widetilde{\mathcal{F}}(U) := \left\{ (s_P)_{P\in U} \in \prod_{P\in U} \mathcal{F}_P \;\middle|\; \begin{array}{l} \text{for every } P \in U \text{ there is an open set } V \\ \text{with } P \in V \subseteq U \text{ and } t \in \mathcal{F}(V) \\ \text{such that } s_Q = t_Q \text{ in } \mathcal{F}_Q \text{ for every } Q \in V \end{array} \right\},$$

together with the natural restriction maps, is called the *sheafification* of $\mathcal{F}$.

The condition in this definition, that the local sections define the same germs in the stalks, is also called the *compatibility condition.*

> **Editorial note - openness of the local neighbourhood.** The source formula writes only $P \in V \subseteq U$, without saying that V is open. Since $\mathcal{F}(V)$ and this local construction use the presheaf, V must be an open neighbourhood. This edition makes that condition explicit.

Lemma 5.2: properties of sheafification

Let $\mathcal{F}$ be a presheaf on a topological space X, and let $\widetilde{\mathcal{F}}$ be its sheafification. The following properties hold.

1. There is a natural presheaf morphism

$$\eta : \mathcal{F} \longrightarrow \widetilde{\mathcal{F}},$$

   given on every open set U by

$$\begin{aligned} \eta_U : \mathcal{F}(U) &\longrightarrow \widetilde{\mathcal{F}}(U), \\ s &\longmapsto (s_P)_{P\in U}. \end{aligned}$$

2. For every $P \in X$, there is a natural isomorphism

$$\widetilde{\mathcal{F}}_P \cong \mathcal{F}_P.$$

3. The sheafification $\widetilde{\mathcal{F}}$ is a sheaf.
4. If $\mathcal{F}$ is already a sheaf, the natural morphism

$$\mathcal{F} \longrightarrow \widetilde{\mathcal{F}}$$

   is an isomorphism.

5. For every presheaf morphism

$$\psi : \mathcal{F} \longrightarrow \mathcal{G}$$

to a sheaf $\mathcal{G}$, there is a unique factorisation

$$\widetilde{\psi} : \widetilde{\mathcal{F}} \longrightarrow \mathcal{G}.$$

Proof

1. An element $s \in \mathcal{F}(U)$ defines a tuple

$$(s_P)_{P \in U},$$

which immediately satisfies the compatibility condition. Thus there is a well-defined map

$$\eta_U : \mathcal{F}(U) \longrightarrow \widetilde{\mathcal{F}}(U).$$

If $V \subseteq U$, we have the commutative diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\eta_U} & \prod\limits_{P \in U} \mathcal{F}_P \\ \rho_{U,V} \downarrow & & \downarrow \\ \mathcal{F}(V) & \xrightarrow{\eta_V} & \prod\limits_{P \in V} \mathcal{F}_P. \end{array}$$

Commutativity follows because the germ of a section in the stalk at a point depends only on the open neighbourhoods of that point.

2. By part (1) and Lemma 3.27, there is a natural map

$$\mathcal{F}_P \longrightarrow \widetilde{\mathcal{F}}_P.$$

To prove surjectivity, take $s \in \widetilde{\mathcal{F}}_P$, represented by some

$$s' \in \widetilde{\mathcal{F}}(U).$$

On an open neighbourhood $V \subseteq U$ of P, this section is represented by an element

$$s'' \in \mathcal{F}(V).$$

The germ $s''_P \in \mathcal{F}_P$ is immediately a preimage of s.

To prove injectivity, let $s, t \in \mathcal{F}_P$ have the same image in $\widetilde{\mathcal{F}}_P$. We may assume that s and t are represented by sections on the same open set, say U. Equality in the stalk of the sheafification means that there is an open neighbourhood $P \in V \subseteq U$ with

$$(s_Q)_{Q \in V} = (t_Q)_{Q \in V}.$$

In particular, the germs of the two sections at P agree, so $s = t$ in $\mathcal{F}_P$.

3. Let

$$U = \bigcup_{i \in I} U_i$$

be an open cover, and let

$$s, t \in \Gamma(U, \widetilde{\mathcal{F}})$$

satisfy

$$s|_{U_i} = t|_{U_i}$$

for every i. Every point $P \in U$ belongs to some U_i, so

$$s_P = t_P$$

for every $P \in U$. Thus the two tuples in the product of stalks agree, and consequently $s = t$ in the sheafification.

Now suppose sections

$$s_i \in \widetilde{\mathcal{F}}(U_i)$$

are given with

$$s_i|_{U_i \cap U_j} = s_j|_{U_i \cap U_j}.$$

For every $P \in U$, one of the s_i with $P \in U_i$ determines a germ s_P. This germ is unique by compatibility on the intersections. The tuple

$$(s_P)_{P \in U}$$

immediately satisfies the compatibility condition in the definition of sheafification. Thus $\widetilde{\mathcal{F}}$ satisfies both sheaf conditions.

4. By part (1), there is a presheaf morphism

$$\mathcal{F} \longrightarrow \widetilde{\mathcal{F}}.$$

By part (2), this morphism is bijective on every stalk. The left-hand side is a sheaf by hypothesis, and the right-hand side is a sheaf by part (3). Lemma 4.6 shows that the morphism is an isomorphism.

5. See Exercise 5.2. □

Editorial note - incorrect article in the source. In item (4), the source prints *die natürliche Morphismus*. The correct German form is *der natürliche Morphismus*. The translation uses the intended mathematical expression, “natural morphism”. (The source heading typo *Garbenmorpismen* is also preserved in the Unit 4 authority notes; the term used here remains “sheaf morphism”.)

Homomorphisms of sheaves of groups

Definition 5.3: homomorphism of sheaves of commutative groups

Let X be a topological space, and let $\mathcal{F}$ and $\mathcal{G}$ be sheaves of commutative groups on X. A sheaf morphism

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

is called a *homomorphism of sheaves of commutative groups* if, for every open set $U \subseteq X$, the map

$$\varphi_U : \mathcal{F}(U) \longrightarrow \mathcal{G}(U)$$

is a group homomorphism.

Example 5.4: homomorphisms induced by topological groups

A continuous group homomorphism

$$\varphi : F \longrightarrow G$$

between topological groups F and G determines a homomorphism of sheaves of groups on every topological space X. On each open set U, it is given by

$$\begin{aligned} C^0(U, F) &\longrightarrow C^0(U, G), \\ f &\longmapsto \varphi \circ f. \end{aligned}$$

Scope note. Definition 5.3 is phrased for sheaves of commutative groups, whereas this source example uses general topological groups. The composition construction above does not require commutativity; this edition therefore calls it a homomorphism of sheaves of groups in the general sense.

Definition 5.5: kernel sheaf

Let X be a topological space and

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

a homomorphism of sheaves of commutative groups. The subsheaf of $\mathcal{F}$ defined by

$$(\ker \varphi)(U) := \ker \varphi_U$$

is called the *kernel sheaf* of φ.

More precisely, it is a subsheaf of commutative groups: for every open set U, its value is a subgroup of $\mathcal{F}(U)$; see Exercise 5.6.

Definition 5.6: image sheaf

Let X be a topological space and

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

a homomorphism of sheaves of commutative groups. The sheafification of the presheaf given by

$$(\operatorname{im} \varphi)(U) := \operatorname{im} \varphi_U$$

is called the *image sheaf* of φ.

By Lemma 5.2(5), the image sheaf is naturally a subsheaf of $\mathcal{G}$, and more precisely a subsheaf of commutative groups. It is denoted by $\operatorname{im} \varphi$.

Example 5.7: homomorphisms of trivial vector bundles

Let X be a topological space and

$$\varphi : X \times \mathbb{R}^n \longrightarrow X \times \mathbb{R}^m$$

a homomorphism between trivial vector bundles. This homomorphism is described by a continuous map

$$M : X \longrightarrow \mathrm{Mat}_{m\times n}(C^0(X,\mathbb{R})),$$

that is, a matrix is assigned continuously to each point, describing at that point a linear map $\mathbb{R}^n \to \mathbb{R}^m$. This can immediately be viewed as a homomorphism of sheaves of groups on X:

$$\begin{aligned} C^0(-,\mathbb{R})^n &\longrightarrow C^0(-,\mathbb{R})^m, \\ \begin{pmatrix} f_1 \\ \vdots \\ f_n \end{pmatrix} &\longmapsto M \begin{pmatrix} f_1 \\ \vdots \\ f_n \end{pmatrix}. \end{aligned}$$

This map is the sheaf morphism at the level of sections of the bundles.

In Example 1.2, for $X = \mathbb{R}^3$, we have the map

$$\begin{aligned} \varphi : \mathbb{R}^3 \times \mathbb{R}^3 &\longrightarrow \mathbb{R}^3 \times \mathbb{R}, \\ (r,s,t;u,v,w) &\longmapsto (r,s,t;ru+sv+tw), \end{aligned}$$

or, equivalently,

$$\begin{aligned} M : \mathbb{R}^3 &\longrightarrow \mathrm{Mat}_{1\times 3}(K), \\ (r,s,t) &\longmapsto (r,s,t). \end{aligned}$$

Editorial note - two type mismatches in the source matrix notation. The source writes $M : X \to \mathrm{Mat}_{m\times n}(C^0(X,\mathbb{R}))$, whereas the following description treats M as a pointwise matrix-valued function. The usual correctly typed expression is $M : X \to \mathrm{Mat}_{m\times n}(\mathbb{R})$, or equivalently $M \in \mathrm{Mat}_{m\times n}(C^0(X,\mathbb{R}))$. In the concrete real example, the source also writes $\mathrm{Mat}_{1\times 3}(K)$, although K is undefined and the context uses $\mathbb{R}$. Both source formulae are retained above so that these discrepancies remain visible.

The kernel sheaf over U is

$$\begin{aligned} (\ker\varphi)(U) &= \left\{ \begin{pmatrix} f_1 \\ \vdots \\ f_n \end{pmatrix} \in C^0(U,\mathbb{R})^n \;\middle|\; M \begin{pmatrix} f_1 \\ \vdots \\ f_n \end{pmatrix} = 0 \right\} \\ &\subseteq C^0(U,\mathbb{R})^n. \end{aligned}$$

The quotient sheaf

Definition 5.8: quotient sheaf

Let $\mathcal{G}$ be a sheaf of commutative groups and

$$\mathcal{F} \subseteq \mathcal{G}$$

a subsheaf of groups. The sheafification of the presheaf

$$U \longmapsto \mathcal{G}(U)/\mathcal{F}(U)$$

is called the *quotient sheaf* of $\mathcal{G}$ by $\mathcal{F}$.

The quotient sheaf is denoted by $\mathcal{G}/\mathcal{F}$. Because its construction uses sheafification, the equality

$$(\mathcal{G}/\mathcal{F})(U) = \mathcal{G}(U)/\mathcal{F}(U).$$

need not hold in general. However, for every point $P \in X$,

$$(\mathcal{G}/\mathcal{F})_P = \mathcal{G}_P/\mathcal{F}_P;$$

see Exercise 5.11.

Lemma 5.9: an explicit description of the quotient sheaf

Let $\mathcal{G}$ be a sheaf of commutative groups and

$$\mathcal{F} \subseteq \mathcal{G}$$

a subsheaf of groups, with quotient sheaf $\mathcal{G}/\mathcal{F}$. The following statements hold.

1. Every element

   $$s \in \Gamma(X, \mathcal{G}/\mathcal{F})$$

   is represented by a family

   $$(U_i, g_i)_{i \in I},$$

   where

   $$X = \bigcup_{i \in I} U_i$$

   is an open cover and the sections

   $$g_i \in \Gamma(U_i, \mathcal{G})$$

   satisfy

   $$g_i|_{U_i \cap U_j} - g_j|_{U_i \cap U_j} \in \Gamma(U_i \cap U_j, \mathcal{F})$$

   for all $i, j \in I$.

   Every such family determines an element of $\Gamma(X, \mathcal{G}/\mathcal{F})$.
2. Two families

   $$(U_i, g_i)_{i \in I} \qquad \text{and} \qquad (U_i, h_i)_{i \in I}$$

   on the same open cover determine the same element of

   $$\Gamma(X, \mathcal{G}/\mathcal{F})$$

   precisely when

   $$g_i - h_i \in \Gamma(U_i, \mathcal{F})$$

   for every i.

3. Two families

$$(U_i, g_i)_{i\in I} \qquad \text{and} \qquad (V_j, h_j)_{j\in J}$$

determine the same element precisely when, on some—and hence every—common refinement of the two covers, the differences of their sections belong to $\mathcal{F}$.

Proof

1. The canonical sheaf homomorphism

$$\mathcal{G} \longrightarrow \mathcal{G}/\mathcal{F}$$

is surjective. Therefore every section

$$s \in \Gamma(X, \mathcal{G}/\mathcal{F})$$

has local preimages. Thus there are an open cover

$$X = \bigcup_{i\in I} U_i$$

and elements

$$g_i \in \Gamma(U_i, \mathcal{G})$$

mapping to $s|_{U_i}$. Hence

$$g_i|_{U_i\cap U_j} - g_j|_{U_i\cap U_j} \in \Gamma(U_i \cap U_j, \mathcal{G})$$

maps to zero, so this difference belongs to the kernel, namely $\mathcal{F}$.

Conversely, a family satisfying this condition determines classes

$$[g_i] \in \Gamma(U_i, \mathcal{G}/\mathcal{F}).$$

On every intersection we have

$$\begin{aligned}[g_i]|_{U_i\cap U_j} - [g_j]|_{U_i\cap U_j} &= [g_i|_{U_i\cap U_j} - g_j|_{U_i\cap U_j}] \\ &= 0.\end{aligned}$$

Thus the classes are compatible and determine a global section of the quotient sheaf.

2. Replacing the two families by their difference, it suffices to consider the case $h_i = 0$. We must show that (U_i, g_i) determines the zero element in the quotient sheaf precisely when every

$$g_i \in \Gamma(U_i, \mathcal{F}).$$

If the family determines the zero element, its image in every stalk is also zero. Thus, for every $P \in U_i$, its germ satisfies

$$(g_i)_P \in \mathcal{F}_P.$$

Membership in a subsheaf can be tested on stalks, so

$$g_i \in \Gamma(U_i, \mathcal{F}).$$

The converse is immediate.

3. Equality of sections of a sheaf can be tested locally on any open cover. The statement follows from part (2) and the fact that membership in a subsheaf can also be tested locally. □

   Editorial note - sections and germs in the source proof. In item (2), the source writes $g_i \in \mathcal{F}_P$, although g_i is a section on U_i and $\mathcal{F}_P$ is a stalk. The correctly typed statement used above is $(g_i)_P \in \mathcal{F}_P$.

Worksheet 5: Sheafification and Quotient Sheaves

Exercise 5.1

Let X be a topological space and $\mathcal{F}$ a presheaf on X. Show that the assignment

$$U \longmapsto \prod_{P \in U} \mathcal{F}_P,$$

the product of all stalks at points of U, together with the natural projections as restriction maps, defines a presheaf. Show also that there is a natural presheaf morphism from $\mathcal{F}$ to this presheaf.

Exercise 5.2

Let $\mathcal{F}$ be a presheaf on a topological space X, and let $\widetilde{\mathcal{F}}$ be its sheafification. Show that, for every presheaf morphism

$$\psi : \mathcal{F} \longrightarrow \mathcal{G}$$

to a sheaf $\mathcal{G}$, there is a unique factorisation

$$\widetilde{\psi} : \widetilde{\mathcal{F}} \longrightarrow \mathcal{G}.$$

The sheafification of a constant presheaf is called a *locally constant sheaf*, and sometimes simply a *constant sheaf*.

Exercise 5.3

Let $\mathcal{F}$ be the constant presheaf with value a set M on a topological space X. Show that the stalk of the sheafification of $\mathcal{F}$ at every point

$$P \in X$$

is equal to M.

Exercise 5.4

Let G be a discrete topological group and X a topological space. Let $\mathcal{G}$ be the constant presheaf with value G on X. Show that the sheafification of $\mathcal{G}$ is equal to

$$C^0(-, G).$$

Exercise 5.5*

Let X be a topological space, $\mathcal{G}$ a sheaf on X, and

$$\mathcal{F} \subseteq \mathcal{G}$$

a subsheaf. Suppose

$$t \in \Gamma(X, \mathcal{G})$$

satisfies

$$t_P \in \mathcal{F}_P$$

for every

$$P \in X.$$

Show that

$$t \in \Gamma(X, \mathcal{F}).$$

Exercise 5.6

Let X be a topological space and

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

a homomorphism of sheaves of commutative groups. Show that the assignment

$$(\ker \varphi)(U) := \ker \varphi_U$$

defines a sheaf of groups on X.

Exercise 5.7

Let X be a topological space and

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

a homomorphism of sheaves of commutative groups. Show that φ is injective precisely when

$$\ker \varphi$$

is the zero sheaf.

Exercise 5.8

Let X be a topological space and

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

a homomorphism of sheaves of commutative groups. Show that φ is surjective precisely when

$$\operatorname{im} \varphi = \mathcal{G}.$$

Exercise 5.9

Let X be a topological space and

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

a homomorphism of sheaves of commutative groups. Show that, for every

$$P \in X,$$

we have

$$(\operatorname{im} \varphi)_P = \operatorname{im}(\varphi_P).$$

Exercise 5.10

Let $\mathcal{G}$ be a sheaf of commutative groups and

$$\mathcal{F} \subseteq \mathcal{G}$$

a subsheaf of groups. Show that there is a canonical surjective homomorphism of sheaves of commutative groups

$$\mathcal{G} \longrightarrow \mathcal{G}/\mathcal{F}.$$

Exercise 5.11

Let $\mathcal{G}$ be a sheaf of commutative groups and

$$\mathcal{F} \subseteq \mathcal{G}$$

a subsheaf of groups, and let $\mathcal{G}/\mathcal{F}$ be their quotient sheaf. Show that

$$(\mathcal{G}/\mathcal{F})_P = \mathcal{G}_P/\mathcal{F}_P$$

for every point

$$P \in X.$$

Public Solutions and Coverage for Worksheet 5

At the frozen revision boundary, the source provides exactly one public solution among the eleven exercises on Worksheet 5, namely the solution to Exercise 5.5. The frozen exercise map and candidate evidence record negative results for Exercises 5.1-5.4 and 5.6-5.11. No new solutions have been created for this edition.

Source solution to Exercise 5.5

Membership in the stalk,

$$t_P \in \mathcal{F}_P$$

means that there are an open neighbourhood

$$P \in U_P$$

and a section

$$s_P \in \mathcal{F}(U_P) \subseteq \mathcal{G}(U_P)$$

whose germ at P is t_P. Thus there is a smaller open neighbourhood

$$P \in V_P \subseteq U_P$$

such that the restrictions of t and s_P, regarded as sections of $\mathcal{G}$, agree on V_P.

Thus there is an open cover

$$X = \bigcup_{i \in I} V_i$$

such that

$$t|_{V_i} \in \mathcal{F}(V_i) \subseteq \mathcal{G}(V_i).$$

These sections are compatible both as sections of $\mathcal{G}$ and as sections of $\mathcal{F}$. Therefore there is a section

$$s \in \mathcal{F}(X)$$

whose restriction to each V_i is $t|_{V_i}$. Since a compatible family in a sheaf has a unique global realisation, we have

$$s = t$$

in $\mathcal{G}(X)$. Hence

$$t \in \mathcal{F}(X).$$

> **Editorial note - sections and germs.** The source says that the local sections "restrict to" the germ t_P. More precisely, the germ at P of the local section s_P equals t_P; the translation uses this correctly typed relation without changing the argument.

Return to Exercise 5.5.

Lecture 6: Exactness, Global Sections, and Pullback and Pushforward of Sheaves

Exactness

Definition 6.1: complex of sheaves

Let X be a topological space, let $\mathcal{F}_n$ be sheaves of commutative groups on X, and let

$$\varphi_n : \mathcal{F}_{n-1} \longrightarrow \mathcal{F}_n$$

be sheaf homomorphisms. We say that these form a *complex of sheaves* if

$$\operatorname{im} \varphi_n \subseteq \ker \varphi_{n+1}$$

holds.

Definition 6.2: exactness

Let X be a topological space and let $\mathcal{F}_\bullet$ be a complex of sheaves of commutative groups on X. The complex is called *exact* if

$$\operatorname{im} \varphi_n = \ker \varphi_{n+1}$$

for every $n \in \mathbb{Z}$.

Lemma 6.3: stalkwise characterisation of exactness

Let X be a topological space and let

$$\mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H}$$

be a complex of sheaves of commutative groups on X. This complex is exact if and only if, for every point $P \in X$, the complex of stalks

$$\mathcal{F}_P \longrightarrow \mathcal{G}_P \longrightarrow \mathcal{H}_P$$

is exact.

Proof Denote the maps in question by

$$\mathcal{F} \xrightarrow{\alpha} \mathcal{G} \xrightarrow{\beta} \mathcal{H}.$$

By Corollary 4.11, this is a complex of sheaves if and only if all the induced maps on stalks form complexes. Suppose the complex is exact, so that

$$\operatorname{im} \alpha = \ker \beta.$$

Fix $P \in X$ and take $s \in \mathcal{G}_P$ with $\beta_P(s) = 0$. There is an open neighbourhood U of P on which s is represented by a section s, and a smaller open neighbourhood

$$P \in V \subseteq U$$

such that

$$\beta_V(s|_V) = 0.$$

The element $s \in \mathcal{G}(V)$ (we again denote the restriction by s) belongs to the kernel of β_V, and hence to the sheaf image of α. Thus there is an open neighbourhood

$$P \in W \subseteq V$$

on which s lies in the image of

$$\alpha_W : \mathcal{F}(W) \longrightarrow \mathcal{G}(W).$$

Consequently, the germ s lies in the image of α_P. This proves exactness of the complex of stalks.

> **Edition note — converse of Lemma 6.3.** The frozen source proves only the forward implication. For completeness, the converse added in this edition is as follows: if $s \in (\ker \beta)(U)$, stalkwise exactness gives a germ lifting s_P at every $P \in U$. Representing that germ and then shrinking its neighbourhood makes its image equal to s there. Thus s belongs locally to the image of α, hence belongs to its sheaf image. The opposite inclusion follows from the complex condition.

Definition 6.4: short exact sequence

An exact complex

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H} \longrightarrow 0$$

of sheaves of commutative groups on a topological space X is called a *short exact sequence.*

In particular, the first map is injective and the last map is surjective as a map of sheaves (that is, locally surjective at every point).

Lemma 6.5: a sheaf sequence from a sequence of topological groups

Let

$$0 \longrightarrow F \longrightarrow G \longrightarrow H \longrightarrow 0$$

be a short exact sequence of commutative topological groups, with continuous group homomorphisms. Suppose that F carries the topology induced by G, and that the surjection

$$p : G \longrightarrow H$$

has the following property: for every $h \in H$ there is an open neighbourhood

$$h \in W \subseteq H$$

and a continuous section of p over W. Then, for every topological space X, the corresponding sequence of sheaves of continuous maps,

$$0 \longrightarrow C^0(-, F) \longrightarrow C^0(-, G) \longrightarrow C^0(-, H) \longrightarrow 0,$$

is also exact.

Proof Clearly this is a complex of sheaves of commutative groups on X. Injectivity on the left is also clear. For exactness in the middle, let $U \subseteq X$ be open and let $\varphi : U \to G$ be continuous with $p \circ \varphi$ the zero map. The image of φ lies in F; since F carries the topology induced by G, the map $\varphi : U \to F$ is continuous as well.

For surjectivity as a sheaf map on the right, take a point $P \in X$ and a continuous map $\psi : V \to H$ defined on an open neighbourhood V of P. Write $\psi(P) = h$. By hypothesis, there is an open neighbourhood

$$h \in W \subseteq H$$

and a section $s : W \to G$ with

$$p \circ s = \mathrm{Id}_W .$$

Set

$$U := V \cap \psi^{-1}(W).$$

Then $s \circ \psi$, restricted to U, is a continuous G-valued section that is mapped to ψ by p.

Example 6.6: the exponential sequence

Consider the short exact sequence

$$0 \longrightarrow 2\pi \mathrm{i}\, \mathbb{Z} \longrightarrow \mathbb{C} \xrightarrow{\exp} \mathbb{C}^{\times} \longrightarrow 0$$

of topological groups. Exactness in the middle follows from Theorem 21.5 (Analysis (Osnabrück 2021–2023), part (2)); the homomorphism property follows from the functional equation of the exponential function. By Theorem 21.6 (Analysis (Osnabrück 2021–2023)), the complex exponential function maps surjectively onto $\mathbb{C} \setminus \{0\}$ and is a covering map (see Example 21.3, Funktionentheorie (Osnabrück 2023–2024)). Since a logarithm exists locally, the hypotheses of Lemma 6.5 are satisfied. Thus, for every topological space X, we obtain a short exact sequence of sheaves

$$0 \longrightarrow C^0(-, \mathbb{Z}) \longrightarrow C^0(-, \mathbb{C}) \longrightarrow C^0(-, \mathbb{C}^{\times}) \longrightarrow 0.$$

This is called the *continuous complex exponential sequence*. On the left is the locally constant sheaf with values in $\mathbb{Z}$; in the middle is the sheaf of complex-valued continuous functions; and on the right is the sheaf of nowhere-zero complex-valued continuous functions. If $X = \mathbb{C}^{\times}$, the induced map on global sections at the right is not surjective, since the identity function is not in its image.

> **Edition note — dates in the source references.** Although this course is entitled 2019–2020, the two source surfaces differ: the terminal PDF cites Analysis (Osnabrück 2014–2016), whereas the current semantic TeX witness cites Analysis 2021–2023 and Funktionentheorie 2023–2024. All dates are retained as identifiers of their respective sources, without implying that the editions have been harmonised.

Global sections

Lemma 6.7: taking global sections preserves complexes

Let X be a topological space and let

$$\mathcal{F} \xrightarrow{d} \mathcal{G} \xrightarrow{d'} \mathcal{H}$$

be a complex of homomorphisms of sheaves of commutative groups on X. Then

$$\Gamma(X, \mathcal{F}) \longrightarrow \Gamma(X, \mathcal{G}) \longrightarrow \Gamma(X, \mathcal{H})$$

is also a complex.

Proof The hypothesis says precisely that $d' \circ d$ is the zero map. Consequently, its evaluation on global sections is also the zero map.

Lemma 6.8: taking global sections is left exact

Let X be a topological space and let

$$0 \longrightarrow \mathcal{F} \xrightarrow{d} \mathcal{G} \xrightarrow{d'} \mathcal{H}$$

be an exact complex of homomorphisms of sheaves of commutative groups on X. Then

$$0 \longrightarrow \Gamma(X, \mathcal{F}) \longrightarrow \Gamma(X, \mathcal{G}) \longrightarrow \Gamma(X, \mathcal{H})$$

is also exact.

Proof By Lemma 6.7, the sequence of global sections is a complex. Exactness means that, at every point $P \in X$,

$$0 \longrightarrow \mathcal{F}_P \longrightarrow \mathcal{G}_P \longrightarrow \mathcal{H}_P$$

is exact on stalks.

Take $s \in \Gamma(X, \mathcal{F})$ with $d(s) = 0$ in $\Gamma(X, \mathcal{G})$. Then $d(s)_P = 0$ at every point. Hence $s_P = 0$ for every P, and Lemma 4.4 gives $s = 0$. The map on the left is injective.

Next, take $t \in \Gamma(X, \mathcal{G})$ with $d'(t) = 0$ in $\Gamma(X, \mathcal{H})$. Exactness on stalks means that, for every P, the germ t_P belongs to $\mathcal{F}_P$. By Exercise 5.5, this implies that t itself is a section of $\mathcal{F}$.

Thus taking global sections of sheaves of abelian groups is an *additive covariant left exact functor*.

Pullback and pushforward

So far we have considered sheaves and their relationships only on a fixed topological space. We now consider topological spaces connected by a continuous map.

Definition 6.9: pushforward presheaf

For a continuous map

$$\varphi : X \longrightarrow Y$$

and a presheaf $\mathcal{F}$ on X, the presheaf on Y given on each open set $U \subseteq Y$ by

$$(\varphi_* \mathcal{F})(U) := \mathcal{F}(\varphi^{-1}(U))$$

is called the *pushforward presheaf* of $\mathcal{F}$ along φ.

If $V \subseteq W$ are open, then

$$\varphi^{-1}(V) \subseteq \varphi^{-1}(W),$$

so there are natural restriction maps, and this does indeed define a presheaf.

Lemma 6.10: the pushforward of a sheaf is a sheaf

For a continuous map $\varphi : X \to Y$ and a sheaf $\mathcal{F}$ on X, the pushforward presheaf $\varphi_* \mathcal{F}$ is a sheaf.

Proof Let

$$V = \bigcup_{i \in I} V_i$$

be an open cover of an open set $V \subseteq Y$. Then $\varphi^{-1}(V_i)$, for $i \in I$, form an open cover of $\varphi^{-1}(V)$. If $s, t \in (\varphi_* \mathcal{F})(V)$ satisfy

$$s|_{V_i \cap V_j} = t|_{V_i \cap V_j} \qquad (i, j \in I),$$

then $s, t \in \mathcal{F}(\varphi^{-1}(V))$, and, interpreted on X,

$$\begin{aligned} s|_{\varphi^{-1}(V_i) \cap \varphi^{-1}(V_j)} &= s|_{\varphi^{-1}(V_i \cap V_j)} \\ &= t|_{\varphi^{-1}(V_i \cap V_j)} \\ &= t|_{\varphi^{-1}(V_i) \cap \varphi^{-1}(V_j)}. \end{aligned}$$

The first sheaf axiom for $\mathcal{F}$ gives $s = t$ in $\mathcal{F}(\varphi^{-1}(V))$, and hence in $(\varphi_* \mathcal{F})(V)$.

Now take sections $s_i \in (\varphi_* \mathcal{F})(V_i)$ with

$$s_i|_{V_i \cap V_j} = s_j|_{V_i \cap V_j} \qquad (i, j \in I).$$

Interpreted on X, these are sections $s_i \in \mathcal{F}(\varphi^{-1}(V_i))$ compatible on all intersections. The gluing axiom for $\mathcal{F}$ yields a section in

$$\mathcal{F}(\varphi^{-1}(V)) = (\varphi_* \mathcal{F})(V).$$

Edition note — the type of the sections in the proof of Lemma 6.10. The source writes $s_i \in \mathcal{F}(V_i)$, although $\mathcal{F}$ is a presheaf on X and $V_i \subseteq Y$. The well-typed expression is $s_i \in \mathcal{F}(\varphi^{-1}(V_i))$; this is used above, and the source discrepancy is recorded rather than concealed.

Lemma 6.11: stalks of the pushforward presheaf

For a continuous map $\varphi : X \to Y$, a point $Q \in Y$, and a presheaf $\mathcal{F}$ on X, the stalk of the pushforward presheaf $\varphi_* \mathcal{F}$ at Q is

$$\operatorname*{colim}_{\substack{V \subseteq Y,\, V \text{ open} \\ Q \in V}} \mathcal{F}(\varphi^{-1}(V)) = \operatorname*{colim}_{\substack{U \subseteq X,\, U \text{ open} \\ \exists \text{ open neighbourhood } V \ni Q \colon \varphi^{-1}(V) \subseteq U}} \mathcal{F}(U).$$

Edition note — the neighbourhood index in the source. In the second colimit index, the source writes “there is an open neighbourhood $Q \in V$”; this is read as “there is an open neighbourhood $V \ni Q$”. The translation displays the explicit, well-typed formulation, including the implicit requirement that U be open, since $\mathcal{F}(U)$ is defined only for open sets.

See Exercise 6.5. Thus the stalk of the pushforward presheaf is a stalk of the original presheaf at a filter (namely, the inverse-image filter of the neighbourhood filter $\mathcal{U}(Q)$), but in general not at a point.

Definition 6.12: pullback presheaf

For a continuous map $\varphi : X \to Y$ and a presheaf $\mathcal{G}$ on Y, the presheaf on X given on an open set $U \subseteq X$ by

$$U \longmapsto \operatorname*{colim}_{\substack{V \subseteq Y,\, V \text{ open} \\ U \subseteq \varphi^{-1}(V)}} \mathcal{G}(V)$$

is called the *pullback presheaf* of $\mathcal{G}$ along φ.

Edition note — the source colimit display. The expanded German TeX interchanges the index and the term, placing $\mathcal{G}(V)$ beneath colim. The formula above restores their intended roles and explicitly restricts V to open sets, as required for a presheaf.

Definition 6.13: pullback sheaf

For a continuous map $\varphi : X \to Y$ and a sheaf $\mathcal{G}$ on Y, the *pullback sheaf* is the sheafification of the pullback presheaf. It is denoted by

$$\varphi^{-1}\mathcal{G}.$$

Lemma 6.14: stalks of the pullback sheaf

For a continuous map $\varphi : X \to Y$ and a sheaf $\mathcal{G}$ on Y, the stalk of the pullback sheaf at a point $P \in X$ is equal to the stalk of $\mathcal{G}$ at $\varphi(P)$.

See Exercise 6.6.

Worksheet 6: Covering Maps, Exactness, and Pullback and Pushforward of Sheaves

Definition: covering map

Let X and Y be topological spaces. A continuous map

$$p : Y \longrightarrow X$$

is called a *covering map* if there is an open cover

$$X = \bigcup_{i \in I} U_i$$

and a family of discrete topological spaces F_i, for $i \in I$, such that $p^{-1}(U_i)$ is homeomorphic to $U_i \times F_i$ with the product topology, and these homeomorphisms are compatible with the maps to U_i.

Exercise 6.1

Show that the map

$$\begin{aligned} \mathbb{R} &\longrightarrow S^1, \\ t &\longmapsto (\cos t, \sin t) \end{aligned}$$

is a covering map.

Exercise 6.2

Show that the map

$$\begin{aligned} \mathbb{C} &\longrightarrow \mathbb{C}^\times = \mathbb{C} \setminus \{0\}, \\ z &\longmapsto \exp z \end{aligned}$$

is a covering map.

Exercise 6.3

Prove that, for every covering map

$$p : Y \longrightarrow X$$

and every point $x \in X$, there is an open neighbourhood

$$x \in U \subseteq X$$

and a continuous section

$$s : U \longrightarrow p^{-1}(U)$$

with $p \circ s = \mathrm{Id}_U$.

> **Edition note — non-empty fibres.** This assertion requires p to be surjective (or, at the chosen point, $p^{-1}(x) \neq \emptyset$). The source definition above does not exclude empty discrete fibres F_i. Read the exercise with this additional hypothesis; no section over x can exist when its fibre is empty.

Exercise 6.4

Let F and H be commutative topological groups, and let

$$G = F \times H$$

be their product group with the product topology. Let

$$0 \longrightarrow F \longrightarrow G \longrightarrow H \longrightarrow 0$$

be the corresponding short exact sequence. Show that, for every topological space X, there is a short exact sequence of sheaves

$$0 \longrightarrow C^0(-, F) \longrightarrow C^0(-, G) \longrightarrow C^0(-, H) \longrightarrow 0$$

whose rightmost map is always surjective on global sections.

Exercise 6.5

Let $\varphi : X \to Y$ be a continuous map, let $Q \in Y$ be a point, and let $\mathcal{F}$ be a presheaf on X. Show that the stalk of the pushforward presheaf $\varphi_* \mathcal{F}$ at Q is equal to

$$\operatorname*{colim}_{\substack{V \subseteq Y,\, V \text{ open} \\ Q \in V}} \mathcal{F}(\varphi^{-1}(V)) = \operatorname*{colim}_{\substack{U \subseteq X,\, U \text{ open} \\ \text{there is an open } V \text{ with } Q \in V \text{ and } \varphi^{-1}(V) \subseteq U}} \mathcal{F}(U).$$

> **Edition note — the colimit index in the source.** The source abbreviates the neighbourhood condition to “there is $Q \in V$”; the intended meaning is that there is an open set $V \ni Q$. This explicit formulation is used above. Both colimits range over open sets; the source leaves this presheaf-domain requirement implicit.

Exercise 6.6

Let $\varphi : X \to Y$ be a continuous map and let $\mathcal{G}$ be a sheaf on Y. Show that the stalk of the pullback sheaf at a point $P \in X$ is equal to the stalk of $\mathcal{G}$ at $\varphi(P)$.

> **Edition note — source grammar.** The source prints the German phrase *einer stetige Abbildung*, with a mismatch between the article and adjective. The translation uses grammatical English without changing the mathematical content.

Exercise 6.7

Let X be a set with two topologies τ_1 and τ_2 such that the identity

$$\varphi : X_1 = (X, \tau_1) \longrightarrow X_2 = (X, \tau_2)$$

is continuous; thus the first topology is finer than the second. Let $\mathcal{F}_1$ be a sheaf on X_1 and let $\mathcal{F}_2$ be a sheaf on X_2. Determine $\varphi_*\mathcal{F}_1$ and $\varphi^{-1}\mathcal{F}_2$. What do they look like when τ_1 is the discrete topology and τ_2 is the indiscrete topology?

Exercise 6.8

Let X be a topological space and let $\varphi : X \to \{P\}$ be the constant map. If $\mathcal{F}$ is a sheaf on X, determine $\varphi_*\mathcal{F}$.

Exercise 6.9

Let X be a topological space, let $P \in X$, and let

$$i : \{P\} \longrightarrow X$$

be the corresponding inclusion. Let $\mathcal{F}$ be a sheaf of commutative groups on $\{P\}$. Describe the sheaf $i_*\mathcal{F}$ on the open sets of X. What do the stalks of $i_*\mathcal{F}$ look like when P is a closed point?

Compare also Exercise 4.9.

Exercise 6.10

Let X be a topological space and let $\varphi : X \to \{P\}$ be the constant map. If $\mathcal{G}$ is a sheaf on $\{P\}$, determine $\varphi^{-1}\mathcal{G}$.

Exercise 6.11

Let $\varphi : X \to Y$ be a continuous map between topological spaces X and Y, and let $\mathcal{F}$ be a sheaf on X. Prove that there is a natural sheaf morphism on X,

$$\varphi^{-1}(\varphi_*\mathcal{F}) \longrightarrow \mathcal{F}.$$

Exercise 6.12

Let $\varphi : X \to Y$ be a continuous map between topological spaces X and Y, and let $\mathcal{G}$ be a sheaf on Y. Prove that there is a natural sheaf morphism on Y,

$$\mathcal{G} \longrightarrow \varphi_*(\varphi^{-1}\mathcal{G}).$$

Exercise 6.13

Let $\varphi : X \to Y$ be a continuous map between topological spaces X and Y. Let $\mathcal{F}$ be a sheaf on X and let $\mathcal{G}$ be a sheaf on Y. Prove that there is a natural bijection between sheaf morphisms on X

$$\psi : \varphi^{-1}\mathcal{G} \longrightarrow \mathcal{F}$$

and sheaf morphisms on Y

$$\theta : \mathcal{G} \longrightarrow \varphi_*\mathcal{F}.$$

Exercise 6.14

Let L_1, L_2, M be sets, and let $p_1 : L_1 \to M$ and $p_2 : L_2 \to M$ be maps. Define

$$L_1 \times_M L_2 := \{(x_1, x_2) \mid p_1(x_1) = p_2(x_2)\} \subseteq L_1 \times L_2.$$

1. Show that there is a commutative diagram

$$\begin{array}{ccc} L_1 \times_M L_2 & \longrightarrow & L_1 \\ \downarrow & & \downarrow \\ L_2 & \longrightarrow & M \end{array}$$

2. Let T be another set, and let $\psi_1 : T \to L_1$ and $\psi_2 : T \to L_2$ be maps with

$$p_1 \circ \psi_1 = p_2 \circ \psi_2.$$

   Show that there is a unique map $\psi : T \to L_1 \times_M L_2$ whose projections to L_1 and L_2 agree with ψ_1 and ψ_2, respectively.

Exercise 6.15

Let L_1, L_2, M be topological spaces, and let $p_1 : L_1 \to M$ and $p_2 : L_2 \to M$ be continuous maps. Define

$$L_1 \times_M L_2 := \{(x_1, x_2) \mid p_1(x_1) = p_2(x_2)\} \subseteq L_1 \times L_2$$

with the induced topology.

1. Show that there is a commutative diagram of continuous maps,

$$\begin{array}{ccc} L_1 \times_M L_2 & \longrightarrow & L_1 \\ \downarrow & & \downarrow \\ L_2 & \longrightarrow & M \end{array}$$

2. Let T be another topological space, and let $\psi_1 : T \to L_1$ and $\psi_2 : T \to L_2$ be continuous maps with

$$p_1 \circ \psi_1 = p_2 \circ \psi_2.$$

   Show that there is a unique continuous map $\psi : T \to L_1 \times_M L_2$ whose projections to L_1 and L_2 agree with ψ_1 and ψ_2, respectively.

   Edition note — source grammar. The source prints the German phrase *eine weiterer topologischer Raum*, with a mismatch between the article and adjective. The translation uses "another topological space" without changing the mathematical content.

   Edition note — fibre-product notation in the source. On some source surfaces, the equality condition is written with the symbols φ_1, φ_2, although the maps just defined are called p_1, p_2. The translation uses p_1, p_2 consistently and preserves the intended mathematical object.

Exercise 6.16

Let X and Y be topological spaces, let $\varphi : Y \to X$ be a continuous map, and let $p : V \to X$ be a vector bundle over X. Prove that

$$Y \times_X V$$

(see Exercise 6.15) is a vector bundle over Y.

Exercise 6.17

Let X be a topological space, and let $p : V \to X$ and $q : W \to X$ be vector bundles over X. Prove that

$$V \times_X W$$

(see Exercise 6.15) is a vector bundle over X that agrees with the direct sum of the vector bundles over X.

Exercise 6.18

Let X, Y, Z be topological spaces, and let $\varphi : Y \to X$ and $p : Z \to X$ be continuous maps. Let

$$p_Y : Y \times_X Z \longrightarrow Y$$

be the natural projection. Show that a continuous section

$$s : Y \longrightarrow Y \times_X Z$$

is the same as a continuous map $t : Y \to Z$ satisfying

$$p \circ t = \varphi.$$

Exercise 6.19

Let X, Y, Z be topological spaces, and let $\varphi : Y \to X$ and $p : Z \to X$ be continuous maps. Let $p_Y : Y \times_X Z \to Y$ be the natural projection. Let $\mathcal{G}$ be the sheaf of continuous sections of p on X. Show that the pullback $\varphi^*\mathcal{G}$ agrees with the sheaf of sections of p_Y.

> **Edition note — meaning of pullback.** The source uses $\varphi^*\mathcal{G}$ here, whereas Definition 6.13 denotes the inverse image of a sheaf by $\varphi^{-1}\mathcal{G}$. If the former is intended to mean the latter, the assertion is false for arbitrary continuous p: for X a point and $Y = Z = \mathbb{R}$, the inverse image consists of locally constant real-valued functions, while sections of p_Y are all continuous real-valued functions. A sufficient additional hypothesis is that p be a local homeomorphism. This qualification is editorial, not an available source solution; the original notation is retained.

Public-Solution Coverage for Worksheet 6

At the frozen revision boundary, the source provides no public solutions for any of the nineteen exercises on Worksheet 6. The candidate-solution query evidence and the frozen exercise map record negative results for Exercises 6.1 through 6.19; all nineteen solution pages have status *missing*.

No new solutions have been created for this edition. This file only documents the source coverage, so that the absence of solutions is not mistaken for content lost during translation. For practice, use the full statements in Worksheet 6 and refer to the corresponding lecture.

Lecture 7: Ringed Spaces and Locally Ringed Spaces

Ringed spaces

Definition 7.1: ringed space

A topological space equipped with a sheaf of commutative rings is called a *ringed space.*

A ringed space is often written in the form

$$(X, \mathcal{O}_X),$$

where X is the underlying space and $\mathcal{O}_X$ is its sheaf of commutative rings. This sheaf is called the *structure sheaf* of the ringed space. The evaluation

$$\Gamma(U, \mathcal{O}_X) = \mathcal{O}_X(U)$$

is also called the *ring of sections* on the open set $U \subseteq X$, and the notation

$$\Gamma(U, \mathcal{O}_X)$$

is called the *ring of global sections* (on U). Following Examples 3.9 and 3.10, we have the following standard examples.

> **Edition note — global sections.** The source repeats $\Gamma(U, \mathcal{O}_X)$ when naming the global ring of sections. This is global on the open subspace U; the global ring of sections of X itself is $\Gamma(X, \mathcal{O}_X)$.

Example 7.2: real-valued continuous functions

Let X be a topological space. For each open set $U \subseteq X$, set

$$\mathcal{C}(U) = C^0(U, \mathbb{R}) = \{f : U \longrightarrow \mathbb{R} \mid f \text{ continuous}\}.$$

This is a commutative ring, and the assignment $U \mapsto \mathcal{C}(U)$, together with the natural restriction maps, is a sheaf. This makes X a ringed space.

Example 7.3: differentiable functions

On a differentiable manifold M, each open set $U \subseteq M$ has the commutative ring

$$C^1(U, \mathbb{R}) = \{f : U \longrightarrow \mathbb{R} \mid f \text{ continuously differentiable}\}.$$

This assignment is a sheaf, making M a ringed space.

Example 7.4: holomorphic functions

On a complex manifold M, for each open set $U \subseteq M$ we have the commutative ring

$$C^1(U, \mathbb{C}) = \{f : U \longrightarrow \mathbb{C} \mid f \text{ holomorphic}\}.$$

This assignment is a sheaf and makes M a ringed space.

> **Edition note — the source notation C^1.** In this example the right-hand side explicitly means holomorphic functions. The source's symbol $C^1(U, \mathbb{C})$ is retained, but does not mean all functions that are merely continuously differentiable in real coordinates.

Example 7.5: a one-point space

Let R be a commutative ring and let $X = \{P\}$ be a topological space with just one point. Setting

$$\Gamma(X, \mathcal{O}_X) := R, \qquad \Gamma(\emptyset, \mathcal{O}_X) := 0$$

makes X a ringed space.

Definition 7.6: stalk of a ringed space

For a point $P \in X$ in a ringed space $(X, \mathcal{O}_X)$, the stalk of the structure sheaf at P is called the *stalk at P*. It is denoted by

$$\mathcal{O}_{X,P} \quad \text{or, for short,} \quad \mathcal{O}_P.$$

Morphisms of ringed spaces

For a continuous map $\varphi : X \to Y$ between topological spaces, each open set $V \subseteq Y$ gives a ring homomorphism

$$\begin{aligned} C^0(V, \mathbb{R}) &\longrightarrow C^0(\varphi^{-1}(V), \mathbb{R}), \\ f &\longmapsto f \circ \varphi. \end{aligned}$$

This pulled-back continuous function is also written $\varphi^* f$. We use the same notation in the following abstract definition.

Definition 7.7: morphism of ringed spaces

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be ringed spaces. A *morphism of ringed spaces* is a continuous map

$$\varphi : X \longrightarrow Y$$

together with a family of ring homomorphisms

$$\varphi_V^* : \Gamma(V, \mathcal{O}_Y) \longrightarrow \Gamma(\varphi^{-1}(V), \mathcal{O}_X)$$

for every open set $V \subseteq Y$, compatible with the restriction maps.

Compatibility means that, for open sets $W \subseteq V \subseteq Y$, the diagram

$$\begin{tikzcd} \Gamma(V, \mathcal{O}_Y) \arrow[r, "\varphi_V^*"] \arrow[d] & \Gamma(\varphi^{-1}(V), \mathcal{O}_X) \arrow[d] \\ \Gamma(W, \mathcal{O}_Y) \arrow[r, "\varphi_W^*"] & \Gamma(\varphi^{-1}(W), \mathcal{O}_X) \end{tikzcd}$$

commutes. A morphism $\varphi : X \to Y$ of ringed spaces induces, for every point $P \in X$, a ring homomorphism on stalks

$$\mathcal{O}_{Y,\varphi(P)} \longrightarrow \mathcal{O}_{X,P}.$$

Explicitly, if $f \in \mathcal{O}_{Y,\varphi(P)}$ is represented by $f \in \Gamma(V, \mathcal{O}_Y)$ on an open neighbourhood $\varphi(P) \in V$, its image is the germ of

$$\varphi^*(f) \in \Gamma(\varphi^{-1}(V), \mathcal{O}_X).$$

Definition 7.8: isomorphism of ringed spaces

A morphism of ringed spaces

$$\varphi : (X, \mathcal{O}_X) \longrightarrow (Y, \mathcal{O}_Y)$$

is called an *isomorphism* if there is a morphism of ringed spaces

$$\psi : (Y, \mathcal{O}_Y) \longrightarrow (X, \mathcal{O}_X)$$

such that

$$\psi \circ \varphi = \mathrm{Id}_X, \qquad \varphi \circ \psi = \mathrm{Id}_Y,$$

where both identities are understood as identities of ringed spaces.

Gluing data for ringed spaces

The following construction extends Lemma 2.6.

Definition 7.9: gluing data

Gluing data for ringed spaces consist of the following.

1. A family of ringed spaces
$$(U_i, \mathcal{O}_{U_i}), \qquad i \in I.$$
2. For each pair (i, j), an open set $U_{ij} \subseteq U_i$, with $U_{ii} = U_i$.
3. For each pair (i, j), an isomorphism of ringed spaces
$$\varphi_{ji} : (U_{ij}, \mathcal{O}_{U_{ij}}) \longrightarrow (U_{ji}, \mathcal{O}_{U_{ji}}),$$
with $\varphi_{ii} = \mathrm{Id}_{(U_i, \mathcal{O}_{U_i})}$.
4. For indices $i, j, k \in I$, the *cocycle condition*
$$\varphi_{kj} \circ \varphi_{ji} = \varphi_{ki}$$
holds as a morphism from $U_{ik} \cap U_{ij}$ to U_k.

Lemma 7.10: existence of the glued space

Suppose gluing data $(U_i, \mathcal{O}_{U_i})_{i \in I}$ for ringed spaces are given. Then there are a ringed space $(X, \mathcal{O}_X)$, an open cover
$$X = \bigcup_{i \in I} V_i,$$
and isomorphisms of ringed spaces
$$\psi_i : U_i \longrightarrow V_i$$
such that
$$\psi_i(U_{ij}) = V_i \cap V_j$$
and
$$\psi_i|_{U_{ij}} = \psi_j|_{U_{ji}} \circ \varphi_{ji}.$$

Proof The underlying space X exists by Lemma 2.6. For an open set $W \subseteq X$, we have the cover
$$W = \bigcup_{i \in I} (W \cap V_i).$$
Define the ring of sections by
$$\Gamma(W, \mathcal{O}_X) := \{(s_i)_{i \in I} \mid s_i \in \Gamma(\psi_i^{-1}(W \cap V_i), \mathcal{O}_{U_i}),$$
$$\varphi_{ij}^*\left(s_i|_{\psi_i^{-1}(W) \cap U_{ij}}\right) = s_j|_{\psi_j^{-1}(W) \cap U_{ji}}\}.$$
This is a sheaf of commutative rings on X which, on V_i, agrees via ψ_i with the given sheaf on U_i.

Edition note — transport of sections. The source writes $\varphi_{ji}(s_i|_{...})$. Since φ_{ji} maps points from U_{ij} to U_{ji}, its action in that direction on sections is the pullback by its inverse φ_{ij}. The formula above writes this as φ_{ij}^*, in accordance with Definition 7.7.

Locally ringed spaces

Definition 7.11: locally ringed space

A ringed space $(X, \mathcal{O}_X)$ is called a *locally ringed space* if, for every point $P \in X$, the stalk $\mathcal{O}_P$ is a local ring.

Example 7.12: continuous functions

A topological space X, together with the sheaf of continuous functions $C^0(-, \mathbb{R})$, is a locally ringed space. For every point $P \in X$ and every continuous function f defined on an open neighbourhood of P,

$$f(P) \neq 0$$

if and only if there is an open neighbourhood on which f is invertible. Consequently, every stalk $\mathcal{O}_P$ is a local ring and X is locally ringed. The same applies to real and complex manifolds.

Edition note — point variable. The source refers here to a neighbourhood of x after introducing P. The translation consistently uses P.

Definition 7.13: residue field

For a locally ringed space $(X, \mathcal{O}_X)$ and a point $P \in X$, the residue field of the local ring $\mathcal{O}_P$ is called the *residue field* of the point P. It is denoted by

$$\kappa(P).$$

The residue field of a topological space equipped with the sheaf of continuous functions is simply $\mathbb{R}$; see Exercise 7.16.

Definition 7.14: evaluation

For a locally ringed space $(X, \mathcal{O}_X)$, a point $x \in X$, and a global function

$$f \in \Gamma(X, \mathcal{O}_X),$$

the value of f in the residue field $\kappa(x)$ is called the *evaluation* of f at x and is denoted by $f(x)$.

In a locally ringed space, for every $f \in \Gamma(X, \mathcal{O}_X)$ and $P \in X$, we have the equivalences

$$f(P) = 0 \text{ in } \kappa(P) \quad \Longleftrightarrow \quad f_P \in \mathfrak{m}_P \quad \Longleftrightarrow \quad f_P \text{ is not a unit in } \mathcal{O}_{X,P}.$$

Edition note — source capitalisation. The source writes “IN einem lokal beringten Raum”. The translation uses normal English capitalisation; the mathematical content and the equivalences are unchanged.

Definition 7.15: morphism of locally ringed spaces

For locally ringed spaces $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$, a *morphism of locally ringed spaces* from X to Y is a morphism of ringed spaces $\varphi : X \to Y$ whose induced ring homomorphism on stalks

$$\varphi_P^* : \mathcal{O}_{Y,\varphi(P)} \longrightarrow \mathcal{O}_{X,P}$$

is a local homomorphism for every point $P \in X$.

The invertibility locus

Lemma 7.16: openness of the invertibility locus

For a locally ringed space $(X, \mathcal{O}_X)$ and a global function $f \in \Gamma(X, \mathcal{O}_X)$, the set

$$X_f := \{P \in X \mid f(P) \neq 0 \text{ in } \kappa(P)\}$$

is open.

Proof First, $f(P) = 0$ in the residue field if and only if $f_P \in \mathfrak{m}_P$ in the local ring $\mathcal{O}_P$, and this holds exactly when f is not invertible in $\mathcal{O}_P$. Take $P \in X_f$. Then f is invertible in $\mathcal{O}_P$, so there is $g \in \mathcal{O}_P$ with

$$gf = 1.$$

There is an open neighbourhood $P \in U \subseteq X$ on which g has a representative

$$g \in \Gamma(U, \mathcal{O}_X),$$

and, possibly after shrinking, an open neighbourhood U' with

$$fg = 1.$$

Thus f is invertible on U' and

$$P \in U' \subseteq X_f.$$

Taking the union of all such open neighbourhoods shows that X_f is open.

In contrast, the set of points at which f, as an element of the stalk $\mathcal{O}_P$, is nonzero need not be open; see Example 11.17.

Definition 7.17: invertibility locus

For a locally ringed space $(X, \mathcal{O}_X)$ and a global function $f \in \Gamma(X, \mathcal{O}_X)$, the set

$$X_f := \{P \in X \mid f(P) \neq 0 \text{ in } \kappa(P)\}$$

is called the *invertibility locus* of f.

By Exercise 7.20, f is a unit in $\Gamma(X_f, \mathcal{O}_X)$.

Lemma 7.18: inverse image of an invertibility locus

Let X and Y be locally ringed spaces, and let $\varphi : X \to Y$ be a morphism of locally ringed spaces. For every

$$f \in \Gamma(Y, \mathcal{O}_Y),$$

we have

$$\varphi^{-1}(Y_f) = X_{\varphi^* f}.$$

Proof The element f is a unit in $\Gamma(Y_f, \mathcal{O}_Y)$. The induced ring homomorphism

$$\Gamma(Y_f, \mathcal{O}_Y) \longrightarrow \Gamma(\varphi^{-1}(Y_f), \mathcal{O}_X)$$

shows that $\varphi^* f$ is a unit in

$$\Gamma(\varphi^{-1}(Y_f), \mathcal{O}_X),$$

so

$$\varphi^{-1}(Y_f) \subseteq X_{\varphi^* f}.$$

Conversely, take $P \in X_{\varphi^* f}$. Then $\varphi^* f$ is a unit in the local ring $\mathcal{O}_{X,P}$. Since the stalk homomorphism

$$\mathcal{O}_{Y,\varphi(P)} \longrightarrow \mathcal{O}_{X,P}$$

is local, $f \in \mathcal{O}_{Y,\varphi(P)}$ must also be a unit. This means $\varphi(P) \in Y_f$, and therefore

$$P \in \varphi^{-1}(Y_f).$$

The two inclusions give the desired equality.

Worksheet 7: Ringed Spaces and Local Rings

Exercise 7.1

Show that every open subset $U \subseteq X$ of a ringed space $(X, \mathcal{O}_X)$ is again a ringed space.

Exercise 7.2

Let $(X, \mathcal{O}_X)$ be a ringed space with

$$\Gamma(X, \mathcal{O}_X) = 0.$$

Show that for every open subset $U \subseteq X$ we also have

$$\Gamma(U, \mathcal{O}_X) = 0.$$

Exercise 7.3

Let X be a topological space equipped with the sheaf of real-valued continuous functions, and let $U \subseteq X$ be a dense open subset. Show that the restriction map

$$\begin{aligned} \Gamma(X, \mathcal{O}_X) &\longrightarrow \Gamma(U, \mathcal{O}_X), \\ f &\longmapsto f\big|_U \end{aligned}$$

is injective.

Exercise 7.4

Let X be a topological space equipped with the sheaf of real-valued continuous functions, and let $U \subseteq X$ be a dense open subset. Show that the restriction map

$$\Gamma(X, \mathcal{O}_X) \longrightarrow \Gamma(U, \mathcal{O}_X), \qquad f \longmapsto f\big|_U,$$

need not be surjective.

> **Edition note — continuity in Exercises 7.3 and 7.4.** Their source prose says only "real-valued functions", while their source titles specify continuous functions. The translation makes continuity explicit; the assertions are not valid for the sheaf of all set-theoretic functions.

Exercise 7.5

Let $(X, \mathcal{O}_X)$ be a ringed space. Show that the assignment taking each open subset $U \subseteq X$ to the unit group

$$(\Gamma(U, \mathcal{O}_X))^\times$$

of the commutative ring $\Gamma(U, \mathcal{O}_X)$, together with the natural restrictions, is a sheaf of commutative groups.

This sheaf has its own name. For a ringed space $(X, \mathcal{O}_X)$, the sheaf defined on open sets $U \subseteq X$ by

$$\Gamma(U, \mathcal{O}_X^\times) := (\Gamma(U, \mathcal{O}_X))^\times$$

is called the *sheaf of units* on X.

Exercise 7.6

Show that the composition of morphisms of ringed spaces is again a morphism of ringed spaces.

Exercise 7.7

Show that, for every open subset $U \subseteq X$ of a ringed space $(X, \mathcal{O}_X)$, there is a morphism of ringed spaces

$$(U, \mathcal{O}_X|_U) \longrightarrow (X, \mathcal{O}_X).$$

Exercise 7.8

Let X and Y be topological spaces, and let $\varphi : X \to Y$ be a continuous map. Show that this induces a morphism of locally ringed spaces.

Exercise 7.9

Let L and M be differentiable manifolds, and let $\varphi : L \to M$ be a differentiable map. Show that this induces a morphism of locally ringed spaces.

Exercise 7.10

Let M be a differentiable manifold. We can make it a ringed space in two ways: using the sheaf of continuous functions $C^0(-, \mathbb{R})$, or using the sheaf of differentiable functions $C^1(-, \mathbb{R})$. Show that there is a morphism of ringed spaces

$$(M, C^0(-, \mathbb{R})) \longrightarrow (M, C^1(-, \mathbb{R}))$$

which is topologically the identity, but is not an isomorphism of ringed spaces.

Edition note — the functions in question. The two sheaves in the source are read as the sheaves of real-valued continuous and differentiable functions on the same open sets; the notation $C^0(-,\mathbb{R})$ and $C^1(-,\mathbb{R})$ is retained. The non-isomorphism assertion requires M to have a positive-dimensional component. For a zero-dimensional manifold the two sheaves coincide; the source omits this exception.

The following exercises focus on local rings.

Exercise 7.11

Let R be a commutative ring. Show that R is a local ring if and only if $a+b$ can be a unit only when a or b is a unit.

Edition note — exclusion of the zero ring. Here assume $R \neq 0$. The source does not state this hypothesis: the zero ring satisfies the displayed unit condition but has no maximal ideal, so is not local in the sense of Exercise 7.12.

Exercise 7.12

Let R be a commutative ring. Show that the following statements are equivalent.

1. R has exactly one maximal ideal.
2. The set of nonunits $R \setminus R^\times$ forms an ideal in R.

Exercise 7.13

Let R be a local ring with residue field K. Show that R and K have the same characteristic if and only if R contains a field.

Exercise 7.14*

Let R be a local ring and let $\mathfrak{a}$ be an ideal of R. Show that the map

$$R^\times \longrightarrow (R/\mathfrak{a})^\times$$

is surjective.

Exercise 7.15

Determine the subrings of the rational numbers $\mathbb{Q}$ that are local.

Exercise 7.16

Let X be a topological space equipped with the sheaf of real-valued continuous functions. Show that the residue field at every point of X is equal to $\mathbb{R}$.

Exercise 7.17

Show that the only field isomorphism

$$\varphi : \mathbb{R} \longrightarrow \mathbb{R}$$

is the identity.

Exercise 7.18

Let X be a topological space equipped with the sheaf of real-valued continuous functions. Regard it as an abstract ringed space: we forget that its elements are functions, but still know the topological space, the rings, and their restriction maps. Can the meaning of the ring elements as functions be reconstructed from these data?

Exercise 7.19

Let X be a topological space equipped with the sheaf of complex-valued continuous functions. Show that the assignment

$$(X, C^0(-,\mathbb{C})) \longrightarrow (X, C^0(-,\mathbb{C})),$$

which is topologically the identity and takes each function on an open set to its complex conjugate, is an automorphism of ringed spaces. Deduce that knowing $(X, C^0(-,\mathbb{C}))$ as an abstract ringed space does not allow one to reconstruct how the ring elements act as functions.

Exercise 7.20

Let $(X, \mathcal{O}_X)$ be a ringed space and let

$$f \in \Gamma(X, \mathcal{O}_X).$$

Show that the following properties are equivalent.

1. f is a unit in $\Gamma(X, \mathcal{O}_X)$.
2. There is an open cover

   $$X = \bigcup_{i \in I} U_i$$

   such that every restriction $f|_{U_i}$ is a unit.
3. The germ $f_P \in \mathcal{O}_{X,P}$ is a unit for every point $P \in X$.

   Edition note — source variable. In the restriction in item (2), the source uses the letter s, although the function introduced is f. The translation writes $f|_{U_i}$ to make the mathematical quantification consistent; no new content is added.

Exercise 7.21

Let $(X, \mathcal{O}_X)$ be a locally ringed space. Show that the assignment

$$\begin{aligned} \Gamma(X, \mathcal{O}_X) &\longrightarrow \tau(X), \\ f &\longmapsto X_f \end{aligned}$$

is a monoid homomorphism from the multiplicative monoid of the ring of global sections to the monoid of open subsets of X, with intersection as the operation.

> **Edition note — underlying space in the source.** The final sentence in the source refers to the monoid of open subsets of M, although the space under consideration is X. The translation corrects the symbol for the underlying space to X; the definition of X_f and the intersection operation remain unchanged.

Public-Solution Coverage for Worksheet 7

At the frozen revision boundary, the source provides exactly one public solution, for Exercise 7.14. The other twenty solution candidates (Exercises 7.1–7.13 and 7.15–7.21) have status *missing* in the query evidence; no new solutions have been created for this edition.

Solution to Exercise 7.14

If $\mathfrak{a} = R$, the quotient ring is the zero ring and the assertion is clear. Thus assume $\mathfrak{a} \subseteq \mathfrak{m}$, where $\mathfrak{m}$ is the unique maximal ideal of the local ring R.

Take $r \in R$ representing a unit in $(R/\mathfrak{a})^\times$, and take $s \in R$ such that

$$rs = 1 \quad \text{in } R/\mathfrak{a}.$$

This means

$$rs - 1 \in \mathfrak{a} \subseteq \mathfrak{m}$$

in R. If r were not a unit, then $rs \in \mathfrak{m}$, and hence

$$1 = (1 - rs) + rs \in \mathfrak{m},$$

a contradiction. Thus r itself is a unit in R, and every unit in $R/\mathfrak{a}$ has a preimage that is a unit. Consequently,

$$R^\times \longrightarrow (R/\mathfrak{a})^\times$$

is surjective.

No public solutions for the other twenty exercises may be supplied or treated as implicit; the exercise map and candidate evidence remain the record of source coverage.

Lecture 8: The Spectrum of a Commutative Ring

So far we have considered ringed spaces whose underlying space was, in a certain sense, given first: an arbitrary topological space, a real manifold, or a complex manifold. From these spaces arose natural sheaves of commutative rings, namely the sheaves of continuous, differentiable, or holomorphic functions. Their individual elements were familiar as functions, but the rings themselves were generally very large and difficult to grasp as a whole.

Conversely, we may ask to what extent every commutative ring can be obtained as the ring of global sections of a ringed space, or whether there is a ringed space that reflects the properties of the ring particularly well and helps us understand it. We shall answer these questions positively in this lecture and the next. The resulting ringed spaces are also the local building blocks of algebraic geometry.

The spectrum of a commutative ring

Definition 8.1: spectrum

For a commutative ring R, the set of all prime ideals of R is called the *spectrum* of R and is denoted by

$$\text{Spek}(R).$$

It is also called an *affine scheme*.

Definition 8.2: Zariski topology

On the spectrum of a commutative ring R, the *Zariski topology* is defined by declaring the sets

$$D(T) := \{\mathfrak{p} \in \text{Spek}(R) \mid T \nsubseteq \mathfrak{p}\}$$

to be open for every subset $T \subseteq R$.

For a one-element subset $T = \{f\}$, we write $D(f)$ instead of $D(\{f\})$.

Lemma 8.3: the Zariski topology is indeed a topology

The Zariski topology on the spectrum $\operatorname{Spek}(R)$ of a commutative ring R is indeed a topology.

Proof We have

$$D(0) = \emptyset, \qquad D(1) = \operatorname{Spek}(R),$$

since every prime ideal contains 0 and no prime ideal contains 1.

For an arbitrary family of subsets $T_i \subseteq R$, $i \in I$, we have

$$\bigcup_{i \in I} D(T_i) = D\left(\bigcup_{i \in I} T_i\right).$$

The inclusion from left to right is clear, since $T_i \subseteq \bigcup_{i \in I} T_i$ and $S \subseteq T$ always implies $D(S) \subseteq D(T)$. For the reverse inclusion, take

$$\mathfrak{p} \in D\left(\bigcup_{i \in I} T_i\right).$$

There is $f \in \bigcup_{i \in I} T_i$ with $f \notin \mathfrak{p}$. Thus there is $i \in I$ with $f \in T_i$, and consequently $\mathfrak{p} \in D(T_i)$.

For a finite family $T_1, \dots, T_n \subseteq R$, we have

$$\bigcap_{i=1}^{n} D(T_i) = D(T_1 \cdots T_n),$$

where $T_1 \cdots T_n$ is the set of all products $f_1 \cdots f_n$ with $f_i \in T_i$. The inclusion from right to left is clear. For the reverse inclusion, suppose $\mathfrak{p} \in D(T_i)$ for every $i = 1, \dots, n$. Then there are $f_i \in T_i$ with $f_i \notin \mathfrak{p}$. Since $\mathfrak{p}$ is prime, $f_1 \cdots f_n \notin \mathfrak{p}$, so $\mathfrak{p} \in D(T_1 \cdots T_n)$.

We always regard the spectrum as a topological space. The prime ideals are the points of this space. To emphasise the geometric viewpoint, we often write

$$X = \operatorname{Spek}(R), \qquad x \in X,$$

and denote the prime ideal represented by x by $\mathfrak{p}_x$.

The complements of the open sets, that is, the closed sets in the Zariski topology, are denoted by

$$V(T) = \{\mathfrak{p} \in \operatorname{Spek}(R) \mid T \subseteq \mathfrak{p}\}.$$

Proposition 8.4: first properties of the Zariski topology

For the spectrum $X = \operatorname{Spek}(R)$ of a commutative ring R, the following properties hold.

1. $D(T) = D(\mathfrak{a})$, where $\mathfrak{a}$ is the ideal generated by T (or its radical). Thus, to describe the open sets, it suffices to consider the radical ideals of R.
2. For a family of ideals $\mathfrak{a}_i \subseteq R$, $i \in I$,

$$\bigcup_{i \in I} D(\mathfrak{a}_i) = D\left(\sum_{i \in I} \mathfrak{a}_i\right).$$

3. For a finite family of ideals $\mathfrak{a}_i \subseteq R$, $i = 1, \dots, n$,

$$\bigcap_{i=1}^{n} D(\mathfrak{a}_i) = D\left(\bigcap_{i=1}^{n} \mathfrak{a}_i\right) = D(\mathfrak{a}_1 \cdots \mathfrak{a}_n).$$

4. $D(\mathfrak{a}) = X$ if and only if $\mathfrak{a}$ is the unit ideal.
5. $D(\mathfrak{a}) \subseteq D(\mathfrak{b})$ if and only if $\mathfrak{a} \subseteq \mathrm{rad}(\mathfrak{b})$.
6. The spectrum is empty if and only if R is the zero ring.
7. $D(\mathfrak{a}) = \emptyset$ if and only if $\mathfrak{a}$ contains only nilpotent elements.
8. The open sets $D(f)$, $f \in R$, form a basis for the topology.
9. A family of open sets $D(\mathfrak{a}_i)$, $i \in I$, covers X if and only if the ideals $\mathfrak{a}_i$ together generate the unit ideal.

Proof

(1) The inclusion $D(T) \subseteq D(\mathfrak{a})$ is clear. For the reverse inclusion, argue by contraposition and suppose $\mathfrak{p} \notin D(T)$. Then $T \subseteq \mathfrak{p}$, and hence

$$\mathfrak{a} \subseteq \mathrm{rad}(\mathfrak{a}) \subseteq \mathfrak{p},$$

since a prime ideal is radical. Thus $\mathfrak{p} \notin D(\mathrm{rad}(\mathfrak{a}))$.

(2) and (3) follow from (1) and the proof of Lemma 8.3.

(3) If $\mathfrak{a}$ is not the unit ideal, then by Exercise 8.1 there is a maximal ideal $\mathfrak{m}$ with $\mathfrak{a} \subseteq \mathfrak{m}$. Consequently, $\mathfrak{m} \notin D(\mathfrak{a})$.

(4) The implication from right to left is clear. For the converse, suppose

$$\mathfrak{a} \not\subseteq \mathrm{rad}(\mathfrak{b}).$$

Then there is $f \in \mathfrak{a}$ with $f^n \notin \mathfrak{b}$ for every $n \in \mathbb{N}$. Applying Exercise 8.5 to the multiplicative system $\{f^n \mid n \in \mathbb{N}\}$ gives a prime ideal $\mathfrak{p} \supseteq \mathfrak{b}$ with $f \notin \mathfrak{p}$. Thus $\mathfrak{p} \in D(\mathfrak{a})$ but $\mathfrak{p} \notin D(\mathfrak{b})$.

Edition note — missing source reference number. The source TeX witness displays ***** at this reference. The frozen HTML links instead to an exercise titled “the radical is the intersection of prime ideals”, whose course reference-number page is missing. Worksheet Exercise 8.5 proves exactly the existence statement required here when applied to $M = \{f^n \mid n \in \mathbb{N}\}$; the internal reference above is therefore an explicit editorial application, not a recovered source number.

(6) The zero ring has no prime ideals. Every nonzero commutative ring has a maximal ideal by Exercise 8.1.

(7) Every prime ideal contains all nilpotent elements, so $V(\mathfrak{a}) = X$ for such an ideal. Conversely, if $\mathfrak{a}$ contains a nonnilpotent element f, then by Exercise 8.5 there is a prime ideal $\mathfrak{p}$ with $f \notin \mathfrak{p}$. Hence $\mathfrak{p} \in D(f) \subseteq D(\mathfrak{a})$.

(8) This follows directly from

$$D(\mathfrak{a}) = \bigcup_{f \in \mathfrak{a}} D(f).$$

(9) follows from (2) and (4).

Edition note — duplicated word in the source. The last sentence of the source proof reads the equivalent of “follows from and (2) and (4)”. The duplicated conjunction is normalised; the mathematical references remain (2) and (4).

Proposition 8.5: closure in the spectrum

For the spectrum $X = \mathrm{Spek}(R)$ of a commutative ring R:

1. the closure of a subset $T \subseteq X$ is

$$V\left(\bigcap_{x \in T} \mathfrak{p}_x\right);$$

2. the closure of a point $x \in X$ is $V(\mathfrak{p}_x)$;

3. a point $x \in \text{Spek}(R)$ is closed if and only if $\mathfrak{p}_x$ is a maximal ideal.

Proof

(1) For $y \in T$, we have

$$y \in V(\mathfrak{p}_y) \subseteq V\left(\bigcap_{x \in T} \mathfrak{p}_x\right),$$

so the set on the right is a closed set containing T. Take a prime ideal $\mathfrak{q}$ with

$$\mathfrak{q} \in V\left(\bigcap_{x \in T} \mathfrak{p}_x\right), \qquad \bigcap_{x \in T} \mathfrak{p}_x \subseteq \mathfrak{q}.$$

To show that $\mathfrak{q}$ belongs to the closure of T, it suffices to show that T meets every open neighbourhood of $\mathfrak{q}$. Suppose $\mathfrak{q} \in D(f)$, that is, $f \notin \mathfrak{q}$. Then $f \notin \bigcap_{x \in T} \mathfrak{p}_x$, so there is $x \in T$ with $f \notin \mathfrak{p}_x$. Thus $\mathfrak{p}_x \in D(f)$ and $T \cap D(f) \neq \emptyset$.

(2) is a special case of (1), and (3) follows from (2).

Corollary 8.6: spectra are quasi-compact

The spectrum $X = \text{Spek}(R)$ of every commutative ring R is quasi-compact.

Proof By Proposition 8.4(9),

$$X = \bigcup_{i \in I} D(\mathfrak{a}_i)$$

if and only if the ideals $\mathfrak{a}_i$, $i \in I$, together generate the unit ideal. The ideal generated by this family consists of all finite sums $f_1 + \cdots + f_n$ with $f_j \in \mathfrak{a}_{i_j}$. Thus, if the unit ideal is generated, there are a finite selection $\{i_1, \ldots, i_n\} \subseteq I$ and elements $f_j \in \mathfrak{a}_{i_j}$ with

$$\sum_{j=1}^{n} f_j = 1.$$

Consequently,

$$X = D(1) = \bigcup_{j=1}^{n} D(f_j) = \bigcup_{j=1}^{n} D(\mathfrak{a}_{i_j}),$$

giving a finite subcover.

A spectrum is Hausdorff only in special circumstances. In general, two points of a spectrum cannot be separated by open neighbourhoods.

Example 8.7: the spectrum of a field

A field has only two ideals: the unit ideal K, which is not prime, and the zero ideal 0, which is prime. Thus the spectrum of a field consists of a single point.

Example 8.8: the spectrum of the integers

The prime ideals of $\mathbb{Z}$ are the maximal ideals (p), where p is a prime number, together with the zero ideal 0. The maximal ideals form the closed points of Spek($\mathbb{Z}$). The zero ideal is an additional, nonclosed point. The only closed set containing this point is the whole space. Apart from the whole space, the closed sets of Spek($\mathbb{Z}$) are the finite subsets of maximal ideals.

We picture Spek($\mathbb{Z}$) as an imagined line: the prime numbers lie discretely along it, while the zero ideal is drawn as a thick point representing the whole line.

> **Source illustration unavailable.** The source declares `File:Spektrum_von_Z._xcf`, but the official Commons API returns it as *missing*, the source HTML displays broken media, and the official PDF contains no image binary. The complete source caption reads: "This is how one imagines the spectrum of $\mathbb{Z}$. The connecting lines are meant to convey that it is a one-dimensional object. The zero ideal is drawn in bold to indicate that it is a dense point." The edition preserves this caption and its accessible descriptive meaning without claiming to have recovered the image. The source names Bocardodarapti as the creator and gives the licence label CC-by-sa 4.0; these declarations are retained independently of the unavailable image binary.

Example 8.9: the spectrum of a polynomial ring

For the polynomial ring

$$R = K[X_1, \dots, X_n]$$

over a field K, the so-called point ideals give a useful geometric picture of Spek(R). A point ideal has the form

$$(X_1 - a_1, X_2 - a_2, \dots, X_n - a_n)$$

for a fixed tuple $a = (a_1, a_2, \dots, a_n) \in K^n$. This ideal is the kernel of the K-algebra homomorphism

$$\begin{aligned} \varphi_a : R &\longrightarrow K, \\ X_i &\longmapsto a_i, \end{aligned}$$

and is therefore maximal. This assignment defines an injective map

$$K^n \longrightarrow \text{Spek}(R).$$

If K is algebraically closed, this map even accounts for all maximal ideals of R. We therefore picture the spectrum of the polynomial ring in n variables as affine space, but it also contains additional, nonclosed points that are harder to visualise. For a polynomial $f \in K[X_1, \dots, X_n]$, the set $V(f) \cap K^n$ has a concrete interpretation:

$$a \in V(f) \cap K^n \quad \Longleftrightarrow \quad f(a_1, \dots, a_n) = 0.$$

Functorial properties

Proposition 8.10: functoriality of the spectrum

Let

$$\varphi : R \longrightarrow S$$

be a ring homomorphism between commutative rings. Then:

1. the assignment

$$\begin{aligned} \varphi^* : \text{Spek}(S) &\longrightarrow \text{Spek}(R), \\ \mathfrak{p} &\longmapsto \varphi^*(\mathfrak{p}) := \varphi^{-1}(\mathfrak{p}) \end{aligned}$$

is well-defined and continuous;

2. for every ideal $\mathfrak{a} \subseteq R$,

$$(\varphi^*)^{-1}(D(\mathfrak{a})) = D(\mathfrak{a}S);$$

3. for another ring homomorphism $\psi : S \to T$,

$$(\psi \circ \varphi)^* = \varphi^* \circ \psi^*.$$

Proof By Exercise 8.9, the map is well-defined. To prove continuity, it suffices to prove (2). We argue using closed sets. For a prime ideal $\mathfrak{q} \in \text{Spek}(S)$, we have

$$\varphi^*(\mathfrak{q}) \in V(\mathfrak{a})$$

if and only if $\mathfrak{a} \subseteq \varphi^{-1}(\mathfrak{q})$. This is equivalent to $\varphi(\mathfrak{a}) \subseteq \mathfrak{q}$, and also to $\mathfrak{a}S \subseteq \mathfrak{q}$. Statement (3) is immediate.

The continuous map introduced above is called the *map on spectra* associated with the given ring homomorphism. For a subring $R \subseteq S$, it is simply

$$\mathfrak{p} \longmapsto \mathfrak{p} \cap R,$$

also called *contraction* of a prime ideal.

Proposition 8.11: closed and open subsets

Let R be a commutative ring. Then:

1. for an ideal $\mathfrak{a} \subseteq R$ and the quotient map

$$q : R \longrightarrow R/\mathfrak{a},$$

the map on spectra

$$q^* : \text{Spek}(R/\mathfrak{a}) \longrightarrow \text{Spek}(R)$$

is a closed embedding with image $V(\mathfrak{a})$;

2. for a multiplicative system $M \subseteq R$, the map associated with the canonical map

$$\iota : R \longrightarrow R_M$$

is an injective map

$$\iota^* : \text{Spek}(R_M) \longrightarrow \text{Spek}(R),$$

whose image consists of the prime ideals of R disjoint from M;

3. for $f \in R$, the map associated with

$$\iota : R \longrightarrow R_f$$

is an open embedding

$$\iota^* : \text{Spek}(R_f) \longrightarrow \text{Spek}(R)$$

with image $D(f)$.

Proof

(1) follows from Exercise 8.6. The prime ideals of $R/\mathfrak{a}$ correspond to the prime ideals of R containing $\mathfrak{a}$ via

$$\mathfrak{p} \longmapsto q^{-1}(\mathfrak{p}).$$

Thus the map is bijective onto the stated image. For an ideal $\mathfrak{b} \subseteq R/\mathfrak{a}$ and a prime ideal $\mathfrak{p} \subseteq R/\mathfrak{a}$, we have $\mathfrak{b} \subseteq \mathfrak{p}$ if and only if, under the quotient correspondence,

$$q^{-1}(\mathfrak{b}) \subseteq q^{-1}(\mathfrak{p})$$

in R. Thus the image of $V(\mathfrak{b})$ is $V(q^{-1}(\mathfrak{b}))$, which is closed.

Edition note — quotient correspondence notation. The source writes $q^{-1}(\mathfrak{p}) = \mathfrak{p} + \mathfrak{a}$ and likewise uses $\mathfrak{b} + \mathfrak{a}$, although $\mathfrak{p}$ and $\mathfrak{b}$ here are ideals of $R/\mathfrak{a}$, while $\mathfrak{a}$ is an ideal of R. The inverse-image notation above states the same correspondence without adding ideals that belong to different rings.

(2) See Exercise 8.7.

(3) For a prime ideal $\mathfrak{p}$ and an element $f \in R$, $f \notin \mathfrak{p}$ holds if and only if $\mathfrak{p}$ is disjoint from the multiplicative system

$$\{f^n \mid n \in \mathbb{N}\}.$$

By (2), the map is injective with image $D(f)$. The same argument, applied to $g \in R$ and $g/1 \in R_f$, shows that the image of

$$D(g) \subseteq \text{Spek}(R_f)$$

is $D(fg)$, and is therefore open.

Lemma 8.12: fibres of the map on spectra

Let $\varphi : R \to S$ be a ring homomorphism between commutative rings, and let

$$\begin{aligned} \varphi^* : \text{Spek}(S) &\longrightarrow \text{Spek}(R), \\ \mathfrak{p} &\longmapsto \varphi^*(\mathfrak{p}) \end{aligned}$$

be the associated map on spectra. Its fibre over a prime ideal $\mathfrak{q} \in \text{Spek}(R)$ is

$$\text{Spek}\Big((S/\mathfrak{q}S)_{\varphi(R \setminus \mathfrak{q})}\Big)\,.$$

In other words, this fibre consists of all prime ideals $\mathfrak{p} \in \text{Spek}(S)$ satisfying

$$\mathfrak{q}S \subseteq \mathfrak{p}, \qquad \mathfrak{p} \cap \varphi(R \setminus \mathfrak{q}) = \emptyset.$$

Proof By Proposition 8.11, it suffices to prove the second formulation. For a prime ideal $\mathfrak{p} \subseteq S$, we have

$$\varphi^{-1}(\mathfrak{p}) = \mathfrak{q}$$

if and only if both

$$\varphi(\mathfrak{q}) \subseteq \mathfrak{p} \quad \text{and} \quad \varphi(R \setminus \mathfrak{q}) \subseteq S \setminus \mathfrak{p}.$$

The first condition is equivalent to $\mathfrak{q}S \subseteq \mathfrak{p}$, and the second is equivalent to

$$\varphi(R \setminus \mathfrak{q}) \cap \mathfrak{p} = \emptyset.$$

In particular, the fibre of a map on spectra over a point is itself the spectrum of a ring. If $\mathfrak{m}$ is maximal, its fibre is

$$\mathrm{Spek}(S/\mathfrak{m}S),$$

since $\mathfrak{m}S \subseteq \mathfrak{p}$ immediately gives $\mathfrak{m} \subseteq \varphi^{-1}(\mathfrak{p})$, and maximality forces equality. If R is an integral domain and the point is the zero ideal, there is no need to consider the extension ideal; the fibre is simply described by

$$\mathrm{Spek}\left(S_{\varphi(R\setminus\{0\})}\right).$$

Corollary 8.13: criterion for an empty fibre

Let $\varphi : R \to S$ be a ring homomorphism between commutative rings, and let $\varphi^* : \mathrm{Spek}(S) \to \mathrm{Spek}(R)$ be the associated map on spectra. The fibre over a prime ideal $\mathfrak{q} \in \mathrm{Spek}(R)$ is empty if and only if

$$\mathfrak{q}S \cap \varphi(R \setminus \mathfrak{q}) \neq \emptyset.$$

Proof This follows from Lemma 8.12 and Proposition 8.4(6).

Worksheet 8: Spectra and Maps on Spectra

Exercise 8.1

Let R be a nonzero commutative ring. Use Zorn's lemma to show that R has a maximal ideal.

Exercise 8.2

Show that every maximal ideal $\mathfrak{m}$ in a commutative ring R is a prime ideal.

Exercise 8.3*

Let R be a commutative ring and let $\mathfrak{p}$ be an ideal. Show that $\mathfrak{p}$ is prime if and only if the quotient ring

$$R/\mathfrak{p}$$

is an integral domain.

Exercise 8.4*

Let $\mathfrak{a}$ be an ideal in a commutative ring R. Show that $\mathfrak{a}$ is prime if and only if it is the kernel of a ring homomorphism

$$\varphi : R \longrightarrow K$$

to a field K.

Exercise 8.5

Let R be a commutative ring, let $\mathfrak{a} \subseteq R$ be an ideal, and let $M \subseteq R$ be a multiplicative system with

$$\mathfrak{a} \cap M = \emptyset.$$

Use Zorn's lemma to show that there is a prime ideal $\mathfrak{p}$ satisfying

$$\mathfrak{a} \subseteq \mathfrak{p}, \qquad \mathfrak{p} \cap M = \emptyset.$$

Exercise 8.6

Let R be a commutative ring, let $\mathfrak{a}$ be an ideal, and let

$$S = R/\mathfrak{a}.$$

Show that the ideals of S correspond bijectively to the ideals of R containing $\mathfrak{a}$.

Exercise 8.7

Let R be a commutative ring and let $S \subseteq R$ be a multiplicative system. Show that the prime ideals in R_S correspond exactly to the prime ideals in R disjoint from S.

Exercise 8.8

Describe the spectrum

$$\text{Spek}(R_{\mathfrak{p}})$$

of the localisation of a commutative ring R at a prime ideal $\mathfrak{p}$.

Exercise 8.9

Let R and S be commutative rings and let $\varphi : R \to S$ be a ring homomorphism. If $\mathfrak{p}$ is a prime ideal in S, show that the inverse image

$$\varphi^{-1}(\mathfrak{p})$$

is a prime ideal in R.

Give an example showing that the inverse image of a maximal ideal need not be maximal.

Exercise 8.10

Let $\varphi : R \to S$ be a ring homomorphism between commutative rings R and S, and let $\mathfrak{p} \in \text{Spek}(S)$ be a prime ideal. Show that there are natural ring homomorphisms

$$R_{\varphi^{-1}(\mathfrak{p})} \longrightarrow S_{\mathfrak{p}}$$

between the localisations, and

$$\kappa(\varphi^{-1}(\mathfrak{p})) \longrightarrow \kappa(\mathfrak{p})$$

between the residue fields.

Exercise 8.11*

Let K be a field, and let R and S be finitely generated K-algebras that are integral domains. Let

$$\varphi : R \longrightarrow S$$

be a K-algebra homomorphism, and let $\mathfrak{n}$ be a maximal ideal in S with

$$\varphi^{-1}(\mathfrak{n}) = \mathfrak{m}.$$

Suppose the map induces an isomorphism

$$R_{\mathfrak{m}} \longrightarrow S_{\mathfrak{n}}.$$

Show that there is $f \in R$ with $f \notin \mathfrak{m}$ such that

$$R_f \longrightarrow S_{\varphi(f)}$$

is an isomorphism.

Exercise 8.12

Show that the map on spectra associated with the reduction

$$R \longrightarrow R/\mathfrak{n}_R$$

of a commutative ring R is a homeomorphism.

Exercise 8.13

Let R be a commutative ring containing a field of positive characteristic

$$p > 0,$$

where p is prime. Show that the map

$$\begin{aligned} R &\longrightarrow R, \\ f &\longmapsto f^p \end{aligned}$$

is a ring homomorphism, called the *Frobenius homomorphism.*

Exercise 8.14

Let R be a commutative ring of positive characteristic $p > 0$. Show that the map on spectra associated with the Frobenius homomorphism

$$\begin{aligned} R &\longrightarrow R, \\ f &\longmapsto f^p \end{aligned}$$

is a homeomorphism.

> **Edition note — characteristic hypothesis.** Here p must be prime, as in Exercise 8.13. A unital ring can have composite positive characteristic, for which $f \mapsto f^p$ need not be a ring homomorphism. The source leaves the word "prime" implicit in this exercise.

Exercise 8.15

Let R_1 and R_2 be commutative rings and let

$$R = R_1 \times R_2$$

be their product ring. Show that there is a natural homeomorphism

$$\text{Spek}(R_1) \uplus \text{Spek}(R_2) \longrightarrow \text{Spek}(R_1 \times R_2).$$

Exercise 8.16

Let R be a commutative ring. Determine the fibres of the map on spectra associated with the ring extension

$$R \subseteq R[X_1, \dots, X_n].$$

Exercise 8.17

Determine the fibres of the map on spectra associated with

$$\mathbb{R}[X] \subseteq \mathbb{C}[X].$$

When the ground field is the complex numbers, the $\mathbb{C}$-spectrum also has a complex topology, which is much finer than the Zariski topology. The following exercises develop this.

Exercise 8.18

Let R be a finitely generated commutative $\mathbb{C}$-algebra. Show that the $\mathbb{C}$-spectrum

$$\mathbb{C}\text{-Spek}(R)$$

has a *natural topology* (or *complex topology*) that, for the polynomial ring $\mathbb{C}[X_1, \dots, X_n]$, agrees with the metric topology on $\mathbb{C}^n$. Show also that, for a $\mathbb{C}$-algebra homomorphism

$$\varphi : R \longrightarrow S$$

between finitely generated $\mathbb{C}$-algebras, the induced map

$$\mathbb{C}\text{-Spek}(S) \longrightarrow \mathbb{C}\text{-Spek}(R)$$

is continuous in the natural topology.

Exercise 8.19

Let $P \in \mathbb{C}[X]$ be a nonconstant polynomial. Show that the function

$$\begin{aligned} \mathbb{C} &\longrightarrow \mathbb{C}, \\ z &\longmapsto P(z) \end{aligned}$$

has the property that the inverse image of every bounded subset $T \subseteq \mathbb{C}$ is bounded.

Exercise 8.20

Let

$$F_1, \dots, F_k \in \mathbb{C}[X_1, \dots, X_n]$$

be polynomials such that the $\mathbb{C}$-algebra homomorphism

$$\begin{aligned} \mathbb{C}[Y_1, \dots, Y_k] &\longrightarrow \mathbb{C}[X_1, \dots, X_n], \\ Y_j &\longmapsto F_j \end{aligned}$$

is integral. Show that the associated map

$$\begin{aligned} \mathbb{C}^n &\longrightarrow \mathbb{C}^k, \\ (x_1, \dots, x_n) &\longmapsto (F_1(x_1, \dots, x_n), \dots, F_k(x_1, \dots, x_n)) \end{aligned}$$

has the property that the inverse image of every bounded subset $T \subseteq \mathbb{C}^k$ is again bounded.

Deduce that, in this situation, the map F is proper, meaning that inverse images of compact subsets are compact, and that F is a closed map.

Exercise 8.21

Determine the fibres of the map on spectra associated with

$$\mathbb{Q}[X] \subseteq \mathbb{R}[X].$$

Which fibres are finite?

Exercise 8.22

Let $\varphi : R \to S$ be a ring homomorphism between commutative rings, and let

$$\begin{aligned} \varphi^* : \mathrm{Spek}(S) &\longrightarrow \mathrm{Spek}(R), \\ \mathfrak{p} &\longmapsto \varphi^*(\mathfrak{p}) \end{aligned}$$

be the associated map on spectra. Show that the fibre over a prime ideal $\mathfrak{p} \in \mathrm{Spek}(R)$ is canonically homeomorphic to

$$\mathrm{Spec}(S \otimes_R \kappa(\mathfrak{p}))\,.$$

Edition note — source notation. The final exercise uses Spec, whereas the lecture and preceding exercises use Spek. This source difference is preserved; both symbols denote the spectrum of a ring.

Public Solutions and Coverage for Worksheet 8

At the frozen revision boundary, the source provides exactly three public solutions among the 22 exercises on Worksheet 8: those for Exercises 8.3, 8.4, and 8.11. The frozen exercise map and candidate evidence record negative results for Exercises 8.1-8.2, 8.5-8.10, and 8.12-8.22. No new solutions have been created for this edition.

Source solution to Exercise 8.3

First, let $\mathfrak{p}$ be a prime ideal. In particular,

$$\mathfrak{p} \subsetneq R,$$

so the quotient ring $R/\mathfrak{p}$ is not the zero ring. Suppose $fg = 0$ in $R/\mathfrak{p}$, where f and g are represented by elements of R. Then $fg \in \mathfrak{p}$, so $f \in \mathfrak{p}$ or $g \in \mathfrak{p}$. In $R/\mathfrak{p}$, this means precisely that $f = 0$ or $g = 0$.

Conversely, suppose $R/\mathfrak{p}$ is an integral domain. It is not the zero ring, so $\mathfrak{p} \neq R$. Take $f, g \notin \mathfrak{p}$. Then $f, g \neq 0$ in $R/\mathfrak{p}$ and, since this ring is an integral domain,

$$fg \neq 0$$

in $R/\mathfrak{p}$. Thus $fg \notin \mathfrak{p}$, proving that $\mathfrak{p}$ is prime.

Source solution to Exercise 8.4

First, let $\mathfrak{a}$ be a prime ideal. Then $R/\mathfrak{a}$ is an integral domain, so its field of fractions

$$Q(R/\mathfrak{a})$$

exists. The canonical projection followed by inclusion into the field of fractions,

$$\begin{aligned} \varphi : R &\longrightarrow Q(R/\mathfrak{a}), \\ x &\longmapsto [x], \end{aligned}$$

is therefore a ring homomorphism to a field with

$$\ker \varphi = \mathfrak{a}.$$

Conversely, the kernel of a ring homomorphism

$$\varphi : R \longrightarrow K$$

is always an ideal. If $ab \in \ker \varphi$, then

$$0 = \varphi(ab) = \varphi(a)\varphi(b).$$

Since the field K has no zero divisors, we obtain $\varphi(a) = 0$ or $\varphi(b) = 0$. This is equivalent to $a \in \ker \varphi$ or $b \in \ker \varphi$. Thus $\ker \varphi$ is a prime ideal.

Source solution to Exercise 8.11

We first show that, for a suitable $f \in R$, the map

$$R_f \longrightarrow S_{\varphi(f)}$$

is surjective. Take a set of K-algebra generators $x_1, \dots, x_n$ for S. Since the local map in the hypothesis is surjective, there are elements

$$y_i = \frac{r_i}{g_i}, \qquad g_i \notin \mathfrak{m},$$

with $\varphi(y_i) = x_i$ in $S_\mathfrak{n}$. This means $y_i g_i = r_i$ for $i = 1, \dots, n$. With

$$f = g_1 \cdots g_n,$$

all the y_i can be written over the common denominator f, so $y_i \in R_f$. The map $R_f \to S_{\varphi(f)}$ is surjective because a set of generators lies in its image and the denominators $\varphi(f)^n$ are the images of f^n.

We claim that this map is also injective. Suppose $q \in R_f$ maps to zero. Then q is also zero in $S_\mathfrak{n}$ and comes from $q \in R_\mathfrak{m}$. Since the local map is an isomorphism, $q = 0$ in $R_\mathfrak{m}$. Since R is an integral domain by hypothesis, this also gives $q = 0$ in R_f. Thus the map is injective and hence an isomorphism.

Lecture 9: Affine Schemes

Let

$$X = \text{Spek}(R)$$

be the spectrum of a commutative ring R, equipped with the Zariski topology. If R is a field, its spectrum consists of just one point, the zero ideal, which is also maximal. Viewed this way, the spectrum alone contains very little information. Thus the contravariant functor

$$\begin{aligned} \{\text{commutative rings}\} &\longrightarrow \{\text{topological spaces}\}, \\ R &\longmapsto \text{Spek}(R) \end{aligned}$$

loses information. We shall enrich the spectrum with additional structure so that the original ring can be reconstructed from it. To do this, we define a structure sheaf on the spectrum. The spectrum together with this structure sheaf is a meaningful geometrisation of the ring: a ringed space.

> **Edition note — two typos in the source introduction.** The source writes `dass Spektrum` and `zu sätztlichen`. The translation normalises these to “the spectrum” and “additional”; no mathematical content is changed.

The structure sheaf on the spectrum

Example 9.1: the presheaf of localisations

Let $X = \text{Spek}(R)$ be the spectrum of a commutative ring R. On X, define a presheaf of commutative rings by setting, for every open set $U \subseteq X$,

$$\mathcal{P}(U) := \operatorname*{colim}_{U \subseteq D(f)} R_f,$$

with the natural ring homomorphisms

$$R_f \longrightarrow R_g$$

for

$$U \subseteq D(g) \subseteq D(f).$$

Together with these natural homomorphisms, this assignment is a presheaf. We have

$$\mathcal{P}(D(f)) = R_f, \qquad \mathcal{P}(X) = R.$$

For the second equality, the directed system has the terminal object $X = D(1)$, so the resulting ring is $R_1 = R$.

> **Edition note — terminal object in the source.** When deducing $\mathcal{P}(X) = R$, the source writes the terminal object as `D(f)`. Here it is written explicitly as $X = D(1)$; the colimit construction is unchanged.

The stalk of this presheaf at a point $\mathfrak{p} \in X$ is

$$\begin{aligned} \operatorname*{colim}_{\mathfrak{p} \in U} \mathcal{P}(U) &= \operatorname*{colim}_{\mathfrak{p} \in D(f)} \mathcal{P}(D(f)) \\ &= \operatorname*{colim}_{f \notin \mathfrak{p}} R_f \\ &= R_{\mathfrak{p}}. \end{aligned}$$

This presheaf is not a sheaf. Its sheafification is the structure sheaf on the spectrum.

Definition 9.2: structure sheaf

Let $X = \operatorname{Spek}(R)$ be the spectrum of a commutative ring R. The *structure sheaf* on X is the assignment taking each open set $U \subseteq X$ to the commutative ring

$$\Gamma(U, \mathcal{O}_X) = \left\{ (s_{\mathfrak{p}})_{\mathfrak{p} \in U} \in \prod_{\mathfrak{p} \in U} R_{\mathfrak{p}} \;\middle|\; \begin{array}{l} \text{for every } \mathfrak{p} \in U \text{ there are } a, b \in R \text{ with} \\ \mathfrak{p} \in D(b) \subseteq U \text{ and } s_{\mathfrak{q}} = \dfrac{a}{b} \text{ in } R_{\mathfrak{q}} \\ \text{for every } \mathfrak{q} \in D(b) \end{array} \right\}.$$

For each inclusion $U \subseteq U'$, the restriction homomorphism is the natural projection from the family indexed by U' to the family indexed by U.

Lemma 9.3: the structure sheaf is indeed a sheaf

The structure sheaf $\mathcal{O}_X$ on the spectrum $X = \operatorname{Spek}(R)$ of a commutative ring R is a sheaf of commutative rings.

Proof The definition above is precisely the sheafification of the presheaf in Example 9.1. The only difference in presentation is that the compatibility condition is formulated using basic neighbourhoods $D(b)$ instead of arbitrary open neighbourhoods.

Definition 9.4: affine scheme

The spectrum

$$X = \operatorname{Spek}(R)$$

of a commutative ring R, together with its structure sheaf $\mathcal{O}_X$, is called the *affine scheme* associated with R.

An element

$$q \in \Gamma(U, \mathcal{O}_X)$$

is called an algebraic function defined on U. The terms *rational function* and *regular function* are also used in this context.

Sections as local functions

Remark 9.5: the case of an integral domain

If R is an integral domain, its structure sheaf has a particularly simple description. For an open set $U \subseteq X$,

$$\Gamma(U, \mathcal{O}_X) = \bigcap_{\mathfrak{p} \in U} R_{\mathfrak{p}},$$

where the intersection is taken in the field of fractions $Q(R)$, in which all the localisations $R_{\mathfrak{p}}$ are subrings. Thus the functions on U are precisely the rational elements of $Q(R)$ defined at every point of U. By Lemma 12.4 in the Commutative Algebra course,

$$\Gamma(X, \mathcal{O}_X) = \bigcap_{\mathfrak{p} \in X} R_{\mathfrak{p}} = R.$$

Similarly,

$$\Gamma(D(f), \mathcal{O}_X) = \bigcap_{\mathfrak{p} \in D(f)} R_{\mathfrak{p}} = R_f.$$

If there is an open cover

$$U = \bigcup_{i \in I} D(f_i),$$

then

$$\Gamma(U, \mathcal{O}_X) = \bigcap_{\mathfrak{p} \in U} R_{\mathfrak{p}} = \bigcap_{i \in I} \left(\bigcap_{\mathfrak{p} \in D(f_i)} R_{\mathfrak{p}} \right) = \bigcap_{i \in I} R_{f_i}.$$

Edition note — the empty open set. The intersection description in this remark assumes $U \neq \emptyset$. For $U = \emptyset$, the sheaf assigns the zero ring, whereas the set-theoretic intersection of the empty family of subrings of $Q(R)$ would be $Q(R)$. The source does not state this exception.

Edition note — cross-reference numbers in different witnesses. The frozen semantic witness refers to Lemma 12.4 in Commutative Algebra, whereas the older terminal PDF prints Lemma 16.4. The edition follows the authoritative semantic revision and records the difference without conflating the witnesses.

Example 9.6: a function on two punctured lines

Consider

$$R = K[X, Y]/(XY)$$

over a field K. On the open set

$$U = D(X, Y) = D(X) \cup D(Y) = \mathrm{Spec}(R) \setminus \{(X, Y)\},$$

the function that takes the value 0 on the punctured line

$$V(X) \cap U = D(Y)$$

and the value 1 on the punctured line

$$V(Y) \cap U = D(X)$$

is an algebraic function. This assignment specifies an element $s_{\mathfrak{p}} \in R_{\mathfrak{p}}$ for every prime ideal $\mathfrak{p} \in U$. If

$$\mathfrak{p} \in V(X) \cap U = D(Y),$$

its fractional representation is

$$0 = \frac{0}{Y};$$

whereas if

$$\mathfrak{p} \in V(Y) \cap U = D(X),$$

its representation is

$$1 = \frac{X}{X}.$$

The source changes from the operator Spek to Spec in this display; both forms are retained as source notation for the spectrum of a ring.

Remark 9.7: denominator ideal

Let R be an integral domain with field of fractions $Q(R)$, and let $q \in Q(R)$ be a rational function. There is a largest open set $U \subseteq \operatorname{Spek}(R)$ on which q is defined. It is

$$U = D(\mathfrak{a}),$$

where the *denominator ideal* is

$$\mathfrak{a} = \{r \in R \mid rq \in R\}.$$

If $q \in R_{\mathfrak{p}}$, then

$$q = \frac{s}{r}$$

with $r \notin \mathfrak{p}$. Since r belongs to the denominator ideal, we obtain $\mathfrak{p} \in D(\mathfrak{a})$. Reading this argument backwards gives the converse implication. In particular, $D(f)$ is the largest domain of definition of $1/f$.

Theorem 9.8: unique factorisation domains

Let R be a unique factorisation domain. Then, for open sets $U \subseteq \operatorname{Spek}(R)$, the assignment

$$U \longmapsto \operatorname*{colim}_{U \subseteq D(f)} R_f$$

agrees with the structure sheaf on $\operatorname{Spek}(R)$.

Proof This assignment is a presheaf of commutative rings whose sheafification is the structure sheaf. It therefore suffices to prove that, in the unique factorisation case, the presheaf is already a sheaf.

Take a nonzero element

$$q \in \Gamma(U, \mathcal{O}_X) \subseteq Q(R).$$

Since R is a unique factorisation domain, there is a reduced representation

$$q = \frac{a}{f}.$$

We claim that $U \subseteq D(f)$. Take $\mathfrak{p} \in U$. Since q is defined on U, Remark 9.5 gives a representation

$$q = \frac{b}{g} = \frac{a}{f}$$

with $g \notin \mathfrak{p}$. Thus

$$fb = ag$$

in R. Every prime factor of f divides ag but not a, so it must divide g. Hence the radical of (f) contains the radical of (g), and

$$\mathfrak{p} \in D(g) \subseteq D(f).$$

Thus the section q already comes from R_f on a principal open set containing U, as required.

This applies in particular to polynomial rings, and hence to affine space.

Example 9.9: a section that appears only after sheafification

Consider the integral domain

$$R = K[X, Y, Z, W]/(WX - ZY)$$

over a field K, and set

$$U := D(X, Y) \subseteq \mathrm{Spek}(R).$$

By Remark 9.5,

$$q = \frac{Z}{X} = \frac{W}{Y}$$

is an algebraic function defined on U, so $q \in \Gamma(U, \mathcal{O}_X)$. However, apart from units, there is no element $f \in R$ with

$$(X, Y) \subseteq (f),$$

since X and Y are irreducible. Consequently, q is not a section over U of the presheaf in Example 9.1, but it is a section of its sheafification.

> **Edition note — symbol for the open set in the source.** The source introduces $D(X, Y)$ as an open set, then uses the symbol U without defining the equality. The edition writes $U := D(X, Y)$ explicitly; no new mathematical object is added.

Local properties and principal open sets

Lemma 9.10: stalks are localisations

Let $(X, \mathcal{O}_X)$ be the affine scheme associated with a commutative ring R, and let $x \in X$ be the point corresponding to a prime ideal $\mathfrak{p}$. Then the stalk of the structure sheaf is

$$\mathcal{O}_x = R_{\mathfrak{p}}.$$

Proof This follows from Example 9.1 and Lemma 5.2(2).

Corollary 9.11: affine schemes are locally ringed

Every affine scheme is a locally ringed space.

Proof This follows immediately from Lemma 9.10 and Theorem 12.3 in the Commutative Algebra course.

> **Edition note — cross-reference numbers in different witnesses.** The semantic witness refers to Theorem 12.3 in Commutative Algebra, whereas the older terminal PDF prints Theorem 16.3. As in Remark 9.5, the edition follows the semantic authority.

Lemma 9.12: sections on principal open sets

Let $(X, \mathcal{O}_X)$ be the affine scheme associated with a commutative ring R, and let $f \in R$. Then

$$\Gamma(D(f), \mathcal{O}_X) = R_f.$$

In particular,

$$\Gamma(X, \mathcal{O}_X) = R.$$

Proof We first prove the special case of X. There is a natural ring homomorphism

$$R \longrightarrow \Gamma(X, \mathcal{O}_X).$$

It is injective because whether an element is zero can be checked locally; compare Appendix Lemma 1.1. To prove surjectivity, take $q \in \Gamma(X, \mathcal{O}_X)$. There are an open cover

$$X = \bigcup_{i \in I} U_i = \bigcup_{i \in I} D(f_i)$$

and elements

$$q_i = \frac{a_i}{f_i^{k_i}}$$

which agree as sections on

$$D(f_i) \cap D(f_j) = D(f_i f_j),$$

that is, as elements of $R_{f_i f_j}$. By Corollary 8.6, we may assume that I is finite. We may also replace all k_i by their maximum k; of course, this changes the local numerators a_i as well.

The compatibility

$$\frac{a_i}{f_i^k} = \frac{a_j}{f_j^k}$$

means that there are equations

$$(f_i f_j)^m a_i f_j^k = (f_i f_j)^m a_j f_i^k$$

in R, where m is chosen as a maximum valid for all pairs. By Proposition 8.4(2),(4), the elements f_i, $i \in I$, generate the unit ideal. The same holds for the f_i^{m+k}, so there are $g_i \in R$ with

$$1 = \sum_{i \in I} g_i f_i^{m+k}.$$

Set

$$a := \sum_{i \in I} g_i a_i f_i^m.$$

Then

$$\begin{aligned} a f_j^{m+k} &= \left(\sum_{i \in I} g_i a_i f_i^m \right) f_j^{m+k} \\ &= \sum_{i \in I} g_i (f_i f_j)^m a_i f_j^k \\ &= \sum_{i \in I} g_i (f_i f_j)^m a_j f_i^k \\ &= a_j f_j^m \left(\sum_{i \in I} g_i f_i^{m+k} \right) \\ &= a_j f_j^m. \end{aligned}$$

Consequently,

$$a = \frac{a_j}{f_j^k} = q_j$$

in R_{f_j}. Thus the section is represented by a single ring element $a \in R$.

The situation on $D(f)$ is the same case with R_f taken as the new ring. Hence $\Gamma(D(f), \mathcal{O}_X) = R_f$.

Lemma 9.13: principal open sets are affine schemes

Let $(X, \mathcal{O}_X)$ be the affine scheme associated with a commutative ring R, and let $f \in R$. Then, via the canonical map on spectra,

$$D(f) = \mathrm{Spec}(R_f)$$

as ringed spaces.

Proof By Proposition 8.11(3), the canonical ring homomorphism

$$R \longrightarrow R_f$$

induces an open embedding

$$\mathrm{Spek}(R_f) \longrightarrow D(f) \subseteq \mathrm{Spek}(R).$$

Edition note (serialization). The missing backslash in the arrow command is restored according to the frozen TeX witness. This corrects command spelling, not the mathematical claim.

By Lemma 9.12, the ring of sections on both sides is R_f. The same holds for every open set $D(g) \subseteq D(f)$. Thus the structure sheaves on both sides are identified, giving an isomorphism of ringed spaces.

> **Edition note — equality via canonical identification.** The source statement writes $D(f) = \mathrm{Spec}(R_f)$, while its proof constructs an open embedding and an isomorphism of ringed spaces. The edition retains the source display and explicitly states that the equality uses the canonical identification. The alternation between `Spek`/`Spec` is also retained.

Worksheet 9: Affine Schemes

None of the 13 exercises has a public solution at the frozen revision boundary. No solution asterisks are therefore used, and this edition creates no new solutions.

Exercise 9.1

Determine the subrings of $\mathbb{Q}$ that occur as rings of sections on $\mathrm{Spek}(\mathbb{Z})$, and those that occur as stalks on $\mathrm{Spek}(\mathbb{Z})$.

Exercise 9.2

Let $T \subseteq \mathbb{P}$ be a subset of the prime numbers. Prove that

$$R_T = \{q \in \mathbb{Q} \mid q \text{ can be written with a denominator containing only primes from } T\}$$

is a subring of $\mathbb{Q}$. What do we obtain for

$$T = \emptyset, \qquad T = \{3\}, \qquad T = \{2, 5\}, \qquad T = \mathbb{P}?$$

Exercise 9.3

Let

$$R = \mathbb{Z}\left[\frac{2}{3}\right]$$

be the subring of $\mathbb{Q}$ generated by $\mathbb{Z}$ and $2/3$. Prove that R contains all rational numbers that can be written with a power of 3 as denominator.

Exercise 9.4

Let R be a commutative ring and let $f \in R$. For the associated localisation R_f, prove the R-algebra isomorphism

$$R_f \cong R[T]/(Tf - 1).$$

Exercise 9.5

Let R be a commutative ring and let $f, g \in R$. Prove that the following properties are equivalent.

1. $D(f) \subseteq D(g)$ in the spectrum of R.
2. $\mathrm{rad}((f)) \subseteq \mathrm{rad}((g))$.
3. $f \in \mathrm{rad}((g))$.
4. There is $n \in \mathbb{N}$ with $f^n \in (g)$.
5. The element g divides a power of f.
6. The element g is a unit in R_f.

7. There is an R-algebra homomorphism

$$R_g \longrightarrow R_f.$$

Exercise 9.6

Let R be a commutative ring, let $f \in R$, and let R_f be the associated localisation. Prove that f is nilpotent if and only if R_f is the zero ring.

Exercise 9.7

Let R be an integral domain and let

$$U \subseteq X = \mathrm{Spek}(R)$$

be an open set. Prove that

$$\Gamma(U, \mathcal{O}_X) = \bigcap_{\substack{f \neq 0 \\ D(f) \subseteq U}} R_f,$$

where the intersection is taken in the field of fractions $Q(R)$.

> **Edition note — nonempty open set.** This assertion requires $U \neq \emptyset$. If $U = \emptyset$, the left-hand side is the zero ring, while no nonzero f in an integral domain has $D(f) \subseteq \emptyset$, so the displayed set-theoretic intersection is the empty-family intersection in $Q(R)$. The source omits this exception.

Exercise 9.8

Let R be a principal ideal domain with field of fractions $Q = Q(R)$. Prove that every intermediate ring

$$R \subseteq S \subseteq Q$$

is a localisation of R.

Exercise 9.9

Prove that the open set U in Example 9.6 satisfies

$$U = D(X + Y).$$

Describe the function considered there using the denominator $X + Y$.

> **Edition note — elliptical predicate in the source.** The source sentence literally ends with the equivalent of "prove that for the open set ... holds". The following display is $U = D(X + Y)$; the edition makes it the predicate of the sentence without changing the task.

Exercise 9.10

Let R be a unique factorisation domain and let $\mathfrak{m}$ be a maximal ideal of height

$$\mathrm{ht}(\mathfrak{m}) \geq 2.$$

Prove that the restriction map

$$R = \Gamma(X, \mathcal{O}_X) \longrightarrow \Gamma(X \setminus \{\mathfrak{m}\}, \mathcal{O}_X)$$

is bijective.

Edition note — implicit spectrum. The source uses X in the restriction map without defining it in this exercise. Here $X = \mathrm{Spek}(R)$, as in the lecture; the task is otherwise unchanged.

Exercise 9.11

For

$$R = K[X, Y, Z]/(XY - Z^n)$$

and

$$U = D(X, Z) \subseteq \mathrm{Spek}(R),$$

find rational functions defined on U that cannot be written with a single optimal denominator.

Edition note — exponent range. The source does not specify n. The intended A-type singularity and the requested phenomenon require $n \geq 2$. For $n = 1$, one has $R \cong K[X, Y]$ and $U = D(X)$, so the requested counterexample does not exist. Read this exercise with $n \geq 2$.

Exercise 9.12

Let

$$X = \mathrm{Spek}(R)$$

be the spectrum of a commutative ring R, and let

$$f \in R = \Gamma(X, \mathcal{O}_X).$$

Prove that $D(f)$ agrees with the invertibility locus X_f.

Exercise 9.13

Let

$$\varphi : R \longrightarrow S$$

be a ring homomorphism between commutative rings. Prove that the map on spectra

$$\varphi^* : \mathrm{Spek}(S) \longrightarrow \mathrm{Spek}(R)$$

can naturally be made into a morphism of locally ringed spaces.

Public-Solution Coverage for Worksheet 9

At the frozen revision boundary, all thirteen candidate solution pages (Exercises 9.1–9.13) have status *missing* in the official query evidence. There are therefore no public solutions to translate for this unit, and the edition neither creates nor implies new solutions.

The ordered exercise map and candidate evidence remain the record of source coverage.

Lecture 10: Schemes and Scheme Morphisms

Schemes

Definition 10.1: scheme

A *scheme* is a ringed space $(X, \mathcal{O}_X)$ for which there is an open cover

$$X = \bigcup_{i \in I} U_i$$

such that, for every i,

$$(U_i, \mathcal{O}_X|_{U_i})$$

is an affine scheme.

Lemma 10.2: affine neighbourhoods inside open neighbourhoods

Let $(X, \mathcal{O}_X)$ be a scheme and let $P \in X$ be a point. For every open neighbourhood $P \in U$, there is an affine open neighbourhood

$$P \in V \subseteq U.$$

Proof Take an affine open neighbourhood

$$P \in W = \mathrm{Spec}(R).$$

Then

$$P \in U \cap W \subseteq \mathrm{Spek}(R)$$

is an open subset of Spek(R), and therefore has the form

$$U \cap W = D(\mathfrak{a})$$

for an ideal $\mathfrak{a} \subseteq R$. Since

$$D(\mathfrak{a}) = \bigcup_{f \in \mathfrak{a}} D(f),$$

there is $f \in \mathfrak{a}$ with

$$P \in D(f) \subseteq D(\mathfrak{a}) \subseteq U.$$

By Lemma 9.13, $D(f)$ is affine.

> **Edition note — source operator notation.** The proof writes $W = \mathrm{Spec}(R)$ and then returns to Spek(R). Both spellings of the operator are retained and refer to the same ring spectrum.

Lemma 10.3: an open subset of a scheme is a scheme

Every open subset $U \subseteq X$ of a scheme $(X, \mathcal{O}_X)$ has a cover by affine open sets and is therefore itself a scheme.

Proof As an open subset of a ringed space, U is also a ringed space. The existence of an affine cover follows immediately from Lemma 10.2.

Definition 10.4: quasi-affine scheme

An open subset

$$U \subseteq X = \mathrm{Spek}(R)$$

of an affine scheme X is called a *quasi-affine scheme*.

Definition 10.5: punctured spectrum

For a local ring $(R, \mathfrak{m})$, the space

$$\mathrm{Spek}(R) \setminus \{\mathfrak{m}\}$$

is called the *punctured spectrum* of R.

As ringed spaces, schemes can be glued along open subsets as in Lemma 7.10. Here are two examples.

Example 10.6: the line with a doubled point

Take two copies of the affine line,

$$U = \mathbb{A}^1_K = \mathrm{Spek}(K[S]), \qquad V = \mathbb{A}^1_K = \mathrm{Spek}(K[T]),$$

with the open subsets

$$U' = \mathbb{A}^1_K \setminus \{(S)\} = \mathrm{Spek}(K[S, S^{-1}]) \subset \mathbb{A}^1_K$$

and

$$V' = \mathbb{A}^1_K \setminus \{(T)\} = \mathrm{Spek}(K[T, T^{-1}]) \subset \mathbb{A}^1_K.$$

Consider the isomorphism

$$\varphi : U' \longrightarrow V'$$

determined by $S \mapsto T$, and glue U and V in the sense of Lemma 7.10. The resulting space is a scheme X called the line with a doubled point. Denote the points of X determined by (S) and (T) by P and Q, respectively.

There is a commutative diagram of restriction homomorphisms

$$\begin{array}{ccc} \Gamma(X, \mathcal{O}_X) & \longrightarrow & \Gamma(U, \mathcal{O}_X) = K[S] \\ \downarrow & & \downarrow \\ \Gamma(V, \mathcal{O}_X) = K[S] & \longrightarrow & \Gamma(U', \mathcal{O}_X) = K[S, S^{-1}], \end{array}$$

where we have made the identification $S = T$. The sheaf condition gives

$$\Gamma(X, \mathcal{O}_X) = K[S],$$

and global functions have the same value at P and Q. A similar argument shows that the stalks also agree:

$$\mathcal{O}_{X,P} = \mathcal{O}_{X,Q} = K[S]_{(S)}.$$

The entire calculation takes place in the function field $K(S)$.

Example 10.7: the projective line by gluing

Again take two copies of the affine line

$$U = \mathbb{A}^1_K = \text{Spek}(K[S]), \qquad V = \mathbb{A}^1_K = \text{Spek}(K[T])$$

with the punctured open subsets

$$U' = \mathbb{A}^1_K \setminus \{(S)\} = \text{Spek}(K[S, S^{-1}]) \subset \mathbb{A}^1_K$$

and

$$V' = \mathbb{A}^1_K \setminus \{(T)\} = \text{Spek}(K[T, T^{-1}]) \subset \mathbb{A}^1_K.$$

Now use the isomorphism

$$\varphi : U' \longrightarrow V', \qquad S \longmapsto T^{-1},$$

to glue U and V in the sense of Lemma 7.10. The resulting space,

$$X = \mathbb{P}^1_K,$$

is a model of the projective line over K. Denote the points determined by (S) and (T) by P and Q, respectively. For $K = \mathbb{R}$ or $K = \mathbb{C}$ with the metric topology, a sequence in U' converging to $P \in U$ necessarily tends to infinity when viewed in V.

The commutative diagram of restriction homomorphisms is

$$\begin{array}{ccc} \Gamma(\mathbb{P}^1_K, \mathcal{O}_{\mathbb{P}^1_K}) & \longrightarrow & \Gamma(U, \mathcal{O}_{\mathbb{P}^1_K}) = K[S] \\ \downarrow & & \downarrow \\ \Gamma(V, \mathcal{O}_{\mathbb{P}^1_K}) = K[T] = K[S^{-1}] & \longrightarrow & \Gamma(U', \mathcal{O}_{\mathbb{P}^1_K}) = K[S, S^{-1}], \end{array}$$

with the identification $S = T^{-1}$. The sheaf condition gives

$$\Gamma(X, \mathcal{O}_X) = K,$$

since only the constant functions belong to both $K[S]$ and $K[S^{-1}]$; the intersection is taken in the function field $K(S)$. Moreover,

$$\mathcal{O}_{X,P} = K[S]_{(S)}, \qquad \mathcal{O}_{X,Q} = K[S^{-1}]_{(S^{-1})}.$$

Scheme morphisms

Definition 10.8: scheme morphism

A *scheme morphism*

$$\varphi : X \longrightarrow Y$$

between schemes X and Y is a morphism of locally ringed spaces.

We first want to make the map on spectra associated with a ring homomorphism $\theta : R \to S$,

$$\text{Spek}(S) \longrightarrow \text{Spek}(R),$$

into a scheme morphism. This is a special case of the following theorem.

Theorem 10.9: morphisms to an affine scheme

Let $(X, \mathcal{O}_X)$ be a locally ringed space and let $Y = \text{Spek}(R)$ be an affine scheme. For every ring homomorphism

$$\theta : R \longrightarrow \Gamma(X, \mathcal{O}_X),$$

there is a unique morphism of locally ringed spaces $X \to Y$ whose homomorphism on global sections is θ.

Proof By Lemma 7.18, for every $x \in X$ we must have

$$\varphi(x) = \{f \in R \mid x \notin X_{\theta(f)}\} = (\rho_x \circ \theta)^{-1}(\mathfrak{m}_x),$$

where

$$\rho_x : \Gamma(X, \mathcal{O}_X) \longrightarrow \mathcal{O}_{X,x}$$

is the restriction homomorphism to the stalk $\mathcal{O}_{X,x}$ and $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$ is its maximal ideal. This formula determines a continuous map, since

$$\varphi^{-1}(D(f)) = X_{\theta(f)};$$

the sets $D(f)$ form a basis by Proposition 8.4(8), and $X_{\theta(f)}$ is open by Lemma 7.16.

For every $f \in R$, there are ring homomorphisms

$$R \xrightarrow{\theta} \Gamma(X, \mathcal{O}_X) \longrightarrow \Gamma(X_{\theta(f)}, \mathcal{O}_X),$$

and $\theta(f)$ becomes a unit in the rightmost ring. By Theorem 11.13 in the Commutative Algebra course, there is a unique homomorphism

$$R_f \longrightarrow \Gamma(X_{\theta(f)}, \mathcal{O}_X)$$

compatible with these homomorphisms. By the sheaf property, for every open set $D(\mathfrak{a})$ we also obtain a unique homomorphism

$$\Gamma(D(\mathfrak{a}), \mathcal{O}_Y) \longrightarrow \Gamma(\varphi^{-1}(D(\mathfrak{a})), \mathcal{O}_X).$$

Indeed, if

$$D(\mathfrak{a}) = \bigcup_{i \in I} D(f_i),$$

then

$$\Gamma(D(\mathfrak{a}), \mathcal{O}_Y) = \left\{ (s_i)_{i \in I} \in \prod_{i \in I} R_{f_i} \;\middle|\; s_i = s_j \text{ in } R_{f_i f_j} \right\}$$

and

$$\Gamma(\varphi^{-1}(D(\mathfrak{a})), \mathcal{O}_X) = \left\{ (t_i)_{i \in I} \in \prod_{i \in I} \Gamma(X_{\theta(f_i)}, \mathcal{O}_X) \;\middle|\; t_i = t_j \text{ in } \Gamma(X_{\theta(f_i f_j)}, \mathcal{O}_X) \right\}.$$

The homomorphisms already defined on R_{f_i} and $R_{f_i f_j}$ respect the compatibility equations, and therefore give a homomorphism from the ring in the first display to the ring in the second. These assignments do indeed yield a morphism of locally ringed spaces.

> **Edition note — two source surfaces.** In the map ρ_x, the semantic witness writes $\Gamma(X, \mathcal{O})$, while the context and surrounding maps use $\mathcal{O}_X$; the edition writes the subscript X explicitly. The semantic witness also refers to Theorem 11.13 in Commutative Algebra, whereas the older official PDF prints Theorem 15.13. The edition follows the semantic authority's numbering and records the difference without conflating the editions of the cited course.

Corollary 10.10: ring homomorphisms give morphisms of spectra

Let R and S be commutative rings, and let $\theta : R \to S$ be a ring homomorphism. There is a unique scheme morphism

$$\text{Spek}(S) \longrightarrow \text{Spek}(R)$$

whose homomorphism on global sections is θ. Topologically, this is the map on spectra.

Proof This follows immediately from Theorem 10.9. The beginning of its proof shows that the underlying topological map is the map on spectra.

Corollary 10.11: the canonical morphism to the spectrum of the integers

For every locally ringed space $(X, \mathcal{O}_X)$, there is a canonical morphism of locally ringed spaces

$$X \longrightarrow \text{Spek}(\mathbb{Z}).$$

It sends a point $x \in X$ to the characteristic of its residue field $\kappa(x)$.

> **Edition clarification — the target point.** A point of Spek($\mathbb{Z}$) is a prime ideal. Thus "the characteristic" here means $\ker(\mathbb{Z} \to \kappa(x))$: it is (0) in characteristic zero and (p) in characteristic $p > 0$.

Proof The canonical ring homomorphism

$$\mathbb{Z} \longrightarrow \Gamma(X, \mathcal{O}_X)$$

determines a unique morphism of locally ringed spaces

$$X \longrightarrow \text{Spek}(\mathbb{Z})$$

by Theorem 10.9.

Corollary 10.12: global functions give morphisms to the affine line

Let $(X, \mathcal{O}_X)$ be a locally ringed space. Every global function

$$f \in \Gamma(X, \mathcal{O}_X)$$

determines a unique morphism of locally ringed spaces

$$X \longrightarrow \mathbb{A}^1_{\mathbb{Z}},$$

which sends the variable of the affine line to f. If $\Gamma(X, \mathcal{O}_X)$ is a K-algebra over a field K, the function f also determines a morphism of locally ringed spaces

$$X \longrightarrow \mathbb{A}^1_K.$$

In this case, a point $x \in X$ is sent to the kernel of the ring homomorphism

$$K[T] \longrightarrow \kappa(x), \qquad T \longmapsto f(x).$$

Proof The ring element $f \in \Gamma(X, \mathcal{O}_X)$ determines a unique substitution homomorphism

$$\mathbb{Z}[T] \longrightarrow \Gamma(X, \mathcal{O}_X).$$

By Theorem 10.9, this homomorphism determines a unique morphism of locally ringed spaces

$$(X, \mathcal{O}_X) \longrightarrow \operatorname{Spek}(\mathbb{Z}[T]) = \mathbb{A}^1_{\mathbb{Z}}.$$

The additional assertion follows in the same way.

Corollary 10.13: tuples of functions give morphisms to affine space

Let $(X, \mathcal{O}_X)$ be a locally ringed space. Every tuple of functions

$$f_1, \dots, f_n \in \Gamma(X, \mathcal{O}_X)$$

determines a unique morphism of locally ringed spaces

$$X \longrightarrow \mathbb{A}^n_{\mathbb{Z}},$$

which sends the variable T_i of affine space to f_i. If $\Gamma(X, \mathcal{O}_X)$ is an R-algebra over a commutative ring R, the functions $f_1, \dots, f_n$ also determine a morphism of locally ringed spaces

$$X \longrightarrow \mathbb{A}^n_R.$$

In this case, a point $x \in X$ is sent to the kernel of the ring homomorphism

$$R[T_1, \dots, T_n] \longrightarrow \kappa(x), \qquad T_i \longmapsto f_i(x).$$

Proof See Exercise 10.3.

Thus a morphism to affine space is nothing other than a tuple of global functions.

If $\varphi : X \to Y$ is a morphism, then for every open subset $V \subseteq Y$, the induced map

$$\varphi^{-1}(V) \longrightarrow V$$

is also a morphism. If V is moreover affine, then by Theorem 10.9 this morphism is given locally on Y by a ring homomorphism. This means that, using an affine cover

$$Y = \bigcup_{i \in I} V_i = \bigcup_{i \in I} \operatorname{Spek}(R_i),$$

the scheme morphism $\varphi : X \to Y$ is essentially determined by the ring homomorphisms

$$R_i \longrightarrow \Gamma(\varphi^{-1}(V_i), \mathcal{O}_X).$$

Schemes over a base scheme

For a commutative K-algebra A over a field K, the canonical ring homomorphism $K \to A$ determines a canonical map on spectra

$$\operatorname{Spek}(A) \longrightarrow \operatorname{Spek}(K).$$

Topologically this is simply the constant map, but it still specifies how the constants from K are to be interpreted. In the context of schemes, the role of a ground ring is taken by a base scheme.

Definition 10.14: scheme over a base

A scheme X together with a fixed morphism

$$p : X \longrightarrow S$$

to another scheme S is called a *scheme over S*. The scheme S is called the *base scheme*.

Often the base scheme is simply the spectrum of a field. By Corollary 10.11, every scheme is uniquely a scheme over Spek($\mathbb{Z}$). A scheme over Spek(R) is also called a scheme over R. The role of algebra homomorphisms is taken by morphisms compatible with the base.

Definition 10.15: scheme morphism over a base

Let X and Y be schemes over a base scheme S. A scheme morphism

$$\varphi : X \longrightarrow Y$$

is called a *scheme morphism over S* if the diagram

$$\begin{tikzcd} X \arrow[r, "\varphi"] \arrow[dr] & Y \arrow[d] \\ & S \end{tikzcd}$$

commutes.

Definition 10.16: morphism of finite type

A scheme morphism

$$\varphi : X \longrightarrow Y$$

is called *of finite type* if there is an affine open cover

$$Y = \bigcup_{i \in I} V_i$$

such that, for every $i \in I$, there is a finite affine cover

$$\varphi^{-1}(V_i) = \bigcup_{j \in J_i} U_{ij}$$

and, for every $j \in J_i$, the ring homomorphism

$$\Gamma(V_i, \mathcal{O}_Y) \longrightarrow \Gamma(U_{ij}, \mathcal{O}_X)$$

is of finite type.

> **Edition note — conflicting indices in the source.** The source writes the cover $Y = \bigcup_{i \in I} V_i$, but then uses $\varphi^{-1}(V_i) = \bigcup_{i \in I_j} U_i$, the condition $i \in I_j$, and a map from $\Gamma(V_j, \mathcal{O}_Y)$. The edition explicitly replaces these dummy indices by $j \in J_i$ and U_{ij}, keeping the base open set V_i fixed; the mathematical definition is not expanded.

Immersions

Definition 10.17: open immersion

A scheme morphism $f : Y \to X$ is called an *open immersion* if f induces an isomorphism onto an open subset of X.

Definition 10.18: closed immersion

A scheme morphism $f : Y \to X$ is called a *closed immersion* if $f(Y)$ is a closed subset of X, there is a homeomorphism

$$Y \longrightarrow f(Y),$$

and the associated sheaf homomorphism

$$\mathcal{O}_X \longrightarrow f_*\mathcal{O}_Y$$

is surjective.

Definition 10.19: immersion

A scheme morphism $f : Y \to X$ is called an *immersion* if there is a factorisation

$$Y \xrightarrow{g} Z \xrightarrow{h} X$$

with g an open immersion and h a closed immersion.

Worksheet 10: Schemes and Scheme Morphisms

None of the six exercises has a public solution at the frozen revision boundary. No solution asterisks are therefore used, and this edition creates no new solutions.

Exercise 10.1

Give an example of a quasi-affine scheme that is not affine.

Exercise 10.2

Give an example of a quasi-affine scheme that is not quasi-compact.

Exercise 10.3

Let $(X, \mathcal{O}_X)$ be a locally ringed space. Prove that every tuple of functions

$$f_1, \dots, f_n \in \Gamma(X, \mathcal{O}_X)$$

determines a unique morphism of locally ringed spaces

$$X \longrightarrow \mathbb{A}^n_{\mathbb{Z}},$$

which sends the variable T_i of affine space to f_i.

Exercise 10.4

Let $(X, \mathcal{O}_X)$ be a scheme. Prove that X is affine if and only if the canonical morphism

$$X \longrightarrow \text{Spek}(\Gamma(X, \mathcal{O}_X))$$

is an isomorphism.

Exercise 10.5

Let X be a differentiable manifold. Prove that the canonical morphism

$$X \longrightarrow \mathrm{Spek}(C^1(X, \mathbb{R}))$$

is injective.

Exercise 10.6

Let R be a commutative ring, and let A, B be commutative R-algebras. Prove that an R-algebra homomorphism

$$\varphi : A \longrightarrow B$$

is the same data as a scheme morphism

$$\psi : \mathrm{Spek}(B) \longrightarrow \mathrm{Spek}(A)$$

over $\mathrm{Spek}(R)$.

Public-Solution Coverage for Worksheet 10

At the frozen revision boundary, all six candidate solution pages (Exercises 10.1-10.6) have status *missing* in the official query evidence. There are therefore no public solutions to translate for this unit, and the edition neither creates nor implies new solutions.

The ordered exercise map and candidate evidence remain the record of source coverage.

Lecture 11: Irreducible Spaces and Noetherian Schemes

Compared with a metric space, a scheme has rather unusual topological properties, which we shall introduce here. We begin with irreducibility.

Irreducible spaces

Definition 11.1: irreducible space

A topological space V is called *irreducible* if $V \neq \emptyset$ and there is no decomposition

$$V = Y \cup Z$$

with both $Y, Z \subsetneq V$ closed.

Lemma 11.2: characterisation by intersections of open sets

A nonempty topological space X is irreducible if and only if, for any two nonempty open subsets $U, V \subseteq X$, the intersection $U \cap V$ is also nonempty.

Proof This follows immediately from the definition. For the closed subsets $Y = X \setminus U$ and $Z = X \setminus V$, the equality $X = Y \cup Z$ holds precisely when $U \cap V = \emptyset$.

A subset $Y \subseteq X$ of a topological space X is called irreducible if Y, equipped with the induced topology, is an irreducible topological space.

Lemma 11.3: irreducible closed subsets of the spectrum

Let R be a commutative ring and $\mathfrak{a} \subseteq R$ an ideal. The closed subset

$$V(\mathfrak{a}) \subseteq \mathrm{Spek}(R)$$

is irreducible if and only if the radical of $\mathfrak{a}$ is a prime ideal.

Proof We may assume at once that $\mathfrak{a}$ is a radical ideal. Moreover, $\mathfrak{a}$ is not the unit ideal. If $V(\mathfrak{a})$ is not irreducible, there is a nontrivial decomposition

$$V(\mathfrak{a}) = Y \cup Z = V(\mathfrak{b}) \cup V(\mathfrak{c}),$$

where we may assume that $\mathfrak{b}$ and $\mathfrak{c}$ are radical. This means that

$$\mathfrak{a} = \mathfrak{b} \cap \mathfrak{c}.$$

Since $V(\mathfrak{b}), V(\mathfrak{c}) \subsetneq V(\mathfrak{a})$, Proposition 8.4(5) gives

$$\mathfrak{a} \subsetneq \mathfrak{b}, \qquad \mathfrak{a} \subsetneq \mathfrak{c}.$$

Thus there are $f \in \mathfrak{b} \setminus \mathfrak{a}$ and $g \in \mathfrak{c} \setminus \mathfrak{a}$. However,

$$fg \in \mathfrak{b} \cap \mathfrak{c} = \mathfrak{a},$$

so $\mathfrak{a}$ is not a prime ideal.

Conversely, if $\mathfrak{a}$ is not a prime ideal, there are $f, g \notin \mathfrak{a}$ with $fg \in \mathfrak{a}$. Then

$$D(fg) \subseteq D(\mathfrak{a}), \qquad D(fg) \cap V(\mathfrak{a}) = \emptyset.$$

Since $\mathfrak{a}$ is radical, $f^n \notin \mathfrak{a}$ for every $n \in \mathbb{N}$. By Exercise 8.5, there is a prime ideal $\mathfrak{p}$ with

$$f \notin \mathfrak{p}, \qquad \mathfrak{a} \subseteq \mathfrak{p}.$$

Hence $D(f) \cap V(\mathfrak{a}) \neq \emptyset$, and the same holds for $D(g)$. Since

$$D(f) \cap D(g) \cap V(\mathfrak{a}) = D(fg) \cap V(\mathfrak{a}) = \emptyset,$$

Lemma 11.2 shows that $V(\mathfrak{a})$ is not irreducible.

Thus the correspondence

$$\mathfrak{p} \longleftrightarrow V(\mathfrak{p})$$

relates prime ideals to irreducible closed subsets of the spectrum. Maximal ideals correspond to individual closed points, whereas minimal prime ideals correspond to the irreducible components of the spectrum discussed below.

Definition 11.4: generic point

Let X be a topological space and $Y \subseteq X$ an irreducible closed subset. A point $\eta \in Y$ is called a *generic point* of Y if, for every open subset $U \subseteq X$,

$$U \cap Y \neq \emptyset \quad \Longleftrightarrow \quad \eta \in U.$$

Lemma 11.5: existence and uniqueness of the generic point

In a scheme X, every irreducible closed subset $Y \subseteq X$ has exactly one generic point.

Proof By assumption, Y is nonempty. Choose $P \in Y$ and an open affine neighbourhood

$$P \in W = \text{Spek}(R).$$

Then $Y \cap W$ is an irreducible closed subset of the affine scheme W. By Lemma 11.3,

$$Y \cap W = V(\mathfrak{p})$$

for some prime ideal $\mathfrak{p} \in W$. We claim that $\mathfrak{p}$ is the generic point of Y. If $U \subseteq X$ is open and $Y \cap U \neq \emptyset$, the irreducibility of Y gives

$$Y \cap U \cap W \neq \emptyset,$$

and hence $\mathfrak{p} \in U$. The generic point is unique because it is uniquely determined as a point of the affine scheme W.

Krull dimension

Definition 11.6: Krull dimension of a topological space

For a topological space X, the maximum length of a chain of irreducible closed subsets

$$X_0 \subsetneq X_1 \subsetneq \cdots \subsetneq X_{n-1} \subsetneq X_n$$

in X is called the *Krull dimension* of the space.

> **Editorial note - unbounded dimension.** The source says “maximum”. More generally, take the supremum of the lengths n of these chains, allowing infinite dimension when the lengths are unbounded.

Lemma 11.7: dimension of a ring and its spectrum

The Krull dimension of a commutative ring R equals the Krull dimension of its spectrum $\text{Spek}(R)$.

Proof The assertion follows from Lemma 11.3 and Proposition 8.4(5).

Noetherian spaces

Definition 11.8: noetherian topological space

A topological space X is called *noetherian* if every ascending chain of open subsets

$$U_1 \subseteq U_2 \subseteq U_3 \subseteq \cdots$$

becomes stationary, that is, there is an n such that

$$U_n = U_{n+1} = U_{n+2} = \cdots.$$

Lemma 11.9: characterisation by quasicompactness

A topological space X is noetherian if and only if every open subset of it is quasicompact.

Proof Every open subset of a noetherian space is itself noetherian. For the forward implication, it therefore suffices to prove that X is quasicompact. Let

$$X = \bigcup_{i \in I} U_i$$

be an open cover, and suppose that it has no finite subcover. We can then construct an infinite strictly ascending chain of open subsets

$$V_n = \bigcup_{i \in I_n} U_i,$$

where each $I_n \subseteq I$ is finite, contradicting noetherianity.

Conversely, suppose that every open subset is quasicompact, and consider an ascending chain $U_k \subseteq U_{k+1}$. The set

$$U = \bigcup_{k \in \mathbb{N}} U_k$$

is open and quasicompact, so this cover has a finite subcover. Thus there is an index n with $U_n = U_k$ for all $k \geq n$.

In a noetherian space, every nonempty collection of open sets (respectively, closed sets) has a maximal (respectively, minimal) element. This gives the proof principle of *noetherian induction*. To prove that a property E holds for all closed subsets, suppose there are closed subsets that fail to satisfy E, and choose a minimal one. This subset must then lead to a contradiction. The principle is valid because an infinite descending chain can be constructed in any nonempty collection without a minimal element.

The proof of the following assertion gives a typical example of this proof principle.

Theorem 11.10: irreducible components

Every noetherian topological space X has a unique irredundant decomposition

$$X = V_1 \cup \cdots \cup V_k$$

into irreducible closed subsets; *irredundant* means that no V_i is contained in V_j for $i \neq j$.

Proof We prove existence by noetherian induction on the closed subsets of X. Suppose that not every closed subset has such a decomposition. Then there is a minimal subset, say $V \subseteq X$, without one. The set V cannot be irreducible, so there is a nontrivial decomposition

$$V = V_1 \cup V_2.$$

Since V_1 and V_2 are proper subsets of V, each has a finite expression as a union of irreducible closed subsets. Combining these two expressions gives a finite decomposition of V, a contradiction.

For uniqueness, let

$$X = V_1 \cup \cdots \cup V_k = W_1 \cup \cdots \cup W_m$$

be two decompositions into irreducible subsets, each without inclusion relations. Then

$$\begin{aligned} V_1 &= V_1 \cap X \\ &= V_1 \cap (W_1 \cup \cdots \cup W_m) \\ &= (V_1 \cap W_1) \cup \cdots \cup (V_1 \cap W_m). \end{aligned}$$

Since V_1 is irreducible, there is a j with $V_1 \subseteq W_j$. By the same argument, there is an i with $W_j \subseteq V_i$. Irredundancy forces $i = 1$ and $V_1 = W_j$. Each of the other V_i occurs in the right-hand decomposition in the same way, so the decomposition is unique.

> **Editorial note - irredundancy condition.** The source statement says only "unique decomposition", while its proof first requires "each without inclusion relations" when comparing two decompositions. This edition brings that necessary condition into the statement; the source components and argument are unchanged.

The subsets occurring in this decomposition are called the *irreducible components* of the space.

Definition 11.11: noetherian scheme

A scheme X is called *noetherian* if it can be covered by finitely many affine schemes associated with noetherian rings.

In particular, the spectrum of a noetherian ring is a noetherian scheme.

Lemma 11.12: topology of a noetherian scheme

A noetherian scheme is a noetherian topological space.

Proof A finite union of noetherian spaces is again noetherian, so it suffices to consider the spectrum of a noetherian ring. By Lemma 11.9, we must show that every open subset

$$U = D(\mathfrak{a}) \subseteq \mathrm{Spek}(R)$$

is quasicompact. Since R is noetherian,

$$\mathfrak{a} = (f_1, \dots, f_n),$$

and Proposition 8.4(2) gives

$$D(\mathfrak{a}) = D(f_1) \cup \cdots \cup D(f_n).$$

Corollary 8.6 together with Proposition 8.11 says that each $D(f_i)$ is quasicompact, so their finite union is quasicompact as well.

These topological methods immediately give the following purely algebraic result.

Lemma 11.13: finiteness of the minimal prime ideals

A noetherian commutative ring has only finitely many minimal prime ideals.

Proof See Exercise 11.15.

Integral schemes

Definition 11.14: reduced ringed space

A ringed space $(X, \mathcal{O}_X)$ is called *reduced* if, for every open subset $U \subseteq X$, the ring $\Gamma(U, \mathcal{O}_X)$ is reduced.

Definition 11.15: integral scheme

A scheme X is called *integral* if it is irreducible and reduced.

Lemma 11.16: restrictions on an integral scheme are injective

In an integral scheme, the restriction maps

$$\Gamma(U, \mathcal{O}_X) \longrightarrow \Gamma(V, \mathcal{O}_X)$$

are injective for all open $\emptyset \neq V \subseteq U \subseteq X$.

Proof Let $0 \neq f \in \Gamma(U, \mathcal{O}_X)$. The set

$$U_f = \{P \in U \mid f(P) \neq 0\}$$

is open by Lemma 7.16 and is nonempty by reducedness. Since X is irreducible, $U_f \cap V$ is nonempty as well. Thus the restriction of f to V is nonzero.

> **Editorial note - domain of the section.** The source calls the set above X_f and writes $P \in X$, although f is given only as a section on U. This edition writes U_f and $P \in U$; the nonvanishing locus and the injectivity argument are unchanged.

Example 11.17: irreducibility alone is not enough

Let K be a field and

$$R = K[X, Y]/(X^2, XY).$$

The ideal $\mathfrak{q} = (X)$ is the only minimal prime ideal of R, so $\operatorname{Spek}(R)$ is irreducible. Since $Y \notin \mathfrak{q}$ and $XY = 0$, the equality $X = 0$ holds in the localisation $R_{\mathfrak{q}}$, and

$$R_{\mathfrak{q}} = K[Y]_{(0)} = K(Y)$$

is a field. The restriction map $R \to R_{\mathfrak{q}}$ is not injective. Moreover,

$$D(X) = \emptyset,$$

but the element X is nonzero in the localisation $R_{(X,Y)}$.

Lemma 11.18: rings of sections are integral domains

In an integral scheme X, for every nonempty open subset $U \subseteq X$, the ring of sections $\Gamma(U, \mathcal{O}_X)$ is an integral domain.

Proof Since U is open and nonempty, there is a nonempty affine open subset

$$V = \operatorname{Spek}(R) \subseteq U.$$

By Lemma 11.16, it suffices to show that R is an integral domain. Let $\mathfrak{a}$ be the nilradical of R. Since V is irreducible as a consequence of the irreducibility of X, Lemma 11.3 says that $\mathfrak{a}$ is a prime ideal. Reducedness is a local property by Exercise 11.18, so $\mathfrak{a} = 0$. Thus the zero ideal is prime, and R is an integral domain.

> **Editorial note - source exercise number.** The source refers to Exercise 10.14, but frozen Worksheet 10 contains only six exercises. Frozen Exercise 11.18 states precisely the equivalence between reducedness of a ringed space and reducedness of all its stalks. This edition corrects the cross-reference to 11.18 and preserves the source's erroneous reference in this note.

Lemma 11.19: the generic stalk is a field

For an integral scheme, the stalk of the structure sheaf at the generic point is a field.

Proof The stalk can be computed from any nonempty affine open subset. Such a subset has the form

$$U = \text{Spek}(R),$$

where R is a commutative ring that is an integral domain by Lemma 11.18. The generic point corresponds to the zero ideal, and localisation at the zero ideal gives the fraction field of R.

Definition 11.20: function field

For an integral scheme X, the stalk of the structure sheaf at the generic point is called the *function field* of X.

In an integral scheme, the ring of sections on every nonempty open subset is a subring of the function field.

Worksheet 11: Irreducible Spaces and Noetherian Schemes

Asterisks mark exactly the three exercises with frozen public solutions: Exercises 11.9, 11.13, and 11.14. The other sixteen exercises have negative candidate results; this edition creates no new solutions.

Exercise 11.1

Prove that, in an irreducible topological space X, every nonempty open subset $U \subseteq X$ is dense.

Exercise 11.2

Prove that a metric space X can be irreducible only if it consists of a single point.

Exercise 11.3

Let X be a topological space and $Y \subseteq X$ a subset with the induced topology. Prove that Y is irreducible if and only if its closure $\overline{Y}$ is irreducible.

A topological space is said to satisfy the T_0 *separation property* if, for any two points $x \neq y$, there is an open set U with $x \in U$ and $y \notin U$, or an open set V with $x \notin V$ and $y \in V$.

A topological space is said to satisfy the T_1 *separation property* if every point $x \in X$ is closed.

Exercise 11.4

Prove that a scheme satisfies the T_0 separation property.

Exercise 11.5

Prove that, for an affine scheme $X = \text{Spek}(R)$, the following properties are equivalent.

1. Every prime ideal of R is maximal.
2. Every point of X is closed.
3. X is a Hausdorff space.

Exercise 11.6

Give an example of a zero-dimensional affine scheme $X = \text{Spek}(R)$ that is not discrete.

Exercise 11.7

Let $Y \subseteq X$ be an irreducible subset of a topological space X, and let $\eta \in Y$. Prove that η is a generic point of Y if and only if

$$\overline{\{\eta\}} = Y.$$

Editorial note - closedness of the subset. The source omits the hypothesis that Y is closed. With closure taken in X, include that hypothesis, as in Definition 11.4. For an arbitrary irreducible subset, the corresponding statement uses closure in the subspace Y instead.

Exercise 11.8

Let $X = \mathrm{Spek}(R)$ be the spectrum of a commutative ring R, and let

$$Y = V(\mathfrak{p}) \subseteq X$$

be the closed subset associated with a prime ideal $\mathfrak{p}$. Prove that $\mathfrak{p}$ is the generic point of $V(\mathfrak{p})$.

Exercise 11.9*

Let $M \neq \emptyset$ be a differentiable manifold of dimension n. Prove that there is a chain of closed submanifolds

$$M_0 \subseteq M_1 \subseteq M_2 \subseteq \cdots \subseteq M_{n-1} \subseteq M_n = M$$

such that the closed submanifold M_i has dimension i.

Exercise 11.10

Let X be a noetherian topological space. Prove that every subset $Y \subseteq X$ with the induced topology is also noetherian.

Exercise 11.11

Let X be a noetherian topological space. Prove that every subset $Y \subseteq X$ with the induced topology is quasicompact.

Exercise 11.12

Prove that the real numbers $\mathbb{R}$ with the metric topology do not form a noetherian topological space.

Exercise 11.13*

Let R be a commutative ring and

$$\mathrm{Spek}(R) = \bigcup_{i \in I} D(f_i), \qquad f_i \in R.$$

Let $\mathfrak{a}$ be an ideal of R such that each extended ideal $\mathfrak{a}R_{f_i}$ is finitely generated. Prove that $\mathfrak{a}$ is finitely generated.

Exercise 11.14*

Let R be a commutative ring and $X = \mathrm{Spek}(R)$ the associated affine scheme. Prove that X is a noetherian scheme if and only if R is a noetherian ring.

Exercise 11.15

Prove that a noetherian commutative ring has only finitely many minimal prime ideals.

Exercise 11.16

Give an example of a non-noetherian ring whose reduction is a field.

Let R be a commutative ring. A multiplicative system $F \subseteq R$ is called an *ultrafilter* if $0 \notin F$ and F is maximal with this property.

Exercise 11.17

Let R be a commutative ring and $F \subset R$ an ultrafilter. Prove that the complement of F is a minimal prime ideal of R.

Exercise 11.18

Let $(X, \mathcal{O}_X)$ be a ringed space. Prove that the following assertions are equivalent.

1. $(X, \mathcal{O}_X)$ is a reduced ringed space.
2. For every point $P \in X$, the stalk $\mathcal{O}_{X,P}$ is reduced.

Exercise 11.19

Prove that integrality of a scheme is not, in general, a local property.

Public Solutions and Coverage of Worksheet 11

At the frozen revision boundary, the source provides exactly three public solutions among the 19 exercises, namely those for Exercises 11.9, 11.13, and 11.14. The exercise map and candidate evidence record negative results for Exercises 11.1-11.8, 11.10-11.12, and 11.15-11.19. No new solutions have been created for this edition.

Source solution to Exercise 11.9

> **Editorial note - dimension zero.** The source's coordinate construction below applies when $n \geq 1$. If $n = 0$, the required chain consists simply of $M_0 = M$.

Choose a point $P \in M$ and an open coordinate neighbourhood $P \in U$ together with a chart

$$\alpha : U \longrightarrow V \subseteq \mathbb{R}^n,$$

where $V = B(0, 3)$ is the open ball centred at 0 with radius 3 and $\alpha(P) = 0$. For $i = 0, \dots, n-1$, set

$$B_i = \left\{ x \in V \,\middle|\, (x_1 - 1)^2 + x_2^2 + \cdots + x_n^2 = 1,\ x_{i+2} = \cdots = x_n = 0 \right\}.$$

Thus B_{n-1} is the sphere centred at $(1, 0, \dots, 0)$ with radius 1, while B_{n-2} is its "equator" defined by $x_n = 0$, and so on. The set B_i is obtained from B_{i+1} by adding the equation $x_{i+2} = 0$. We therefore have a descending chain of closed subsets

$$B_{n-1} \supseteq B_{n-2} \supseteq \cdots \supseteq B_1 \supseteq B_0,$$

and

$$B_0 = \{(0, 0, \dots, 0), (2, 0, \dots, 0)\}.$$

We can regard B_i as the fibre over the origin of the map

$$\begin{aligned} \varphi_i : V &\longrightarrow \mathbb{R}^{n-i}, \\ (x_1, \dots, x_n) &\longmapsto ((x_1 - 1)^2 + x_2^2 + \cdots + x_n^2 - 1, x_{i+2}, \dots, x_n). \end{aligned}$$

Its Jacobian matrix is

$$D\varphi_i(x) = \begin{pmatrix} 2x_1 - 2 & 2x_2 & \cdots & 2x_{i+1} & 2x_{i+2} & \cdots & 2x_n \\ 0 & 0 & \cdots & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & \cdots & 1 \end{pmatrix}.$$

The rank of this matrix is less than $n - i$ only if

$$x_1 = 1, \qquad x_2 = \cdots = x_{i+1} = 0,$$

and such a point does not lie on B_i. Thus φ_i is regular along the fibre B_i. By the implicit function theorem, B_i is a closed submanifold of V of dimension i.

Now set

$$M_i = \alpha^{-1}(B_i) \quad (0 \leq i \leq n-1), \qquad M_n = M.$$

Since each B_i is compact, M_i is also closed in M. Being a closed submanifold is a local property, so all the M_i are closed submanifolds of M of the required dimensions.

Editorial note - index bounds and the source's two points. The source solution initially defines B_i and M_i only for $i = 1, \dots, n-1$, but then uses B_0 and a chain requiring M_0. This edition includes $i = 0$ in both ranges. The source also states that B_0 consists of the points $(\pm 1, 0, \dots, 0)$; substitution into the equation of the sphere centred at $(1, 0, \dots, 0)$ gives the correct points $(0, 0, \dots, 0)$ and $(2, 0, \dots, 0)$. The formulae, rank calculation, and submanifold construction are otherwise preserved.

Source solution to Exercise 11.13

Since the spectrum is quasicompact, we may assume that I is finite. By Proposition 8.4(9), the elements f_i generate the unit ideal.

Let $(a_j)_{j\in J}$ be a generating system for the ideal $\mathfrak{a}$. Viewed in R_{f_i}, it also generates $\mathfrak{a}R_{f_i}$. For each i, a finite subsystem suffices; since I is finite, the union of the required index sets is a finite subset $J_0 \subseteq J$. Thus $(a_j)_{j\in J_0}$ generates every $\mathfrak{a}R_{f_i}$.

We claim that these elements already generate $\mathfrak{a}$. Let $b \in \mathfrak{a}$. For each i, there is an equality in R_{f_i}

$$b = \sum_{j\in J_0} \frac{c_{ij}}{f_i^{n_{ij}}} a_j.$$

Clearing denominators gives an equality in R of the form

$$f_i^{m_i} b = \sum_{j\in J_0} d_{ij} a_j.$$

Since the sets $D(f_i^{m_i}) = D(f_i)$ still cover the spectrum, there are $g_i \in R$ with

$$\sum_{i\in I} g_i f_i^{m_i} = 1.$$

Consequently,

$$\begin{aligned} b &= b\left(\sum_{i\in I} g_i f_i^{m_i}\right) \\ &= \sum_{i\in I} g_i b f_i^{m_i} \\ &= \sum_{i\in I} g_i \left(\sum_{j\in J_0} d_{ij} a_j\right) \\ &= \sum_{j\in J_0} \left(\sum_{i\in I} g_i d_{ij}\right) a_j. \end{aligned}$$

Thus every $b \in \mathfrak{a}$ is a linear combination of $(a_j)_{j\in J_0}$, so $\mathfrak{a}$ is finitely generated.

Source solution to Exercise 11.14

Clearly, if R is a noetherian ring, then $X = \mathrm{Spek}(R)$ is a noetherian scheme.

Conversely, suppose that X is a noetherian scheme. Choose a finite affine cover

$$X = \bigcup_{i\in I} U_i, \qquad U_i = \mathrm{Spek}(R_i),$$

with each R_i a noetherian ring. Since U_i is open in $X = \mathrm{Spek}(R)$ and quasicompact, for each i there are finitely many $f_{ij} \in R$ such that

$$U_i = \bigcup_{j\in J_i} D_X(f_{ij}).$$

If

$$\rho_i : R = \Gamma(X, \mathcal{O}_X) \longrightarrow R_i = \Gamma(U_i, \mathcal{O}_X)$$

is the restriction, then

$$D_X(f_{ij}) = D_{U_i}(\rho_i(f_{ij})).$$

The ring of sections on this principal open subset is therefore

$$R_{f_{ij}} \cong (R_i)_{\rho_i(f_{ij})},$$

which is noetherian as a localisation of a noetherian ring. Combining all i and j, we obtain a finite principal open cover

$$X = \bigcup_{k=1}^{N} D(g_k)$$

with every R_{g_k} noetherian. For any ideal $\mathfrak{a} \subseteq R$, each ideal $\mathfrak{a}R_{g_k}$ is finitely generated. Exercise 11.13 now shows that $\mathfrak{a}$ is finitely generated. Hence every ideal of R is finitely generated, and R is noetherian.

> **Editorial note - notation in the source solution.** The source solution writes $U_i = \bigcup_{j \in J_i} D(f_j)$, with the index i subsequently disappearing, uses $f_j \in R$, then places $\rho(f_j)$ in the ring of sections of an undefined set U and writes a localisation with only the subscript f. This edition explicitly writes f_{ij}, the restriction $\rho_i : R \to R_i$, and the isomorphism $R_{f_{ij}} \cong (R_i)_{\rho_i(f_{ij})}$. These are the data used in the source argument; no new hypothesis is introduced.

Lecture 12: The Projective Spectrum of a Graded Ring

The projective spectrum

The projective spectrum is usually introduced for $\mathbb{N}$-graded rings. Its construction, however, uses localisations at homogeneous elements, where negative degrees also occur. It is therefore more natural to work with $\mathbb{Z}$-gradings from the outset.

Definition 12.1: irrelevant ideal

For a $\mathbb{Z}$-graded ring, the ideal generated by all homogeneous elements of nonzero degree is called the *irrelevant ideal.* It is denoted by R_+.

In the positively graded case, this is simply the ideal

$$\bigoplus_{n \geq 1} R_n.$$

If negative degrees occur, the irrelevant ideal can even be the unit ideal.

Definition 12.2: the underlying set of the projective spectrum

Let R be a $\mathbb{Z}$-graded commutative ring. The set of all homogeneous prime ideals of R that do not contain R_+ is called the *projective spectrum* of R and is denoted by

$$\text{Proj}(R).$$

Definition 12.3: topology on the projective spectrum

Let R be a $\mathbb{Z}$-graded commutative ring. The projective spectrum of R as a topological space is the set of homogeneous prime ideals of R that do not contain R_+, with the subsets

$$D_+(\mathfrak{a}) = \{\mathfrak{p} \in \mathrm{Proj}(R) \mid \mathfrak{a} \not\subseteq \mathfrak{p}\}$$

declared to be open.

These subsets do indeed define a topology. We have

$$D_+(\mathfrak{a}) = \bigcup_{\substack{f \in \mathfrak{a} \\ f \text{ homogeneous}}} D_+(f), \qquad D_+(f) = \{\mathfrak{p} \in \mathrm{Proj}(R) \mid f \notin \mathfrak{p}\}.$$

The sets $D_+(f)$ form a basis for this topology.

> **Editorial note - homogeneous ideals.** In this definition and the union formula, $\mathfrak{a}$ ranges over homogeneous ideals, as explicitly stipulated in Exercise 12.6. The source definition leaves this qualification implicit.

Definition 12.4: projective space

The projective spectrum of the polynomial ring

$$R[X_0, X_1, \dots, X_n]$$

is called *projective n-space over R*.

For a $\mathbb{Z}$-graded ring S, the ring in degree zero is denoted by S_0. If R is $\mathbb{N}$-graded, R_0 often has a simple form, for example a field. But if $f \in R$ is homogeneous of positive degree, the localisation R_f is naturally $\mathbb{Z}$-graded, and $(R_f)_0$ can be arbitrarily complicated.

For a homogeneous prime ideal $\mathfrak{p}$, the intersection of the multiplicative system $R \setminus \mathfrak{p}$ with the set of all homogeneous elements is again a multiplicative system. The degree-zero ring of this localisation plays a special role and is denoted by

$$R_{(\mathfrak{p})} = \left(R_{(R\setminus\mathfrak{p}) \cap \{\text{homogeneous elements}\}}\right)_0.$$

Lemma 12.5: the local ring at a homogeneous prime ideal

For a homogeneous prime ideal $\mathfrak{p}$ in an $\mathbb{N}$-graded ring R, the ring $R_{(\mathfrak{p})}$ is local.

Proof Let $r, s \in R_{(\mathfrak{p})}$ be nonunits. We can write

$$r = \frac{a}{f}, \qquad s = \frac{b}{g},$$

with homogeneous $f, g \notin \mathfrak{p}$ and homogeneous $a, b \in R$, each of the same degree as its denominator. We must have $a, b \in \mathfrak{p}$; otherwise the corresponding fraction would be a unit. Thus $ag + bf \in \mathfrak{p}$, so

$$\frac{a}{f} + \frac{b}{g} = \frac{ag + bf}{fg}$$

is also a nonunit in $R_{(\mathfrak{p})}$. Hence the nonunits are closed under addition, giving a unique maximal ideal.

The structure sheaf

Definition 12.6: the structure sheaf on the projective spectrum

Let R be a $\mathbb{Z}$-graded ring and $Y = \mathrm{Proj}(R)$ its projective spectrum with the Zariski topology. The *structure sheaf* on Y assigns to each open subset $U \subseteq Y$ the commutative ring

$$\Gamma(U, \mathcal{O}_Y) = \Big\{ (s_\mathfrak{p})_{\mathfrak{p}\in U} \in \prod_{\mathfrak{p}\in U} R_{(\mathfrak{p})} \;\Big|\; \text{for every } \mathfrak{p} \in U \text{ there are homogeneous elements } a, b \in R \text{ with } \mathfrak{p} \in D_+(b) \subseteq U, \text{ and } s_\mathfrak{q} = a/b \text{ in } R_{(\mathfrak{q})} \text{ for all } \mathfrak{q} \in D_+(b) \Big\}.$$

To each inclusion $U \subseteq V$ it assigns the natural restriction projection.

In particular, this definition requires $\deg(a) - \deg(b) = 0$ in every representation. It can be viewed as the sheafification of the presheaf

$$D_+(\mathfrak{a}) \longmapsto \operatorname*{colim}_{D_+(\mathfrak{a}) \subseteq D_+(f)} (R_f)_0,$$

which also shows that we do indeed obtain a sheaf of commutative rings.

Definition 12.7: the projective spectrum as a ringed space

For a $\mathbb{Z}$-graded ring R, the *projective spectrum*

$$Y = \mathrm{Proj}(R)$$

means the projective spectrum equipped with its Zariski topology and structure sheaf.

Lemma 12.8: a homogeneous unit and the spectrum of the degree-zero component

Let R be a $\mathbb{Z}$-graded ring with at least one homogeneous unit of positive degree. The map

$$\mathrm{Proj}(R) \longrightarrow \mathrm{Spek}(R_0), \qquad \mathfrak{p} \longmapsto \mathfrak{p} \cap R_0,$$

is a bijection, a homeomorphism for the Zariski topologies, and an isomorphism with respect to the structure sheaves. Moreover,

$$R_{(\mathfrak{p})} = (R_0)_{\mathfrak{p}\cap R_0}.$$

Proof First,

$$\mathfrak{p}_0 := \mathfrak{p} \cap R_0$$

is a prime ideal, so the map is well defined. Let g be a homogeneous unit of positive degree. We shall reconstruct $\mathfrak{p}$ from $\mathfrak{p}_0$. If H denotes the set of homogeneous elements, then

$$\mathfrak{p} \cap H = \{ h \text{ homogeneous} \mid \text{there exist } i \in \mathbb{Z}, j \in \mathbb{N}_+ \text{ with } g^i h^j \in \mathfrak{p}_0 \}.$$

The inclusion from left to right is clear: i and j can be chosen so that $\deg(g^i h^j) = 0$. Conversely, if $g^i h^j \in \mathfrak{p}_0$, then $h^j \in \mathfrak{p}$ because g is a unit, and then $h \in \mathfrak{p}$ because $\mathfrak{p}$ is prime. Thus the map is injective.

For surjectivity, take a prime ideal $\mathfrak{q} \subseteq R_0$ and form the ideal generated by

$$\{ h \text{ homogeneous} \mid \text{there exist } i \in \mathbb{Z}, j \in \mathbb{N}_+ \text{ with } g^i h^j \in \mathfrak{q} \}.$$

This ideal contains no unit, and therefore does not contain all of R_+; in particular it does not contain g, since $\mathfrak{q}$ does not contain 1. If $h_1 h_2$ belongs to the set above, then for some i, j,

$$g^i h_1^j h_2^j \in \mathfrak{q}.$$

For some power, we can write

$$(g^i h_1^j h_2^j)^n = (g^a h_1^b)(g^c h_2^d)$$

with both factors of degree zero. One factor must belong to $\mathfrak{q}$, so h_1 or h_2 belongs to the stated set. The generated ideal is therefore prime and gives the inverse map.

For the homeomorphism, consider the bases formed by $D_+(f)$ on the projective side and $D(f)$ on the degree-zero side. The inverse image of $D(f)$ is $D_+(f)$, and a homogeneous element can be multiplied by a power of g without changing its projective open set:

$$D_+(f) = D_+(fg^j).$$

The identification of the local rings and compatibility of the sheaves follow from the same description of degree-zero localisations.

> **Editorial note - degree adjustment.** The source's last displayed identity preserves the open set, but multiplying f by a power of g alone need not give degree zero. If $\deg(g) = e > 0$ and $\deg(f) = d$, use $f^e g^{-d}$: it has degree zero and $D_+(f) = D_+(f^e g^{-d})$. This supplies the degree-zero basis used in the argument.

Lemma 12.9: principal projective open subsets are affine

For a $\mathbb{Z}$-graded ring R and a homogeneous element $f \in R$ of nonzero degree,

$$D_+(f) \cong \mathrm{Spek}((R_f)_0)$$

as ringed spaces. In particular, the projective spectrum $\mathrm{Proj}(R)$ is a scheme.

Proof Apply Lemma 12.8 to R_f, noting that homogeneous prime ideals in $D_+(f)$ correspond to homogeneous prime ideals of R_f. The scheme property follows because the sets $D_+(f)$ for $f \in R_+$ cover the projective spectrum.

Example 12.10: the standard affine cover of projective space

Projective space, the projective spectrum of the standard-graded polynomial ring $R[X_0, X_1, \dots, X_n]$, is covered by the $D_+(X_i)$. This is called the *standard affine cover of projective space.* We have

$$\begin{aligned}\Gamma(D_+(X_i), \mathcal{O}_{\mathbb{P}^n_R}) &= (R[X_0, X_1, \dots, X_n]_{X_i})_0 \\ &= R\left[\frac{X_0}{X_i}, \dots, \frac{X_{i-1}}{X_i}, \frac{X_{i+1}}{X_i}, \dots, \frac{X_n}{X_i}\right].\end{aligned}$$

This is a polynomial ring in n variables. With $Y_j = X_j/X_i$ for $j \neq i$, we obtain

$$D_+(X_i) = \mathrm{Spek}(R[Y_0, \dots, Y_{i-1}, Y_{i+1}, \dots, Y_n]) = \mathbb{A}^n_R.$$

Thus projective n-space is covered by $n+1$ affine spaces.

Morphisms and projective schemes

Theorem 12.11: the morphism induced by a homogeneous homomorphism

Let A and B be $\mathbb{Z}$-graded rings over a commutative ring $R = A_0$, and let

$$\theta : A \longrightarrow B$$

be a homogeneous ring homomorphism. There is a natural scheme morphism

$$D_+(\theta(A_+)B) \longrightarrow \mathrm{Proj}(A), \qquad \mathfrak{p} \longmapsto \theta^{-1}(\mathfrak{p}).$$

Proof The inverse image of a prime ideal under a ring homomorphism is again prime, and the inverse image of a homogeneous ideal is again homogeneous. If

$$\mathfrak{p} \in D_+(\theta(A_+)B),$$

there is a homogeneous element $f \in A_+$ with $\theta(f) \notin \mathfrak{p}$. Then $A_+ \not\subseteq \theta^{-1}(\mathfrak{p})$, so

$$\theta^{-1}(\mathfrak{p}) \in \mathrm{Proj}(A).$$

We therefore have a map

$$\varphi : D_+(\theta(A_+)B) \longrightarrow \mathrm{Proj}(A).$$

For a homogeneous element $f \in A_+$,

$$\varphi^{-1}(D_+(f)) = D_+(\theta(f)),$$

as for the map on spectra. Thus φ is continuous. By Corollary 10.10, θ induces a unique morphism of affine schemes $\mathrm{Spek}(B) \to \mathrm{Spek}(A)$. On the affine principal open $D(f) \subseteq \mathrm{Spek}(A)$, this morphism is given by the homogeneous homomorphism

$$\theta_f : A_f \longrightarrow B_{\theta(f)},$$

which induces a homomorphism on the degree-zero components

$$(\theta_f)_0 : (A_f)_0 \longrightarrow (B_{\theta(f)})_0.$$

By Lemma 12.9, this is a homomorphism between the rings of sections on $D_+(f)$ and $D_+(\theta(f))$. These homomorphisms are compatible with restrictions, and the diagram

$$\begin{array}{ccc} D_+(\theta(f)) & \xrightarrow{\varphi=\theta^{-1}} & D_+(f) \\ \downarrow & & \downarrow \\ \mathrm{Spek}((B_{\theta(f)})_0) & \xrightarrow{(\theta_f)_0^{-1}} & \mathrm{Spek}((A_f)_0) \end{array}$$

commutes. Hence φ is a morphism of locally ringed spaces.

> **Editorial note - domains in the source proof.** The source statement correctly gives the domain $D_+(\theta(A_+)B)$, but the first line of its proof prints $D_+(\theta(A_+)B) = \mathrm{Proj}(B)$. This equality does not hold without an additional hypothesis. This edition retains the open domain from the statement. The later $D(f)$ belongs to the intermediate affine-spectrum argument; passing to degree-zero components then gives the projective charts $D_+(f)$ and $D_+(\theta(f))$. Uniqueness for the affine morphism is relative to the specified ring homomorphism, not to its underlying continuous map alone.

Example 12.12: projection away from a point

The subring inclusion

$$K[X_1, \dots, X_n] \subseteq K[X_0, X_1, \dots, X_n]$$

gives, by Theorem 12.11, a scheme morphism

$$\mathbb{P}^n_K \supset D_+(X_1, \dots, X_n) \longrightarrow \mathbb{P}^{n-1}_K.$$

A K-point with homogeneous coordinates $(x_0, x_1, \dots, x_n)$ is sent to $(x_1, \dots, x_n)$. This map is defined only on the displayed open subset and has no meaningful extension to the point $(1, 0, \dots, 0)$. It is called *projection away from a point*. Interpreted in the affine spaces $\mathbb{A}^{n+1}_K$ and $\mathbb{A}^n_K$, it projects each line onto a hyperplane. For the line "perpendicular" to the hyperplane, the construction is not well defined because the projection does not give a line.

We shall use the following definition especially over a field.

> **Editorial note - projection to a point.** The nonextension assertion in Example 12.12 presupposes $n \geq 2$. For $n = 1$, the target is $\mathbb{P}^0_K$, and the constant morphism does extend to all of $\mathbb{P}^1_K$.

Definition 12.13: projective scheme

A scheme $(X, \mathcal{O}_X)$ over a commutative ring R is called *projective* if there is a factorisation

$$X \xrightarrow{i} \mathbb{P}^n_R \longrightarrow \mathrm{Spek}(R)$$

in which i is a closed immersion.

Lemma 12.14: Proj of a standard-graded ring is projective

For a standard-graded ring A, the projective spectrum $\mathrm{Proj}(A)$ is a projective scheme over $\mathrm{Spek}(A_0)$.

Proof We can write

$$A = A_0[X_0, X_1, \dots, X_n]/\mathfrak{a}$$

with a homogeneous ideal $\mathfrak{a} \subseteq A_0[X_0, X_1, \dots, X_n]$. By Theorem 12.11, the quotient map

$$\theta : A_0[X_0, X_1, \dots, X_n] \longrightarrow A$$

gives a scheme morphism

$$i : \mathrm{Proj}(A) \longrightarrow \mathbb{P}^n_{A_0}.$$

Just as the map on spectra

$$\mathrm{Spek}(A) \longrightarrow V(\mathfrak{a}) \subseteq \mathbb{A}^{n+1}_{A_0}, \qquad \mathfrak{p} \longmapsto \theta^{-1}(\mathfrak{p}),$$

is a homeomorphism, this projective version is a homeomorphism onto $V_+(\mathfrak{a})$. Thus $\mathrm{Proj}(A)$ naturally corresponds to a closed subset of projective space over A_0.

It remains to show that the sheaf morphism

$$\mathcal{O}_{\mathbb{P}^n_{A_0}} \longrightarrow i_* \mathcal{O}_{\mathrm{Proj}(A)}$$

is surjective. On $D_+(f)$ for a homogeneous element $f \in A_0[X_0, X_1, \dots, X_n]_+$, this is the map

$$(A_0[X_0, X_1, \dots, X_n]_f)_0 \longrightarrow (A_f)_0,$$

which is surjective.

In the assertion above, the ring homomorphisms are surjective only on sets of the form $D_+(f)$, not on all open sets. Since these sets form a basis for the topology, the maps on stalks are also surjective. We therefore obtain a surjective sheaf morphism, and i is a closed immersion.

Projective hypersurfaces

Definition 12.15: projective hypersurface

For a homogeneous polynomial

$$F \in K[X_0, X_1, \dots, X_n]$$

over a field K, the set

$$V_+(F) = \mathrm{Proj}(K[X_0, X_1, \dots, X_n]/(F)) \subset \mathbb{P}^n_K$$

is called the *projective hypersurface* defined by F.

> **Editorial note - nondegenerate equation.** In Definitions 12.15 and 12.16, a hypersurface equation is understood to be nonzero and of positive degree. The source does not explicitly exclude the zero polynomial or nonzero constants, which instead give the whole projective space or the empty subscheme.

Definition 12.16: degree of a hypersurface

For a projective hypersurface

$$V_+(F) = \mathrm{Proj}(K[X_0, X_1, \dots, X_n]/(F)) \subset \mathbb{P}^n_K,$$

the degree of the homogeneous polynomial F is also called the *degree of the hypersurface.*

A hypersurface of degree 1 is called a *hyperplane*; these are projective linear subspaces of codimension 1.

Lemma 12.17: affine description by dehomogenisation

Let

$$R = K[X_0, X_1, \dots, X_d]/(f_1, \dots, f_s)$$

be a standard-graded ring, with homogeneous generators f_j of degrees d_j. Then

$$(R_{X_i})_0 = K\left[\frac{X_0}{X_i}, \dots, \frac{X_{i-1}}{X_i}, \frac{X_{i+1}}{X_i}, \dots, \frac{X_d}{X_i}\right] \Big/ \left(\frac{f_1}{X_i^{d_1}}, \dots, \frac{f_s}{X_i^{d_s}}\right).$$

This quotient ring is described by the dehomogenisations of the f_ℓ with respect to the variable X_i.

Proof We have

$$\begin{aligned}
R_{X_i} &= K[X_0, \dots, X_d, X_i^{-1}]/(f_1, \dots, f_s) \\
&= K\left[\frac{X_0}{X_i}, \dots, \frac{X_{i-1}}{X_i}, \frac{X_{i+1}}{X_i}, \dots, \frac{X_d}{X_i}, X_i, X_i^{-1}\right] \Big/ \left(\frac{f_1}{X_i^{d_1}}, \dots, \frac{f_s}{X_i^{d_s}}\right) \\
&= \left(K\left[\frac{X_0}{X_i}, \dots, \frac{X_{i-1}}{X_i}, \frac{X_{i+1}}{X_i}, \dots, \frac{X_d}{X_i}\right] \Big/ \left(\frac{f_1}{X_i^{d_1}}, \dots, \frac{f_s}{X_i^{d_s}}\right)\right)[X_i, X_i^{-1}].
\end{aligned}$$

In the final description, the degree-zero component can be read off directly. If

$$Y_k = \frac{X_k}{X_i}, \qquad Y_i = 1,$$

then, writing $f_\ell = \sum_\nu a_\nu X^\nu$,

$$\frac{f_\ell}{X_i^{d_\ell}} = \frac{\sum_\nu a_\nu X^\nu}{X_i^{d_\ell}} = \sum_\nu a_\nu Y^\nu.$$

This is the dehomogenisation of f_ℓ with respect to the variable X_i.

> **Editorial note - indices and degrees.** The source uses d as the index of the last variable in $X_0, \dots, X_d$, whereas d_j and d_ℓ denote polynomial degrees. This edition preserves all the symbols but explicitly distinguishes their two roles to avoid confusion.

Worksheet 12: The Projective Spectrum of a Graded Ring

Asterisks mark exactly the two exercises with frozen public solutions, Exercises 12.5 and 12.10. The other seventeen exercises have negative candidate results; this edition creates no new solutions.

Exercise 12.1

Let R be a commutative ring, D a commutative group, and A a D-graded R-algebra. Prove that

$$1 \in A_0,$$

and deduce that A_0 is an R-subalgebra of A.

Exercise 12.2

Let

$$A = \bigoplus_{d \in D} A_d$$

be a graded commutative ring, and suppose that the component A_e contains a unit. Prove that A_e is isomorphic to A_0 as an A_0-module.

Exercise 12.3

Let R be a commutative ring, D a commutative group, and A a D-graded commutative R-algebra. Let $\mathfrak{a} \subseteq A$ be a homogeneous ideal. Prove that the quotient ring $A/\mathfrak{a}$ is also D-graded.

Exercise 12.4

Prove that a polynomial ring in n variables contains exactly

$$\binom{d+n-1}{n-1}$$

monomials of degree d.

Exercise 12.5 ★

Give an example of two monomial ideals $\mathfrak{a}$ and $\mathfrak{b}$ in a polynomial ring and a natural number d such that the product ideal $\mathfrak{a}\mathfrak{b}$ has a generating system consisting of monomials of degree at most d, but neither of the two ideals has such a generating system.

Exercise 12.6

Let R be a $\mathbb{Z}$-graded ring. Prove that the subsets

$$D_+(\mathfrak{a}) \subseteq \text{Proj}(R),$$

for homogeneous ideals $\mathfrak{a} \subseteq R$, do indeed define a topology on the projective spectrum $\text{Proj}(R)$.

Exercise 12.7

Let R be a $\mathbb{Z}$-graded ring. Prove that the open subsets

$$D_+(f) \subseteq \text{Proj}(R),$$

for homogeneous elements $f \in R_+$, form a basis for the topology of the projective spectrum.

Exercise 12.8

Determine the projective spectrum associated with the coordinate cross

$$\text{Spek}(K[X,Y]/(XY))$$

with its standard grading.

Exercise 12.9

Sketch the projective spectrum associated with the union of coordinate planes

$$\text{Spek}(K[X,Y,Z]/(XYZ))$$

with its standard grading.

Exercise 12.10 ★

Determine the intersection point of the two lines

$$L = V_+(6X - 8Y + 3Z)$$

and

$$M = V_+(2X + 9Y - 5Z)$$

in the projective plane.

Editorial note - base field. The source solution uses division by 70, so its displayed affine coordinates assume a field of characteristic different from 2, 5, and 7. Over those exceptional characteristics, the same exercise can be treated in homogeneous coordinates without that division.

Exercise 12.11

Prove that two distinct points P and Q in the projective plane uniquely determine a projective line containing both. How is its equation computed from the coordinates of the two points?

Editorial note - rational points. Here "points" means K-rational points of $\mathbb{P}^2_K$, as required by the coordinate formulation. The assertion is not about arbitrary points of the underlying scheme.

Exercise 12.12

Prove that the ring of global sections of projective space is the base ring:

$$\Gamma(\mathbb{P}^n_R, \mathcal{O}_{\mathbb{P}^n_R}) = R.$$

Exercise 12.13

Prove that the projective line $\mathbb{P}^1_K$ constructed by gluing in Example 10.7 agrees with the projective line in the sense of Example 12.10, namely

$$\operatorname{Proj}(K[X, Y]).$$

Exercise 12.14

Let L be a homogeneous linear form in $K[X_0, \dots, X_n]$, and let

$$D_+(L) \cong \mathbb{A}^n_K \subseteq \mathbb{P}^n_K.$$

Prove that the Zariski topology on projective space induces the Zariski topology on this affine space.

Exercise 12.15

Let

$$P = (a_0, \dots, a_n) \in \mathbb{P}^n_K.$$

Prove that there is an affine open neighbourhood

$$U \cong \mathbb{A}^n_K \subset \mathbb{P}^n_K$$

such that P corresponds to the origin in this affine space.

Exercise 12.16

Let $\mathbb{P}^n_K$ be projective n-space over a field K, and let

$$D_+(X_i) \cong \mathbb{A}^n_K, \qquad D_+(X_j) \cong \mathbb{A}^n_K$$

be two affine open subsets of $\mathbb{P}^n_K$. Describe the transition map from $D_+(X_i)$ to $D_+(X_j)$, which is not defined everywhere.

Exercise 12.17

Suppose that $m+1$ homogeneous polynomials

$$F_0, \dots, F_m$$

in $n+1$ variables are given, all of the same degree d. Prove that there is an open subset $U \subseteq \mathbb{P}^n_K$ on which these polynomials define a morphism

$$\mathbb{P}^n_K \supseteq U \longrightarrow \mathbb{P}^m_K.$$

Exercise 12.18

Let S be a $\mathbb{Z}$-graded ring with a homogeneous unit of degree one, and let $\mathfrak{a} \subseteq S$ be a homogeneous ideal. For $n \in \mathbb{Z}$, prove the equality of $(S/\mathfrak{a})_0$-modules

$$(S/\mathfrak{a})_n = (S/\mathfrak{a})_0 \otimes_{S_0} S_n.$$

Exercise 12.19

Let R be a standard-graded ring, $\mathfrak{a} \subseteq R$ a homogeneous ideal, and $f \in R$ a homogeneous element of degree 1. For $n \in \mathbb{Z}$, prove the equality of $(R_f/\mathfrak{a}_f)_0$-modules

$$((R/\mathfrak{a})_f)_n = (R_f/\mathfrak{a}_f)_n = (R_f/\mathfrak{a}_f)_0 \otimes_{(R_f)_0} (R_f)_n.$$

Public Solutions and Coverage of Worksheet 12

At the frozen revision boundary, the source provides exactly two public solutions among the 19 exercises, namely those for Exercises 12.5 and 12.10. The exercise map and candidate evidence record negative results for Exercises 12.1–12.4, 12.6–12.9, and 12.11–12.19. The absence of a public solution page is not replaced with a newly written solution.

Solution to Exercise 12.5

We choose the polynomial ring

$$K[X_1, X_2, X_3, X_4, X_5, Y_1, Y_2, Y_3, Y_4, Y_5]$$

over any base field (or, more generally, any nonzero base ring). Consider the ideals

$$\mathfrak{a} = (X_1X_2X_3X_4X_5,\ \text{all monomials of degree 2 except } X_iX_j \text{ with } i \neq j)$$

and

$$\mathfrak{b} = (Y_1Y_2Y_3Y_4Y_5,\ \text{all monomials of degree 2 except } Y_iY_j \text{ with } i \neq j).$$

The two groups of variables are interchanged. For each ideal, the maximum degree of a generator is 5, since $X_1X_2X_3X_4X_5$ (and, symmetrically, the product of all the Y_i) cannot be expressed using the other generating monomials.

We claim that the product ideal is generated by monomials of degree

$$d = 4.$$

By the symmetry of the situation, this follows from the following factorisations, where Q is in each case the appropriate quotient monomial:

$$X_1 \cdots X_5 Y_1 \cdots Y_5 = (X_1Y_1)(X_2Y_2)Q,$$

$$X_i^2 Y_1 \cdots Y_5 = (X_iY_1)(X_iY_2)Q,$$

$$X_iY_jY_1 \cdots Y_5 = (Y_1Y_2)(X_iY_j)Q,$$

and

$$Y_iY_jY_1 \cdots Y_5 = (Y_i^2)(Y_j^2)Q.$$

Each pair of factors on the right consists of one generator from $\mathfrak{a}$ and one from $\mathfrak{b}$, both of degree 2. Thus $\mathfrak{a}\mathfrak{b}$ has generators of degree at most 4, whereas $\mathfrak{a}$ and $\mathfrak{b}$ themselves require generators of degree 5.

> **Editorial note - repeated indices.** The source's last factorisation requires $i \neq j$. For the omitted case $i = j$, choose $k \neq i$ and use $Y_i^2 Y_1 \cdots Y_5 = (Y_iY_k)(Y_i^2) \prod_{\ell \neq i,k} Y_\ell$. The first factor is in $\mathfrak{a}$, the second in $\mathfrak{b}$. This explicitly fills the repeated-index case in the existing source argument.

Solution to Exercise 12.10

We must find a nontrivial solution of the linear system

$$6X - 8Y + 3Z = 0,$$

$$2X + 9Y - 5Z = 0.$$

Eliminating X gives the condition

$$-35Y + 18Z = 0.$$

Set $Z = 1$. Then

$$Y = \frac{18}{35}$$

and

$$\begin{aligned} X &= -\frac{9}{2}Y + \frac{5}{2}Z \\ &= -\frac{9}{2} \cdot \frac{18}{35} + \frac{5}{2} \\ &= \frac{-162 + 175}{70} \\ &= \frac{13}{70}. \end{aligned}$$

Consequently,

$$\left(\frac{13}{70}, \frac{18}{35}, 1\right)$$

is the intersection point of the two lines in the projective plane.

> **Editorial note - characteristic.** The source's divisions require characteristic different from 2, 5, and 7. The homogeneous representative $(13, 36, 70)$ satisfies both equations over every field and is nonzero in every characteristic; it provides the same intersection point without division. This characteristic qualification and homogeneous representative are editorial additions.

Frozen negative results

There is no public solution page at the frozen revision for Exercise 12.1, 12.2, 12.3, 12.4, 12.6, 12.7, 12.8, 12.9, 12.11, 12.12, 12.13, 12.14, 12.15, 12.16, 12.17, 12.18, or 12.19. This is a result of the candidate check, not a claim that mathematical solutions do not exist.

Lecture 13: The Cone Map, Sheaves of Modules, and Invertible Sheaves

The cone map

Definition 13.1: homogenisation of an ideal

Let R be a $\mathbb{Z}$-graded ring. For an ideal $\mathfrak{a}$, the ideal generated by all homogeneous elements of $\mathfrak{a}$ is called the *homogenisation* of $\mathfrak{a}$. It is denoted by $\mathfrak{a}^h$.

The homogenisation is again an ideal and is contained in the original ideal. The homogenisation of a prime ideal is a prime ideal.

Definition 13.2: cone map

Let R be a $\mathbb{Z}$-graded ring. The *cone map* is the scheme morphism

$$\begin{aligned} D(R_+) &\longrightarrow \mathrm{Proj}(R), \\ \mathfrak{p} &\longmapsto \mathfrak{p}^h. \end{aligned}$$

On the open subsets associated with homogeneous elements $f \in R_+$, it is given by the map on spectra associated with the homomorphism

$$(R_f)_0 \longrightarrow R_f.$$

Theorem 13.3: the cone map is a scheme morphism

Let R be a $\mathbb{Z}$-graded ring. Then the cone map is indeed a scheme morphism.

Proof The map is well defined because the homogenisation of a prime ideal is again prime. For a homogeneous element $f \in R_+$ and a prime ideal $\mathfrak{p}$,

$$f \in \mathfrak{p} \quad \Longleftrightarrow \quad f \in \mathfrak{p}^h.$$

Thus the inverse image of $D_+(f)$ is $D(f)$, and the map is continuous. The diagram of maps

$$\begin{array}{ccc} D(f) & \longrightarrow & D_+(f) \\ \downarrow & & \downarrow \\ \mathrm{Spek}(R_f) & \longrightarrow & \mathrm{Spek}((R_f)_0) \end{array}$$

commutes. The top row is the restriction of the cone map, the bottom row is the natural map on spectra, the left side is the identification from Lemma 9.13, and the right side is the identification from Lemma 12.8.

To prove commutativity, for a prime ideal $\mathfrak{p} \in D(f)$ we must show the equality

$$\mathfrak{p}^h \cap (R_f)_0 = \mathfrak{p} \cap (R_f)_0,$$

where the prime ideals are regarded as ideals in R_f. This equality rests on the argument for Lemma 12.8.

The morphism is thereby determined scheme-theoretically on $D(f)$. The morphisms

$$\mathrm{Spek}(R_f) \longrightarrow \mathrm{Spek}((R_f)_0)$$

for different f are compatible with one another and determine a global scheme morphism.

Example 13.4: the cone map for projective space

For the polynomial ring $K[X_0, X_1, \dots, X_n]$ in $n+1$ variables with the standard grading over a field K, the cone map is

$$\begin{aligned} \mathbb{A}_K^{n+1} \supset D(X_0, X_1, \dots, X_n) &\longrightarrow \mathbb{P}_K^n, \\ \mathfrak{p} &\longmapsto \mathfrak{p}^h. \end{aligned}$$

On K-points, this map is simply given by

$$\begin{aligned} K^{n+1} \setminus \{0\} &\longrightarrow \mathbb{P}^n(K), \\ (x_0, x_1, \dots, x_n) &\longmapsto (x_0, x_1, \dots, x_n). \end{aligned}$$

It sends each nonzero point to the line through that point and the origin.

Modules on a ringed space

Just as it is important to understand R-modules for a commutative ring R, for a ringed space we must understand the $\mathcal{O}_X$-modules on it.

Definition 13.5: module on a ringed space

A sheaf $\mathcal{M}$ on a ringed space $(X, \mathcal{O}_X)$ is called an *$\mathcal{O}_X$-module* if, for every open subset $U \subseteq X$, a $\Gamma(U, \mathcal{O}_X)$-module structure is given on $\Gamma(U, \mathcal{M})$, compatible with the restriction maps for $U \subseteq V$.

The compatibility condition means that, for open subsets $U \subseteq V$, the diagram

$$\begin{array}{ccc} \Gamma(V, \mathcal{O}_X) \times \Gamma(V, \mathcal{M}) & \longrightarrow & \Gamma(V, \mathcal{M}) \\ \downarrow & & \downarrow \\ \Gamma(U, \mathcal{O}_X) \times \Gamma(U, \mathcal{M}) & \longrightarrow & \Gamma(U, \mathcal{M}) \end{array}$$

commutes. In particular, the structure sheaf $\mathcal{O}_X$ is an $\mathcal{O}_X$-module. Every $\mathcal{O}_X$-module is, in particular, a sheaf of abelian groups.

Definition 13.6: subsheaf of modules

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{M}$ an $\mathcal{O}_X$-module. A subsheaf

$$\mathcal{N} \subseteq \mathcal{M}$$

such that, for every open subset $U \subseteq X$, $\Gamma(U, \mathcal{N})$ is a submodule of $\Gamma(U, \mathcal{M})$ over $\Gamma(U, \mathcal{O}_X)$ is called an *$\mathcal{O}_X$-submodule* of $\mathcal{M}$.

Definition 13.7: ideal sheaf

Let $(X, \mathcal{O}_X)$ be a ringed space. An $\mathcal{O}_X$-submodule

$$\mathcal{I} \subseteq \mathcal{O}_X$$

is called an *ideal sheaf*.

Definition 13.8: fibre of a module at a point

Let $(X, \mathcal{O}_X)$ be a locally ringed space and $\mathcal{M}$ an $\mathcal{O}_X$-module. For a point $P \in X$, the space

$$\mathcal{M}(P) := \mathcal{M}_P \otimes_{\mathcal{O}_{X,P}} \kappa(P)$$

is called the *fibre* of $\mathcal{M}$ at P.

In particular, this fibre is a vector space over the residue field $\kappa(P)$.

Definition 13.9: homomorphism of sheaves of modules

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{M}, \mathcal{N}$ be $\mathcal{O}_X$-modules on X. A sheaf morphism

$$\varphi : \mathcal{M} \longrightarrow \mathcal{N}$$

is called an *$\mathcal{O}_X$-module homomorphism* if, for every open subset $U \subseteq X$, the map

$$\Gamma(U, \mathcal{M}) \longrightarrow \Gamma(U, \mathcal{N})$$

is a $\Gamma(U, \mathcal{O}_X)$-module homomorphism.

An $\mathcal{O}_X$-module homomorphism is, in particular, a homomorphism of sheaves of abelian groups.

The following assertion is a version of the theorem from linear algebra that a linear map is determined by its values on a basis.

Theorem 13.10: global sections determine a module homomorphism

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{M}$ an $\mathcal{O}_X$-module on X. Global sections

$$s_1, \dots, s_n \in \Gamma(X, \mathcal{M})$$

uniquely determine a module homomorphism

$$\begin{aligned} \varphi : \mathcal{O}_X^n &\longrightarrow \mathcal{M}, \\ e_i &\longmapsto s_i. \end{aligned}$$

Proof For every open subset $U \subseteq X$,

$$\Gamma(U, \mathcal{O}_X^n) = \Gamma(U, \mathcal{O}_X)^n.$$

This is the free $\Gamma(U, \mathcal{O}_X)$-module with basis $e_1, \dots, e_n$. The global sections s_i give restrictions

$$s_i|_U \in \Gamma(U, \mathcal{M}).$$

By the theorem on specifying a map on a basis, these restrictions uniquely determine a $\Gamma(U, \mathcal{O}_X)$-module homomorphism

$$\begin{aligned} \Gamma(U, \mathcal{O}_X)^n &\longrightarrow \Gamma(U, \mathcal{M}), \\ e_i &\longmapsto s_i|_U. \end{aligned}$$

These module homomorphisms are compatible with restrictions to $V \subseteq U$ and therefore give a morphism of sheaves of modules.

Lemma 13.11: global sections as module homomorphisms

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{M}$ an $\mathcal{O}_X$-module on X. A global section

$$s \in \Gamma(X, \mathcal{M})$$

corresponds uniquely to a module homomorphism

$$\begin{aligned} \varphi : \mathcal{O}_X &\longrightarrow \mathcal{M}, \\ 1 &\longmapsto s. \end{aligned}$$

Proof This is a special case of Theorem 13.10.

Constructions for sheaves of modules

Definition 13.12: global homomorphism module

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{M}, \mathcal{N}$ sheaves of modules on X. The space

$$\mathrm{Hom}(\mathcal{M}, \mathcal{N}) = \{\varphi : \mathcal{M} \longrightarrow \mathcal{N} \mid \varphi \text{ is a module homomorphism}\},$$

with its natural $\Gamma(X, \mathcal{O}_X)$-module structure, is called the *global homomorphism module* from $\mathcal{M}$ to $\mathcal{N}$.

Essentially every construction for R-modules has an analogous construction for $\mathcal{O}_X$-modules. The guiding idea is that the constructed objects should again have the “right properties” within the category of all $\mathcal{O}_X$-modules. We should therefore expect that the definition above is not the last word and must be extended to a sheaf version.

Definition 13.13: homomorphism sheaf

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{M}, \mathcal{N}$ sheaves of modules on X. The assignment

$$U \longmapsto \mathrm{Hom}(\mathcal{M}|_U, \mathcal{N}|_U) = \{\varphi : \mathcal{M}|_U \longrightarrow \mathcal{N}|_U \mid \varphi \text{ is a module homomorphism}\}$$

is called the *homomorphism sheaf* from $\mathcal{M}$ to $\mathcal{N}$. It is denoted by

$$\mathcal{H}om(\mathcal{M}, \mathcal{N}).$$

Thus,

$$\mathcal{H}om(\mathcal{M}, \mathcal{N})(U) = \mathrm{Hom}(\mathcal{M}|_U, \mathcal{N}|_U).$$

Lemma 13.14: the homomorphism sheaf is a sheaf of modules

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{M}, \mathcal{N}$ sheaves of modules on X. Then the homomorphism sheaf

$$\mathcal{H}om(\mathcal{M}, \mathcal{N})$$

is a sheaf of $\mathcal{O}_X$-modules on X.

Proof We have the relation

$$\begin{aligned}\mathcal{H}om(\mathcal{M}, \mathcal{N})(U) &= \mathrm{Hom}(\mathcal{M}|_U, \mathcal{N}|_U) \\ &\subseteq \mathrm{Mor}(\mathcal{M}|_U, \mathcal{N}|_U).\end{aligned}$$

By Lemma 4.9, the right-hand side is a sheaf. The homomorphism property, that is, compatibility with addition and scalar multiplication, can be tested locally; see Exercise 13.11. Thus the left-hand side is a subsheaf. The $\mathcal{O}_X$-structure on $\mathcal{H}om(\mathcal{M}, \mathcal{N})$ is given by addition and scalar multiplication in the second component, and these operations are compatible with restrictions.

Definition 13.15: dual module

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{M}$ a sheaf of modules on X. The sheaf

$$\mathcal{M}^* = \mathcal{H}om(\mathcal{M}, \mathcal{O}_X),$$

with its natural $\mathcal{O}_X$-module structure, is called the *dual module* of $\mathcal{M}$.

Definition 13.16: tensor product of sheaves of modules

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{F}, \mathcal{G}$ sheaves of modules on X. The sheafification of the presheaf

$$U \longmapsto \Gamma(U, \mathcal{F}) \otimes_{\Gamma(U, \mathcal{O}_X)} \Gamma(U, \mathcal{G})$$

is called the *tensor product* of the two modules. It is denoted by

$$\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}.$$

By the universal property of sheafification, there is a canonical map

$$\Gamma(U, \mathcal{F}) \otimes_{\Gamma(U, \mathcal{O}_X)} \Gamma(U, \mathcal{G}) \longrightarrow \Gamma(U, \mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G})$$

which is an isomorphism on stalks. The stalk is the tensor product of the two stalks; see Exercise 13.13.

Invertible sheaves

Definition 13.17: invertible sheaf

An $\mathcal{O}_X$-module $\mathcal{L}$ on a ringed space $(X, \mathcal{O}_X)$ is called *invertible* if there is an open cover

$$X = \bigcup_{i \in I} U_i$$

such that the restrictions

$$\mathcal{L}|_{U_i}$$

are isomorphic to $\mathcal{O}_X|_{U_i}$.

An invertible sheaf $\mathcal{L}$ on a ringed space $(X, \mathcal{O}_X)$ is called *trivial* if it is isomorphic to the structure sheaf $\mathcal{O}_X$.

Example 13.18: the sheaf of sections of a line bundle

Let X be a topological space equipped with the sheaf of continuous functions $C(-, \mathbb{R})$, and let

$$L \longrightarrow X$$

be a real line bundle on X. Then the sheaf S of continuous sections in the sense of Example 3.12 is an invertible $C(-, \mathbb{R})$-module. Indeed, for an open subset $U \subseteq X$ with a trivialisation

$$L|_U \cong \mathbb{R} \times U,$$

we have

$$S(U, L|_U) \cong C(U, \mathbb{R}).$$

Example 13.19: invertible sheaves on projective space

> **Editorial note - domains and dimension.** In the formulae below, take $U = D_+(\mathfrak{a})$ nonempty. The empty open set has the zero module of sections. The global formula $(K[X_0, \dots, X_n])_\ell$ for every integer ℓ, and the nonisomorphism conclusion, require $n \geq 1$. For $n = 0$, every twist is trivial and its global sections form a one-dimensional K-vector space, also for negative ℓ. These qualifications are implicit or missing in the source.

Let K be a field and $\mathbb{P}^n_K$ projective space over K. For every open subset U, the structure sheaf gives a subset of the function field

$$\Gamma(U, \mathcal{O}_{\mathbb{P}^n_K}) \subseteq K\left(\frac{X_1}{X_0}, \dots, \frac{X_n}{X_0}\right).$$

Since the polynomial ring is factorial, for every homogeneous ideal $\mathfrak{a}$ there is a homogeneous polynomial f of maximal degree without repeated factors, uniquely determined up to multiplication by a scalar, such that

$$D_+(\mathfrak{a}) \subseteq D_+(f).$$

Here $f = 1$ is allowed, although in that case the notation $D_+(f)$ is not used. By Theorem 9.8, the ring of global sections is

$$\begin{aligned}\Gamma(U, \mathcal{O}_{\mathbb{P}^n_K}) &= \Gamma(D_+(f), \mathcal{O}_{\mathbb{P}^n_K}) \\ &= (K[X_0, X_1, \dots, X_n]_f)_0 \\ &\subseteq K\left(\frac{X_1}{X_0}, \dots, \frac{X_n}{X_0}\right).\end{aligned}$$

Fix $\ell \in \mathbb{Z}$. We define a sheaf $\mathcal{O}_{\mathbb{P}^n_K}(\ell)$ by

$$\begin{aligned}\Gamma(U, \mathcal{O}_{\mathbb{P}^n_K}(\ell)) &= \Gamma(D_+(f), \mathcal{O}_{\mathbb{P}^n_K}(\ell)) \\ &:= (K[X_0, X_1, \dots, X_n]_f)_\ell.\end{aligned}$$

This is an invertible sheaf. On $D_+(X_0)$, and likewise on $D_+(X_i)$, the map

$$\begin{aligned}(K[X_0, X_1, \dots, X_n]_{X_0})_0 &\longrightarrow (K[X_0, X_1, \dots, X_n]_{X_0})_\ell, \\ \frac{F}{G} &\longmapsto X_0^\ell \frac{F}{G}\end{aligned}$$

is an isomorphism of modules over $(K[X_0, X_1, \dots, X_n]_{X_0})_0$ that carries over to smaller open subsets. Evaluation on the whole projective space is simply

$$(K[X_0, X_1, \dots, X_n])_\ell.$$

This shows that, for $n \geq 1$, these invertible sheaves are pairwise nonisomorphic for $\ell \geq 0$. In fact, this holds for all ℓ.

Definition 13.20: invertibility locus of a section

Let $(X, \mathcal{O}_X)$ be a locally ringed space, $\mathcal{L}$ an invertible sheaf on X, and

$$s \in \Gamma(X, \mathcal{L})$$

a global section. The set

$$X_s := \{P \in X \mid s_P \notin \mathfrak{m}_P \mathcal{L}_P\}$$

is called the *invertibility locus* of s.

Its complement,

$$X \setminus X_s,$$

the zero locus of the section, is denoted by $Z(s)$.

Lemma 13.21: the invertibility locus is open

Let $(X, \mathcal{O}_X)$ be a locally ringed space, $\mathcal{L}$ an invertible sheaf on X, and

$$s \in \Gamma(X, \mathcal{L})$$

a global section. Then the invertibility locus

$$X_s \subseteq X$$

is open.

Proof The assertion follows from Lemma 7.16 by a local argument.

Lemma 13.22: trivialisation on the invertibility locus

Let $(X, \mathcal{O}_X)$ be a locally ringed space and $\mathcal{L}$ an invertible sheaf on X. Let

$$s \in \Gamma(X, \mathcal{L})$$

be a global section with invertibility locus X_s. Then the restriction

$$\mathcal{L}|_{X_s}$$

is trivial.

Proof By Lemma 13.11, the global section s determines a module homomorphism

$$\begin{aligned} \mathcal{O}_X &\longrightarrow \mathcal{L}, \\ 1 &\longmapsto s, \end{aligned}$$

and, in particular, a module homomorphism

$$\varphi : \mathcal{O}_X|_{X_s} \longrightarrow \mathcal{L}|_{X_s}.$$

At every point $P \in X_s$, this map is an isomorphism. Thus φ is also an isomorphism by Lemma 4.6.

Worksheet 13: The Cone Map, Sheaves of Modules, and Invertible Sheaves

The frozen source marks none of the exercises with an asterisk. Separately, the frozen candidate evidence records negative results for all 23 exercises. This edition creates no new solutions.

Exercise 13.1

Give an example showing that the cone map

$$\mathbb{A}_K^{n+1} \setminus \{0\} \longrightarrow \mathbb{P}_K^n$$

need not be a closed map.

Exercise 13.2

Let R be a $\mathbb{Z}$-graded ring and $\mathfrak{p}$ a prime ideal of R. Prove that the homogenisation $\mathfrak{p}^h$ is also a prime ideal.

Exercise 13.3

Let K be a field and

$$R = K[X_0, X_1, \ldots, X_n]/\mathfrak{a}$$

a standard-graded K-algebra. Prove that the diagram

$$\begin{array}{ccccc} \mathrm{Spek}(R) \supseteq D(R_+) & \longrightarrow & V(\mathfrak{a}) \cap D(X_0, X_1, \ldots, X_n) & \longrightarrow & \mathbb{A}_K^{n+1} \supseteq D(X_0, X_1, \ldots, X_n) \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Proj}(R) & \longrightarrow & V_+(\mathfrak{a}) & \longrightarrow & \mathbb{P}_K^n \end{array}$$

of scheme morphisms commutes. Here the vertical maps on the left and right are cone maps, and the horizontal maps are isomorphisms and the natural closed immersions.

Exercise 13.4

Discuss the relationship between the cone map

$$\mathbb{A}^2_{\mathbb{C}} \supset \mathbb{A}^2_{\mathbb{C}} \setminus \{(0,0)\} \longrightarrow \mathbb{P}^1_{\mathbb{C}}$$

and the Hopf fibration

$$S^3 \longrightarrow S^2.$$

Exercise 13.5

Let $\mathcal{F}$ be an $\mathcal{O}_X$-module on a ringed space $(X, \mathcal{O}_X)$. Prove that, for every point $P \in X$, the stalk $\mathcal{F}_P$ is an $\mathcal{O}_{X,P}$-module.

Exercise 13.6

Let $\mathcal{F}$ and $\mathcal{G}$ be $\mathcal{O}_X$-modules on a ringed space $(X, \mathcal{O}_X)$. Prove that the direct sum

$$\mathcal{F} \oplus \mathcal{G}$$

is also an $\mathcal{O}_X$-module.

Exercise 13.7

Let $\mathcal{F}$ be an $\mathcal{O}_X$-module on a ringed space $(X, \mathcal{O}_X)$, and let

$$\mathcal{G} \subseteq \mathcal{F}$$

be an $\mathcal{O}_X$-submodule. Prove that the quotient sheaf

$$\mathcal{F}/\mathcal{G}$$

is naturally an $\mathcal{O}_X$-module.

Exercise 13.8

Let $(X, \mathcal{O}_X)$ be a ringed space. Prove that

$$s \in \Gamma(X, \mathcal{O}_X)$$

is a unit if and only if the associated $\mathcal{O}_X$-module homomorphism

$$\mathcal{O}_X \longrightarrow \mathcal{O}_X$$

is an isomorphism.

Exercise 13.9

Let $(X, \mathcal{O}_X)$ be a ringed space. Let

$$s_1, \dots, s_n \in \Gamma(X, \mathcal{O}_X)$$

be global sections generating the unit ideal in $\Gamma(X, \mathcal{O}_X)$. Prove that the associated $\mathcal{O}_X$-module homomorphism

$$\begin{aligned} \mathcal{O}_X^n &\longrightarrow \mathcal{O}_X, \\ e_i &\longmapsto s_i, \end{aligned}$$

is surjective.

The converse of this assertion does not hold; see Exercise 14.10.

Exercise 13.10

Let $(X, \mathcal{O}_X)$ be a ringed space. Let

$$s_1, \dots, s_n \in \Gamma(X, \mathcal{O}_X)^n$$

be global sections with

$$s_i = (s_{i1}, \dots, s_{in}).$$

Prove that the determinant of the matrix

$$(s_{ij})_{1 \le i,j \le n}$$

is a unit in $\Gamma(X, \mathcal{O}_X)$ if and only if the associated $\mathcal{O}_X$-module homomorphism

$$\begin{aligned} \mathcal{O}_X^n &\longrightarrow \mathcal{O}_X^n, \\ e_i &\longmapsto s_i, \end{aligned}$$

is an isomorphism.

Exercise 13.11

Let $(X, \mathcal{O}_X)$ be a ringed space, $\mathcal{M}$ and $\mathcal{N}$ sheaves of modules on X, and

$$\varphi : \mathcal{M} \longrightarrow \mathcal{N}$$

a sheaf morphism. Also let

$$X = \bigcup_{i \in I} U_i$$

be an open cover. Prove the following assertions.

1. If, for every i, the map

   $$\varphi_{U_i} : \Gamma(U_i, \mathcal{M}) \longrightarrow \Gamma(U_i, \mathcal{N})$$

   is compatible with addition, then the same holds for

   $$\varphi_X : \Gamma(X, \mathcal{M}) \longrightarrow \Gamma(X, \mathcal{N}).$$

2. If, for every i, the map

$$\varphi_{U_i} : \Gamma(U_i, \mathcal{M}) \longrightarrow \Gamma(U_i, \mathcal{N})$$

is compatible with scalar multiplication by $\Gamma(U_i, \mathcal{O}_X)$, then the same holds for φ_X.

3. If, for every i, the map

$$\varphi|_{U_i} : \mathcal{M}|_{U_i} \longrightarrow \mathcal{N}|_{U_i}$$

is an $\mathcal{O}_X|_{U_i}$-module homomorphism, then φ is also an $\mathcal{O}_X$-module homomorphism.

Exercise 13.12

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{F}$ a sheaf of modules on X. Prove that there is a natural $\mathcal{O}_X$-module homomorphism

$$\mathcal{F} \longrightarrow \mathcal{F}^{**}.$$

Exercise 13.13

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{F}, \mathcal{G}$ sheaves of modules on X. Prove that the stalk at a point $P \in X$ of the presheaf

$$U \longmapsto \Gamma(U, \mathcal{F}) \otimes_{\Gamma(U, \mathcal{O}_X)} \Gamma(U, \mathcal{G})$$

equals

$$\begin{aligned}
&\operatorname{colim}_{P \in U} \left(\Gamma(U, \mathcal{F}) \otimes_{\Gamma(U, \mathcal{O}_X)} \Gamma(U, \mathcal{G})\right) \\
&\quad = (\operatorname{colim}_{P \in U} \Gamma(U, \mathcal{F})) \otimes_{(\operatorname{colim}_{P \in U} \Gamma(U, \mathcal{O}_X))} (\operatorname{colim}_{P \in U} \Gamma(U, \mathcal{G})) \\
&\quad = \mathcal{F}_P \otimes_{\mathcal{O}_{X,P}} \mathcal{G}_P.
\end{aligned}$$

Editorial note - colimit notation. The source interchanges each colimit's index and argument. This edition places $P \in U$ in the subscript and the section module in the argument. The index runs over open neighbourhoods of P, with maps given by restriction to smaller neighbourhoods; the asserted tensor-product identity is unchanged.

Exercise 13.14

Let $\mathcal{F}$ be a sheaf of modules on a ringed space $(X, \mathcal{O}_X)$, and let $\mathcal{F}^*$ be its dual sheaf. Prove that there is a natural $\mathcal{O}_X$-module homomorphism

$$\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{F}^* \longrightarrow \mathcal{O}_X.$$

Exercise 13.15

Let $(X, \mathcal{O}_X)$ be a locally ringed space and $\mathcal{L}$ an invertible sheaf on X. Let $U \subseteq X$ be an open subset such that the restriction $\mathcal{L}|_U$ is trivial, and let

$$\varphi : \mathcal{L}|_U \longrightarrow \mathcal{O}_X|_U$$

be an isomorphism. Let

$$s \in \Gamma(X, \mathcal{L})$$

be a global section with invertibility locus X_s. Prove that

$$X_s \cap U = U_{\varphi(s)},$$

where the right-hand side denotes the invertibility locus of $\varphi(s) \in \Gamma(U, \mathcal{O}_X|_U)$.

Exercise 13.16

Let $\mathcal{L}$ be an invertible sheaf on a ringed space $(X, \mathcal{O}_X)$. Prove that the dual sheaf

$$\mathcal{L}^*$$

is also invertible.

Exercise 13.17

Let $\mathcal{L}$ and $\mathcal{M}$ be invertible sheaves on a ringed space $(X, \mathcal{O}_X)$. Prove that the tensor product

$$\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{M}$$

is also invertible.

Exercise 13.18

Let $\mathcal{L}$ and $\mathcal{M}$ be invertible sheaves on a locally ringed space $(X, \mathcal{O}_X)$. Let

$$s \in \Gamma(X, \mathcal{L}), \qquad t \in \Gamma(X, \mathcal{M}),$$

and

$$st \in \Gamma(X, \mathcal{L} \otimes \mathcal{M}).$$

Prove that their invertibility loci satisfy

$$X_{st} = X_s \cap X_t.$$

Exercise 13.19

Prove that an invertible sheaf $\mathcal{L}$ on a ringed space $(X, \mathcal{O}_X)$ is naturally isomorphic to its bidual $\mathcal{L}^{**}$.

Exercise 13.20

Let $\mathcal{L}$ be an invertible sheaf on a ringed space $(X, \mathcal{O}_X)$ and $\mathcal{L}^*$ its dual sheaf. Prove that there is a natural $\mathcal{O}_X$-module isomorphism

$$\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{L}^* \longrightarrow \mathcal{O}_X.$$

Exercise 13.21

Consider projective space over a field K and the invertible sheaf

$$\mathcal{O}_{\mathbb{P}^n_K}(m)$$

for $m > 0$. Let

$$f \in \Gamma\Big(\mathbb{P}^n_K, \mathcal{O}_{\mathbb{P}^n_K}(m)\Big) = K[X_0, X_1, \ldots, X_n]_m.$$

Prove the following equality of invertibility loci:

$$(\mathbb{P}^n_K)_f = D_+(f).$$

Exercise 13.22

Consider projective space over a field K and the invertible sheaves $\mathcal{O}_{\mathbb{P}^n_K}(\ell)$. Prove that

$$\mathcal{O}_{\mathbb{P}^n_K}(\ell) \otimes \mathcal{O}_{\mathbb{P}^n_K}(m) \cong \mathcal{O}_{\mathbb{P}^n_K}(\ell + m).$$

Exercise 13.23

Let

$$0 \neq F \in K[X_0, X_1, \ldots, X_n]$$

be a homogeneous polynomial of degree d. Prove that it determines a short exact sequence of sheaves

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^n_K}(-d) \xrightarrow{\cdot F} \mathcal{O}_{\mathbb{P}^n_K} \longrightarrow \mathcal{O}_{V_+(F)} \longrightarrow 0$$

on projective space. Here the structure sheaf of the projective hypersurface $V_+(F)$ is regarded as a sheaf on projective space.

Public Solutions and Coverage of Worksheet 13

At the frozen revision boundary, the source provides no public solution page for any of the 23 exercises. The exercise map and candidate evidence record negative results for Exercises 13.1–13.23. The absence of a public solution page is not replaced with a newly written solution.

Frozen negative results

There is no public solution page at the frozen revision for Exercise 13.1, 13.2, 13.3, 13.4, 13.5, 13.6, 13.7, 13.8, 13.9, 13.10, 13.11, 13.12, 13.13, 13.14, 13.15, 13.16, 13.17, 13.18, 13.19, 13.20, 13.21, 13.22, or 13.23. This is a result of the candidate check, not a claim that mathematical solutions do not exist.

Lecture 14: Quasicoherent Modules

Quasicoherent modules on affine schemes

For a commutative ring R, its R-modules are important objects that characterise the ring: for example, ideals, quotient rings, projective modules, and the module of Kähler differentials. We want to recover these modules in the context of the spectrum, that is, in a geometrised form. The construction runs parallel to the introduction of the structure sheaf on the spectrum.

Example 14.1: the presheaf arising from a module

Let M be an R-module over a commutative ring R, and write $X = \text{Spek}(R)$. We can define a presheaf of modules by setting, for each open subset $U \subseteq X$,

$$\mathcal{P}(U) = \operatorname*{colim}_{U \subseteq D(f)} M_f.$$

These are modules over the ring

$$\operatorname*{colim}_{U \subseteq D(f)} R_f,$$

and there are natural restriction homomorphisms compatible with the module structures. The stalk of this presheaf at a prime ideal $\mathfrak{p}$ is $M_\mathfrak{p}$.

Definition 14.2: the sheaf of modules on the spectrum

Let

$$(X, \mathcal{O}_X) = \text{Spek}(R)$$

be the affine scheme of a commutative ring R, and let M be an R-module. The *$\mathcal{O}_X$-module associated with M*, denoted by $\widetilde{M}$, is the assignment that associates to each open subset $U \subseteq X$ the commutative group

$$\begin{aligned}\Gamma(U, \widetilde{M}) = \Bigg\{ (s_\mathfrak{p})_{\mathfrak{p} \in U} \in \prod_{\mathfrak{p} \in U} M_\mathfrak{p} \;\Bigg|\; & \\ & \text{for every } \mathfrak{p} \in U \text{ there are } m \in M \text{ and } f \in R \text{ with } \mathfrak{p} \in D(f) \subseteq U, \\ & \text{and } s_\mathfrak{q} = m/f \text{ in } M_\mathfrak{q} \text{ for all } \mathfrak{q} \in D(f) \Bigg\},\end{aligned}$$

together with scalar multiplication

$$\begin{aligned}\Gamma(U, \mathcal{O}_X) \times \Gamma(U, \widetilde{M}) &\longrightarrow \Gamma(U, \widetilde{M}), \\ ((g_\mathfrak{p})_{\mathfrak{p} \in U}, (s_\mathfrak{p})_{\mathfrak{p} \in U}) &\longmapsto (g_\mathfrak{p} s_\mathfrak{p})_{\mathfrak{p} \in U}.\end{aligned}$$

To each inclusion $U \subseteq V$ it assigns the natural projection.

Starting the construction with the ring R itself gives the structure sheaf.

Lemma 14.3: the construction gives an $\mathcal{O}_X$-module

For an R-module M over a commutative ring R, $\widetilde{M}$ is an $\mathcal{O}_X$-module on the affine scheme $X = \mathrm{Spek}(R)$.

Proof This follows from the fact that $\widetilde{M}$ is defined as the sheafification of the presheaf

$$U \longmapsto \operatorname*{colim}_{U \subseteq D(f)} M_f,$$

and the module structure is inherited by its sheafification.

Lemma 14.4: stalks are localisations

Let $(X, \mathcal{O}_X)$ be the affine scheme of a commutative ring R, and let $x \in X$ be the point corresponding to a prime ideal $\mathfrak{p}$. If M is an R-module with associated sheaf of modules $\widetilde{M}$, then its stalk is

$$\widetilde{M}_x = M_{\mathfrak{p}}.$$

Proof This follows from Example 14.1 and Lemma 5.2(2).

Lemma 14.5: sections on principal open subsets

Let $(X, \mathcal{O}_X)$ be the affine scheme of a commutative ring R, and let M be an R-module with associated $\mathcal{O}_X$-module $\widetilde{M}$. For $f \in R$,

$$\Gamma(D(f), \widetilde{M}) = M_f.$$

In particular, the module of global sections is

$$\Gamma(X, \widetilde{M}) = M.$$

Proof We first prove the special case stated last. There is a natural R-module homomorphism

$$M \longrightarrow \Gamma(X, \widetilde{M}).$$

It is injective because the vanishing of an element can be tested locally; compare Appendix Lemma 1.1. For surjectivity, let $s \in \Gamma(X, \widetilde{M})$ be a global element. This means that there is an open cover

$$X = \bigcup_{i \in I} U_i = \bigcup_{i \in I} D(f_i)$$

and elements

$$s_i = \frac{a_i}{f_i^{k_i}}, \qquad a_i \in M,$$

which agree as sections on

$$D(f_i) \cap D(f_j) = D(f_i f_j),$$

that is, as elements of $M_{f_i f_j}$. By Corollary 8.6, we may assume that I is finite. We may also replace all k_i by their maximum k; naturally, this also changes the local numerators a_i. The compatibility

$$\frac{a_i}{f_i^k} = \frac{a_j}{f_j^k}$$

means that there are equations

$$(f_i f_j)^m a_i f_j^k = (f_i f_j)^m a_j f_i^k$$

in M, where m is chosen as a maximum valid for all pairs. By Proposition 8.4(2),(4), the elements f_i, $i \in I$, generate the unit ideal. The same holds for f_i^{m+k}, so there are $g_i \in R$ with

$$1 = \sum_{i \in I} g_i f_i^{m+k}.$$

Set

$$a := \sum_{i \in I} g_i a_i f_i^m.$$

Then, for each j,

$$\begin{aligned} a f_j^{m+k} &= \sum_{i \in I} g_i a_i f_i^m f_j^{m+k} \\ &= \sum_{i \in I} g_i (f_i f_j)^m a_i f_j^k \\ &= \sum_{i \in I} g_i (f_i f_j)^m a_j f_i^k \\ &= a_j f_j^m \left(\sum_{i \in I} g_i f_i^{m+k} \right) \\ &= a_j f_j^m. \end{aligned}$$

This means that

$$\frac{a}{1} = \frac{a_j}{f_j^k} = s_j$$

in M_{f_j}, so the section is represented by a single module element.

Now consider the situation on $D(f)$. It is the case just treated, with R_f as the new ring and M_f as the new module.

Example 14.6: a nonprincipal ideal giving an invertible sheaf

In the quadratic number ring

$$R = A_{-5} = \mathbb{Z}[\sqrt{-5}] = \mathbb{Z}[T]/(T^2 + 5)$$

we have the equality

$$2 \cdot 3 = 6 = (1 + \sqrt{5}\,\mathrm{i})(1 - \sqrt{5}\,\mathrm{i}).$$

Consider the ideal

$$I = (2, 1 + \sqrt{-5}),$$

which is prime but not principal, and its associated ideal sheaf $\tilde{I}$ on $X = \mathrm{Spek}(R)$. The spectrum is covered by the two open sets $D(2)$ and $D(3)$. We have

$$\tilde{I}|_{D(2)} \cong \mathcal{O}_X|_{D(2)},$$

since $2 \in I$, so the ideal becomes the unit ideal in the localisation R_2. In R_3, that is, on $D(3)$,

$$2 = \frac{1-\sqrt{5}\,\mathrm{i}}{3}(1+\sqrt{5}\,\mathrm{i}),$$

so I_3 is principal with generator $1+\sqrt{5}\,\mathrm{i}$. Hence

$$\tilde{I}|_{D(3)} \cong \mathcal{O}_X|_{D(3)},$$

and $\tilde{I}$ is an invertible sheaf.

Example 14.7: an invertible ideal on a punctured singularity

Consider the ideal $I = (X, Z)$ in the A_{n-1} singularity

$$R = K[X, Y, Z]/(XY - Z^n).$$

It defines an ideal sheaf $\tilde{I}$ on $\mathrm{Spek}(R)$, and hence also a restricted ideal sheaf $\tilde{I}|_U$ on the quasiaffine scheme

$$U = D(X, Y, Z) = D(X, Y) = \mathrm{Spek}(R) \setminus \{(X, Y, Z)\} \subset \mathrm{Spek}(R).$$

This restricted sheaf is invertible on U. Indeed, $X \in I$, and in R_Y,

$$X = \frac{Z^{n-1}}{Y} Z.$$

There are therefore isomorphisms

$$\tilde{I}|_{D(X)} \cong \mathcal{O}_{\mathrm{Spek}(R)}|_{D(X)} \qquad \text{and} \qquad \tilde{I}|_{D(Y)} \cong \mathcal{O}_{\mathrm{Spek}(R)}|_{D(Y)}.$$

By contrast, $\tilde{I}$ is not invertible on the whole spectrum, since the ideal I in the localisation $R_{(X,Y,Z)}$ is not principal.

> **Editorial note - singularity parameter.** This noninvertibility assertion requires $n \geq 2$. For $n = 1$, the relation is $Z = XY$, and $I = (X, Z) = (X)$ is principal. The source leaves this lower bound implicit in the singularity terminology.

Lemma 14.8: module homomorphisms induce sheaf morphisms

Let $(X, \mathcal{O}_X)$ be the affine scheme of a commutative ring R, and let

$$\varphi : M \longrightarrow N$$

be an R-module homomorphism. There is exactly one $\mathcal{O}_X$-module homomorphism

$$\widetilde{M} \longrightarrow \widetilde{N}$$

that agrees globally with φ.

Proof For each $f \in R$, compatibility with restrictions requires the following diagram to commute:

$$\begin{array}{ccc} \Gamma(X, \widetilde{M}) = M & \xrightarrow{\varphi} & \Gamma(X, \widetilde{N}) = N \\ \downarrow & & \downarrow \\ \Gamma(D(f), \widetilde{M}) = M_f & \longrightarrow & \Gamma(D(f), \widetilde{N}) = N_f. \end{array}$$

The diagram uniquely determines the bottom map. These assignments then determine a unique presheaf morphism and, by sheafification, a unique sheaf morphism.

Lemma 14.9: short exact sequences pass to sheaves

Let R be a commutative ring and

$$0 \longrightarrow L \longrightarrow M \longrightarrow N \longrightarrow 0$$

a short exact sequence of R-modules. On the affine scheme $(X, \mathcal{O}_X) = \text{Spek}(R)$ there is a short exact sequence of sheaves

$$0 \longrightarrow \widetilde{L} \longrightarrow \widetilde{M} \longrightarrow \widetilde{N} \longrightarrow 0$$

consisting of quasicoherent $\mathcal{O}_X$-modules.

Proof By Appendix Lemma 2.2, for every prime ideal $\mathfrak{p}$, the original short exact sequence gives a short exact sequence

$$0 \longrightarrow L_{\mathfrak{p}} \longrightarrow M_{\mathfrak{p}} \longrightarrow N_{\mathfrak{p}} \longrightarrow 0.$$

By Lemma 14.4, this is the stalkwise version of the module homomorphisms between $\widetilde{L}$, $\widetilde{M}$, and $\widetilde{N}$ at the point $\mathfrak{p}$. By Lemma 6.3, this says precisely that the complex of sheaves is exact.

Lemma 14.10: tensor products of affine sheaves of modules

Let R be a commutative ring, M and N be R-modules, and $\widetilde{M}$ and $\widetilde{N}$ the associated sheaves of modules on $X = \text{Spek}(R)$. There is a canonical isomorphism

$$\widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N} \cong \widetilde{M \otimes_R N}.$$

Proof We have

$$M_f \otimes_{R_f} N_f = (M \otimes_R N)_f.$$

Consider the presheaf

$$U \longmapsto \underset{U \subseteq D(f)}{\text{colim}} (M_f \otimes_{R_f} N_f) = \underset{U \subseteq D(f)}{\text{colim}} (M \otimes_R N)_f.$$

By definition, sheafification of the right-hand side gives the quasicoherent sheaf $\widetilde{M \otimes_R N}$. For open subsets $U \subseteq D(f)$, there are canonical module homomorphisms

$$M_f \otimes_{R_f} N_f \longrightarrow \left(\underset{U \subseteq D(f)}{\text{colim}} M_f \right) \otimes_{\text{colim}_{U \subseteq D(f)} R_f} \left(\underset{U \subseteq D(f)}{\text{colim}} N_f \right).$$

Taking colimits gives, for every open subset, a module homomorphism

$$\operatorname*{colim}_{U\subseteq D(f)} (M_f \otimes_{R_f} N_f) \longrightarrow \left(\operatorname*{colim}_{U\subseteq D(f)} M_f\right) \otimes_{\operatorname{colim}_{U\subseteq D(f)} R_f} \left(\operatorname*{colim}_{U\subseteq D(f)} N_f\right).$$

This is a map from the first presheaf to the tensor product of the two presheaves of modules. These homomorphisms are compatible with restrictions, so they form a presheaf morphism. By Lemma 5.2(1),(5), it passes to the associated sheaves. By the initial observation, the sheafification on the left is $\widetilde{M \otimes_R N}$, while by definition the sheafification on the right is $\widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N}$. Since this homomorphism is an isomorphism on every stalk, Lemma 4.6 shows that it is an isomorphism of sheaves.

Quasicoherent modules

For arbitrary schemes, the most important sheaves of modules are those that look like $\widetilde{M}$ on affine pieces.

Definition 14.11: quasicoherent module

An $\mathcal{O}_X$-module $\mathcal{M}$ on a scheme $(X, \mathcal{O}_X)$ is called *quasicoherent* if there is an affine open cover

$$X = \bigcup_{i\in I} U_i, \qquad U_i = \operatorname{Spek}(R_i),$$

and R_i-modules M_i such that

$$\mathcal{M}|_{U_i} = \widetilde{M_i}.$$

In particular, the structure sheaf of a scheme is quasicoherent, since on an affine open subset $U = \operatorname{Spek}(R)$ it agrees with $\widetilde{R}$. Invertible sheaves are also quasicoherent.

One can prove that, for a quasicoherent sheaf, its restriction to every affine open subset $U \subseteq X$ already equals the sheaf of modules associated with a module over the ring $\Gamma(U, \mathcal{O}_X)$.

Definition 14.12: coherent module

A quasicoherent $\mathcal{O}_X$-module $\mathcal{M}$ on a scheme $(X, \mathcal{O}_X)$ is called *coherent* if there is an affine open cover

$$X = \bigcup_{i\in I} U_i$$

such that $\Gamma(U_i, \mathcal{M})$ is a finitely generated module over $\Gamma(U_i, \mathcal{O}_X)$.

> **Editorial note - terminology outside the noetherian case.** This is the source's convention for "coherent". On a locally noetherian scheme it agrees with the usual definition. On an arbitrary scheme the displayed condition means quasicoherent of finite type; usual coherence additionally imposes finite-type relations, so the two notions must not be identified without further hypotheses.

On an affine scheme $\operatorname{Spek}(R)$, quasicoherent modules and R-modules correspond. In particular, on an affine scheme a quasicoherent module $\mathcal{M}$ has "many" global sections, which can be used to understand and reconstruct $\mathcal{M}$. This is by no means true in general for quasicoherent sheaves on nonaffine schemes, and in particular it often fails on projective schemes. There it is even common for a complicated quasicoherent module to have the zero module as its global evaluation.

In such a situation, suitable invertible sheaves can be used to "twist" the module so that the twisted version has global sections. The following general theorem provides guidance. Note that elements

$$g \in \Gamma(X, \mathcal{L}), \qquad s \in \Gamma(X, \mathcal{M}),$$

with $\mathcal{L}$ invertible, define via sheafification elements

$$g^n s \in \Gamma(X, \mathcal{L}^n \otimes \mathcal{M}),$$

where $\mathcal{L}^n$ denotes the nth tensor power of $\mathcal{L}$.

Theorem 14.13: global extension from the invertibility locus

Let $\mathcal{M}$ be a quasicoherent $\mathcal{O}_X$-module on a noetherian scheme $(X, \mathcal{O}_X)$. Let $\mathcal{L}$ be an invertible sheaf on X and

$$g \in \Gamma(X, \mathcal{L})$$

a global section with invertibility locus X_g. Then the following assertions hold.

1. For a global section $r \in \Gamma(X, \mathcal{M})$ with $r|_{X_g} = 0$, there is an $m \in \mathbb{N}$ such that

   $$g^m r = 0$$

   in $\Gamma(X, \mathcal{L}^m \otimes \mathcal{M})$.
2. For a section $s \in \Gamma(X_g, \mathcal{M})$, there is an $n \in \mathbb{N}$ such that

   $$g^n s \in \Gamma(X_g, \mathcal{L}^n \otimes \mathcal{M})$$

   comes from a global section in $\Gamma(X, \mathcal{L}^n \otimes \mathcal{M})$.

Proof Choose a finite affine open cover

$$X = \bigcup_{i \in I} U_i$$

such that the restriction of $\mathcal{L}$ to every U_i is trivial. Consider

$$V_i = X_g \cap U_i \subseteq U_i,$$

an open subset of the affine scheme $U_i = \mathrm{Spek}(R_i)$. There is an R_i-module M_i with

$$\mathcal{M}|_{U_i} = \widetilde{M_i}.$$

Under the isomorphism $\mathcal{L}|_{U_i} \cong \mathcal{O}_{U_i}$, the restriction of g to U_i corresponds to a function $f_i \in R_i$, and its invertibility locus satisfies

$$X_g \cap U_i = D(f_i).$$

Thus,

$$\Gamma(V_i, \mathcal{M}) \cong (M_i)_{f_i}.$$

1. Let $r_i = r|_{U_i} \in M_i$. By assumption, its restriction to V_i is zero, so there is an $m_i \in \mathbb{N}$ with

   $$f_i^{m_i} r_i = 0$$

   in M_i; the equation continues to hold for every larger exponent. Translated into $\mathcal{L}^{m_i} \otimes \mathcal{M}$, this means that the global element $g^{m_i} r$ restricts to zero on U_i. Hence, setting

   $$m = \max(m_i \mid i \in I),$$

   we obtain an m such that $g^m r$ vanishes on every U_i. By the sheaf property, $g^m r = 0$ on X.

2. The given section $s \in \Gamma(X_g, \mathcal{M})$ yields, by restriction, sections

$$s_i \in \Gamma(V_i, \mathcal{M}) = \Gamma(D(f_i), \widetilde{M_i}) = (M_i)_{f_i}.$$

Thus

$$s_i = \frac{t_i}{f_i^{\ell_i}}$$

with $t_i \in M_i$. The exponents ℓ_i can be increased, so we may assume that such a representation holds for every i with a common exponent ℓ. This means that the restriction of

$$g^{\ell} s \in \Gamma(X_g, \mathcal{L}^{\ell} \otimes \mathcal{M})$$

to $X_g \cap U_i$ comes from an element

$$t_i \in \Gamma(U_i, \mathcal{L}^{\ell} \otimes \mathcal{M}).$$

In general, these elements t_i are not yet compatible. However, the restriction of $t_i - t_j$ to $X_g \cap U_i \cap U_j$ is zero. Applying the first part to

$$X_g \cap U_i \cap U_j \subseteq U_i \cap U_j,$$

gives an m_{ij} such that

$$g^{m_{ij}}(t_i - t_j) = 0$$

in $\Gamma(U_i \cap U_j, \mathcal{L}^{\ell + m_{ij}} \otimes \mathcal{M})$. Multiply the entire situation by g^m, where m is the maximum of all the m_{ij}. The local elements are then compatible, so for $n = \ell + m$ we obtain a global extension of $g^n s$.

Editorial note - proof notation. In part 1 the source introduces n_i but uses m_i throughout the equation and maximum; this edition consistently uses m_i. In part 2 the source writes $\Gamma(D(f_i), M_i)$ for sections of the associated sheaf; the tilde is made explicit here. Neither correction changes the argument.

Worksheet 14: Quasicoherent Modules

At the frozen revision boundary, none of the 26 exercises has a public solution page. This edition preserves those negative candidate results and creates no new solutions.

Exercise 14.1

Let $\mathfrak{a} \subseteq R$ be an ideal of a commutative ring R, and let $X = \text{Spek}(R)$. Prove that

$$\tilde{\mathfrak{a}}|_{D(\mathfrak{a})} \cong \mathcal{O}_X|_{D(\mathfrak{a})}.$$

Exercise 14.2

Let M be an R-module over a commutative ring R, and let $f \in R$. Prove that

$$\widetilde{M}|_{D(f)} \cong \widetilde{M_f},$$

where the right-hand side denotes the sheaf of modules associated with the R_f-module M_f.

Exercise 14.3

Let R be a commutative ring and M a finitely generated R-module. Let $\mathfrak{p} \in \operatorname{Spek}(R)$ be a prime ideal with $M_{\mathfrak{p}} = 0$. Prove that there is an $f \notin \mathfrak{p}$ with

$$M_f = 0.$$

Exercise 14.4

Let R be a commutative ring and

$$\varphi : M \longrightarrow N$$

an R-module homomorphism between finitely generated R-modules. Let $\mathfrak{p} \in \operatorname{Spek}(R)$ be a prime ideal such that the induced homomorphism

$$\varphi : M_{\mathfrak{p}} \longrightarrow N_{\mathfrak{p}}$$

is surjective. Prove that there is an $f \notin \mathfrak{p}$ such that

$$\varphi : M_f \longrightarrow N_f$$

is surjective.

Exercise 14.5

Let R be a noetherian commutative ring and

$$\varphi : M \longrightarrow N$$

an R-module homomorphism between finitely generated R-modules. Let $\mathfrak{p} \in \operatorname{Spek}(R)$ be a prime ideal such that the induced homomorphism

$$\varphi : M_{\mathfrak{p}} \longrightarrow N_{\mathfrak{p}}$$

is injective. Prove that there is an $f \notin \mathfrak{p}$ such that

$$\varphi : M_f \longrightarrow N_f$$

is injective.

Exercise 14.6

Using $R = \mathbb{Z}$ and $M = \mathbb{Q}/\mathbb{Z}$, show that the assertions in Exercises 14.3, 14.4, and 14.5 are false without the assumption of finite generation.

Exercise 14.7

Let K be a field,

$$R = K[X_n, Y_n,\ n \in \mathbb{N}]/(X_n Y_n,\ n \in \mathbb{N}),$$

and

$$\mathfrak{p} = (X_n,\ n \in \mathbb{N}) \subseteq R.$$

1. Prove that $\mathfrak{p}$ is a prime ideal.
2. Prove that $\mathfrak{p}_{\mathfrak{p}} = 0$.
3. Prove that $\mathfrak{p}_f \neq 0$ for every $f \notin \mathfrak{p}$.
4. Prove that

$$\varphi : R \longrightarrow R/\mathfrak{p}$$

   becomes injective after localisation at $\mathfrak{p}$ (and hence also bijective), but that no localisation of this homomorphism at a single element $f \notin \mathfrak{p}$ is injective.

Exercise 14.8

Let R be an integral domain with fraction field $Q(R)$. Prove that

$$\widetilde{Q(R)}$$

is a constant sheaf on the spectrum $\operatorname{Spek}(R)$.

Exercise 14.9

Let R be a commutative ring and M, N be R-modules. Prove that

$$\widetilde{\operatorname{Hom}(M, N)} \cong \mathcal{H}om(\widetilde{M}, \widetilde{N}).$$

> **Editorial note - missing finiteness hypothesis.** The source states this for arbitrary modules, but the comparison is not an isomorphism in that generality. Add the hypothesis that M is finitely presented (with N arbitrary), so that localisation commutes with $\operatorname{Hom}_R(M, -)$. This is a correction to the exercise, not an added source solution.

Exercise 14.10

Let

$$U = (\mathbb{A}^2_K \setminus \{(0,0)\}, \mathcal{O}_X)$$

be the punctured affine plane. Give an example of global sections

$$s_1, s_2 \in \Gamma(U, \mathcal{O}_X)$$

such that (s_1, s_2) is not the unit ideal, but the associated $\mathcal{O}_X$-module homomorphism

$$\mathcal{O}_U^2 \longrightarrow \mathcal{O}_U, \qquad e_i \longmapsto s_i,$$

is surjective.

Exercise 14.11

For $R = K[X, Y]$, consider the short exact sequence

$$0 \longrightarrow R \longrightarrow R^2 \longrightarrow (X, Y) = \mathfrak{m} \longrightarrow 0,$$

where the right-hand map sends the standard basis vectors to the ideal generators, while the left-hand map sends 1 to $(Y, -X)$. By Lemma 14.9, this gives an exact sequence of sheaves

$$0 \longrightarrow \mathcal{O}_{\text{Spek}(R)} \longrightarrow \mathcal{O}^2_{\text{Spek}(R)} \longrightarrow \widetilde{\mathfrak{m}} \longrightarrow 0.$$

Let $U = D(X, Y)$. Prove the following assertions.

1. $\Gamma(U, \mathcal{O}_{\text{Spek}(R)}) = R$.
2. $\Gamma(U, \widetilde{\mathfrak{m}}) = R$.
3. Evaluation of this exact sequence of sheaves on U gives

$$0 \longrightarrow R \longrightarrow R^2 \longrightarrow R,$$

   and the right-hand map is not surjective.

Exercise 14.12

Let R be a commutative ring and $I \subseteq R$ an ideal with associated short exact sequence

$$0 \longrightarrow I \longrightarrow R \longrightarrow R/I \longrightarrow 0.$$

Interpret the corresponding short exact sequence of sheaves

$$0 \longrightarrow \tilde{I} \longrightarrow \widetilde{R} \longrightarrow \widetilde{R/I} \longrightarrow 0$$

on the spectrum of R. On which open subsets and at which points do the objects—their evaluations and stalks, respectively—become zero, and the homomorphisms become isomorphisms?

Exercise 14.13

Let $\theta : A \to B$ be a ring homomorphism between commutative rings A and B, and let

$$\varphi : \text{Spek}(B) \longrightarrow \text{Spek}(A)$$

be the associated map on spectra. Let N be a B-module with associated sheaf of modules $\widetilde{N}$ on Spek(B). Prove that

$$\varphi_*(\widetilde{N}) = \widetilde{N'},$$

where N' is simply the B-module N regarded as an A-module.

Exercise 14.14

Let $\theta : A \to B$ be a ring homomorphism between commutative rings A and B, and let

$$\varphi : \text{Spek}(B) \longrightarrow \text{Spek}(A)$$

be the associated map on spectra. Let M be an A-module with associated sheaf of modules $\widetilde{M}$ on Spek(A). Prove that

$$\varphi^*(\widetilde{M}) = \widetilde{M \otimes_A B}$$

on Spek(B).

The following exercise describes the ring-theoretic version of Appendix Lemma 4.3. Together with the preceding two exercises, it recovers the spectrum version of that assertion.

Exercise 14.15

Let $\theta : A \to B$ be a homomorphism between commutative rings A and B. Let M be an A-module and N a B-module. Prove that there is a natural group isomorphism

$$\operatorname{Hom}_B(M \otimes_A B, N) = \operatorname{Hom}_A(M, N'),$$

where N' denotes the B-module N regarded as an A-module.

Exercise 14.16

Let R be a commutative ring and $\mathcal{M}$ a quasicoherent module on Spek(R). Prove that

$$\mathcal{M} \cong \widetilde{M}$$

for some R-module M.

Exercise 14.17

Let $\mathcal{F}$ and $\mathcal{G}$ be quasicoherent modules on a scheme $(X, \mathcal{O}_X)$. Prove that the direct sum

$$\mathcal{F} \oplus \mathcal{G}$$

is also quasicoherent.

Exercise 14.18

Let $\mathcal{F}$ and $\mathcal{G}$ be coherent modules on a scheme $(X, \mathcal{O}_X)$. Prove that the direct sum

$$\mathcal{F} \oplus \mathcal{G}$$

is also coherent.

Exercise 14.19

Let $\mathcal{F}$ and $\mathcal{G}$ be quasicoherent modules on a scheme $(X, \mathcal{O}_X)$, and let

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

be a homomorphism. Prove that the kernel $\ker \varphi$ is also quasicoherent.

Exercise 14.20

Let $\mathcal{F}$ and $\mathcal{G}$ be coherent modules on a noetherian scheme $(X, \mathcal{O}_X)$, and let

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

be a homomorphism. Prove that the kernel $\ker \varphi$ is also coherent.

Exercise 14.21

Let $\mathcal{F}$ and $\mathcal{G}$ be quasicoherent modules on a scheme $(X, \mathcal{O}_X)$, and let

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

be a homomorphism. Prove that the cokernel $\operatorname{coker} \varphi$ is also quasicoherent.

Exercise 14.22

Let $(X, \mathcal{O}_X)$ be a noetherian scheme and

$$f \in \Gamma(X, \mathcal{O}_X)$$

a global function with invertibility locus X_f. Let $\mathcal{M}$ be a quasicoherent $\mathcal{O}_X$-module on X. Prove that

$$\Gamma(X_f, \mathcal{M}) = \Gamma(X, \mathcal{M})_f.$$

> **Editorial note - base space.** The source writes $\mathcal{O}_U$ although the module is on X and no U has been introduced. This edition corrects it to $\mathcal{O}_X$.

Exercise 14.23

Let $(X, \mathcal{O}_X)$ be a noetherian scheme and $U \subseteq X$ an open subset. Let $\mathcal{M}$ be a quasicoherent $\mathcal{O}_U$-module on U. Prove that the direct image

$$i_* \mathcal{M}$$

is a quasicoherent module on X.

Hint: first consider the case where X is affine.

Exercise 14.24

Consider the invertible sheaf $\mathcal{O}_{\mathbb{P}^n_K}(1)$ on projective space over a field K, together with the global section

$$X_0 \in \Gamma(\mathbb{P}^n_K, \mathcal{O}_{\mathbb{P}^n_K}(1))$$

and its invertibility locus

$$(\mathbb{P}^n_K)_{X_0} = D_+(X_0).$$

Let

$$f \in \Gamma(D_+(X_0), \mathcal{O}_{\mathbb{P}^n_K})$$

be a function defined on $D_+(X_0) \subseteq \mathbb{P}^n_K$. Prove directly that there is an $m \in \mathbb{N}$ such that

$$X_0^m f \in \Gamma\Big(D_+(X_0), (\mathcal{O}_{\mathbb{P}^n_K}(1))^m \otimes \mathcal{O}_{\mathbb{P}^n_K}\Big)$$

comes from a global element in

$$\Gamma\Big(\mathbb{P}^n_K, (\mathcal{O}_{\mathbb{P}^n_K}(1))^m \otimes \mathcal{O}_{\mathbb{P}^n_K}\Big).$$

> **Editorial note - local section before extension.** The source places $X_0^m f$ in the global section module before asking it to extend globally. Here its initial domain is corrected to $D_+(X_0)$, where f is given; the requested global target is unchanged.

Exercise 14.25

Consider the invertible sheaf $\mathcal{O}_{\mathbb{P}^1_K}(1)$ on the projective line

$$\mathbb{P}^1_K = \text{Proj}(K[X,Y])$$

over a field K, together with the global section

$$X \in \Gamma(\mathbb{P}^1_K, \mathcal{O}_{\mathbb{P}^1_K}(1))$$

and its invertibility locus

$$(\mathbb{P}^1_K)_X = D_+(X).$$

For each of the following functions f in $\Gamma(D_+(X), \mathcal{O}_{\mathbb{P}^1_K})$, find a suitable n such that

$$X^n f \in \Gamma(D_+(X), \mathcal{O}_{\mathbb{P}^1_K}(n))$$

comes from an element—which one?—of $\Gamma(\mathbb{P}^1_K, \mathcal{O}_{\mathbb{P}^1_K}(n))$.

1. $\dfrac{Y}{X}$,
2. $\dfrac{2Y^3 - 3Y^2X + 4X^3}{X^3}$,
3. $\dfrac{Y^{17} + X^{17}}{X^{17}}$.

   Editorial note - degree of the input. The source puts the listed degree-zero fractions in $\mathcal{O}(1)$. They are regular functions, hence sections of $\mathcal{O}$; multiplying by X^n then has degree n, as required by the unchanged target $\mathcal{O}(n)$. This edition removes the erroneous input twist.

Exercise 14.26

Specialise Theorem 14.13 to the case where $\mathcal{M}$ is the structure sheaf of X.

Public Solutions and Coverage of Worksheet 14

At the frozen revision boundary, the source provides no public solution page for any of the 26 exercises in Worksheet 14. The exercise map and candidate evidence record negative results for Exercises 14.1–14.26. The absence of a public solution page is not replaced with a newly written solution.

Frozen negative results

There is no public solution page at the frozen revision for Exercise 14.1, 14.2, 14.3, 14.4, 14.5, 14.6, 14.7, 14.8, 14.9, 14.10, 14.11, 14.12, 14.13, 14.14, 14.15, 14.16, 14.17, 14.18, 14.19, 14.20, 14.21, 14.22, 14.23, 14.24, 14.25, or 14.26. This is a result of the candidate check, not a claim that mathematical solutions do not exist.

Lecture 15: Modules on Projective Schemes

Quasicoherent modules on projective schemes

Graded modules over a graded ring R give rise to quasicoherent modules on

$$\mathrm{Proj}(R).$$

> **Editorial note - terminology.** The source says "quasiprojective modules", but the construction and Lemma 15.3 concern quasicoherent modules. This edition corrects that slip.

Lemma 15.1: gradings on homogeneous open subsets

Let R be a $\mathbb{Z}$-graded commutative ring and M a $\mathbb{Z}$-graded R-module. The associated $\mathcal{O}_X$-module $\widetilde{M}$ on $X = \mathrm{Spek}(R)$ has the following property: for every open subset

$$U = D(\mathfrak{a}) \subseteq \mathrm{Spek}(R)$$

arising from a homogeneous ideal $\mathfrak{a}$, the $\Gamma(U, \mathcal{O}_X)$-module $\Gamma(U, \widetilde{M})$ has a $\mathbb{Z}$-grading compatible with the restriction maps.

Proof For $M = R$, the assertion first means that the structure sheaf has a grading on the open subsets arising from homogeneous ideals. This is clear for $D(f)$ with f homogeneous, and follows from this for $D(\mathfrak{a})$ with any homogeneous ideal $\mathfrak{a}$. The module case follows in the same way.

It makes no sense to say that $\widetilde{M}$ is graded as a whole, since the grading is not defined on arbitrary open subsets that do not arise from homogeneous ideals. However, the grading on homogeneous subsets allows us to define a sheaf of modules on the projective spectrum associated with R.

Definition 15.2: the sheaf of modules on the projective spectrum

Let R be a $\mathbb{Z}$-graded commutative ring, M a $\mathbb{Z}$-graded R-module, and

$$Y = \mathrm{Proj}(R)$$

the projective spectrum of R. The sheaf of $\mathcal{O}_Y$-modules associated with M, denoted by $\widehat{M}$, is specified as follows. For every open subset

$$V = D_+(\mathfrak{a}) \subseteq Y$$

arising from a homogeneous ideal $\mathfrak{a}$, set

$$\Gamma(V, \widehat{M}) := \Gamma(D(\mathfrak{a}), \widetilde{M})_0,$$

and equip this with the natural restriction maps and the natural $\mathcal{O}_Y$-module structure.

For a graded R-module M and a homogeneous prime ideal $\mathfrak{p}$, we set

$$M_{(\mathfrak{p})} = \left(M_{\{h \text{ homogeneous} | h \notin \mathfrak{p}\}}\right)_0.$$

Lemma 15.3: properties of the projective sheaf of modules

Let R be a $\mathbb{Z}$-graded commutative ring, M a $\mathbb{Z}$-graded R-module, $Y = \mathrm{Proj}(R)$ the projective spectrum of R, and $\widehat{M}$ the associated $\mathcal{O}_Y$-module. Then the following properties hold.

1. $\widehat{M}$ is a quasicoherent module.
2. For a homogeneous element $f \in R_+$,

$$\Gamma(D_+(f), \widehat{M}) = (M_f)_0.$$

   Moreover, the restriction of $\widehat{M}$ to $D_+(f)$ equals the affine sheaf associated with $(M_f)_0$ on

$$D_+(f) = \mathrm{Spek}((R_f)_0).$$

3. For a homogeneous prime ideal $\mathfrak{p}$ with $R_+ \not\subseteq \mathfrak{p}$,

$$\widehat{M}_{\mathfrak{p}} = M_{(\mathfrak{p})}.$$

4. We have

$$\Gamma(Y, \widehat{M}) = \left(\Gamma(D(R_+), \widetilde{M})\right)_0.$$

Proof

1. The sheaf property follows from that of $\widetilde{M}$. For quasicoherence, see part (2).
2. For homogeneous $f \in R_+$, Lemma 14.5 gives

$$\Gamma(D_+(f), \widehat{M}) = \Gamma(D(f), \widetilde{M})_0 = (M_f)_0.$$

   Thus, on the whole of $D_+(f)$, the sheaf $\widehat{M}|_{D_+(f)}$ agrees with $\widetilde{(M_f)_0}$. The corresponding equalities hold for open subsets $D(g) \subseteq D(f)$, and these identifications are compatible with restrictions. Hence the sheaves agree, and quasicoherence follows.
3. This follows from (2) via

$$\begin{aligned}
\widehat{M}_{\mathfrak{p}} &= \operatorname*{colim}_{\mathfrak{p} \in D_+(f)} \Gamma(D_+(f), \widehat{M}) \\
&= \operatorname*{colim}_{\mathfrak{p} \in D_+(f)} (M_f)_0 \\
&= \left(\operatorname*{colim}_{\mathfrak{p} \in D_+(f)} M_f\right)_0 \\
&= \left(M_{\{h \in R | h \text{ homogeneous and } h \notin \mathfrak{p}\}}\right)_0 \\
&= M_{(\mathfrak{p})}.
\end{aligned}$$

   Editorial note - localisation set. In this line the source writes $R \setminus \mathfrak{p} \cap H$ without defining H or parenthesising the set operations. The explicit set above is the multiplicative system of homogeneous elements outside $\mathfrak{p}$ used in Definition 15.2.

4. This is a special case of the general definition.

The last assertion means that, in general, the module of global sections of $\widehat{M}$ on Y cannot be computed directly from M.

Definition 15.4: twisted structure sheaf

Let R be a $\mathbb{Z}$-graded commutative ring and let $R(n)$ be the graded R-module obtained by shifting R by n. The associated $\mathcal{O}_Y$-module on $Y = \mathrm{Proj}(R)$, denoted by

$$\mathcal{O}_Y(n) := \widetilde{R(n)}$$

is called a *twisted structure sheaf.*

> **Editorial note - module rather than ring.** The source calls $R(n)$ a shifted graded ring. With the shifted grading it is a graded R-module and, as Exercise 15.11 notes, is a graded ring only when $n = 0$.

Example 15.5: twisted structure sheaves on projective space

For the standard-graded polynomial ring $K[X_0, X_1, \dots, X_d]$, with $d \geq 1$, we have

$$\Gamma\left(\mathbb{P}^d_K, \mathcal{O}_{\mathbb{P}^d_K}(\ell)\right) = K[X_0, X_1, \dots, X_d]_\ell,$$

the space of polynomials of degree ℓ in $d+1$ variables. For $\ell < 0$, this is the zero space; for $\ell = 0$ (the structure sheaf), it equals K; for $\ell = 1$, it consists of all linear forms; and so on. For the open subsets $D_+(X_i)$,

$$\begin{aligned}\Gamma\left(D_+(X_i), \mathcal{O}_{\mathbb{P}^d_K}(\ell)\right) &= \left(K[X_0, X_1, \dots, X_d]_{X_i}\right)_\ell \\ &= K\left[\frac{X_0}{X_i}, \dots, \frac{X_{i-1}}{X_i}, \frac{X_{i+1}}{X_i}, \dots, \frac{X_d}{X_i}\right] \cdot X_i^\ell.\end{aligned}$$

For projective space, we already saw in Example 13.19 that these sheaves are invertible. This also holds in general.

Lemma 15.6: twisted structure sheaves are invertible

Let R be a standard-graded commutative ring. Then the twisted structure sheaves $\mathcal{O}_Y(n)$ on $Y = \mathrm{Proj}(R)$ are invertible.

Proof Write

$$R = R_0[x_1, \dots, x_d] = R_0[X_1, \dots, X_d]/\mathfrak{a},$$

where the x_i have degree 1. The elements x_i also generate the irrelevant ideal, so there is an affine open cover

$$\mathrm{Proj}(R) = \bigcup_{i=1}^{d} D_+(x_i).$$

Let x be one of the x_i. By Lemma 15.3(2),

$$\mathcal{O}_Y(n)|_{D_+(x)} = \mathcal{L},$$

where $\mathcal{L}$ is the affine sheaf associated with the $(R_x)_0$-module

$$L = (R_x(n))_0 = (R_x)_n$$

on

$$D_+(x) = \mathrm{Spek}((R_x)_0).$$

In this situation,

$$(R_x)_0 \longrightarrow (R_x)_n, \qquad h \longmapsto hx^n,$$

is an isomorphism of $(R_x)_0$-modules. There is therefore an isomorphism of $\mathcal{O}_Y|_{D_+(x)}$-modules

$$\mathcal{O}_Y|_{D_+(x)} \longrightarrow \mathcal{O}_Y(n)|_{D_+(x)}.$$

The twisted structure sheaves $\mathcal{O}_Y(n)$ are distinguished invertible sheaves associated with the projective scheme $Y = \text{Proj}(R)$, although they depend on the graded ring R.

Lemma 15.7: shifting a module and tensoring with a twisted sheaf

Let R be a standard-graded commutative ring, M a graded R-module, and $n \in \mathbb{Z}$. There is a natural $\mathcal{O}_Y$-isomorphism

$$\widehat{M} \otimes_{\mathcal{O}_Y} \mathcal{O}_Y(n) \longrightarrow \widehat{M(n)}$$

on $Y = \text{Proj}(R)$, where $M(n)$ denotes the module obtained by shifting M by n.

Proof For a homogeneous element $f \in R_+$, there is an $(R_f)_0$-module homomorphism

$$(M_f)_0 \otimes_{(R_f)_0} (R_f)_n \longrightarrow (M_f)_n, \qquad \frac{m}{f^k} \otimes \frac{r}{f^\ell} \longmapsto \frac{rm}{f^{k+\ell}},$$

arising directly from the homogeneous module multiplication $M \times R \to M$. For every open subset $U \subseteq \text{Proj}(R)$, these homomorphisms induce a module homomorphism

$$\operatorname*{colim}_{U \subseteq D_+(f)} \Big((M_f)_0 \otimes_{(R_f)_0} (R_f)_n \Big) \longrightarrow \operatorname*{colim}_{U \subseteq D_+(f)} (M_f)_n,$$

which together form a presheaf morphism. Since

$$(M(n)_f)_0 = (M_f)_n,$$

the sheafification of the presheaf on the right is $\widehat{M(n)}$. Sheafifying the left-hand side, in two steps, gives

$$\widehat{M} \otimes_{\mathcal{O}_{\text{Proj}(R)}} \mathcal{O}_Y(n),$$

so we obtain a module homomorphism

$$\widehat{M} \otimes_{\mathcal{O}_{\text{Proj}(R)}} \mathcal{O}_Y(n) \longrightarrow \widehat{M(n)}.$$

That this is an isomorphism can be proved on an affine cover. If f is homogeneous, then by Lemma 14.10 and Lemma 15.3(2), the $(R_f)_0$-module homomorphism above,

$$(M_f)_0 \otimes_{(R_f)_0} (R_f)_n \longrightarrow (M_f)_n,$$

equals the evaluation of the sheafified homomorphism. If f has degree 1—and the corresponding open subsets $D_+(f)$ cover Y—then it is an isomorphism. By Exercise 12.2,

$$(R_f)_0 \cong (R_f)_n$$

via $1 \mapsto f^n$, so the module on the left is isomorphic to $(M_f)_0$. With this identification, the map is given by

$$\frac{m}{f^k} \longmapsto f^n \cdot \frac{m}{f^k},$$

and it is bijective because f is a unit.

Definition 15.8: twist of a quasicoherent module

Let R be a standard-graded ring, $\mathcal{F}$ a quasicoherent module on $Y = \mathrm{Proj}(R)$, and $n \in \mathbb{Z}$. The module

$$\mathcal{F}(n) := \mathcal{F} \otimes_{\mathcal{O}_Y} \mathcal{O}_Y(n)$$

is called the *nth twist* of $\mathcal{F}$.

Thus the sheaf of modules $\widehat{M(n)}$ agrees with the nth twist of $\widehat{M}$.

Global generation

Definition 15.9: generated by global sections

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{M}$ an $\mathcal{O}_X$-module on X. We say that $\mathcal{M}$ is *generated by global sections* if there is a family

$$s_i \in \Gamma(X, \mathcal{M}), \qquad i \in I,$$

such that, for every point $x \in X$, the stalk $\mathcal{M}_x$ is generated as an $\mathcal{O}_{X,x}$-module by the restrictions of the s_i.

Proposition 15.10: properties of global generation

Let $(X, \mathcal{O}_X)$ be a scheme. Then the following assertions hold.

1. The structure sheaf $\mathcal{O}_X$ is generated by global sections.
2. A quasicoherent module $\mathcal{M}$ is generated by global sections if and only if there is a surjective module homomorphism

$$\mathcal{O}_X^{(I)} \longrightarrow \mathcal{M}.$$

3. On an affine scheme, every quasicoherent module is generated by global sections.
4. If $\mathcal{M}$ is generated by global sections and $\mathcal{M} \to \mathcal{N}$ is surjective, then $\mathcal{N}$ is also generated by global sections.

Proof See Exercise 15.17.

Lemma 15.11: global generation of twisted structure sheaves

On projective space $\mathbb{P}^d_R$ over a commutative ring R, the twisted structure sheaf $\mathcal{O}_{\mathbb{P}^d_R}(k)$ is generated by global sections for $k \geq 0$, and is not generated by global sections for $k < 0$ and $d \geq 1$.

Proof See Exercise 15.18.

Theorem 15.12: a positive twist is eventually globally generated

Let $\mathbb{P}^d_R$ be projective space over a noetherian ring R, and let $\mathcal{G}$ be a coherent sheaf on $\mathbb{P}^d_R$. Then there is an $\ell \in \mathbb{N}_+$ such that $\mathcal{G}(\ell)$ is generated by global sections.

Proof There is an isomorphism

$$\mathcal{G}|_{D_+(X_i)} \cong \widetilde{M}_i,$$

where M_i is a finitely generated module over the polynomial ring R_i associated with $D_+(X_i)$. For the invertible sheaf $\mathcal{O}_{\mathbb{P}^d_R}(1)$, the invertibility locus of the global section X_i is $D_+(X_i)$ by Exercise 13.21. For a finite generating system s_{ij}, $j \in J_i$, of the R_i-module M_i, Theorem 14.13(2) gives a common exponent n such that the $X_i^n s_{ij}$ come from global elements in

$$\Gamma(\mathbb{P}^d_R, \mathcal{G}(n)).$$

This can be done for every i, giving an ℓ such that global sections in $\Gamma(\mathbb{P}^d_R, \mathcal{G}(\ell))$ generate the modules on the affine open cover. They therefore also generate every stalk, so global generation follows.

Theorem 15.13: a surjective presentation by twisted structure sheaves

Let $\mathbb{P}^d_R$ be projective space over a noetherian ring R, and let $\mathcal{G}$ be a coherent sheaf on $\mathbb{P}^d_R$. Then there is a finite direct sum

$$\bigoplus_{j \in J} \mathcal{O}_{\mathbb{P}^d_R}(\ell_j)$$

and a surjective module homomorphism

$$\bigoplus_{j \in J} \mathcal{O}_{\mathbb{P}^d_R}(\ell_j) \longrightarrow \mathcal{G}.$$

Proof By Theorem 15.12, there is an ℓ such that $\mathcal{G}(\ell)$ is generated by finitely many global sections. By Proposition 15.10(2), there is a surjective module homomorphism

$$\mathcal{O}^r_{\mathbb{P}^d_R} \longrightarrow \mathcal{G}(\ell).$$

Tensoring this map with $\mathcal{O}_{\mathbb{P}^d_R}(-\ell)$ gives a surjection

$$\left(\mathcal{O}_{\mathbb{P}^d_R}(-\ell)\right)^r \longrightarrow \mathcal{G}.$$

Worksheet 15: Modules on Projective Schemes

The frozen candidate map records exactly 20 exercises and finds no public solution page for any of them. There are therefore no asterisks on this worksheet, and this edition creates no new solutions.

Exercise 15.1

Let R be a $\mathbb{Z}$-graded ring, M a graded module over R, and $\widetilde{M}$ the associated sheaf of modules on the spectrum $X = \operatorname{Spek}(R)$. Let $\mathfrak{a} \subseteq R$ be a homogeneous ideal. Prove that

$$\Gamma(D(\mathfrak{a}), \widetilde{M})_\ell = \left\{ s \in \Gamma(D(\mathfrak{a}), \widetilde{M}) \;\middle|\; \begin{array}{l} \text{there is an open cover } D(\mathfrak{a}) = \bigcup_{i \in I} D(f_i) \text{ with } f_i \text{ homogeneous,} \\ \text{such that } s|_{D(f_i)} \in M_{f_i} \text{ has degree } \ell \text{ for every } i \end{array} \right\}$$

defines a grading on $\Gamma(D(\mathfrak{a}), \widetilde{M})$ for which the natural restriction homomorphisms are homogeneous.

> **Editorial note - open cover and restrictions.** The source calls the covered open set U although the left-hand side is defined on $D(\mathfrak{a})$, and it leaves the restriction of s to each $D(f_i)$ implicit. Both are made explicit above.

Exercise 15.2

Using $R = K[X]$ and $f = X + 1$, explain why there is no grading on the localisation

$$K[X]_{X+1}$$

that meaningfully extends the standard grading on the polynomial ring.

Exercise 15.3

Let R be a $\mathbb{Z}$-graded ring and M a graded module over R. Prove that the assignment

$$U = D_+(\mathfrak{a}) \longmapsto \operatorname*{colim}_{U \subseteq D_+(f)} (M_f)_0$$

is a presheaf of commutative groups on $\mathrm{Proj}(R)$ whose sheafification agrees with $\widehat{M}$.

Exercise 15.4

Let R be a $\mathbb{Z}$-graded ring and M a graded module over R. Prove that the associated sheaf of $\mathcal{O}_Y$-modules $\widehat{M}$ on $Y = \mathrm{Proj}(R)$ is given by

$$\Gamma(U, \widehat{M}) = \Bigg\{ (s_\mathfrak{p})_{\mathfrak{p} \in U} \in \prod_{\mathfrak{p} \in U} M_{(\mathfrak{p})} \;\Bigg|\; \begin{array}{l} \text{for every } \mathfrak{p} \in U \text{ there are homogeneous elements } b \in R \text{ and } t \in M \text{ with} \\ \mathfrak{p} \in D_+(b) \subseteq U \text{ and } s_\mathfrak{q} = \dfrac{t}{b} \text{ in } M_{(\mathfrak{q})} \text{ for every } \mathfrak{q} \in D_+(b) \end{array} \Bigg\}.$$

Exercise 15.5

Let R be a $\mathbb{Z}$-graded integral domain, let H be the multiplicative system generated by all nonzero homogeneous elements of nonzero degree, and set

$$M = R_H.$$

Prove that $\widehat{M}$ is the constant sheaf with value $(R_H)_0$, the function field of the integral scheme $\mathrm{Proj}(R)$.

> **Editorial note - localisation and function field.** The source describes H only as “all homogeneous elements of degree $\neq 0$”, without excluding zero or specifying multiplicative closure, and identifies a sheaf directly with a field. The formulation above makes the localisation well-defined and identifies its degree-zero field as the value of the associated sheaf.

Exercise 15.6

Let

$$R = K[X_0, X_1, \dots, X_d]/\mathfrak{a}$$

be a standard-graded ring and

$$(a_0, a_1, \dots, a_d) \in V(\mathfrak{a}) \subseteq \mathbb{A}_K^{d+1}$$

a nonzero K-point of $V(\mathfrak{a})$, with associated homogeneous prime ideal

$$\mathfrak{p} = (a_i X_j - a_j X_i \mid 0 \leq i, j \leq d) \subseteq R,$$

which is a closed point in

$$\mathrm{Proj}(R) = V_+(\mathfrak{a}) \subseteq \mathbb{P}^d_K.$$

Prove that $\widehat{R/\mathfrak{p}}$ is a quasicoherent module on Proj(R) (and on $\mathbb{P}^d_K$) whose support equals $\{\mathfrak{p}\}$.

> **Editorial note - ambient variety and point ideal.** The source prints $V(\mathfrak{a}) = \mathbb{A}^{d+1}_K$, calls the tuple a point of the ring, and leaves a trailing comma instead of the index range in the generators of $\mathfrak{p}$. The corrected notation above states the intended closed subvariety, point, and homogeneous point ideal.

Exercise 15.7

Let R be a $\mathbb{Z}$-graded ring and $Y = \mathrm{Proj}(R)$. Prove that nonisomorphic graded R-modules M and N can give rise to isomorphic $\mathcal{O}_Y$-modules $\widehat{M}$ and $\widehat{N}$.

Exercise 15.8

Let

$$R = K[X_0, X_1, \dots, X_d]/\mathfrak{a}$$

with $\mathfrak{a}$ a homogeneous ideal, and let

$$Y = \mathrm{Proj}(R) \subseteq \mathbb{P}^d_K.$$

Writing $i : Y \hookrightarrow \mathbb{P}^d_K$ for the inclusion, prove that

$$i_* \mathcal{O}_Y = \widehat{R}$$

on $\mathbb{P}^d_K$.

> **Editorial note - inclusion map.** The source uses i_* without naming i. The inclusion whose direct image is intended is made explicit above.

Exercise 15.9

Let R be a $\mathbb{Z}$-graded ring and

$$0 \longrightarrow L \longrightarrow M \longrightarrow N \longrightarrow 0$$

a short exact sequence of $\mathbb{Z}$-graded R-modules with homogeneous homomorphisms. Prove that, in each degree, there is a short exact sequence of R_0-modules

$$0 \longrightarrow L_k \longrightarrow M_k \longrightarrow N_k \longrightarrow 0.$$

Exercise 15.10

Let R be a $\mathbb{Z}$-graded ring, and let L, M, N be $\mathbb{Z}$-graded R-modules with homogeneous homomorphisms

$$\varphi : L \longrightarrow M, \qquad \psi : M \longrightarrow N.$$

For every prime ideal $\mathfrak{p}$ with $R_+ \not\subseteq \mathfrak{p}$, suppose that the sequence

$$0 \longrightarrow L_{\mathfrak{p}} \xrightarrow{\varphi} M_{\mathfrak{p}} \xrightarrow{\psi} N_{\mathfrak{p}} \longrightarrow 0$$

is exact. Prove that there is a short exact sequence

$$0 \longrightarrow \widehat{L} \longrightarrow \widehat{M} \longrightarrow \widehat{N} \longrightarrow 0$$

on $Y = \operatorname{Proj}(R)$.

Exercise 15.11

Let R be a $\mathbb{Z}$-graded ring. Prove that the shifted R-module $R(n)$ is a graded ring only for $n = 0$.

Exercise 15.12

Give an explicit basis for

$$\Gamma\left(\mathbb{P}^2_K, \mathcal{O}_{\mathbb{P}^2_K}(3)\right)$$

over a field K, and determine its dimension.

Exercise 15.13

Let $Y = \operatorname{Proj}(R)$ be the projective spectrum of a standard-graded ring R. Prove that the twisted structure sheaves $\mathcal{O}_Y(\ell)$ satisfy

$$\mathcal{O}_Y(\ell) \otimes_{\mathcal{O}_Y} \mathcal{O}_Y(m) \cong \mathcal{O}_Y(\ell + m).$$

Exercise 15.14

Let $Y = \operatorname{Proj}(R)$ be the projective spectrum of a standard-graded ring R. Prove that the twisted structure sheaves satisfy

$$\mathcal{H}om(\mathcal{O}_Y(\ell), \mathcal{O}_Y(m)) \cong \mathcal{O}_Y(m - \ell).$$

Exercise 15.15

Let $Y = \operatorname{Proj}(R)$ be the projective spectrum of a standard-graded ring R. Prove that the twisted structure sheaves $\mathcal{O}_Y(\ell)$ for $\ell \leq 0$ can be realised as ideal sheaves on Y. Also prove that, in general, there is more than one way to do this.

Exercise 15.16

Let

$$R = \bigoplus_{d \in \mathbb{Z}} R_d$$

be a $\mathbb{Z}$-graded commutative ring, and let

$$M = \bigoplus_{d \in \mathbb{Z}} M_d$$

be a finitely generated $\mathbb{Z}$-graded module over R. Prove that M is also generated by finitely many homogeneous elements and that there is a surjective homogeneous module homomorphism of the form

$$\bigoplus_{i=1}^{k} R(-d_i) \longrightarrow M.$$

Exercise 15.17

Let $(X, \mathcal{O}_X)$ be a scheme. Prove the following assertions.

1. The structure sheaf $\mathcal{O}_X$ is generated by global sections.
2. A quasicoherent module $\mathcal{M}$ is generated by global sections if and only if there is a surjective module homomorphism

$$\mathcal{O}_X^{(I)} \longrightarrow \mathcal{M}.$$

3. On an affine scheme, every quasicoherent module is generated by global sections.
4. If $\mathcal{M}$ is generated by global sections and $\mathcal{M} \to \mathcal{N}$ is surjective, then $\mathcal{N}$ is also generated by global sections.

Exercise 15.18

Prove that, on projective space $\mathbb{P}^d_R$ over a commutative ring R, the twisted structure sheaf $\mathcal{O}_{\mathbb{P}^d_R}(k)$ is generated by global sections for $k \geq 0$, and is not generated by global sections for $k < 0$ and $d \geq 1$.

Exercise 15.19

Let $(X, \mathcal{O}_X)$ be a scheme and $\mathcal{M}$ a quasicoherent module on X. Prove that $\mathcal{M}$ is generated by global sections if and only if there is an affine open cover

$$X = \bigcup_{i \in I} U_i$$

and sections $s_j \in \Gamma(X, \mathcal{M})$ for $j \in J$ such that the restrictions

$$\rho_{U_i}(s_j) \in \Gamma(U_i, \mathcal{M})$$

form a generating system for $\Gamma(U_i, \mathcal{M})$ over $\Gamma(U_i, \mathcal{O}_X)$.

Exercise 15.20

Let $Y = \text{Proj}(R)$ be the projective spectrum of a standard-graded ring R. Prove that the twisted structure sheaves $\mathcal{O}_Y(\ell)$ for $\ell \geq 0$ are generated by global sections.

Public Solutions and Coverage of Worksheet 15

At the frozen revision boundary, the source provides no public solution page for any of the 20 exercises in Worksheet 15. The exercise map and candidate evidence record negative results for Exercises 15.1–15.20. The absence of a public solution page is not replaced with a newly written solution.

Frozen negative results

There is no public solution page at the frozen revision for Exercise 15.1, 15.2, 15.3, 15.4, 15.5, 15.6, 15.7, 15.8, 15.9, 15.10, 15.11, 15.12, 15.13, 15.14, 15.15, 15.16, 15.17, 15.18, 15.19, or 15.20. This is the result of the exact candidate check, not a claim that mathematical solutions do not exist.

Lecture 16: Locally Free Sheaves

Locally free sheaves

Definition 16.1: locally free sheaves

An $\mathcal{O}_X$-module $\mathcal{F}$ on a ringed space X is called *locally free* of *rank r* if there is an open cover

$$X = \bigcup_{i \in I} U_i$$

and $\mathcal{O}_{U_i}$-module isomorphisms

$$\mathcal{F}|_{U_i} \cong (\mathcal{O}_{U_i})^r$$

for every $i \in I$.

For $r = 1$, we obtain the invertible sheaves: these are precisely the locally free sheaves of rank 1. The simplest locally free sheaves are the *free sheaves*

$$\mathcal{O}_X^r, \qquad r \in \mathbb{N}.$$

> **Editorial note — the rank variable in the source.** Both the frozen semantic page and the official course PDF print "rank r" in the sentence after setting $r = 1$. This edition states the mathematical consequence, namely rank 1, and explicitly records the change.

By definition, a locally free sheaf is free locally, that is, on a cover by open sets. Thus free sheaves and locally free sheaves cannot be distinguished locally. Locally free sheaves therefore reflect global properties of the ringed space X.

We consider locally free sheaves on schemes, where there are close connections with projective and flat modules. In particular, locally free sheaves are coherent modules. Over a local ring, all locally free sheaves are free, because its spectrum has only one closed point, whose only open neighbourhood is the whole space. However, if we consider the punctured spectrum

$$U = D(\mathfrak{m}) = \text{Spek}(R) \setminus \{\mathfrak{m}\}$$

of a local ring, there are generally many nontrivial (nonfree) locally free sheaves on it, reflecting properties of the local ring, or of the singularity. Since every scheme is covered by affine schemes, we must first understand locally free sheaves on an affine scheme.

Theorem 16.2: local characterisations of local freeness

Let R be a commutative Noetherian ring, let M be a finitely generated R-module, and let $r \in \mathbb{N}$. The following properties are equivalent.

1. The localisations $M_{\mathfrak{p}}$ are free of rank r for every prime ideal $\mathfrak{p} \in \text{Spek}(R)$.
2. The localisations $M_{\mathfrak{m}}$ are free of rank r for every maximal ideal $\mathfrak{m}$ of R.

3. There are elements $f_1, \dots, f_k \in R$ generating the unit ideal such that the localisations M_{f_j} are free of rank r for every $j = 1, \dots, k$.
4. The coherent sheaf $\widetilde{M}$ associated with M on $\operatorname{Spek}(R)$ is locally free of rank r.

Proof $(1) \Rightarrow (2)$. This is a special case.

$(2) \Rightarrow (3)$. Fix a maximal ideal $\mathfrak{m}$. By assumption, there is an $R_{\mathfrak{m}}$-module isomorphism

$$\varphi : (R_{\mathfrak{m}})^r \longrightarrow M_{\mathfrak{m}}.$$

Write the image of the ith standard vector e_i as

$$\varphi(e_i) = \frac{m_i}{g_i},$$

with $m_i \in M$ and $g_i \in R \setminus \mathfrak{m}$. Let $g = g_1 \cdots g_r$ be the product of the denominators. We consider the situation over $D(g)$. The isomorphism φ is defined over $D(g)$, that is, on R_g, so we have an R_g-module homomorphism

$$\psi : (R_g)^r \longrightarrow M_g$$

which induces the isomorphism φ after localisation at $\mathfrak{m}$. In general, however, ψ need not be an isomorphism.

Let $v_1, \dots, v_s$ be a generating system for the module M. Since ψ induces a surjection over $R_{\mathfrak{m}}$, there are elements

$$u_j = \frac{a_j}{h_j} \in (R_{\mathfrak{m}})^r$$

mapping to v_j. The denominators h_j do not belong to $\mathfrak{m}$. We can therefore replace g by $h = gh_1 \cdots h_s$ and obtain

$$\psi : (R_h)^r \longrightarrow M_h.$$

Here there are elements $u_j \in (R_h)^r$ such that $\psi(u_j)$ and the generators v_j have the same restrictions in $M_{\mathfrak{m}}$. This means that there are elements $p_j \notin \mathfrak{m}$ with

$$p_j \psi(u_j) = p_j v_j$$

in M_h. Replacing h by $p = hp_1 \cdots p_s$ makes ψ surjective as well.

> **Editorial note — the relation symbol in the source.** The source prints $u_j = (R_h)^r$ in the sentence above. Since u_j is an element and $(R_h)^r$ is the module containing it, this edition uses the membership relation $u_j \in (R_h)^r$ and explicitly records the change.

Let N be the kernel of this new ψ. Since φ is injective, $N_{\mathfrak{m}} = 0$. As R is Noetherian, Lemma 23.2 (Commutative Algebra) implies that N is finitely generated. Hence there is again an element $f \notin \mathfrak{m}$ with $N_f = 0$. Shrinking the open set once more gives an isomorphism

$$\psi : (R_f)^r \longrightarrow M_f$$

for some $f \notin \mathfrak{m}$.

Thus every maximal ideal $\mathfrak{m}$ has an open neighbourhood

$$\mathfrak{m} \in D(f_{\mathfrak{m}})$$

such that $M_{f_{\mathfrak{m}}}$ is free of rank r. Consequently,

$$\bigcup_{\mathfrak{m} \text{ maximal ideal}} D(f_{\mathfrak{m}})$$

contains all maximal ideals and also all prime ideals, and hence is an open cover of Spek(R). By Proposition 8.4 (4), the ideal

$$(f_{\mathfrak{m}} : \mathfrak{m} \text{ maximal ideal})$$

is the unit ideal, and it is already generated by finitely many of the elements $f_{\mathfrak{m}}$.

$(3) \Rightarrow (4)$. Since the elements generate the unit ideal, the open sets $D(f_j)$, $j = 1, \dots, k$, cover Spek(R). Since the M_{f_j} are free R_{f_j}-modules of rank r, there are $\mathcal{O}_X|_{D(f_j)}$-module isomorphisms

$$\widetilde{M}|_{D(f_j)} \cong \widetilde{(R_{f_j})^r} \cong \mathcal{O}^r_{D(f_j)}.$$

Thus $\widetilde{M}$ is locally free.

$(4) \Rightarrow (1)$. Let $\mathfrak{p} \in$ Spek(R) be a prime ideal. Local freeness means that there is an open cover

$$X = \bigcup_{i \in I} U_i$$

such that the $\widetilde{M}|_{U_i}$ are free of rank r. Hence there is an index i with $\mathfrak{p} \in U_i$. Passing to a possibly smaller open neighbourhood, we may take $U_i = D(f)$ with $f \notin \mathfrak{p}$. There,

$$\widetilde{M_f} \cong \widetilde{M}|_{D(f)}$$

is free of rank r. Its localisation $M_{\mathfrak{p}}$ is therefore also free of rank r.

The example in Exercise 14.7 shows that, over a non-Noetherian ring R, there may be a module M with

$$M_{\mathfrak{p}} = R_{\mathfrak{p}}$$

without this isomorphism extending to an open neighbourhood.

We now relate locally free modules to projective modules.

Definition 16.3: projective modules

Let R be a commutative ring and let M be an R-module. The module M is called *projective* if, for every surjective R-module homomorphism

$$\theta : A \longrightarrow B$$

and every module homomorphism

$$\varphi : M \longrightarrow B,$$

there is a module homomorphism

$$\psi : M \longrightarrow A$$

with

$$\varphi = \theta \circ \psi.$$

A module is projective if and only if it is a direct summand of a free module.

Lemma 16.4: finitely generated projective modules over a local ring

Let R be a commutative local ring and let M be a finitely generated R-module. Then M is free if and only if M is a projective module.

Proof That free modules are projective was proved in Lemma 47.2 (Commutative Algebra). Thus suppose that M is projective. Choose a minimal generating system $m_1, \dots, m_n$ of M, and let

$$p : R^n \longrightarrow M$$

be the corresponding surjective module homomorphism. By minimality, the map

$$(R/\mathfrak{m})^n \longrightarrow M/\mathfrak{m}M$$

is a bijective $R/\mathfrak{m}$-linear map. Since M is projective, there is a module homomorphism $i : M \to R^n$ with

$$p \circ i = \mathrm{Id}_M .$$

Then

$$R^n \cong M \oplus N,$$

with $N = \operatorname{kern} p$, where we identify M with $i(M)$. Now consider

$$R^n \xrightarrow{\cong} M \oplus N \longrightarrow M$$

and the induced $R/\mathfrak{m}$-linear maps

$$(R/\mathfrak{m})^n \longrightarrow M/\mathfrak{m}M \oplus N/\mathfrak{m}N \longrightarrow M/\mathfrak{m}M.$$

Both the map on the left and the composite are bijective. Hence $N/\mathfrak{m}N = 0$. Lemma 29.5 (Commutative Algebra) gives $N = 0$, so $R^n = M$ is free.

Lemma 16.5: local freeness and projectivity

Let R be a commutative Noetherian ring and let M be a finitely generated R-module. Then M is locally free if and only if M is a projective module.

Proof One direction follows directly from Lemma 16.4, taking Exercise 16.16 into account. To prove the converse, let

$$p : L \longrightarrow M$$

be a surjective module homomorphism, with L a finitely generated free R-module. We must show that there is a homomorphism $i : M \to L$ with

$$p \circ i = \mathrm{Id}_M .$$

In particular, this is assured if the natural homomorphism

$$\operatorname{Hom}_R(M, L) \longrightarrow \operatorname{Hom}_R(M, M), \qquad \varphi \longmapsto p \circ \varphi,$$

is surjective, since then its image contains the identity. By Appendix Theorem 1.4, surjectivity can be tested locally. Under the given finiteness assumptions, the homomorphism modules satisfy

$$(\operatorname{Hom}_R(M, L))_{\mathfrak{p}} = \operatorname{Hom}_{R_{\mathfrak{p}}}(M_{\mathfrak{p}}, L_{\mathfrak{p}}).$$

For every prime ideal $\mathfrak{p}$, surjectivity of the map

$$\operatorname{Hom}_{R_{\mathfrak{p}}}(M_{\mathfrak{p}}, L_{\mathfrak{p}}) \longrightarrow \operatorname{Hom}_{R_{\mathfrak{p}}}(M_{\mathfrak{p}}, M_{\mathfrak{p}})$$

follows from the freeness of $M_{\mathfrak{p}}$ and Lemma 47.2 (Commutative Algebra).

> **Editorial note — unresolved reference.** Immediately before the localisation identity for homomorphism modules, the semantic source page displays the reference `Fakt *****` to `Nenneraufnahme/Homomorphismenmodul/Fakt`. This edition retains the mathematical identity without inventing an unavailable reference number.

The following theorem also holds; we do not prove it.

Theorem 16.6: locally free, projective, and flat

Let R be a commutative Noetherian ring and let M be a finitely generated R-module. The following statements are equivalent.

1. M is locally free.
2. M is a projective module.
3. M is a flat module.

The following theorem produces many locally free sheaves that are generally nontrivial.

Theorem 16.7: the kernel of a surjection of locally free sheaves

Let X be a Noetherian scheme and let

$$\theta : \mathcal{F} \longrightarrow G$$

be a surjective sheaf homomorphism between locally free sheaves on X. Then the kernel of θ is also locally free.

Proof Since local freeness is a local property, we may assume at once that

$$X = \text{Spek}(R)$$

is the affine scheme of a Noetherian ring R and, after shrinking the open set further, that we have a surjective module homomorphism

$$\theta : R^r \longrightarrow R^s.$$

By Theorem 19.11 (Commutative Algebra), there is a map $\varphi : R^s \to R^r$ with

$$\theta \circ \varphi = \operatorname{Id}_{R^s}.$$

Thus there is a direct sum decomposition

$$R^r = \operatorname{kern} \theta \oplus R^s,$$

and θ is the projection onto the summand R^s. Hence, by Lemma 47.3 (Commutative Algebra), $\operatorname{kern} \theta$ is a projective R-module, and by Lemma 16.5 it is locally free.

Remark 16.8: syzygy sheaves

Elements $f_1, \dots, f_n \in R$ in a commutative ring R give a module homomorphism

$$R^n \longrightarrow R, \qquad e_i \longmapsto f_i.$$

Its image is the ideal generated by the f_i. In particular, this map is surjective only if the f_i generate the unit ideal. The corresponding homomorphism of sheaves of modules

$$\mathcal{O}_X^n \longrightarrow \mathcal{O}_X$$

is also generally not surjective, and its kernel is generally not locally free. However, consider the restriction of this sheaf homomorphism to the open subset

$$U = \bigcup_{i=1}^{n} D(f_i),$$

namely

$$\mathcal{O}_U^n \longrightarrow \mathcal{O}_U.$$

We obtain a surjective sheaf homomorphism, since on each $D(f_i)$ we have

$$\frac{1}{f_i} e_i \longmapsto 1.$$

By Theorem 16.7, its kernel is a locally free sheaf on the quasiaffine scheme U. This kernel is denoted by

$$\mathrm{Syz}(f_1, \dots, f_n)$$

and is called the *syzygy sheaf* or *kernel sheaf.* If R is a local ring and the f_i generate an ideal primary to the maximal ideal $\mathfrak{m}$ — in other words, the f_i geometrically cut out the closed point — then the syzygy sheaf is a locally free sheaf on the punctured spectrum $D(\mathfrak{m})$.

Example 16.9: the syzygy sheaf of the variables

The variables

$$X_1, \dots, X_n \in K[X_1, \dots, X_n] = R$$

define the maximal ideal $(X_1, \dots, X_n)$ and the short exact sequence

$$0 \longrightarrow \mathrm{Syz}(X_1, \dots, X_n) \longrightarrow R^n \longrightarrow (X_1, \dots, X_n) \longrightarrow 0$$

of R-modules, where the ith standard vector e_i is sent to X_i. By Lemma 14.9, this induces a short exact sequence of quasicoherent modules

$$0 \longrightarrow \mathrm{Syz}(\widetilde{X_1, \dots}, X_n) \longrightarrow \mathcal{O}_{\mathbb{A}_K^n}^n \longrightarrow (X_1, \widetilde{\dots,} X_n) \longrightarrow 0$$

on affine space $\mathbb{A}_K^n$. The middle sheaf is free, whereas the sheaves on the left and right are not locally free, except for small n. If we restrict this sequence to the punctured spectrum

$$U = D(X_1, \dots, X_n) \subseteq \mathbb{A}_K^n,$$

then, by Exercise 14.1, the maximal ideal on the right becomes the structure sheaf. We thus obtain the situation of Remark 16.8,

$$0 \longrightarrow \mathrm{Syz}(X_1, \dots, X_n) \longrightarrow \mathcal{O}_U^n \longrightarrow \mathcal{O}_U \longrightarrow 0,$$

with the locally free syzygy sheaf on the left. For $n = 3$, this is the sheaf version of Example 1.2.

Determinant sheaves

Definition 16.10: determinant sheaves

Let $\mathcal{G}$ be a locally free sheaf of rank r on the ringed space $(X, \mathcal{O}_X)$. The sheafification of the presheaf

$$U \longmapsto \bigwedge^r \Gamma(U, \mathcal{G})$$

is called the *determinant sheaf* of $\mathcal{G}$. It is denoted by

$$\operatorname{Det} \mathcal{G}.$$

Theorem 16.11: the determinant of a short exact sequence

Let $(X, \mathcal{O}_X)$ be a ringed space and let

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H} \longrightarrow 0$$

be a short exact sequence of locally free sheaves on X. Then there is a canonical isomorphism

$$\operatorname{Det} G \cong \operatorname{Det} F \otimes \operatorname{Det} H.$$

Proof Let r be the rank of $\mathcal{F}$ and s the rank of $\mathcal{H}$. Consider open subsets $U \subseteq X$ on which all three sheaves are trivial and on which the sheaf surjection $\mathcal{G} \to \mathcal{H}$ has a section. Such open sets cover X. On each U we have the situation

$$0 \longrightarrow \mathcal{O}_U^r \longrightarrow \mathcal{O}_U^{r+s} \longrightarrow \mathcal{O}_U^s \longrightarrow 0,$$

and let

$$\theta : \mathcal{O}_U^s \longrightarrow \mathcal{O}_U^{r+s}$$

be a section. Define

$$\Psi : \bigwedge^r \mathcal{O}_U^r \times \bigwedge^s \mathcal{O}_U^s \longrightarrow \bigwedge^{r+s} \mathcal{O}_U^{r+s}$$

by

$$\begin{aligned} &\Psi(u_1 \wedge \cdots \wedge u_r, w_1 \wedge \cdots \wedge w_s) \\ &\qquad := u_1 \wedge \cdots \wedge u_r \wedge \theta(w_1) \wedge \cdots \wedge \theta(w_s). \end{aligned}$$

This map is independent of the chosen section θ. For another section θ', the difference $\theta - \theta'$ takes values in $\mathcal{F}$. Then

$$\begin{aligned} &u_1 \wedge \cdots \wedge u_r \wedge \theta'(w_1) \wedge \cdots \wedge \theta'(w_s) \\ &= u_1 \wedge \cdots \wedge u_r \wedge (\theta(w_1) + u_1') \wedge \cdots \wedge (\theta(w_s) + u_s') \\ &= u_1 \wedge \cdots \wedge u_r \wedge \theta(w_1) \wedge \cdots \wedge \theta(w_s), \end{aligned}$$

because the $r+1$ vectors $u_1, \dots, u_r, u'_j$ are always linearly dependent, so the corresponding wedge products are 0. The map Ψ is bilinear and therefore defines a linear map

$$\widetilde{\Psi} : \bigwedge^{r} \mathcal{O}_U^r \otimes \bigwedge^{s} \mathcal{O}_U^s \longrightarrow \bigwedge^{r+s} \mathcal{O}_U^{r+s}.$$

Since these maps are canonical, their restrictions to smaller open subsets always give the same map. By Corollary 4.10, they therefore glue to a sheaf homomorphism

$$\bigwedge^{r} \mathcal{F} \otimes \bigwedge^{s} \mathcal{H} \longrightarrow \bigwedge^{r+s} \mathcal{G}.$$

By its explicit description, this homomorphism is locally an isomorphism, and hence, by Lemma 4.6, it is also a global isomorphism.

> **Editorial note — the surjection in the source proof.** The opening sentence of the source proof prints the surjection $\mathcal{F} \to \mathcal{H}$. The displayed exact sequence and the section $\theta : \mathcal{O}_U^s \to \mathcal{O}_U^{r+s}$ show that the surjection used in the construction is $\mathcal{G} \to \mathcal{H}$. This edition uses that surjection in the sentence above and explicitly records the correction.

Corollary 16.12: the determinant of a direct sum of invertible sheaves

Let $(X, \mathcal{O}_X)$ be a ringed space and let

$$\mathcal{F} = \mathcal{L}_1 \oplus \cdots \oplus \mathcal{L}_r$$

be a direct sum of invertible sheaves. Then

$$\operatorname{Det} F \cong \mathcal{L}_1 \otimes \cdots \otimes \mathcal{L}_r.$$

Proof See Exercise 16.22.

Worksheet 16: Locally Free Sheaves

The star marks the only exercise with a frozen public solution, Exercise 16.12. The other twenty-two exercises have negative candidate results; this edition does not invent new solutions.

Exercise 16.1

Let $\mathcal{F}$ and $\mathcal{G}$ be locally free sheaves on a ringed space, of ranks r and s respectively. Prove that the direct sum $\mathcal{F} \oplus \mathcal{G}$ is locally free of rank $r+s$.

Exercise 16.2

Let $\mathcal{F}$ be a locally free sheaf of rank r on the ringed space $(X, \mathcal{O}_X)$. Prove that the dual sheaf $\mathcal{F}^*$ is also locally free of rank r.

Exercise 16.3

Prove that a locally free sheaf $\mathcal{F}$ on a ringed space $(X, \mathcal{O}_X)$ is naturally isomorphic to its bidual $\mathcal{F}^{**}$.

Exercise 16.4

Let $\mathcal{F}$ and $\mathcal{G}$ be locally free sheaves on a ringed space, and let

$$\varphi : \mathcal{F} \longrightarrow \mathcal{G}$$

be a surjective module homomorphism. Prove that the kernel $\operatorname{kern} \varphi$ is also locally free.

Exercise 16.5

Prove that the rank of locally free sheaves on a ringed space $(X, \mathcal{O}_X)$ is additive in short exact sequences. That is, if there is a short exact sequence of locally free sheaves

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H} \longrightarrow 0,$$

then

$$\operatorname{rank} \mathcal{G} = \operatorname{rank} \mathcal{F} + \operatorname{rank} \mathcal{H}.$$

Exercise 16.6

Let $\mathcal{F}$ and $\mathcal{G}$ be locally free sheaves on a ringed space, of ranks r and s respectively. Prove that the tensor product

$$\mathcal{F} \otimes_{\mathcal{O}_X} \mathcal{G}$$

is locally free of rank rs.

Exercise 16.7

Let $\mathcal{F}$ and $\mathcal{G}$ be locally free sheaves on a ringed space, and let $\varphi : \mathcal{F} \to \mathcal{G}$ be an injective module homomorphism. Show that the quotient sheaf $\mathcal{G}/\mathcal{F}$ is generally not locally free.

Exercise 16.8

Let $\mathcal{F}$ be a coherent module on the Noetherian scheme $(X, \mathcal{O}_X)$, and let $r \in \mathbb{N}$. Prove that the following properties are equivalent.

1. $\mathcal{F}$ is locally free of rank r.
2. For every point $P \in X$, the stalk $\mathcal{F}_P$ is a free $\mathcal{O}_{X,P}$-module of rank r.

Exercise 16.9

Let $(X, \mathcal{O}_X)$ be a Noetherian integral scheme and let $\mathcal{F}$ be a coherent module on X. Prove that there is a nonempty open subset $U \subseteq X$ such that $\mathcal{F}|_U$ is free.

Exercise 16.10

Let $(X, \mathcal{O}_X)$ be a Noetherian scheme and let $\varphi : \mathcal{F} \to \mathcal{G}$ be a homomorphism of coherent modules $\mathcal{F}$ and $\mathcal{G}$ on X. Let $P \in X$ be a point such that

$$\varphi_P : \mathcal{F}_P \longrightarrow \mathcal{G}_P$$

is an isomorphism. Prove that there is an open neighbourhood $P \in U \subseteq X$ such that

$$\varphi : \mathcal{F}|_U \longrightarrow \mathcal{G}|_U$$

is an isomorphism.

Exercise 16.11

Let R be a commutative ring, let M be a flat R-module, and let S be an R-algebra. Prove that $M \otimes_R S$ is a flat S-module.

Exercise 16.12 ★

Let R be a commutative ring and let M be an R-module. Prove that M is a projective module if and only if there is another module N such that the direct sum $M \oplus N$ is free.

Exercise 16.13

Let K be a field and let $R = K^n$ be the product ring. Prove that every R-module M is projective.

Exercise 16.14

Give an example of an Artinian ring and a finitely generated R-module M that is not projective.

Exercise 16.15

Prove that for the surjective group homomorphism

$$p : \mathbb{Z}^{(\mathbb{N}_+)} \longrightarrow \mathbb{Q}, \qquad e_n \longmapsto \frac{1}{n},$$

there is no group homomorphism

$$i : \mathbb{Q} \longrightarrow \mathbb{Z}^{(\mathbb{N}_+)}$$

with $p \circ i = \mathrm{Id}_{\mathbb{Q}}$. Deduce that $\mathbb{Q}$ is not projective as a $\mathbb{Z}$-module.

Exercise 16.16

Let R be a commutative ring, let M be a projective R-module, and let $T \subseteq R$ be a multiplicative system. Prove that M_T is a projective R_T-module.

Exercise 16.17

Let R be a commutative ring, let M be a projective R-module, and let S be an R-algebra. Prove that $M \otimes_R S$ is a projective S-module.

Exercise 16.18

Let R be a commutative ring and let $f_1, \dots, f_n \in R$. Let

$$\mathcal{S} = \mathrm{Syz}(f_1, \dots, f_n)$$

be the corresponding syzygy sheaf on

$$U = D(f_1, \dots, f_n) \subseteq \mathrm{Spek}(R).$$

Give explicit trivialisations of $\mathcal{S}|_{D(f_i)}$.

Exercise 16.19

Let R be a $\mathbb{Z}$-graded ring and let $f_1, \dots, f_n \in R$ be homogeneous elements of degrees d_i. Suppose that the ideal I generated by the f_i and the irrelevant ideal R_+ have the same radical. Put $Y = \mathrm{Proj}(R)$, and assume that each twisting sheaf $\mathcal{O}_Y(-d_i)$ is invertible; this holds, for example, when R is standard graded. Prove the following statements.

1. There is a short exact sequence

$$0 \longrightarrow \mathrm{Syz}(f_1, \dots, f_n) \longrightarrow \bigoplus_{i=1}^{n} R(-d_i) \longrightarrow I \longrightarrow 0$$

   of graded R-modules with homogeneous homomorphisms.

2. On Y there is a short exact sequence

$$0 \longrightarrow \mathrm{Syz}(f_1, \dots, f_n) \longrightarrow \bigoplus_{i=1}^{n} \mathcal{O}_Y(-d_i) \longrightarrow \mathcal{O}_Y \longrightarrow 0$$

   of locally free sheaves.

3. On $D_+(f_i)$, the restriction of the locally free sheaf $\mathrm{Syz}(f_1, \dots, f_n)$ is isomorphic to a direct sum of twisted structure sheaves.

   Editorial note — the grading hypothesis. The source assumes only that R is $\mathbb{Z}$-graded, but under that hypothesis the sheaves $\mathcal{O}_Y(-d_i)$ need not be invertible, so the sequence in part (2) need not consist of locally free sheaves. This edition states the precise invertibility hypothesis used by parts (2) and (3); standard grading is a familiar sufficient condition.

Exercise 16.20

Let $\mathcal{F}$ be a locally free sheaf on a ringed space. Prove that its determinant sheaf $\mathrm{Det}\, F$ is invertible.

Exercise 16.21

Let $\mathcal{F}$ be a locally free sheaf on a ringed space, with dual sheaf $\mathcal{F}^*$. Prove the following relation between their determinant sheaves:

$$(\mathrm{Det}\, F)^* = \mathrm{Det}(F^*).$$

Exercise 16.22

Let $(X, \mathcal{O}_X)$ be a ringed space and let

$$\mathcal{F} = \mathcal{L}_1 \oplus \cdots \oplus \mathcal{L}_r$$

be a direct sum of invertible sheaves. Prove that

$$\mathrm{Det}\, F \cong \mathcal{L}_1 \otimes \cdots \otimes \mathcal{L}_r.$$

Exercise 16.23

Prove that in the situation of Exercise 16.19, the determinant sheaf of the locally free sheaf $\mathrm{Syz}(f_1, \dots, f_n)$ on $Y = \mathrm{Proj}(R)$ is

$$\mathrm{Det}(\mathrm{Syz}(f_1, \dots, f_n)) = \mathcal{O}_Y\left(-\sum_{i=1}^{n} d_i\right).$$

Public Solutions and Coverage of Worksheet 16

At the frozen revision boundary, the source provides exactly one public solution among the 23 exercises: the solution to Exercise 16.12. The exercise map and candidate evidence record negative results for Exercises 16.1–16.11 and 16.13–16.23. Missing public solution pages are not replaced by invented solutions.

Solution to Exercise 16.12

First suppose that M is projective. Since M has a generating system v_i, $i \in I$, there is a surjective R-module homomorphism

$$\theta : R^{(I)} \longrightarrow M.$$

Projectivity, applied to the identity

$$\mathrm{Id} : M \longrightarrow M,$$

shows that there is a module homomorphism

$$\psi : M \longrightarrow R^{(I)}$$

with

$$\theta \circ \psi = \mathrm{Id}_M .$$

This means that

$$R^{(I)} = M \oplus \operatorname{kern} \theta.$$

Conversely, suppose that

$$M \oplus N = F$$

is free. Given a surjective R-module homomorphism

$$\theta : A \longrightarrow B$$

and a module homomorphism

$$\varphi : M \longrightarrow B,$$

apply the result for free modules to

$$\varphi \circ p_M : F \longrightarrow B.$$

We obtain a homomorphism

$$\rho : F \longrightarrow A$$

with

$$\varphi \circ p_M = \theta \circ \rho.$$

The restriction of ρ to M has the required property, since

$$\begin{aligned}\theta \circ (\rho \circ i_M) &= (\theta \circ \rho) \circ i_M \\ &= (\varphi \circ p_M) \circ i_M \\ &= \varphi \circ (p_M \circ i_M) \\ &= \varphi.\end{aligned}$$

Frozen negative results

There is no public solution page at the frozen revision for Exercises 16.1, 16.2, 16.3, 16.4, 16.5, 16.6, 16.7, 16.8, 16.9, 16.10, 16.11, 16.13, 16.14, 16.15, 16.16, 16.17, 16.18, 16.19, 16.20, 16.21, 16.22, or 16.23. This records the outcome of checking candidates; it does not claim that mathematical solutions do not exist.

Lecture 17: Geometric Vector Bundles

Geometric vector bundles

For an affine scheme

$$U = \mathrm{Spek}(R),$$

the scheme

$$\mathbb{A}^r_U := \mathrm{Spek}(R[T_1, \dots, T_r])$$

together with its natural projection to $\mathrm{Spek}(R)$, namely the spectrum map arising from

$$R \longrightarrow R[T_1, \dots, T_r],$$

is called the *trivial bundle of rank r over U*. For a point $P \in \mathrm{Spek}(R)$ corresponding to the ring homomorphism

$$R \longrightarrow \kappa(P),$$

the fibre of $\mathbb{A}^r_U$ over P is $\mathbb{A}^r_{\kappa(P)}$, that is, affine space of dimension r over the residue field $\kappa(P)$.

For an arbitrary scheme X with an affine cover

$$X = \bigcup_{i \in I} U_i,$$

we define $\mathbb{A}^r_X$ by gluing the $\mathbb{A}^r_{U_i}$ as prescribed by the gluing of the U_i inside X, in the sense of Lemma 7.10. This trivial bundle of rank r comes with a projection

$$\mathbb{A}^r_X \longrightarrow X.$$

It is also called the *affine cylinder of rank r over X* and is written $X \times \mathbb{A}^r$. These trivial bundles are the local building blocks for the concept of a geometric vector bundle over a scheme.

Definition 17.1: geometric vector bundles

Let X be a scheme. A scheme V together with a morphism

$$p : V \longrightarrow X$$

is called a *geometric vector bundle of rank r over X* if there is an open cover

$$X = \bigcup_{i \in I} U_i$$

and U_i-isomorphisms

$$\psi_i : U_i \times \mathbb{A}^r = \mathbb{A}^r_{U_i} \longrightarrow V|_{U_i} = p^{-1}(U_i)$$

such that, for every affine open subset $U \subseteq U_i \cap U_j$, the transition maps

$$\psi_j^{-1} \circ \psi_i : \mathbb{A}^r_{U_i}|_U \longrightarrow \mathbb{A}^r_{U_j}|_U$$

are linear automorphisms. At the ring level, this means that they are induced by automorphisms of the polynomial ring $\Gamma(U, \mathcal{O}_X)[T_1, \ldots, T_r]$ of the form

$$T_i \longmapsto \sum_{j=1}^{r} a_{ij} T_j.$$

The maps ψ_i are called *trivialisations* of the vector bundle.

The following example continues Example 14.6.

Example 17.2: a line bundle from the standard ideal

Consider the quadratic number ring

$$R = \mathbb{Z}[\sqrt{-5}],$$

in which the equality

$$2 \cdot 3 = 6 = (1 + \sqrt{5}\, i)(1 - \sqrt{5}\, i),$$

holds, and the R-algebra

$$A = R[X, Y]/(3X - (1 - i\sqrt{5})Y)$$

with its associated spectrum map Spek$(A) \to$ Spek(R). We claim that this is a geometric line bundle. To see this, use the open cover

$$\text{Spek}(R) = D(2) \cup D(3).$$

We have

$$A_2 = R_2[X, Y]/(3X - (1 - i\sqrt{5})Y) \cong R_2[S]$$

with $X \mapsto 2S$ and $Y \mapsto (1 + i\sqrt{5})S$. Indeed, $S = X/2$, hence $X = 2S$, and

$$Y = \frac{3}{1 - i\sqrt{5}} X = \frac{1 + i\sqrt{5}}{2} X = (1 + i\sqrt{5})S.$$

Similarly,

$$A_3 = R_3[X, Y]/(3X - (1 - i\sqrt{5})Y) \cong R_3[T]$$

with $X \mapsto (1 - i\sqrt{5})T$ and $Y \mapsto 3T$. Indeed, $T = Y/3$, hence $Y = 3T$ and

$$X = \frac{1 - i\sqrt{5}}{3} Y = (1 - i\sqrt{5})T.$$

On $D(6) = D(2) \cap D(3)$, the transition map is given by

$$S = \frac{X}{2} = \frac{1 - i\sqrt{5}}{2} T,$$

and is therefore linear.

> **Editorial note — two source defects in this example.** The source prints the malformed factor $1 - i\sqrt{5}|$; the displayed substitution $Y = 3T$ gives $(1 - i\sqrt{5})$, used above. More substantially, the algebra A as stated is not a line bundle over all of Spek(R). At the maximal ideal $\mathfrak{p} = (3, 1 - \sqrt{-5})$, both coefficients of its defining relation vanish, so its fibre is Spek$(\mathbb{F}_3[X, Y])$, an affine plane. This point lies in $D(2)$, and the claimed isomorphism $A_2 \cong R_2[S]$ therefore fails. The source construction is retained here with this warning; no replacement algebra is attributed to the author.

Example 17.3: a syzygy bundle on punctured affine space

Consider the ring homomorphism

$$K[X, Y, Z] \longrightarrow K[X, Y, Z][U, V, W]/(XU + YV + ZW),$$

the associated spectrum map

$$\text{Spek}\big(K[X, Y, Z][U, V, W]/(XU + YV + ZW)\big) \longrightarrow \mathbb{A}^3_K,$$

and its restriction

$$\begin{aligned} \varphi: \quad \text{Spek}\big(K[X, Y, Z][U, V, W]/(XU + YV + ZW)\big) &\supseteq D(X) \cup D(Y) \cup D(Z) \\ &\longrightarrow \mathbb{A}^3_K \setminus \{(0, 0, 0)\} = D(X) \cup D(Y) \cup D(Z). \end{aligned}$$

The latter is a geometric vector bundle of rank 2 over punctured affine space. Natural trivialisations are given on $D(X)$, $D(Y)$, and $D(Z)$; compare Example 1.2. For example,

$$\big(K[X, Y, Z][U, V, W]/(XU + YV + ZW)\big)_X \cong K[X, Y, Z]_X[V, W],$$

because U can be expressed as

$$U = -\frac{YV + ZW}{X}.$$

Example 17.4: geometric realisation of twisted line bundles

Consider projective space

$$\mathbb{P}^n_K = \text{Proj}(K[X_0, X_1, \dots, X_n])$$

and the projective spectrum

$$W_k = \text{Proj}(K[X_0, X_1, \dots, X_n, Y]),$$

with $\deg(X_i) = 1$ and $\deg(Y) = k$. By Theorem 12.11, the homogeneous inclusion

$$K[X_0, X_1, \dots, X_n] \subset K[X_0, X_1, \dots, X_n, Y]$$

induces a scheme morphism

$$p : W_k \supset D_+(X_0, X_1, \dots, X_n) = V_k \longrightarrow \mathbb{P}^n_K.$$

Over $D_+(X_i)$, the map takes the form

$$D_+(X_i) \cong \text{Spek}\, K\left[\frac{X_j}{X_i},\ j \neq i,\ \frac{Y}{X_i^k}\right] \longrightarrow D_+(X_i) \cong \text{Spek}\, K\left[\frac{X_j}{X_i},\ j \neq i\right],$$

and is thus a trivial line bundle. For $D_+(X_i)$ and $D_+(X_j)$, the transition maps over

$$\Gamma(D_+(X_i X_j), \mathcal{O}_{\mathbb{P}^n_K}) = K\left[\frac{X_r}{X_i},\ r \neq i, \frac{X_s}{X_j},\ s \neq j\right]$$

are given by

$$\frac{Y}{X_i^k} \longmapsto \frac{Y}{X_j^k} \frac{X_j^k}{X_i^k}.$$

These maps are linear, so we obtain a line bundle over projective space.

A geometric vector bundle has additional structures. First consider the case

$$\mathbb{A}^r_{\text{Spek}(R)} = \text{Spek}(R[T_1, \dots, T_r]) \longrightarrow \text{Spek}(R).$$

The R-algebra homomorphism

$$R[T_1, \dots, T_r] \longrightarrow R, \qquad T_i \longmapsto 0,$$

gives a spectrum map

$$\text{Spek}(R) \longrightarrow \text{Spek}(R[T_1, \dots, T_r]),$$

which is a closed embedding, called the *zero section.* The R-algebra homomorphism

$$R[T_1, \dots, T_r] \longrightarrow R[S_1, \dots, S_r, T_1, \dots, T_r], \qquad T_i \longmapsto S_i + T_i,$$

gives a spectrum map

$$\alpha : \mathbb{A}^{r+r}_R \cong \mathbb{A}^r_R \times_{\text{Spek}(R)} \mathbb{A}^r_R \longrightarrow \mathbb{A}^r_R,$$

called addition on the vector bundle. Furthermore, the R-algebra homomorphism

$$R[T_1, \dots, T_r] \longrightarrow R[Z, T_1, \dots, T_r], \qquad T_i \longmapsto ZT_i,$$

gives a spectrum map

$$\mathbb{A}^{r+1}_R \cong \mathbb{A}^1_R \times_{\text{Spek}(R)} \mathbb{A}^r_R \longrightarrow \mathbb{A}^r_R,$$

called scalar multiplication.

Lemma 17.5: operations on geometric vector bundles

A geometric vector bundle $p : V \to X$ over a scheme X has a zero section

$$X \longrightarrow V,$$

an addition map

$$V \times_X V \longrightarrow V,$$

and a scalar multiplication

$$\mathbb{A}^1_X \times_X V \longrightarrow V.$$

Proof On an affine subset $U = \mathrm{Spek}(R) \subseteq X$ with a trivialisation

$$V|_U \cong \mathbb{A}^r_U \cong \mathrm{Spek}(R[T_1, \dots, T_r]),$$

there is a zero section given by $T_i \mapsto 0$. Since the transition maps are linear, on the intersection $U_i \cap U_j$ this section is independent of the chosen affine subset, and is therefore well-defined.

The existence of addition rests essentially on the fact that, for a linear R-algebra isomorphism

$$\theta : R[T_1, \dots, T_r] \longrightarrow R[U_1, \dots, U_r],$$

the diagram

$$\begin{array}{cccccc} & R[T_1, \dots, T_r] & \xrightarrow{\alpha^*} & R[T_1, \dots, T_r, S_1, \dots, S_r] & \\ \theta & \downarrow & & \downarrow & \theta\times\theta \\ & R[U_1, \dots, U_r] & \xrightarrow{\alpha^*} & R[U_1, \dots, U_r, V_1, \dots, V_r] & \end{array}$$

commutes. Scalar multiplication is obtained similarly.

Vector bundle homomorphisms

Definition 17.6: vector bundle homomorphisms

Let V and W be vector bundles over a scheme X. A *vector bundle homomorphism* $\varphi : V \to W$ is a scheme morphism from V to W over X with the following property. For every point $P \in X$, there is an affine open neighbourhood $P \in U \subseteq X$ refining the given trivialising neighbourhoods of both bundles, that is, $U \subseteq U_i, U'_j$ for suitable i, j, such that the composite

$$\mathbb{A}^r_U \xrightarrow{\psi_i|_U} V|_U \xrightarrow{\varphi|_U} W|_U \xrightarrow{\theta_j^{-1}|_U} \mathbb{A}^s_U$$

is given at the ring level by a linear substitution homomorphism.

In the following statement, the kernel means the inverse image of the zero section, hence the inverse image of the zero point in each fibre. For a vector bundle homomorphism, the fibre map over every point $P \in X$ is given by a matrix over its residue field $\kappa(P)$. However, affine spaces over this field contain points with very different residue fields, so concepts from linear algebra must be applied with some care.

Lemma 17.7: the kernel of a surjective homomorphism

Let V and W be vector bundles over a scheme X, and let $\varphi : V \to W$ be a surjective vector bundle homomorphism. Then its pointwise kernel is a vector bundle over X.

Proof See Exercise 17.18.

Without the surjectivity condition, the kernel of a vector bundle homomorphism need not be a vector bundle. In Example 17.3, the pointwise-defined kernel is a vector bundle only on the punctured spectrum; at the origin, the kernel degenerates to a three-dimensional vector space.

Vector bundles and locally free sheaves

Definition 17.8: the sheaf of sections

For a geometric vector bundle $p : V \to X$ on a scheme X, the sheaf $\mathcal{F}$ defined on an open subset $U \subseteq X$ by

$$\Gamma(U, \mathcal{F}) = \{s : U \to V|_U \text{ scheme morphism} \mid p \circ s = \mathrm{Id}_U\}$$

is called the *sheaf of sections of* V.

Lemma 17.9: the sheaf of sections is locally free

For a geometric vector bundle $p : V \to X$, the sheaf of sections $\mathcal{F}$ is a locally free sheaf.

Proof The addition

$$V \times_X V \longrightarrow V,$$

gives, by Exercise 17.19, a well-defined addition on the sheaf of sections. By Exercise 17.11, this makes $\mathcal{F}$ a sheaf of commutative groups. Scalar multiplication

$$\mathbb{A}^1_X \times_X V \longrightarrow V$$

gives $\mathcal{F}$ an $\mathcal{O}_X$-module structure. For an open set $U \subseteq X$ with $V|_U \cong \mathbb{A}^r_U$, we have

$$\mathcal{F}|_U \cong (\mathcal{O}_X|_U)^r,$$

so $\mathcal{F}$ is locally free.

Geometric vector bundles and locally free sheaves are essentially equivalent objects.

Theorem 17.10: the correspondence between vector bundles and locally free sheaves

On a scheme, geometric vector bundles correspond to locally free sheaves. Moreover, vector bundle homomorphisms correspond to $\mathcal{O}_X$-module homomorphisms.

A geometric vector bundle V over X is assigned its sheaf of sections $\mathcal{S}_V$, which is locally free by Lemma 17.9. A vector bundle homomorphism $\varphi : V \to W$ is assigned the module homomorphism $\mathcal{S}_V \to \mathcal{S}_W$ sending a section $s : U \to V|_U$ to the section $\varphi \circ s : U \to W|_U$.

Proof First we show that every locally free sheaf is isomorphic to the sheaf of sections of a vector bundle. A locally free sheaf $\mathcal{F}$ of rank r is given by an open cover

$$X = \bigcup_{i \in I} U_i,$$

where the U_i may be chosen affine, together with isomorphisms

$$\varphi_i : \mathcal{O}^r_{U_i} \longrightarrow \mathcal{F}|_{U_i}.$$

By Theorem 13.10, the composite

$$\mathcal{O}^r_{U_i}|_{U_i\cap U_j} = \mathcal{O}^r_{U_i\cap U_j} \xrightarrow{\varphi_i|_{U_i\cap U_j}} \mathcal{F}|_{U_i\cap U_j} \xrightarrow{\varphi_j^{-1}|_{U_i\cap U_j}} \mathcal{O}^r_{U_i\cap U_j} = \mathcal{O}^r_{U_j}|_{U_i\cap U_j}$$

is given by $e_k \mapsto f_k$ with

$$f_k \in \Gamma(U_i \cap U_j, \mathcal{O}^r_{U_i\cap U_j}).$$

Here $f_k = (f_{k\ell})_{1\le \ell\le r}$ with

$$f_{k\ell} \in \Gamma(U_i \cap U_j, \mathcal{O}_X).$$

The determinant of the matrix $(f_{k\ell})_{k\ell}$ is a unit in $\Gamma(U_i \cap U_j, \mathcal{O}_X)$. Via

$$T_k \longmapsto \sum_{\ell=1}^{r} f_{k\ell} S_\ell,$$

these data define a linear $\Gamma(U_i \cap U_j, \mathcal{O}_X)$-algebra isomorphism

$$\Gamma(U_i \cap U_j, \mathcal{O}_X)[T_1, \dots, T_r] \longrightarrow \Gamma(U_i \cap U_j, \mathcal{O}_X)[S_1, \dots, S_r]$$

and a scheme isomorphism

$$\varphi_{ji} : \mathbb{A}^r_{U_i\cap U_j} \longrightarrow \mathbb{A}^r_{U_i\cap U_j},$$

of the form required in the definition of a geometric vector bundle.

Consider the gluing data for ringed spaces

$$(W_i = \mathbb{A}^r_{U_i},\ W_{ij} = \mathbb{A}^r_{U_i}|_{U_j} \subseteq W_i,\ \varphi_{ji} : W_{ij} \longrightarrow W_{ji}).$$

The cocycle condition holds because these data come from the global object $\mathcal{F}$. By Lemma 7.10, there is a scheme W realising these gluing data. The local projections

$$W_i = \mathbb{A}^r_{U_i} \longrightarrow U_i$$

glue to a scheme morphism

$$W \longrightarrow X.$$

By construction, this is a geometric vector bundle over X. Let $\mathcal{S}$ be the sheaf of sections of W. We claim that there is a natural isomorphism

$$\mathcal{F} \longrightarrow \mathcal{S}.$$

The construction gives natural sheaf isomorphisms

$$\mathcal{F}|_{U_i} \longrightarrow \mathcal{S}|_{U_i}$$

for every U_i, whose restrictions to the intersections $U_i \cap U_j$ agree. By Corollary 4.10, there is a global sheaf homomorphism, and by Lemma 4.6 it is an isomorphism. The assignment is injective because a vector bundle can be reconstructed, up to isomorphism, from its sheaf of sections by the construction above. For the statements about homomorphisms, see Exercises 17.21, 17.22, and 17.23.

Under this equivalence, the free sheaf of rank r corresponds to affine space $\mathbb{A}^r_X$ over X.

Example 17.11: injectivity for bundles and for sheaves

Let R be a commutative ring and let $f \in R$. Via

$$\mathrm{Spek}(R[T]) = \mathbb{A}^1_R \longrightarrow \mathrm{Spek}(R[T]) = \mathbb{A}^1_R, \qquad T \longmapsto fT,$$

the element f defines a vector bundle homomorphism. On the fibres over points $P \in \mathrm{Spek}(R)$ where f is a unit, namely the points of $D(f)$, this map is bijective; over the other points it is the zero map. Thus this map is injective — and at the same time surjective and bijective — only if f is a unit.

However, multiplication by f also defines a homomorphism of the structure sheaf

$$\mathcal{O}_X \longrightarrow \mathcal{O}_X, \qquad 1 \longmapsto f.$$

Thus, on every open subset $U \subseteq X = \mathrm{Spek}(R)$, there is a $\Gamma(U, \mathcal{O}_X)$-module homomorphism

$$\Gamma(U, \mathcal{O}_X) \longrightarrow \Gamma(U, \mathcal{O}_X), \qquad r \longmapsto rf.$$

This sheaf homomorphism is injective exactly when f is a non-zero-divisor in R, and bijective exactly when f is a unit. Thus the notions of injectivity for vector bundles and locally free sheaves do not coincide.

Worksheet 17: Geometric Vector Bundles

At the frozen authority boundary, the source provides no public solution page for any of the 24 exercises. The exercise map records negative candidate results for Exercises 17.1–17.24; this edition does not invent new solutions.

Exercise 17.1

Compare Definition 17.1 of a geometric vector bundle over a scheme with Definition 1.4 of a real vector bundle over a topological space.

Exercise 17.2

Let $p : V \to X$ be a geometric vector bundle of rank r over a scheme X. Prove that the fibre of p over a point $P \in X$ is isomorphic to $\mathbb{A}^r_{\kappa(P)}$.

Exercise 17.3

What is a vector bundle of rank 0 over a scheme X?

Exercise 17.4

Discuss the trivial line bundle

$$\mathrm{Spek}(\mathbb{Z}[X]) = \mathbb{A}^1_{\mathbb{Z}} \longrightarrow \mathrm{Spek}(\mathbb{Z}).$$

What can be said about its fibres, its sections, and the closed subsets $Y \subseteq \mathrm{Spek}(\mathbb{Z}[X])$ and their images in $\mathrm{Spek}(\mathbb{Z})$?

Exercise 17.5

Determine the trivialisations and transition maps in Example 17.3 explicitly.

Exercise 17.6

Prove that for $k = 0$ in Example 17.4 one obtains the trivial line bundle

$$\mathbb{A}^1_{\mathbb{P}^n_K} = \mathbb{A}^1_K \times \mathbb{P}^n_K \longrightarrow \mathbb{P}^n_K.$$

Exercise 17.7

Prove that for $k = 1$ in Example 17.4 one obtains the so-called projection away from a point,

$$\mathbb{P}^{n+1}_K \supset \mathbb{P}^{n+1}_K \setminus (0, 0, \dots, 0, 1) = V_1 \longrightarrow \mathbb{P}^n_K.$$

Exercise 17.8

Let X be a scheme, and let $p : V \to X$ and $W \to X$ be vector bundles over X of ranks r and s respectively. Define their direct sum $V \times_X W$ in terms of simultaneous trivialisations

$$V|_U \cong \mathbb{A}^r_U, \qquad W|_U \cong \mathbb{A}^s_U.$$

Thus one should have

$$(V \times_X W)_U \cong V|_U \times_U W|_U \cong \mathbb{A}^{r+s}_U.$$

Exercise 17.9

Define constructions from linear algebra, such as the direct sum, dual, tensor product, and exterior product, for geometric vector bundles over a scheme.

Use Lecture 3 as a guide.

Exercise 17.10

Prove that the zero section of a geometric vector bundle $V \to X$ is a closed embedding.

Exercise 17.11

Let $p : V \to X$ be a vector bundle over X. Prove that the addition $\alpha : V \times_X V \to V$ has the following properties. Prove also that the displayed morphisms exist.

1. The diagram

$$\begin{array}{ccccc} & V & \xrightarrow{\mathrm{Id}\times(N\circ p)} & V \times_X V & \\ \mathrm{Id} & \downarrow & & \downarrow & \alpha \\ & V & = & V & \end{array}$$

   commutes, where $N : X \to V$ denotes the zero section.

2. The diagram

$$\begin{array}{ccccc} & V \times_X V & \xrightarrow{\pi} & V \times_X V & \\ \alpha & \downarrow & & \downarrow & \alpha \\ & V & = & V & \end{array}$$

   commutes, where π interchanges the two factors.

3. The diagram

$$\begin{array}{ccccc} & V \times_X V \times_X V & \xrightarrow{\alpha\times\mathrm{Id}} & V \times_X V & \\ \mathrm{Id}\times\alpha & \downarrow & & \downarrow & \alpha \\ & V \times_X V & \xrightarrow{\alpha} & V & \end{array}$$

   commutes.

For the following exercise, compare Exercise 1.6.

Exercise 17.12

Let K be a field. Consider

$$A = K[X, Y, U, V]/(XU + YV - 1),$$

the spectrum map

$$\text{Spek}(A) \longrightarrow \text{Spek}(K[X, Y]) = \mathbb{A}^2_K,$$

and the restriction

$$p : \text{Spek}(A) = D(X, Y) = V \longrightarrow U = D(X, Y) \subseteq \mathbb{A}^2_K.$$

What justifies the equality on the left? Prove the following statements.

1. On $D(X)$ and on $D(Y)$, the map p is isomorphic to the affine line over the base.
2. The map $p : V \to U$ has no section.
3. V is not a line bundle.

Exercise 17.13

Let R be a commutative algebra of finite type over a field K. Let $f_1, \dots, f_n, f$ be elements of R, and let

$$A = R[T_1, \dots, T_n]/(f_1 T_1 + \cdots + f_n T_n + f)$$

be the forcing algebra for these data. Let

$$p : \text{Spek}(A) \supseteq V = D(f_1, \dots, f_n) \longrightarrow U = D(f_1, \dots, f_n) \subseteq \text{Spek}(R)$$

be the restricted spectrum map. Prove that there is an affine open cover

$$U = \bigcup_{i \in I} U_i$$

such that $p^{-1}(U_i)$ is isomorphic to $U_i \times \mathbb{A}^{n-1}$ and the transition maps are affine-linear.

Exercise 17.14

Prove that a homomorphism of trivial vector bundles

$$\varphi : \mathbb{A}^r_{\text{Spek}(R)} \longrightarrow \mathbb{A}^s_{\text{Spek}(R)}$$

over the affine scheme $\text{Spek}(R)$ is given by an $s \times r$ matrix over R.

Exercise 17.15

Prove that a vector bundle homomorphism

$$\varphi : V \longrightarrow W$$

over X maps the zero section of V, regarded as a closed subscheme, into the zero section of W.

Exercise 17.16

Let $\varphi : V \to W$ be a homomorphism of vector bundles V, W over X. Prove that the diagram

$$\begin{array}{ccccc} & V \times_X V & \xrightarrow{\varphi\times\varphi} & W \times_X W & \\ \alpha & \downarrow & & \downarrow & \alpha \\ & V & \xrightarrow{\varphi} & W & \end{array}$$

commutes. In other words, a vector bundle homomorphism is compatible with addition.

Exercise 17.17

Consider the homomorphism of trivial vector bundles

$$\varphi : \mathbb{A}^2_{\mathrm{Spek}(\mathbb{Z})} \longrightarrow \mathbb{A}^2_{\mathrm{Spek}(\mathbb{Z})}$$

over Spek($\mathbb{Z}$) given by the matrix

$$\begin{pmatrix} 5 & 6 \\ 7 & 4 \end{pmatrix}.$$

Determine the points $P \in$ Spek($\mathbb{Z}$) where the corresponding fibre map is injective, and those where it is surjective.

Exercise 17.18

Let V and W be vector bundles over a scheme X, and let $\varphi : V \to W$ be a surjective vector bundle homomorphism. Prove that its pointwise kernel is a vector bundle over X.

Exercise 17.19

Let V and W be vector bundles over a scheme X. Prove that the sheaf of sections of the direct sum $V \times_X W$ is the direct sum of the sheaves of sections of V and W.

Exercise 17.20

Prove that the sheaf of sections of the line bundle

$$V_k \longrightarrow \mathbb{P}^n_K$$

from Example 17.4 is the twisted structure sheaf $\mathcal{O}_{\mathbb{P}^n_K}(k)$.

Exercise 17.21

Let $p : Y \to X$ and $q : Z \to X$ be schemes over a scheme X, and let $\varphi : Y \to Z$ be a scheme morphism over X. Prove that the associated map between the sheaves of sections in the category of schemes,

$$\mathcal{S}_Y \longrightarrow \mathcal{S}_Z,$$

sending a section $s : U \to p^{-1}(U)$ to the section $\varphi \circ s : U \to q^{-1}(U)$, is a sheaf morphism.

Exercise 17.22

Let $\varphi : V \to W$ be a vector bundle homomorphism between vector bundles V and W over X, and let

$$\psi : \mathcal{S}_V \longrightarrow \mathcal{S}_W$$

be the associated morphism between their sheaves of sections. Prove that ψ is an $\mathcal{O}_X$-module homomorphism.

Exercise 17.23

Let V and W be vector bundles over a scheme X, and let $\mathcal{S}_V, \mathcal{S}_W$ be the associated sheaves of sections. Prove that the map

$$\operatorname{Hom}(V, W) \longrightarrow \operatorname{Hom}(\mathcal{S}_V, \mathcal{S}_W),$$

sending a vector bundle homomorphism to the associated $\mathcal{O}_X$-module homomorphism is a bijection. Prove also that isomorphisms correspond to isomorphisms under this correspondence.

Exercise 17.24

Let V and W be vector bundles over a scheme X, and let $\mathcal{F}$ and $\mathcal{G}$ be their associated locally free sheaves of sections. Let $\varphi : V \to W$ be a vector bundle homomorphism and let $\psi : \mathcal{F} \to \mathcal{G}$ be the associated sheaf homomorphism. Prove that the following statements are equivalent.

1. φ is a surjective scheme morphism.
2. At every point $P \in X$, the fibre map $V(P) \to W(P)$ is surjective.
3. The homomorphism $\psi : \mathcal{F} \to \mathcal{G}$ is surjective.
4. There is an open cover $X = \bigcup_{i \in I} U_i$ and local sections, given by vector bundle homomorphisms

   $$s_i : W|_{U_i} \longrightarrow V|_{U_i}$$

   of φ.
5. There is an open cover $X = \bigcup_{i \in I} U_i$ and local sections, given by module homomorphisms

   $$t_i : \mathcal{G}|_{U_i} \longrightarrow \mathcal{F}|_{U_i}$$

   of ψ.

   Editorial note — the fibre label. The source quantifies the point P but writes $V(x) \to W(x)$. The same point must index the fibres; this edition writes $V(P) \to W(P)$.

Public Solutions and Coverage of Worksheet 17

At the frozen authority boundary, the source provides no public solution page for any of the 24 exercises in Worksheet 17. The exercise map and candidate evidence record negative results for Exercises 17.1–17.24. Missing public solution pages are not replaced by invented solutions.

Frozen negative results

There is no public solution page at the frozen authority boundary for Exercises 17.1–17.24. This records the outcome of checking candidates; it does not claim that mathematical solutions do not exist.

Lecture 18: Kähler Differentials

The module of Kähler differentials

On a manifold M there is a tangent bundle

$$TM \longrightarrow M.$$

Over a point $P \in M$, this consists of the tangent space T_PM, given by equivalence classes of differentiable curves

$$[-\epsilon, \epsilon] \longrightarrow M$$

through P. The tangent bundle is a real vector bundle over M that is characteristic of the manifold and allows many invariants of the manifold to be defined. We wish to define a corresponding object for a scheme, say of finite type over a field. A direct transfer of the analytic concept is impossible, since there is no direct replacement for differentiable curves. We therefore approach the tangent bundle from another perspective.

A continuous or differentiable section of the tangent bundle over an open set $U \subseteq M$ is called a continuous or differentiable vector field. A vector field F assigns to every point $P \in U$ a tangent vector

$$F(P) \in T_PM.$$

Differentiable functions on M can be differentiated along a vector field F, again giving a function. We set

$$(D_F(f))(P) := (D_{F(P)}f)(P),$$

where $(D_{F(P)}f)(P)$ denotes the directional derivative of f at P in the direction $F(P)$. This directional derivative can be calculated in any chart; by the chain rule, the result does not depend on the chosen chart. If M and F are infinitely differentiable, we obtain a map

$$D_F : C^\infty(U, \mathbb{R}) \longrightarrow C^\infty(U, \mathbb{R}), \qquad f \longmapsto D_F(f).$$

> **Editorial note — the vector field.** The source prints $F(P) = T_PM$; the edition uses the required membership relation $F(P) \in T_PM$. For the displayed map to take values in C^∞, the vector field F, as well as M, must be smooth; this hypothesis is made explicit above.

This map is an $\mathbb{R}$-linear derivation in the sense of the following purely algebraic definition. Starting from derivations, we shall introduce the module of Kähler differentials and, dually, develop a tangent sheaf in the scheme-theoretic setting. This sheaf is locally free when the scheme has no singularities. In this lecture we consider the affine situation and omit proofs.

Definition 18.1: algebraic derivations

Let R be a commutative ring, let A be a commutative R-algebra, and let M be an A-module. An R-linear map

$$\delta : A \longrightarrow M$$

satisfying

$$\delta(ab) = a\delta(b) + b\delta(a)$$

for all $a, b \in A$ is called an *R-derivation* with values in M.

The rule used here is called the *Leibniz rule.* Often $M = A$. For example, for the polynomial ring

$$A = R[X_1, \dots, X_n],$$

the ith formal partial derivative

$$\frac{\partial}{\partial X_i}$$

is an R-derivation from A to A. The set of derivations from A to M is naturally an A-module, denoted by

$$\mathrm{Der}_R(A, M).$$

Definition 18.2: the module of Kähler differentials

Let R be a commutative ring and let A be a commutative R-algebra. The A-module generated by all symbols $d(a)$, $a \in A$, subject to the identifications

$$d(ab) = ad(b) + bd(a) \qquad \text{for all } a, b \in A$$

and

$$d(ra + sb) = rd(a) + sd(b) \qquad \text{for all } r, s \in R \text{ and } a, b \in A,$$

is called the *module of Kähler differentials* of A over R. It is denoted by

$$\Omega_{A/R}.$$

In this construction, we start with the free A-module F with basis da, $a \in A$, and take its quotient by the submodule generated by the elements

$$d(ab) - ad(b) - bd(a) \qquad (a, b \in A)$$

and

$$d(ra + sb) - rd(a) - sd(b) \qquad (r, s \in R \text{ and } a, b \in A).$$

The map

$$d : A \longrightarrow \Omega_{A/R}, \qquad a \longmapsto d(a) = da,$$

is called the *universal derivation.* One checks immediately that it is indeed an R-derivation. The elements of $\Omega_{A/R}$ are called *algebraic differential forms.*

Lemma 18.3: the universal property

Let R be a commutative ring and let A be a commutative R-algebra. The module of Kähler differentials $\Omega_{A/R}$ has the following universal property. For every A-module M and every R-derivation

$$\delta : A \longrightarrow M,$$

there is a unique A-linear map

$$\epsilon : \Omega_{A/R} \longrightarrow M$$

satisfying $\epsilon \circ d = \delta$.

Proof This proof was not presented in the lecture.

For every da, $a \in A$, we must have $\epsilon(da) = \delta(a)$. Since the elements da generate $\Omega_{A/R}$ as an A-module, there can be at most one such homomorphism.

Let F be the free module with basis da, $a \in A$. The assignment

$$\tilde{\epsilon}(da) = \delta(a)$$

by the theorem on specifying a homomorphism on a basis, determines an A-module homomorphism

$$\tilde{\epsilon} : F \longrightarrow M.$$

We have $\Omega_{A/R} = F/U$, where U is the submodule generated by the elements expressing the Leibniz rule and linearity. Since δ is a derivation, $\tilde{\epsilon}$ maps U to 0. The homomorphism theorem therefore gives a unique A-linear map

$$\epsilon : \Omega_{A/R} \cong F/U \longrightarrow M$$

with

$$\epsilon(da) = \tilde{\epsilon}(da) = \delta(a).$$

Equivalently, this statement gives a natural A-module isomorphism

$$\mathrm{Der}_R(A, M) \cong \mathrm{Hom}_A(\Omega_{A/R}, M).$$

In particular,

$$\mathrm{Der}_R(A, A) \cong \mathrm{Hom}_A(\Omega_{A/R}, A) = \Omega^*_{A/R},$$

where the right-hand side is the dual module.

Lemma 18.4: elementary properties

Let R be a commutative ring, let A be a commutative R-algebra, and let $\Omega_{A/R}$ be the module of Kähler differentials. The following properties hold.

1. $dr = 0$ for all $r \in R$.
2. $\Omega_{A/R}$ can be described as the quotient of the free A-module with basis da, $a \in A$, by the submodule generated by the additivity relations $d(a + b) - da - db$, the Leibniz relations, and dr, $r \in R$.
3. If $A = R[x_1, \dots, x_n]$, then dx_i, $i = 1, \dots, n$, form an A-module generating system for $\Omega_{A/R}$.

4. Let

$$A = R[x_1, \dots, x_n] = R[X_1, \dots, X_n]/\mathfrak{a}.$$

For a polynomial $F \in R[X_1, \dots, X_n]$ and the associated element $f = F(x_1, \dots, x_n) \in A$, the following relation holds in $\Omega_{A/R}$:

$$df = \frac{\partial F}{\partial x_1}(x_1, \dots, x_n)dx_1 + \dots + \frac{\partial F}{\partial x_n}(x_1, \dots, x_n)dx_n,$$

where $\partial F/\partial x_i$ denotes the ith partial derivation.

5. For a commutative diagram

$$\begin{array}{ccc} R & \longrightarrow & S \\ \downarrow & & \downarrow \\ A & \xrightarrow{\varphi} & B, \end{array}$$

whose arrows are ring homomorphisms, there is a unique A-linear map

$$\Omega_{A/R} \longrightarrow \Omega_{B/S}, \qquad da \longmapsto d\varphi(a).$$

Editorial note — a missing family of relations. The source lists only the Leibniz relations and dr in part (2), but its proof immediately uses $d(ra + sb) = d(ra) + d(sb)$. Additivity is not supplied by those listed relations. This edition includes the additivity relations, as required by Definition 18.2 and by the proof.

Proof This proof was not presented in the lecture.

1. Let $r \in R$. By R-linearity, $d(r1) = rd(1)$. By the product rule,

$$d(1) = d(1 \cdot 1) = 1d(1) + 1d(1),$$

so subtraction gives $d(1) = 0$.

2. We show that the submodule V in question equals the submodule U generated by all Leibniz and linearity relations. By part (1), the inclusion $V \subseteq U$ is clear. For $a, b \in A$ and $r, s \in R$, modulo V we have

$$\begin{aligned} d(ra + sb) &= d(ra) + d(sb) \\ &= rda + adr + sdb + bds \\ &= rda + sdb, \end{aligned}$$

so the linearity relations also belong to V.

3. This follows from linearity and the Leibniz rule.

4. Both sides are R-linear, so it suffices to prove the assertion for monomials. For monomials, it follows by induction on total degree.

5. Since B is an A-algebra via $\varphi : A \to B$, the module $\Omega_{B/S}$ is also an A-module. The composite

$$A \xrightarrow{\varphi} B \xrightarrow{d} \Omega_{B/S}$$

is an R-derivation, as a direct calculation shows. By the universal property of $\Omega_{A/R}$, there is a unique A-linear map

$$\widetilde{\varphi} : \Omega_{A/R} \longrightarrow \Omega_{B/S}$$

with $d\varphi(a) = \widetilde{\varphi}(da)$.

Lemma 18.5: differentials of a polynomial ring

Let R be a commutative ring and let

$$A = R[X_1, \dots, X_n]$$

be the polynomial ring in n variables over R. Then the module of Kähler differentials is the free A-module with basis

$$dX_1, dX_2, \dots, dX_n.$$

With respect to this basis, the universal derivation is given by

$$\begin{aligned} A &\longrightarrow AdX_1 \oplus \cdots \oplus AdX_n, \\ F &\longmapsto dF = \frac{\partial F}{\partial X_1} dX_1 + \cdots + \frac{\partial F}{\partial X_n} dX_n. \end{aligned}$$

Proof This proof was not presented in the lecture.

Let G be the free A-module generated by the symbols dX_i. The map

$$\varphi : G \longrightarrow \Omega_{A/R}$$

sending the basis element dX_i to the differential dX_i is surjective by Lemma 18.4(3). The ith partial derivative

$$\frac{\partial}{\partial X_i} : A \longrightarrow A, \qquad F \longmapsto \frac{\partial F}{\partial X_i},$$

is an R-derivation. The universal property of the module of differential forms therefore gives an A-linear map

$$p_i : \Omega_{A/R} \longrightarrow A$$

with $p_i \circ d = \partial/\partial X_i$. Here $p_i(dX_i) = 1$ and $p_i(dX_j) = 0$ for $j \neq i$. Together these maps give an A-linear map

$$p = p_1 \times \cdots \times p_n : \Omega_{A/R} \longrightarrow A^n \cong G$$

satisfying $p \circ \varphi = \mathrm{Id}_G$. Thus φ is also injective.

In general, the module of Kähler differentials is not free.

Lemma 18.6: differentials and localisation

Let R be a commutative ring, let A be a commutative R-algebra, and let $S \subseteq A$ be a multiplicative system. Then

$$\Omega_{A_S/R} \cong (\Omega_{A/R})_S.$$

Proof See Exercise 18.19.

Lemma 18.7: the sequence of relative differentials

Let R be a commutative ring, let A and B be commutative R-algebras, and let

$$\varphi : A \longrightarrow B$$

be an R-algebra homomorphism. Then the sequence of B-modules

$$\Omega_{A/R} \otimes_A B \longrightarrow \Omega_{B/R} \longrightarrow \Omega_{B/A} \longrightarrow 0$$

is exact. Here $da \otimes b$ maps to $bd\varphi(a)$, while db in $\Omega_{B/R}$ maps to db in $\Omega_{B/A}$.

Proof This proof was not presented in the lecture.

Surjectivity on the right is clear. For exactness at the second position, use the description in Lemma 18.4(2). The modules $\Omega_{B/A}$ and $\Omega_{B/R}$ have the same generating system and the same Leibniz relations. The module $\Omega_{B/A}$ is obtained from $\Omega_{B/R}$ precisely by killing the B-submodule generated by da, $a \in A$, making it 0. This submodule is exactly the image of the map on the left.

Kähler differentials and the Jacobian matrix

Lemma 18.8: the conormal sequence

Let R be a commutative ring, let A be a commutative R-algebra, and let $I \subseteq A$ be an ideal with quotient ring $B = A/I$. Then the sequence of B-modules

$$I/I^2 \longrightarrow \Omega_{A/R} \otimes_A B \longrightarrow \Omega_{B/R} \longrightarrow 0$$

is exact. Here $a \in I$ maps to $da \otimes 1$, while $da \otimes b$ maps to bda.

Proof This proof was not presented in the lecture.

The R-linear map

$$A \longrightarrow \Omega_{A/R}, \qquad a \longmapsto da,$$

can be restricted to the ideal $I \subseteq A$. Tensoring with A/I and using Proposition 16.9 (Invariant Theory (Osnabrück 2025–2026)), part (2), gives the A/I-linear map

$$I/I^2 \cong I \otimes_A A/I \longrightarrow \Omega_{A/R} \otimes_A A/I.$$

Surjectivity of the map on the right is clear, since the B-module $\Omega_{B/R}$ is generated by db, $b \in B$, and these elements come from da, $a \in A$. An element $a \in I$ maps to da and then to 0 in $\Omega_{B/R}$, since a itself becomes 0 in B.

Now suppose that

$$\omega \in \Omega_{A/R} \otimes_A B$$

maps to 0 in $\Omega_{B/R}$. We can write

$$\omega = \sum_{i=1}^{n} da_i \otimes b_i = \sum_{i=1}^{n} da_i \otimes \bar{c}_i$$

with $a_i, c_i \in A$. Since this element maps to 0 in $\Omega_{B/R}$, the free B-module generated by the symbols db, $b \in B$, satisfies the relation

$$\sum_{i=1}^{n} c_i \, da_i = \sum_{j=1}^{m} h_j \omega_j,$$

where $h_j \in B$ and the ω_j generate the relations for the module of Kähler differentials, namely relations of the form

$$d(fg) = f \, dg + g \, df$$

for $f, g \in B$, or

$$d(rf + sg) = r \, df + s \, dg$$

for $r, s \in R$ and $f, g \in B$. This free B-module is obtained from the free A-module generated by da, $a \in A$, by making the coefficients in I and all dI equal to 0. Thus, in this free A-module,

$$\sum_{i=1}^{n} c_i \, da_i - \sum_{j=1}^{m} h_j \omega_j = \sum_{k=1}^{\ell} m_k \, dn_k + du$$

with $m_k \in I$, $n_k \in B$, and $u \in I$. In $\Omega_{A/R} \otimes_A B$, the term $\sum_{k=1}^{\ell} m_k \, dn_k$ becomes 0 after tensoring. Hence there we indeed have

$$\sum_{i=1}^{n} c_i \, da_i = du.$$

Editorial note — gaps in the source proof. The source's final tensor product is printed over R; it must be over A, as in the statement, and is corrected above. There are also gaps that this symbol change alone does not repair: the restriction of the R-linear derivation cannot simply be tensored as an A-linear map; the free-module discussion must identify symbols with the same image in B, not merely kill dI; and elements of B used in the free A-module require lifts to A.

A precise editorial justification is as follows. Put $E = \Omega_{A/R} \otimes_A B$. The map $I \to E$, $i \mapsto di \otimes 1$, is A-linear by the Leibniz rule, since the terms with coefficient in I vanish in E; it vanishes on I^2, giving $I/I^2 \to E$. Let N be its image. The formula $\bar{a} \mapsto da \otimes 1$ defines an R-derivation $B \to E/N$, independent of the chosen lift. By the universal property, it induces an inverse to the natural map $E/N \to \Omega_{B/R}$. Thus $E/N \cong \Omega_{B/R}$, proving the stated exactness. This paragraph supplements, rather than silently replaces, the source argument.

Corollary 18.9: differentials of a quotient algebra

Let R be a commutative ring and let A be a finitely generated commutative R-algebra, presented as

$$A = R[X_1, \dots, X_n]/(F_1, \dots, F_k).$$

Then

$$\Omega_{A/R} = \bigoplus_{i=1}^{n} A dX_i / (dF_1, \dots, dF_k).$$

Proof This proof was not presented in the lecture.

This follows from Lemma 18.5 and Lemma 18.8.

Remark 18.10: presentation as the cokernel of the Jacobian matrix

Let R be a commutative ring and let A be a finitely generated commutative R-algebra, presented as

$$A = R[X_1, \dots, X_n]/(F_1, \dots, F_k).$$

By Lemma 18.4(4),

$$dF_j = \frac{\partial F_j}{\partial X_1} dX_1 + \dots + \frac{\partial F_j}{\partial X_n} dX_n,$$

and by Corollary 18.9 there is an exact sequence

$$A^k \xrightarrow{M} A^n \longrightarrow \Omega_{A/R} \longrightarrow 0,$$

where

$$M = \begin{pmatrix} \frac{\partial F_1}{\partial X_1} & \dots & \frac{\partial F_k}{\partial X_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial F_1}{\partial X_n} & \dots & \frac{\partial F_k}{\partial X_n} \end{pmatrix}$$

is the transpose of the Jacobian matrix, without evaluation at a point. The standard vectors e_j map to dX_j, and the column vectors

$$\begin{pmatrix} \frac{\partial F_j}{\partial X_1} \\ \vdots \\ \frac{\partial F_j}{\partial X_n} \end{pmatrix},$$

representing the zero elements dF_j are the images of the map defined by the matrix.

Smoothness and regularity

Definition 18.11: smooth points

Let K be an algebraically closed field and let

$$F_1, \dots, F_s \in K[X_1, \dots, X_n]$$

be polynomials with associated affine algebraic set

$$Y = V(F_1, \dots, F_s) \subseteq \mathbb{A}^n_K.$$

Let $P \in Y$ be a point such that Y has dimension d at P. The point P is called a *smooth point* of Y if the rank of the matrix

$$\left(\frac{\partial F_i}{\partial X_j}\right)_{i,j}$$

at P is at least $n - d$. Otherwise, the point is called *singular*.

Editorial note — equations versus the reduced algebraic set. This criterion concerns the scheme presented by the given equations. If Y denotes the reduced algebraic set, the equations must generate its vanishing ideal; the source does not state this qualification. For instance, X^2 and X have the same zero set but different Jacobian ranks at 0. The same distinction applies to the local ring in Theorem 18.16.

For a K-algebra

$$A = K[X_1, \dots, X_n]/(f_1, \dots, f_m)$$

and a point

$$P = (a_1, \dots, a_n) \in V = V(f_1, \dots, f_m)$$

with associated maximal ideal $\mathfrak{m}_P \subseteq A$ and localisation

$$R = A_{\mathfrak{m}_P} = \mathcal{O}_{V,P},$$

we have

$$\Omega_{R/K} = \Omega_{A/K} \otimes_A R.$$

The tensor product

$$\Omega_{R/K} \otimes_R K = \Omega_{A/K} \otimes_A K$$

associated with evaluation in the residue field

$$A \longrightarrow A_{\mathfrak{m}_P} \longrightarrow A_{\mathfrak{m}_P}/\mathfrak{m}_P A_{\mathfrak{m}_P} = K$$

plays a special role. There is a direct connection with the dual of the extrinsic tangent space of V at P. In other words, $\Omega_{R/K} \otimes_R K$ is naturally the cotangent space at P.

Lemma 18.12: the extrinsic cotangent space

Let K be a field, let

$$A = K[X_1, \dots, X_n]/(f_1, \dots, f_m)$$

be a finitely generated K-algebra, and let

$$P = (a_1, \dots, a_n) \in V = V(f_1, \dots, f_m)$$

be a point of the associated zero locus, with maximal ideal $\mathfrak{m}_P \subseteq A$ and localisation

$$R = A_{\mathfrak{m}_P}.$$

Then the tangent space to V at P is canonically the dual vector space of $\Omega_{R/K} \otimes_R K$.

Editorial note — the maximal-ideal subscript. The source introduces the maximal ideal $\mathfrak{m}_P$ but then writes $A_{\mathfrak{m}}$ in the residue-field display and in this lemma, without defining $\mathfrak{m}$. This edition consistently uses the already defined $\mathfrak{m}_P$.

Proof This proof was not presented in the lecture.

By Remark 18.10, there is an exact sequence

$$A^m \xrightarrow{M} A^n \longrightarrow \Omega_{A/K} \longrightarrow 0,$$

where M is the transpose of the Jacobian matrix of the f_i. Tensoring with the residue field K gives an exact sequence of finite-dimensional K-vector spaces

$$K^m \xrightarrow{M(P)} K^n \longrightarrow \Omega_{A/K} \otimes_A K \longrightarrow 0.$$

Its dual sequence is

$$0 \longrightarrow (\Omega_{A/K} \otimes_A K)^* \longrightarrow K^n \xrightarrow{\mathrm{Jac}(P)} K^m,$$

and is also exact. By the definition of the tangent space, the kernel of the Jacobian matrix at P is the tangent space to V at P.

> **Editorial note — unresolved source reference.** The source gives `Definition .` at this point. This edition does not invent a reference number and instead names the definition of the tangent space used in the argument.

Definition 18.13: regular local rings

A Noetherian local ring $(R, \mathfrak{m})$ of dimension n is called *regular* if there are n elements

$$f_1, \dots, f_n \in \mathfrak{m}$$

generating the maximal ideal $\mathfrak{m}$.

Lemma 18.14: the maximal ideal and the cotangent space

Let K be a field and let R be a local commutative K-algebra, such that the composite map

$$K \longrightarrow R \longrightarrow R/\mathfrak{m}$$

is an isomorphism. Then the map

$$\mathfrak{m}/\mathfrak{m}^2 \longrightarrow \Omega_{R/K} \otimes_R R/\mathfrak{m}, \qquad [f] \longmapsto df \otimes 1,$$

is an $R/\mathfrak{m}$-module isomorphism.

Proof This proof was not presented in the lecture.

By Lemma 18.8, there is an exact sequence of $R/\mathfrak{m}$-module homomorphisms

$$\mathfrak{m}/\mathfrak{m}^2 \longrightarrow \Omega_{R/K} \otimes_R R/\mathfrak{m} \longrightarrow \Omega_{(R/\mathfrak{m})/K} \longrightarrow 0.$$

By assumption, $R/\mathfrak{m} = K$, hence $\Omega_{(R/\mathfrak{m})/K} = 0$. Thus the stated map is surjective. To prove injectivity, consider its $R/\mathfrak{m}$-dual map, namely

$$\begin{aligned} &\mathrm{Hom}_{R/\mathfrak{m}}(\Omega_{R/K} \otimes_R R/\mathfrak{m}, R/\mathfrak{m}) \\ &\qquad \longrightarrow \mathrm{Hom}_{R/\mathfrak{m}}(\mathfrak{m}/\mathfrak{m}^2, R/\mathfrak{m}), \qquad \varphi \longmapsto \varphi \circ d, \end{aligned}$$

and show that this is surjective, since we are dealing with vector spaces.

By Lemma 32.9 (Commutative Algebra) and Lemma 18.3, the homomorphism module on the left is isomorphic to

$$\operatorname{Hom}_R(\Omega_{R/K}, R/\mathfrak{m}) \cong \operatorname{Der}_K(R, R/\mathfrak{m}).$$

The composite assigns to a K-derivation $\delta : R \to R/\mathfrak{m}$ the map

$$\mathfrak{m}/\mathfrak{m}^2 \longrightarrow R/\mathfrak{m}, \qquad [f] \longmapsto \delta(f).$$

Now let

$$\mathfrak{m}/\mathfrak{m}^2 \longrightarrow R/\mathfrak{m}, \qquad [f] \longmapsto \epsilon(f)$$

be an $R/\mathfrak{m}$-module homomorphism. We must show that it comes from a derivation. For this, consider the map

$$\delta : R \longrightarrow R/\mathfrak{m}, \qquad f \longmapsto \epsilon(f - \bar{f}),$$

where $\bar{f}$ is the value of f in the residue field $R/\mathfrak{m}$, regarded again as an element of R through the identification $K = R/\mathfrak{m}$. Thus $f - \bar{f} \in \mathfrak{m}$, and the map is well-defined. A direct verification, similar to the proof of Theorem 23.2 (Algebraic Curves (Osnabrück 2025–2026)), shows that it is a derivation. This derivation maps to ϵ.

Without the assumption that the natural map from the base field to the residue field is an isomorphism, this assertion does not hold; see Exercise 18.23.

Remark 18.15: the tangent space and the maximal ideal

In the situation of Lemma 18.12, we can directly relate the extrinsic tangent space, given as the kernel of the Jacobian matrix, to the dual of $\mathfrak{m}_P/\mathfrak{m}_P^2$. Let

$$v = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \in T_P V = \ker(\operatorname{Jac}(f_1, \dots, f_m)_P).$$

This vector defines a map

$$\begin{aligned} \mathfrak{m}_P &\longrightarrow K, \\ g &\longmapsto (dg)_P \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = v_1 \partial_1 g(P) + \dots + v_n \partial_n g(P). \end{aligned}$$

In analytic language, a function g is sent to the value at P of its directional derivative in the direction v. The kernel condition ensures that functions in the ideal $(f_1, \dots, f_m)$ map to 0, so the map is well-defined on the maximal ideal of the quotient ring. By the product rule, $\mathfrak{m}_P^2$ also maps to 0. Thus we obtain a K-linear map

$$\mathfrak{m}_P/\mathfrak{m}_P^2 \longrightarrow K.$$

Theorem 18.16: smooth points and regular local rings

Let K be an algebraically closed field and let

$$P \in V(\mathfrak{a}) \subseteq \mathbb{A}_K^n$$

be a point of the affine algebraic set defined by the ideal $\mathfrak{a} = (f_1, \dots, f_m)$, with local ring

$$R = (K[X_1, \dots, X_n]/\mathfrak{a})_{\mathfrak{m}_P}.$$

Then P is smooth if and only if R is regular.

Proof This proof was not presented in the lecture.

Without loss of generality, let P be the origin. The corresponding maximal ideal in the polynomial ring is

$$\mathfrak{n} = (X_1, \dots, X_n),$$

the associated maximal ideal in $K[X_1, \dots, X_n]/\mathfrak{a}$ is

$$\mathfrak{r} = \mathfrak{n}/\mathfrak{a},$$

and the associated maximal ideal in R is

$$\mathfrak{m} = \mathfrak{m}_P = \mathfrak{r}\big(K[X_1, \dots, X_n]/\mathfrak{a}\big)_{\mathfrak{r}}.$$

Consider the K-linear map

$$K[X_1, \dots, X_n] \longrightarrow K^n, \qquad g \longmapsto ((\partial_1 g)(P), \dots, (\partial_n g)(P)).$$

The variables X_i map to the standard vectors e_i, so this map is surjective. An element

$$g = c_0 + c_1 X_1 + \cdots + c_n X_n + \text{higher-degree terms}$$

maps to $(c_1, \dots, c_n)$. A homogeneous element

$$g \in \mathfrak{n}^2 = K[X_1, \dots, X_n]_{\geq 2}$$

has degree at least 2 and therefore maps to 0: partial differentiation reduces the degree by 1, leaving an element of positive degree, which becomes 0 on substituting the origin. This induces a K-linear map

$$\mathfrak{n}/\mathfrak{n}^2 \longrightarrow K^n$$

which is bijective because the two spaces have the same vector space dimension.

By Lemma 22.4 (Algebraic Curves (Osnabrück 2025–2026)), we have

$$\mathfrak{r}/\mathfrak{r}^2 \cong \mathfrak{m}/\mathfrak{m}^2.$$

Under the surjective map

$$\mathfrak{n} \longrightarrow \mathfrak{r} \longrightarrow \mathfrak{r}/\mathfrak{r}^2,$$

both $\mathfrak{n}^2$ and $\mathfrak{a}$ map to 0, and its kernel is exactly $\mathfrak{n}^2 + \mathfrak{a}$. There is therefore a K-linear bijection

$$\mathfrak{n}/(\mathfrak{n}^2 + \mathfrak{a}) \longrightarrow \mathfrak{r}/\mathfrak{r}^2.$$

Consider the maps

$$K^m \xrightarrow{\text{Jac}} K^n \cong \mathfrak{n}/\mathfrak{n}^2 \longrightarrow \mathfrak{n}/(\mathfrak{n}^2 + \mathfrak{a}).$$

An element $[g] \in \mathfrak{n}/\mathfrak{n}^2$ maps to 0 on the right exactly when the linear part of g belongs to $\mathfrak{n}^2 + \mathfrak{a}$. This means that, modulo $\mathfrak{n}^2$, there is an equation

$$g = \sum_{i=1}^{m} h_i f_i.$$

Only the constant terms of the h_i matter, so this is equivalent to the linear equation

$$\begin{pmatrix} (\partial_1 g)(P) \\ \vdots \\ (\partial_n g)(P) \end{pmatrix} = \sum_{i=1}^{m} h_i \begin{pmatrix} (\partial_1 f_i)(P) \\ \vdots \\ (\partial_n f_i)(P) \end{pmatrix}.$$

This holds exactly when the vector on the left lies in the image of the Jacobian matrix. Thus the image of the Jacobian matrix equals the kernel of the surjective map on the right. The dimension formula now gives

$$\begin{aligned} n &= \operatorname{rank}(\mathrm{Jac}) + \dim_K(\mathfrak{n}/(\mathfrak{n}^2 + \mathfrak{a})) \\ &= \operatorname{rank}(\mathrm{Jac}) + \dim_K(\mathfrak{m}/\mathfrak{m}^2). \end{aligned}$$

Let d be the dimension of $V(\mathfrak{a})$ at P, equal to the dimension of the local ring R. By definition, P is nonsingular exactly when

$$n = \operatorname{rank}(\mathrm{Jac}) + d.$$

Thus this condition is equivalent to

$$\dim_K(\mathfrak{m}/\mathfrak{m}^2) = d,$$

which is the definition of a regular ring.

Theorem 18.17: regularity and freeness of the module of differentials

Let K be a perfect field and let $(R, \mathfrak{m})$ be a local ring obtained by localising a finitely generated K-algebra. Suppose that the natural map $K \to R/\mathfrak{m}$ is an isomorphism. Then R is regular if and only if the module of Kähler differentials $\Omega_{R/K}$ is free and its rank equals the dimension of the ring.

> **Editorial note — hypotheses used in the proof.** The source says "a localisation" and that the residue field "is isomorphic" to K. Locality and the natural residue-field identification are made explicit here: these are the hypotheses needed for the stated use of Lemma 18.14 and Nakayama's lemma.

Proof This proof was not presented in the lecture.

We use the natural $R/\mathfrak{m}$-isomorphism of Lemma 18.14,

$$\mathfrak{m}/\mathfrak{m}^2 \longrightarrow \Omega_{R/K} \otimes_R R/\mathfrak{m}, \qquad [f] \longmapsto df \otimes 1.$$

If $\Omega_{R/K}$ is a free R-module whose rank equals the dimension d, the same holds for the $R/\mathfrak{m}$-module $\Omega_{R/K} \otimes_R R/\mathfrak{m}$. In particular, $\mathfrak{m}/\mathfrak{m}^2$ is a $R/\mathfrak{m}$-vector space of dimension d. By definition, this means that R is regular.

Conversely, regularity implies that $\mathfrak{m}/\mathfrak{m}^2$, and hence $\Omega_{R/K} \otimes_R R/\mathfrak{m}$, is a vector space of dimension d. By Nakayama's lemma, $\Omega_{R/K}$ is generated as an R-module by d elements. By Theorem 21.5 (Singularity Theory (Osnabrück 2019)), the ring R is an integral domain; denote its field of fractions by $Q(R)$. The transcendence degree of $Q(R)$ over K equals the dimension of R by Theorem 19.7 (Singularity Theory (Osnabrück 2019)). Since the module of Kähler differentials is compatible with localisation,

$$\Omega_{R/K} \otimes_R Q(R) = \Omega_{Q(R)/K}.$$

Since K is perfect, the field extension $K \subseteq Q(R)$ is separably generated, although not finite. Thus $\Omega_{Q(R)/K}$ is a free $Q(R)$-module whose rank equals the transcendence degree.

In summary, the R-module $\Omega_{R/K}$ is generated by d elements $\omega_1, \ldots, \omega_d$, and its tensor product with $Q(R)$ is a $Q(R)$-vector space of dimension d. Since these elements are linearly independent over $Q(R)$, they are also linearly independent over R. Hence they form a basis, and $\Omega_{R/K}$ is free of rank d.

Editorial note — module label and unresolved reference. In the summary sentence the source names $\Omega_{Q(R)/K}$ as the R-module generated by the d elements. The preceding application of Nakayama concerns $\Omega_{R/K}$, which is the module written here. The source also displays `Fakt *****` for the assertion about the separably generated extension; no unavailable reference number is invented.

Without the assumption that the base field is perfect, this assertion is false; see Exercise 18.24.

Editorial note — the cited exercise. Exercise 18.24 has free differentials of rank 1 over a field of dimension 0, contrary to its printed nonfreeness claim. Moreover, its structural map from K to the residue field is not an isomorphism. It therefore does not establish the source's preceding claim after removing only perfectness while retaining the other hypotheses. See the explicit diagnosis accompanying that exercise.

Corollary 18.18: differentials on smooth varieties

Let $V \subseteq \mathbb{A}^n_K$ be a connected smooth variety over a perfect field K, and let R be the affine coordinate ring of V. Then the module of Kähler differentials $\Omega_{R/K}$ is locally free of constant rank $\dim(R)$ and, in particular, is a projective module.

Proof This proof was not presented in the lecture.

This follows from Theorem 18.16, Theorem 18.17, Lemma 18.6, and Lemma 16.5.

Example 18.19: the two-dimensional sphere

Consider the real sphere

$$S^2 = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\} \subseteq \mathbb{R}^3$$

with affine coordinate ring

$$R = \mathbb{R}[X, Y, Z]/(X^2 + Y^2 + Z^2 - 1).$$

By Corollary 18.9, the R-module of Kähler differentials is

$$\Omega_{R/\mathbb{R}} = RdX \oplus RdY \oplus RdZ/(XdX + YdY + ZdZ).$$

A direct check shows that this real sphere is smooth. By Theorem 18.17, $\Omega_{R/\mathbb{R}}$ is therefore locally free of constant rank 2. This can also be deduced directly from the presentation above; see Exercise 18.25. However, $\Omega_{R/\mathbb{R}}$ is not free. This is an algebraic version of the hairy ball theorem: the hairs on a sphere cannot be combed smoothly so that they all lie tangent to the sphere without forming a whorl.

For a polynomial map

$$f : \mathbb{A}^n_K \longrightarrow \mathbb{A}^m_K$$

with zero locus

$$V = V(f_1, \dots, f_m) \subseteq \mathbb{A}^n_K,$$

the tangent space at a point $P \in V$ is

$$\begin{aligned} T_P V &:= \ker(\mathrm{Jac}(f_1, \dots, f_m)_P) \\ &= \{v \in \mathbb{A}^n_K \mid \mathrm{Jac}(f_1, \dots, f_m)_P(v) = 0\}. \end{aligned}$$

If P is a regular point of the map and the implicit function theorem applies, this is a linear subspace whose dimension equals the manifold dimension of V. This construction is extrinsic: it depends on the embedding of V into affine space. We seek an intrinsic version of the tangent space depending only on V, or equivalently on its affine coordinate ring. For this purpose we introduce the module of Kähler differentials, which provides a dual version of the tangent space for every R-algebra A.

Worksheet 18: Kähler Differentials

Stars mark exactly three exercises with frozen public solutions: Exercises 18.6, 18.17, and 18.18. The other twenty-two exercises have negative candidate results; this edition does not invent new solutions.

Exercise 18.1

Let R be a commutative ring, let A be a commutative R-algebra, let M be an A-module, and let

$$D : A \longrightarrow M$$

be an R-derivation. Prove that

$$D(f^n) = nf^{n-1}D(f)$$

for every $f \in A$.

> **Editorial note — the module in Exercises 18.1–18.3.** The source calls M an R-module. The products by elements of A in the Leibniz rule require an A-module structure, as in Definition 18.1. This edition makes that required structure explicit in all three exercises.

Exercise 18.2

Let R be a commutative ring, let A be a commutative R-algebra, let M be an A-module, and let $D : A \to M$ be an R-derivation. Prove that

$$\begin{aligned} D(f_1 \cdots f_r) = {} & f_2 \cdots f_r D(f_1) + f_1 f_3 \cdots f_r D(f_2) \\ & + \cdots + f_1 \cdots f_{r-1} D(f_r) \end{aligned}$$

for $f_1, \ldots, f_r \in A$.

Exercise 18.3

Let R be a commutative ring, let A be a commutative R-algebra, let M be an A-module, and let $D : A \to M$ be an R-derivation. Let

$$x_1^{n_1} \cdots x_r^{n_r} \in A.$$

Prove that

$$\begin{aligned} D(x_1^{n_1} \cdots x_r^{n_r}) = {} & n_1 x_1^{n_1-1} x_2^{n_2} \cdots x_{r-1}^{n_{r-1}} x_r^{n_r} D(x_1) \\ & + \cdots + n_r x_1^{n_1} \cdots x_{r-1}^{n_{r-1}} x_r^{n_r-1} D(x_r). \end{aligned}$$

Exercise 18.4

Let A be a commutative R-algebra and let M be an A-module. Prove that the set of derivations from A to M becomes an A-module if $f\delta$ is defined by

$$(f\delta)(a) = f\delta(a).$$

Exercise 18.5

Let R be a commutative K-algebra, let $W \subseteq R$ be a multiplicative system, and let $D : R \to R$ be a K-derivation. Prove that the formula

$$D\left(\frac{f}{g}\right) := \frac{gD(f) - fD(g)}{g^2}$$

defines a derivation on the localisation R_W extending D.

Exercise 18.6 ★

Let A be a commutative R-algebra over a commutative ring R. For $f \in A$, write the R-linear map given by multiplication by f as

$$\mu_f : A \longrightarrow A, \qquad x \longmapsto fx,$$

and for two R-linear maps

$$\varphi_1, \varphi_2 : A \longrightarrow A$$

write

$$[\varphi_1, \varphi_2] = \varphi_1 \circ \varphi_2 - \varphi_2 \circ \varphi_1.$$

Let $\delta : A \to A$ be an R-derivation. Prove that for every $g \in A$, the map $[\delta, \mu_g]$ is a multiplication map.

Exercise 18.7

Let R be a commutative ring, let A be a commutative R-algebra, and let $\Omega_{A/R}$ be the module of Kähler differentials. Prove that the universal derivation

$$A \longrightarrow \Omega_{A/R}, \qquad f \longmapsto df,$$

is a derivation.

Exercise 18.8

Determine $\Omega_{\mathbb{C}/\mathbb{R}}$.

Exercise 18.9

Let $K \subseteq L$ be a finite separable field extension. Prove that

$$\Omega_{L/K} = 0.$$

Exercise 18.10

Determine $\Omega_{\mathbb{Z}[i]/\mathbb{Z}}$.

Exercise 18.11

Let R be a commutative ring and let

$$A = R[X_1, \dots, X_n]/(X_n - f(X_1, \dots, X_{n-1}))$$

with $f \in R[X_1, \dots, X_{n-1}]$, so that the zero locus is the graph of f. Prove in two different ways that $\Omega_{A/R}$ is a free A-module of rank $n-1$.

The following exercises concern tensor products of modules and algebras; see also the appendix.

Exercise 18.12

Calculate $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/(5)$.

Exercise 18.13

Calculate the tensor product

$$(\mathbb{Z}^3 \oplus (\mathbb{Z}/(2))^2 \oplus \mathbb{Z}/(3)) \otimes_{\mathbb{Z}} (\mathbb{Z}^2 \oplus \mathbb{Z}/(2) \oplus \mathbb{Z}/(4)).$$

Exercise 18.14

Let R be a commutative ring. Prove the R-module isomorphism

$$R^n \otimes_R R^m \cong R^{nm}.$$

Exercise 18.15

Let R be a commutative ring and let $\mathfrak{a}, \mathfrak{b} \subseteq R$ be ideals. Prove the R-algebra isomorphism

$$R/\mathfrak{a} \otimes_R R/\mathfrak{b} = R/(\mathfrak{a} + \mathfrak{b}).$$

Exercise 18.16

Let R be a commutative ring and let $S, T \subseteq R$ be multiplicative systems. Prove the R-algebra isomorphism

$$R_S \otimes_R R_T = R_{S \cdot T}.$$

Exercise 18.17 ★

Let M and N be commutative monoids and let R be a commutative ring. Prove the R-algebra isomorphism

$$R[M \times N] \cong R[M] \otimes_R R[N].$$

Exercise 18.18 ★

Let A be a commutative ring and let $S \subseteq A$ be a multiplicative system. Prove that

$$\Omega_{A_S/A} = 0.$$

Exercise 18.19

Let R be a commutative ring, let A be a commutative R-algebra, and let $S \subseteq A$ be a multiplicative system. Prove that

$$\Omega_{A_S/R} \cong (\Omega_{A/R})_S.$$

Exercise 18.20

Discuss Lemma 18.8 in the case where $R = K$ is a field of positive characteristic p, $A = K[X]$, and $I = (X^p)$.

Exercise 18.21

For

$$A = \mathbb{C}[X, Y, Z]/(X^2 + Y^3 + Z^5),$$

describe the module of Kähler differentials by generators and relations.

Exercise 18.22

Determine $\Omega_{\mathbb{C}/\mathbb{R}}$ using Corollary 18.9.

Exercise 18.23

Let p be a prime number. Consider the field extension given by

$$K = \mathbb{Z}/(p)(U) \subseteq \mathbb{Z}/(p)(Y) = L, \qquad U \longmapsto Y^p.$$

Prove that Lemma 18.14 does not hold in this situation.

Exercise 18.24

Let p be a prime number and let

$$K = \mathbb{Z}/(p)(U) \subseteq R = K[Y]/(Y^p - U).$$

Prove that $R \cong K(Y)$, that R is regular, and that the module of Kähler differentials $\Omega_{R/K}$ is not free.

> **Editorial note — inconsistent source conclusion.** The final claim is false for the displayed ring. If y is the residue class of Y, then $R = K(y) \cong \mathbb{F}_p(y)$ is a field with $U = y^p$, hence is regular of dimension 0. Corollary 18.9 gives $\Omega_{R/K} = R\,dy$, since $d(Y^p - U) = 0$ relative to K. It is therefore free of rank 1, not nonfree. Here $K(y)$ means the field generated by the algebraic element y, not a rational function field in an independent variable over K. The source exercise is retained with this diagnosis; no replacement exercise or public source solution is asserted. Its rank differs from $\dim R$, and the natural map $K \to R/\mathfrak{m} = R$ is not an isomorphism, so it also fails the residue-field hypothesis used in Theorem 18.17.

Exercise 18.25

Let

$$R = \mathbb{R}[X, Y, Z]/(X^2 + Y^2 + Z^2 - 1).$$

Prove that the R-module of Kähler differentials

$$\Omega_{R/\mathbb{R}} = RdX \oplus RdY \oplus RdZ/(XdX + YdY + ZdZ)$$

becomes free when restricted to the open sets $D(X), D(Y), D(Z)$, so that, for example, $(\Omega_{R/\mathbb{R}})_X = \Omega_{R_X/\mathbb{R}}$, and deduce that $\Omega_{R/\mathbb{R}}$ is locally free.

Public Solutions and Coverage of Worksheet 18

At the frozen revision boundary, the source provides exactly three public solutions among the 25 exercises: the solutions to Exercises 18.6, 18.17, and 18.18. The exercise map and candidate evidence record negative results for Exercises 18.1–18.5, 18.7–18.16, and 18.19–18.25. Missing public solution pages are not replaced by invented solutions.

Solution to Exercise 18.6

We have

$$\begin{aligned}[\delta, \mu_g](x) &= (\delta \circ \mu_g - \mu_g \circ \delta)(x) \\ &= \delta(gx) - g\delta(x) \\ &= g\delta(x) + x\delta(g) - g\delta(x) \\ &= x\delta(g).\end{aligned}$$

Thus $[\delta, \mu_g]$ is multiplication by $\delta(g)$.

Solution to Exercise 18.17

The monoid ring $R[M]$ has R-basis

$$T^m, \qquad m \in M.$$

The tensor product of free modules has as a basis all tensor products of elements of the two bases. Thus

$$T^{(m,n)}, \qquad (m,n) \in M \times N,$$

is a basis of $R[M \times N]$, while

$$T^m \otimes T^n, \qquad m \in M,\ n \in N,$$

is a basis of $R[M] \otimes R[N]$. Hence the assignment on bases

$$T^{(m,n)} \longleftrightarrow T^m \otimes T^n$$

directly gives an R-module isomorphism. Under this assignment,

$$T^{(0,0)} = 1$$

corresponds to

$$T^0 \otimes T^0 = 1 \otimes 1 = 1.$$

Moreover,

$$T^{(m_1,n_1)} \cdot T^{(m_2,n_2)} = T^{(m_1+m_2,n_1+n_2)}$$

corresponds to the element

$$\begin{aligned}T^{m_1} \otimes T^{n_1} \cdot T^{m_2} \otimes T^{n_2} &= (T^{m_1} \cdot T^{m_2}) \otimes (T^{n_1} \cdot T^{n_2}) \\ &= T^{m_1+m_2} \otimes T^{n_1+n_2}.\end{aligned}$$

Thus the assignment also respects multiplication and is an R-algebra isomorphism.

Solution to Exercise 18.18

For $a \in A$ and $s \in S$, we have

$$\begin{aligned} 0 = da &= d\left(\frac{a}{s} \cdot s\right) \\ &= \frac{a}{s}\, ds + s\, d\left(\frac{a}{s}\right) \\ &= s\, d\left(\frac{a}{s}\right). \end{aligned}$$

Since s is a unit in A_S, it follows that

$$d\left(\frac{a}{s}\right) = 0.$$

Frozen negative results

There is no public solution page at the frozen revision for Exercises 18.1, 18.2, 18.3, 18.4, 18.5, 18.7, 18.8, 18.9, 18.10, 18.11, 18.12, 18.13, 18.14, 18.15, 18.16, 18.19, 18.20, 18.21, 18.22, 18.23, 18.24, or 18.25. This records the outcome of checking candidates; it does not claim that mathematical solutions do not exist.

Lecture 19: Tangent Bundles

The sheaf of Kähler differentials on a scheme

Let X be a scheme over a base scheme S. We wish to define a sheaf version of the module of Kähler differentials.

Lemma 19.1: localisation of the affine sheaf of Kähler differentials

Let A be a commutative algebra over a commutative ring R. For every $f \in A$ we have

$$\left(\Gamma(D(f), \widetilde{\Omega_{A|R}}), \tilde{d}\right) = \left(\Omega_{A_f|R}, d\right),$$

and for every prime ideal $\mathfrak{p} \in \mathrm{Spek}(A)$ we have

$$\left((\widetilde{\Omega_{A|R}})_{\mathfrak{p}}, d_{\mathfrak{p}}\right) = \left(\Omega_{A_{\mathfrak{p}}|R}, d\right).$$

Proof This follows from Lemma 18.6 together with Lemma 14.5.

Definition 19.2: the sheaf of Kähler differentials

Let $p : X \to S$ be a scheme over a base scheme S. The *sheaf of Kähler differentials* $\Omega_{X|S}$ is the quasicoherent $\mathcal{O}_X$-module on X, together with a derivation over $p^{-1}\mathcal{O}_S$,

$$d : \mathcal{O}_X \longrightarrow \Omega_{X|S},$$

such that for every point $P \in X$,

$$((\Omega_{X|S})_P, d_P) = (\Omega_{\mathcal{O}_{X,P}|\mathcal{O}_{S,p(P)}}, d).$$

We must show that such an object exists and is unique. By quasicoherence, for every affine open subset $V = \mathrm{Spek}(R) \subseteq S$ and every affine open subset $U = \mathrm{Spek}(A) \subseteq X$ with $U \subseteq p^{-1}(V)$, the module on U must agree with $\widetilde{\Omega_{A|R}}$. In the affine case, Lemma 19.1 shows that $\widetilde{\Omega_{A|R}}$ is indeed the correct model.

If (Ω, d) and (Ω', d') are two models, the universal property, first on affine pieces and then in general, gives an $\mathcal{O}_X$-module homomorphism

$$\widetilde{\Omega_{A|R}} \longrightarrow \Omega'.$$

Since this is an isomorphism at every point, it is an isomorphism. Thus there can be only one such sheaf.

Given an affine cover

$$S = \bigcup_{j \in J} V_j$$

and a corresponding affine cover

$$X = \bigcup_{i \in I} U_i,$$

with $U_i \subseteq p^{-1}(V_j)$ for some $j = j(i)$, the sheaves $\Omega_{U_i|V_{j(i)}}$ can be glued together. This is because their restrictions to affine pieces $U \subseteq U_i \cap U_{i'}$ over $V \subseteq V_{j(i)} \cap V_{j(i')}$ are uniquely determined.

Definition 19.3: the tangent sheaf

Let $p : X \to S$ be a scheme over a base scheme S. The *tangent sheaf* $\mathcal{T}_{X,S}$ is the dual module

$$\mathcal{T}_{X,S} = \Omega^*_{X|S}.$$

Thus

$$\mathcal{T}_{X,S} = \Omega^*_{X|S} = \mathcal{H}om(\Omega_{X|S}, \mathcal{O}_X) = \mathcal{D}er_{p^{-1}\mathcal{O}_S}(\mathcal{O}_X, \mathcal{O}_X),$$

where the last equality follows from the universal property of Kähler differentials. Accordingly, the sheaf of Kähler differentials is also called the *cotangent sheaf.*

> **Editorial note — derivations relative to the base.** The source writes $\mathcal{D}er(\mathcal{O}_X, \mathcal{O}_X)$ without a base subscript. Since $\mathcal{T}_{X,S}$ is the tangent sheaf relative to S, the universal property identifies it with derivations over $p^{-1}\mathcal{O}_S$, as written here.

We now formulate, for a scheme over a base scheme, the general versions of the statements about Kähler differentials in the affine case from the previous lecture.

Lemma 19.4: the sequence of relative differentials

Let $\varphi : X \to Y$ be a scheme morphism over a base scheme S. Then the sequence of quasicoherent $\mathcal{O}_X$-modules

$$\varphi^*\Omega_{Y|S} \longrightarrow \Omega_{X|S} \longrightarrow \Omega_{X|Y} \longrightarrow 0$$

is exact.

Proof This follows from Lemma 18.7.

Lemma 19.5: the conormal sequence

Let X be a scheme over a base scheme S, and let $\mathcal{I} \subseteq \mathcal{O}_X$ be an ideal sheaf on X with associated closed subscheme $j : Y \to X$. Then the sequence of quasicoherent $\mathcal{O}_Y$-modules

$$\mathcal{I}/\mathcal{I}^2 \longrightarrow j^*\Omega_{X|S} \longrightarrow \Omega_{Y|S} \longrightarrow 0$$

is exact.

Proof This follows directly from Lemma 18.8.

Corollary 19.6: smoothness and local freeness of Kähler differentials

Let K be an algebraically closed field and let X be a connected scheme of finite type over K. Then X is smooth if and only if the module of Kähler differentials $\Omega_{X|K}$ is locally free of constant rank $\dim(X)$.

Proof This follows from Theorem 18.16 and Theorem 18.17.

Definition 19.7: the canonical sheaf

Let K be an algebraically closed field and let X be a connected smooth scheme of finite type over K, of dimension d. The sheaf

$$\omega_X := \det \Omega_{X|K} = \bigwedge^d \Omega_{X|K}$$

is called the *canonical sheaf* of X.

The tangent bundle on projective space

Theorem 19.8: the Euler sequence for Kähler differentials

Let

$$\mathbb{P}^n_R = \operatorname{Proj}(R[X_0, X_1, \dots, X_n])$$

be projective space over a commutative ring R. The $\mathcal{O}_{\mathbb{P}^n_R}$-module of Kähler differentials $\Omega_{\mathbb{P}^n_R|R}$ is described by the short exact sequence

$$0 \longrightarrow \Omega_{\mathbb{P}^n_R|R} \longrightarrow \mathcal{O}_{\mathbb{P}^n_R}(-1)^{\oplus(n+1)} \xrightarrow{X_0,\dots,X_n} \mathcal{O}_{\mathbb{P}^n_R} \longrightarrow 0,$$

together with the universal derivation which, on every open set $U \subseteq \mathbb{P}^n_R$, maps a function $f \in \Gamma(U, \mathcal{O}_{\mathbb{P}^n_R})$ to

$$df = \left(\frac{\partial f}{\partial X_0}, \dots, \frac{\partial f}{\partial X_n}\right).$$

Proof Denote the kernel sheaf on the left, which we wish to identify as the Kähler module, by

$$\mathcal{S} = \operatorname{Syz}(X_0, \dots, X_n).$$

The displayed map d, arising from the universal derivation on $n+1$-dimensional space, turns a function of degree 0 into one of degree -1, as can be checked directly for rational monomials. Thus there is an R-linear map

$$d : \Gamma(U, \mathcal{O}_{\mathbb{P}^n_R}) \longrightarrow \Gamma\big(U, \mathcal{O}_{\mathbb{P}^n_R}(-1)^{\oplus(n+1)}\big).$$

The Leibniz rule carries over because partial derivatives satisfy it. We must show that the image of d lies in the kernel of the last map. For a monomial X^ν of degree 0, we have

$$\sum_{j=0}^n X_j \frac{\partial X^\nu}{\partial X_j} = \sum_{j=0}^n \nu_j X^\nu = \left(\sum_{j=0}^n \nu_j\right) X^\nu = 0.$$

Now consider the situation on $D_+(X_0)$ and set $Y_j = X_j / X_0$. Using Example 12.10 and Example 15.5 gives

$$
\begin{aligned}
\Gamma(D_+(X_0),\mathcal{S}) &= \ker\left(\Gamma(D_+(X_0),\mathcal{O}_{\mathbb{P}^n_R}(-1)^{\oplus(n+1)}) \xrightarrow{X_0,X_1,\dots,X_n} \Gamma(D_+(X_0),\mathcal{O}_{\mathbb{P}^n_R})\right)\\
&= \ker\left((R[X_0,X_1,\dots,X_n]_{X_0})^{\oplus(n+1)}_{-1} \xrightarrow{X_0,X_1,\dots,X_n} (R[X_0,X_1,\dots,X_n]_{X_0})_0\right)\\
&= \ker\left(R[Y_1,\dots,Y_n]X_0^{-1}\oplus\dots\oplus R[Y_1,\dots,Y_n]X_0^{-1} \xrightarrow{X_0,X_0Y_1,\dots,X_0Y_n} R[Y_1,\dots,Y_n]\right)\\
&\cong R[Y_1,\dots,Y_n]\oplus\dots\oplus R[Y_1,\dots,Y_n],
\end{aligned}
$$

with n summands in the last line. Under this isomorphism, the tuple $(P_1,\dots,P_n)$ corresponds to the kernel tuple

$$
\left(-\sum_{i=1}^{n} P_iY_iX_0^{-1}, P_1X_0^{-1},\dots,P_nX_0^{-1}\right).
$$

Under the map d, the monomial

$$
Y^\mu = \left(\frac{X_1}{X_0}\right)^{\mu_1}\cdots\left(\frac{X_n}{X_0}\right)^{\mu_n} = X_0^{-\sum_{j=1}^n \mu_j}X_1^{\mu_1}\cdots X_n^{\mu_n} \in \Gamma(D_+(X_0),\mathcal{O}_{\mathbb{P}^n_R})
$$

maps to the element

$$
\begin{aligned}
\Bigg(&-\Big(\sum_{j=1}^n \mu_j\Big)X_0^{-\sum_{j=1}^n \mu_j - 1}X_1^{\mu_1}\cdots X_n^{\mu_n},\\
&\mu_1X_0^{-\sum_{j=1}^n \mu_j}X_1^{\mu_1-1}X_2^{\mu_2}\cdots X_n^{\mu_n},\dots,\mu_nX_0^{-\sum_{j=1}^n \mu_j}X_1^{\mu_1}\cdots X_n^{\mu_n-1}\Bigg).
\end{aligned}
$$

Under the identification above, namely omitting the first component and multiplying by X_0, this becomes the tuple of derivatives with respect to the variables Y_j. Thus, by Lemma 18.5, we obtain the universal derivation of the polynomial ring $R[Y_1,\dots,Y_n]$.

In particular, the module of Kähler differentials on projective space is locally free.

Corollary 19.9: the Euler sequence for the tangent sheaf

Let

$$
\mathbb{P}^n_R = \mathrm{Proj}(R[X_0,X_1,\dots,X_n])
$$

be projective space over a commutative ring R. The tangent sheaf on $\mathbb{P}^n_R$ is described by the short exact sequence

$$
0 \longrightarrow \mathcal{O}_{\mathbb{P}^n_R} \xrightarrow{X_0,\dots,X_n} \mathcal{O}_{\mathbb{P}^n_R}(1)^{\oplus(n+1)} \longrightarrow \mathcal{T}_{\mathbb{P}^n_R,R} \longrightarrow 0.
$$

At the right-hand end, the global element X_ie_j in the jth component maps to the global derivation

$$
X_i\frac{\partial}{\partial X_j}.
$$

Proof This follows directly from Theorem 19.8 by dualising. The additional assertion also follows from Theorem 19.8: under duality, the element X_ie_j in the jth component of $\mathcal{O}_{\mathbb{P}^n_R}(1)^{\oplus(n+1)}$ corresponds to the map

$$
X_i\circ p_j : \mathcal{O}_{\mathbb{P}^n_R}(-1)^{\oplus(n+1)} \longrightarrow \mathcal{O}_{\mathbb{P}^n_R},
$$

that is, projection onto the jth component followed by multiplication by X_i. Viewed as a linear form on the module of Kähler differential forms $\Omega_{\mathbb{P}^n_R|R} \subseteq \mathcal{O}_{\mathbb{P}^n_R}(-1)^{\oplus(n+1)}$, this corresponds to the linear form associated with the derivation

$$f \longmapsto X_i \frac{\partial f}{\partial X_j}.$$

Compared with other projective varieties, projective space has the special feature of possessing many global vector fields.

Corollary 19.10: the canonical sheaf of projective space

Let

$$\mathbb{P}^n_R = \operatorname{Proj}(R[X_0, X_1, \dots, X_n])$$

be projective space over a commutative ring R. Then its canonical sheaf is

$$\omega_{\mathbb{P}^n_R|R} = \det \Omega_{\mathbb{P}^n_R|R} \cong \mathcal{O}_{\mathbb{P}^n_R}(-n-1).$$

Proof This follows from Theorem 19.8, Theorem 16.11, and Corollary 16.12.

Thus the anticanonical sheaf, the dual of the canonical sheaf, on projective space equals $\mathcal{O}_{\mathbb{P}^n_R}(n+1)$ and has many global sections.

> **Editorial note — the base ring.** The source changes the subscript from the theorem's base ring R to an undefined K in this sentence. This edition keeps R.

Hypersurfaces in projective space

Theorem 19.11: characterisations of smooth projective hypersurfaces

Let K be an algebraically closed field and let

$$F \in K[X_0, X_1, \dots, X_n]$$

be a homogeneous polynomial of degree d. The following statements are equivalent.

1. The affine hypersurface

$$V(F) = \operatorname{Spek}(K[X_0, X_1, \dots, X_n]/(F)) \subseteq \mathbb{A}^{n+1}_K$$

   is smooth away from the origin.

2. The projective hypersurface

$$Y = V_+(F) = \operatorname{Proj}(K[X_0, X_1, \dots, X_n]/(F)) \subseteq \mathbb{P}^n_K$$

   is smooth.

3. For every variable X_i, the algebra

$$K[X_0, \dots, X_{i-1}, X_{i+1}, \dots, X_n]/(\widetilde{F}_i)$$

   is smooth, where

$$\widetilde{F}_i = \frac{F}{X_i^d}$$

   denotes the dehomogenisation of F with respect to X_i.

Editorial note — dehomogenisation and degree notation. The source prints F/X_i here, although a degree-d homogeneous polynomial gives the degree-zero element F/X_i^d on $D_+(X_i)$. It also changes the degree symbol from d to δ three times in the proof and uses R there without defining it. This edition uses F/X_i^d, keeps the stated degree d, and names the homogeneous coordinate ring below.

4. The module of Kähler differentials $\Omega_{Y|K}$ is locally free.
5. There is a short exact sequence of locally free sheaves on Y,

$$0 \longrightarrow \mathcal{O}_Y(-d) \longrightarrow j_Y^* \Omega_{\mathbb{P}^n_K|K} \longrightarrow \Omega_{Y|K} \longrightarrow 0.$$

Proof The equivalence of (2) and (3) is clear from Lemma 12.17 and the fact that smoothness is local. The equivalence of (2) and (4) follows from Corollary 19.6. Write

$$R = K[X_0, X_1, \dots, X_n]/(F).$$

The equivalence of (1) and (3) rests on the fact that, locally over $D_+(X_i)$, the cone map is given by

$$D(X_i) = \mathrm{Spek}(R_{X_i}) = \mathrm{Spek}((R_{X_i})_0[X_i, X_i^{-1}]) \longrightarrow D_+(X_i) = \mathrm{Spek}((R_{X_i})_0).$$

Thus the cone map is locally a punctured affine cylinder over the base.

For the implication from (4), or (2), to (5), Lemma 19.5 gives the exact sequence

$$\mathcal{I}/\mathcal{I}^2 \longrightarrow j_Y^* \Omega_{\mathbb{P}^n_K|K} \longrightarrow \Omega_{Y|K} \longrightarrow 0.$$

Here $\mathcal{I}$ is the principal ideal generated by F, and $\mathcal{I} \cong \mathcal{O}_{\mathbb{P}^n_K}(-d)$ via the map

$$\mathcal{O}_{\mathbb{P}^n_K} \longrightarrow \mathcal{O}_{\mathbb{P}^n_K}(d), \qquad 1 \longmapsto F.$$

Furthermore, the restriction of this ideal sheaf to Y is

$$\mathcal{O}_Y(-d) = \mathcal{I} \otimes \mathcal{O}_Y = \mathcal{I} \otimes \mathcal{O}_{\mathbb{P}^n_K}/\mathcal{I} = \mathcal{I}/\mathcal{I}^2.$$

As the restriction of an invertible sheaf, it is again invertible. Locally, the map on the left is given, as in Remark 18.10, by the Jacobian matrix of the dehomogenisation of F. By smoothness, this map is injective, even after passing to residue fields. The implication from (5) to (4) is immediate by restricting the assertion.

Corollary 19.12: the canonical sheaf of a smooth hypersurface

Let K be an algebraically closed field and let $F \in K[X_0, X_1, \dots, X_n]$ be a homogeneous polynomial of degree d such that the projective hypersurface $Y = V_+(F)$ is smooth. Then

$$\omega_Y \cong \mathcal{O}_Y(d - n - 1).$$

Proof Apply the short exact sequence of locally free sheaves on Y from Theorem 19.11,

$$0 \longrightarrow \mathcal{O}_Y(-d) \longrightarrow j_Y^* \Omega_{\mathbb{P}^n_K|K} \longrightarrow \Omega_{Y|K} \longrightarrow 0.$$

By Theorem 16.11 and Corollary 19.10,

$$
\begin{aligned}
\mathcal{O}_Y(-d) \otimes \omega_{Y|K} &= \mathcal{O}_Y(-d) \otimes \det \Omega_{Y|K} \\
&= \det j_Y^* \Omega_{\mathbb{P}^n_K|K} \\
&= j_Y^* \det \Omega_{\mathbb{P}^n_K|K} \\
&= j_Y^* \mathcal{O}_{\mathbb{P}^n_K}(-n-1) \\
&= \mathcal{O}_Y(-n-1).
\end{aligned}
$$

The last equality follows from Appendix Lemma 4.7. Tensoring with $\mathcal{O}_Y(d)$ proves the claim.

Remark 19.13: a rough classification of smooth hypersurfaces

Corollary 19.12 permits a rough classification of smooth hypersurfaces

$$Y = V_+(F) \subseteq \mathbb{P}^n_K$$

in projective space according to whether the twist $d-n-1$ in $\omega_Y \cong \mathcal{O}_Y(d-n-1)$ is negative, equal to 0, or positive.

For $n=2$, that is, curves in the projective plane, $d=1,2$ gives a projective line; $d=3$, when the canonical sheaf is trivial, gives an elliptic curve; and $d \geq 4$ gives a curve of general type.

For $n=3$, that is, surfaces in projective space, $d=1$ gives a projective plane; $d=2$ gives a surface isomorphic to $\mathbb{P}^1_K \times \mathbb{P}^1_K$; and $d=3$ gives a surface isomorphic to a projective plane blown up at six points. In any case, for $d \leq 3$ one obtains a so-called *rational surface*, whose function field is the rational function field in two variables. For $d=4$, when the canonical sheaf is trivial, one obtains a so-called $K3$ surface. For $d \geq 5$ one obtains a surface of general type.

Worksheet 19: Tangent Bundles

The star marks exactly one exercise with a frozen public solution, Exercise 19.10. The other eleven exercises have negative candidate results; this edition does not invent new solutions.

Exercise 19.1

Let

$$R = K[X,Y]/(XY).$$

Show that the module of Kähler differentials $\Omega_{R|K}$ is not free at the origin.

Exercise 19.2

Show that there is a smooth scheme of finite type over a field whose module of Kähler differentials does not have constant rank.

Exercise 19.3

Let R be a commutative ring, let A be a commutative algebra over R, let M be an A-module, and let $\delta : A \to M$ be an R-derivation. Show that on every open set $U \subseteq \operatorname{Spek}(A)$ there is an R-derivation

$$\delta_U : \Gamma(U, \mathcal{O}_U) \longrightarrow \Gamma(U, \widetilde{M})$$

commuting with δ.

Hint. First consider the open sets $D(f)$.

Exercise 19.4

Let $p : X \to S$ be a dominant morphism of integral schemes. Show that a derivation over $p^{-1}\mathcal{O}_S$ defined on an open set $U \subseteq X$,

$$\delta : \mathcal{O}_U \longrightarrow \mathcal{O}_U,$$

defines a $Q(S)$-derivation

$$Q(X) \longrightarrow Q(X).$$

> **Editorial note — dominance.** The source does not assume that p is dominant. Without dominance there is generally no induced embedding $Q(S) \hookrightarrow Q(X)$, so the claimed $Q(S)$-derivation is not defined. This edition adds the required hypothesis.

Exercise 19.5

Let X be a scheme of finite type over a locally Noetherian base scheme S. Show that $\Omega_{X|S}$ is a coherent $\mathcal{O}_X$-module.

> **Editorial note — coherence.** The source allows an arbitrary base scheme. Finite type alone makes $\Omega_{X|S}$ finite type, but does not ensure coherence over a non-Noetherian base. The locally Noetherian hypothesis stated here makes X locally Noetherian and gives the claimed coherence.

Exercise 19.6

Let $p : X \to S$ be a scheme over a scheme S. Show that the sheaf of Kähler differentials $\Omega_{X|S}$ on X is the sheafification of the presheaf

$$U \longmapsto \operatorname*{colim}_{\substack{V \subseteq S \text{ open} \\ U \subseteq p^{-1}(V)}} \Omega_{\Gamma(U,\mathcal{O}_X)|\Gamma(V,\mathcal{O}_S)}.$$

Exercise 19.7

Show that the module of Kähler differentials $\Omega_{\mathbb{P}^1_R|R}$ on the projective line $\mathbb{P}^1_R$ over a commutative ring R is isomorphic to the twisted structure sheaf $\mathcal{O}_{\mathbb{P}^1_R}(-2)$.

Exercise 19.8

Consider the tangent sheaf $\mathcal{T}_{\mathbb{P}^1_R,R}$ on the projective line $\mathbb{P}^1_R$ over a commutative ring R, with the isomorphism

$$\mathcal{T}_{\mathbb{P}^1_R,R} \cong \mathcal{O}_{\mathbb{P}^1_R}(2).$$

Determine the global sections of $\mathcal{O}_{\mathbb{P}^1_R}(2)$ corresponding to the global derivations

$$X\frac{\partial}{\partial X}, \qquad Y\frac{\partial}{\partial X}, \qquad X\frac{\partial}{\partial Y}, \qquad Y\frac{\partial}{\partial Y}.$$

Exercise 19.9

On the projective plane

$$\mathbb{P}^2_K = \operatorname{Proj}(K[X,Y,Z]),$$

determine the derivative

$$Y\frac{\partial f}{\partial X}$$

of the rational function

$$f = \frac{XY - YZ + 3Z^2 - X^2}{4X^2 - YZ}.$$

On which open subset are f and $Y\frac{\partial f}{\partial X}$ defined?

Exercise 19.10 ★

Consider the curve

$$C = V_+(X^3 + Y^3 + Z^3) \subseteq \mathbb{P}^2_K$$

over a field of characteristic different from 3. Show that the differential forms

$$\frac{X^2}{Y^2}\, d\Big(\frac{Z}{X}\Big) \quad \text{on } D_+(XY),$$

$$\frac{Y^2}{Z^2}\, d\Big(\frac{X}{Y}\Big) \quad \text{on } D_+(YZ),$$

and

$$\frac{Z^2}{X^2}\, d\Big(\frac{Y}{Z}\Big) \quad \text{on } D_+(XZ)$$

agree on the intersections and therefore define a nontrivial differential form on the curve C.

Exercise 19.11

Express the restrictions of the global derivations

$$X_i \frac{\partial}{\partial X_j}$$

of projective space

$$\mathbb{P}^n_R = \text{Proj}(R[X_0, X_1, \dots, X_n])$$

to the open subset

$$D_+(X_0) = \text{Spek}(R[Y_1, \dots, Y_n]) \subseteq \mathbb{P}^n_R,$$

with $Y_k = X_k/X_0$, as linear combinations of the form

$$\sum_{k=1}^{n} g_k \frac{\partial}{\partial Y_k}, \qquad g_k \in R[Y_1, \dots, Y_n].$$

Exercise 19.12

Consider the Fermat cubic in four variables,

$$V = V_+(X^3 + Y^3 + Z^3 + W^3) \subseteq \mathbb{P}^3_K,$$

and the affine piece

$$U = D_+(W) \cap V = \mathrm{Spek}(K[X, Y, Z]/(X^3 + Y^3 + Z^3 + 1))$$

over an algebraically closed field K of characteristic different from 3. Show that the formulae

$$x(s,t) = \frac{3t - \frac{1}{3}(s^2 + st + t^2)^2}{t(s^2 + st + t^2) - 3},$$

$$y(s,t) = \frac{3s + 3t + \frac{1}{3}(s^2 + st + t^2)^2}{t(s^2 + st + t^2) - 3},$$

and

$$z(s,t) = \frac{-3 - (s^2 + st + t^2)(s + t)}{t(s^2 + st + t^2) - 3}$$

give a rational parametrisation

$$K^2 \dashrightarrow U.$$

Editorial note — the rational map. The source omits a closing parenthesis in the coordinate ring and prints an ordinary arrow $K^2 \to U$. The coordinate-ring parenthesis is supplied above, and a dashed arrow is used because the displayed rational functions are defined only where $t(s^2 + st + t^2) - 3 \neq 0$.

Public Solutions and Coverage of Worksheet 19

At the frozen source boundary, there is exactly one public solution among the 12 exercises: the solution to Exercise 19.10. The exercise map and candidate evidence record negative results for Exercises 19.1–19.9, 19.11, and 19.12. Missing public solution pages are not replaced by invented solutions.

Solution to Exercise 19.10

Using the curve equation and the relation

$$X^2\,dX + Y^2\,dY + Z^2\,dZ = 0,$$

we obtain

$$\begin{aligned}
\frac{X^2}{Y^2}\,d\left(\frac{Z}{X}\right) &= \frac{X^2}{Y^2}\cdot\frac{X\,dZ - Z\,dX}{X^2} \\
&= \frac{X^3Z^2\,dZ - X^2Z^3\,dX}{X^2Y^2Z^2} \\
&= \frac{-X^3(X^2\,dX + Y^2\,dY) + X^2(X^3+Y^3)\,dX}{X^2Y^2Z^2} \\
&= \frac{-X^3Y^2\,dY + X^2Y^3\,dX}{X^2Y^2Z^2} \\
&= \frac{Y^2}{Z^2}\cdot\frac{-X\,dY + Y\,dX}{Y^2} \\
&= \frac{Y^2}{Z^2}\,d\left(\frac{X}{Y}\right).
\end{aligned}$$

By symmetry, the remaining equalities hold as well.

> **Editorial note — the source-solution boundary.** The frozen public solution proves that the local forms agree, but it contains no separate argument that the resulting global form is nonzero, although Exercise 19.10 also asks for nontriviality. That missing step is disclosed here; no continuation is invented.

Frozen negative results

There is no public solution page at the frozen source boundary for Exercises 19.1, 19.2, 19.3, 19.4, 19.5, 19.6, 19.7, 19.8, 19.9, 19.11, or 19.12. This records the outcome of checking candidates; it does not claim that mathematical solutions do not exist.

Lecture 20: The Picard Group

The Picard group

Definition 20.1: the Picard group

For a ringed space $(X, \mathcal{O}_X)$, the set of isomorphism classes of invertible sheaves on X, with tensor product as the operation, the dual sheaf as inverse, and the structure sheaf as identity, is called the *Picard group* of X. It is denoted by

$$\mathrm{Pic}(X).$$

The following discussion of the gluing data of an invertible sheaf connects with Exercise 2.19 on the one hand and anticipates Čech cohomology on the other.

Remark 20.2: transition data as a cocycle

Let $(X, \mathcal{O}_X)$ be a ringed space and let $\mathcal{L}$ be an invertible sheaf on X. This means that there is an open cover

$$X = \bigcup_{i \in I} U_i$$

and trivialisations

$$\varphi_i : \mathcal{L}|_{U_i} \longrightarrow \mathcal{O}_X|_{U_i}.$$

For two open sets U_i, U_j, the transition maps on $U_i \cap U_j$ are

$$\varphi_i|_{U_i \cap U_j} \circ \varphi_j^{-1}|_{U_i \cap U_j} : \mathcal{O}_X|_{U_i \cap U_j} \longrightarrow \mathcal{O}_X|_{U_i \cap U_j}.$$

These isomorphisms are given, compare Exercise 13.8, by multiplication by units

$$r_{ij} \in \Gamma(U_i \cap U_j, \mathcal{O}_X^\times).$$

Since these data come from a single invertible sheaf $\mathcal{L}$, the cocycle condition holds:

$$r_{kj} r_{ji} = r_{ki},$$

which can also be written as

$$r_{kj} r_{ki}^{-1} r_{ji} = 1.$$

Conversely, such a collection of data determines an invertible sheaf by gluing.

If the invertible sheaf is trivial, there is a global $\mathcal{O}_X$-module isomorphism

$$\psi : \mathcal{O}_X \longrightarrow \mathcal{L}.$$

On U_i we have the isomorphisms

$$\mathcal{O}_X|_{U_i} \xrightarrow{\psi|_{U_i}} \mathcal{L}|_{U_i} \xrightarrow{\varphi_i} \mathcal{O}_X|_{U_i},$$

which are entirely determined by units

$$s_i \in \Gamma(U_i, \mathcal{O}_X^\times).$$

These units satisfy

$$s_i s_j^{-1} = (\varphi_i \circ \psi) \circ (\varphi_j \circ \psi)^{-1} = r_{ij}$$

for all i, j. Conversely, if units s_i realising this relation are given, then

$$\psi|_{U_i} := \varphi_i^{-1} \circ s_i$$

defines compatible module isomorphisms on U_i, which therefore glue to a global isomorphism between $\mathcal{O}_X$ and $\mathcal{L}$.

Thus an invertible sheaf can be identified with a collection of data (U_i, r_{ij}) satisfying the conditions above, called a *cocycle*; such data are regarded as trivial if there are units s_i with

$$r_{ij} = s_i s_j^{-1}.$$

Remark 20.3: the sign issue in the cocycle description

The identification in Remark 20.2 between invertible sheaves and cocycles in the sheaf of units is not canonical, owing to a sign issue. It depends on whether the local trivialisations of the invertible sheaf with the structure sheaf are taken in the direction

$$\varphi_i : \mathcal{L}|_{U_i} \longrightarrow \mathcal{O}_X|_{U_i}$$

or in the opposite direction, and on how the index set is ordered.

Remark 20.4: tensor products of transition data

Tensoring invertible sheaves $\mathcal{L}$ and $\mathcal{L}'$ can be carried out at the level of the data in Remark 20.2. Pass to a common refinement of the covers, so that both sheaves have trivialisations with respect to one cover U_i, $i \in I$. The data

$$r_{ij} \cdot r'_{ij}$$

then describe the tensor product.

> **Editorial note — the second transition datum.** The source prints $r_{i'j}$ in this formula. The transition datum of the second invertible sheaf must instead be r'_{ij}: the cover indices remain i, j, while the prime distinguishes the second sheaf. This edition uses that notation explicitly.

From now on, we restrict our attention to schemes.

Lemma 20.5: the Picard group of a local ring

For a local ring R, the Picard group of Spek(R) is trivial.

Proof This is trivial.

Lemma 20.6: embedding in the function field sheaf

Let $(X, \mathcal{O}_X)$ be an integral scheme. Every invertible sheaf on X is isomorphic to an $\mathcal{O}_X$-submodule of the constant function field sheaf.

Proof Let K be the function field of X and let $\mathcal{K}$ be the associated sheaf. For an invertible sheaf $\mathcal{L}$, the stalk at the generic point η is a one-dimensional vector space over K. Fix a K-isomorphism

$$\mathcal{L}_\eta = \mathcal{K}_\eta = K.$$

For every open set $U \subseteq X$, there is a natural map

$$\Gamma(U, \mathcal{L}) \longrightarrow \mathcal{L}_\eta \longrightarrow K.$$

These maps are injective, compare the proof of Lemma 11.16, and define a submodule of $\mathcal{K}$.

Remark 20.7: describing invertible submodules

An invertible submodule $\mathcal{L}$ of the constant function field sheaf is given by an open cover U_i, $i \in I$, of X, together with nonzero elements $q_i \in Q(X)$ satisfying

$$\frac{q_i}{q_j} \in \left(\Gamma(U_i \cap U_j, \mathcal{O}_X)\right)^\times .$$

Using a trivialising cover U_i, we have

$$\mathcal{L}|_{U_i} \cong q_i \mathcal{O}_{U_i},$$

and the transition maps on intersections imply that the quotient q_i/q_j must be a unit. Conversely, if such data (U_i, q_i) are given, then

$$q_i \mathcal{O}_{U_i} \subseteq \mathcal{Q}_{U_i}$$

is a trivial subsheaf that determines an invertible subsheaf on X.

Another perspective is provided by the exact sequence of sheaves

$$0 \longrightarrow \mathcal{O}_X^\times \longrightarrow \mathcal{Q}^\times \longrightarrow \mathcal{Q}^\times/\mathcal{O}_X^\times \longrightarrow 0.$$

By Lemma 5.9, the data described above are the global sections of the quotient sheaf $\mathcal{Q}^\times/\mathcal{O}_X^\times$.

> **Editorial note — the function-field symbol.** The source writes $q_i \in Q$ after denoting the function field by K in the preceding proof. This edition uses the unambiguous notation $Q(X)$, consistent with the function-field sheaf $\mathcal{Q}$.

Lemma 20.8: realisation as an ideal sheaf

Let R be an integral domain. Every invertible sheaf on

$$X = \mathrm{Spek}(R)$$

is isomorphic to an ideal sheaf.

Proof By Lemma 20.6, we may assume at once that we have an invertible submodule

$$L \subseteq K$$

of the field of fractions $K = Q(R)$. By Theorem 16.2, invertibility means that there is a family

$$f_1, \dots, f_k \in R$$

such that

$$L_{f_i} \cong R_{f_i} \cdot q_i$$

with $q_i \in K \setminus \{0\}$. Let b be a common denominator for all the q_i. The multiplication map

$$K \longrightarrow K, \qquad q \longmapsto qb,$$

which is an R-module isomorphism of K, sends the submodule L to an isomorphic submodule L'. On the given cover, L' is a submodule of the structure sheaf, and hence an ideal.

In general, there are many ways to realise an invertible sheaf as a subsheaf of the function field sheaf. Indeed, a new realisation is obtained from a given one simply by multiplying by an element $f \in K$.

Example 20.9: twisted structure sheaves inside the function field sheaf

On projective space $\mathbb{P}_K^d$ over a field K, a twisted structure sheaf $\mathcal{O}_{\mathbb{P}_K^d}(\ell)$ can be embedded in the function field sheaf $\mathcal{Q}$ as follows. Let

$$G \in K(X_0, \dots, X_d)_{-\ell}$$

be a homogeneous element of degree $-\ell$. On every open set $U \subseteq \mathbb{P}_K^d$, the natural map

$$\Gamma\Big(U, \mathcal{O}_{\mathbb{P}^d_K}(\ell)\Big) \longrightarrow Q(\mathbb{P}^d_K), \qquad s \longmapsto sG,$$

is a realisation as a submodule.

> **Editorial note — the space and variable indices.** The source names the space $\mathbb{P}^d_K$ but writes the rational function field with variables $X_0, \dots, X_n$. This edition uses $X_0, \dots, X_d$ to match the stated projective space.

Lemma 20.10: tensor products and products of subsheaves

Let X be an integral scheme and let

$$\mathcal{L}, \mathcal{M} \subseteq \mathcal{Q}$$

be invertible subsheaves of the constant sheaf $\mathcal{Q}$ associated with the function field $Q(X)$. Then

$$\mathcal{L} \otimes \mathcal{M} \cong \mathcal{L} \cdot \mathcal{M},$$

where $\mathcal{L} \cdot \mathcal{M}$ denotes the subsheaf of $\mathcal{Q}$ whose stalk at each point $P \in X$ is generated by all products fg with $f \in \mathcal{L}_P$ and $g \in \mathcal{M}_P$.

Proof For the field of fractions Q of an integral domain R, natural multiplication gives

$$Q \otimes_R Q = Q.$$

Hence on an integral scheme there is an isomorphism

$$\mathcal{Q} \otimes_{\mathcal{O}_X} \mathcal{Q} \cong \mathcal{Q}.$$

Thus there is a natural homomorphism

$$\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{M} \longrightarrow \mathcal{Q}$$

given by multiplication. Since $\mathcal{L}$ and $\mathcal{M}$ are invertible, this map is locally, and hence globally, an isomorphism onto its image sheaf.

The Picard group in the factorial case

Lemma 20.11: reading prime exponents after localisation

Let R be a unique factorisation domain. The following statements hold.

1. For $f \in R$, $f \neq 0$, we have

$$(f)R_{(p)} = (p^s)$$

   if and only if p occurs with exponent s in the prime factorisation of f.
2. Two principal ideals (f) and (g) agree if and only if, for every prime element p, the ideals

$$(f)R_{(p)} \quad \text{and} \quad (g)R_{(p)}$$

   agree in the localisation $R_{(p)}$.

Proof See Exercise 20.4.

Theorem 20.12: the Picard group of a unique factorisation domain

The Picard group of a unique factorisation domain is trivial.

Proof Let $I \subseteq R$ be an invertible ideal, and let

$$X = \bigcup_{i=1}^{n} D(f_i)$$

be an open cover such that IR_{f_i} is a principal ideal. In particular, for every prime element p, the ideal

$$I_{(p)} \subseteq R_{(p)}$$

is principal and hence has the form (p^{s_p}), since $R_{(p)}$ is a discrete valuation ring. Only finitely many of the s_p are nonzero. Indeed, an element $g \in I$, $g \neq 0$, has only finitely many prime divisors, while for every other prime element q, the element g is a unit in $R_{(q)}$.

> **Editorial note — element versus ideal.** The source says that the ideal $I_{(p)}$ has the form p^{s_p}. Parentheses are supplied here because the object is the principal ideal (p^{s_p}), not the element p^{s_p}.

We claim that I equals the principal ideal generated by

$$h = \prod_{p} p^{s_p}.$$

Since equality of ideals can be tested locally on a cover, we may argue in R_{f_i}. The assertion then follows from Lemma 20.11.

Lemma 20.13: the Picard group of an open set in the factorial case

Let R be a Noetherian unique factorisation domain and let $U \subseteq \operatorname{Spek}(R)$ be an open subset. Then the Picard group of U is trivial.

Proof Write

$$U = D(f_1, \dots, f_n).$$

We proceed by induction on n; the base case follows from Theorem 20.12. Thus we may assume that $\mathcal{L}$ is trivial on

$$D(f_1, \dots, f_{n-1}).$$

By Remark 20.2, the invertible sheaf is determined by a unit over

$$D(f_1, \dots, f_{n-1}) \cap D(f_n).$$

By Theorem 9.8, in the factorial case the structure sheaf, and hence also the sheaf of units, is particularly simple: an element $h \in R$ is a unit on U exactly when

$$U \subseteq D(h).$$

More generally, after viewing sections inside $Q(R)^{\times}$, units on open sets have the form

$$h = u p_1^{r_1} \cdots p_s^{r_s},$$

where the $p_j \in R$ are prime elements, u is a unit in R, and $r_j \in \mathbb{Z}$. The element h is a unit on

$$D(f_1, \dots, f_{n-1}) \cap D(f_n) = D(f_1 f_n, \dots, f_{n-1} f_n)$$

exactly when the p_j that occur, namely those with exponent $r_j \neq 0$, divide the $f_i f_n$. This means that p_j divides f_n or divides all the elements $f_1, \dots, f_{n-1}$. In either case, h can be written as a product of a unit on $D(f_1, \dots, f_{n-1})$ and a unit on $D(f_n)$. These units can be used to trivialise the sheaf.

> **Editorial note — the ambient group of units.** The source first declares $h \in R$ and then allows negative prime exponents. Such expressions belong in the fraction field. This edition explicitly views the units as elements of $Q(R)^{\times}$; the factorisation argument is otherwise unchanged.

Corollary 20.14: extending invertible sheaves

Let $(X, \mathcal{O}_X)$ be a Noetherian integral scheme and let $U \subseteq X$ be an open subset. Suppose that for every point $P \in X \setminus U$, the local ring $\mathcal{O}_{X,P}$ is factorial. Then every invertible sheaf $\mathcal{L}$ on U extends to an invertible sheaf on X.

Proof If U is empty, the structure sheaf on X is an extension, so assume that U is nonempty. Let $\mathcal{L}$ be an invertible sheaf on U and let $P \in X \setminus U$. Choose an affine open neighbourhood

$$W = \mathrm{Spek}(R)$$

of P, where P corresponds to the prime ideal $\mathfrak{p}$. By hypothesis, $R_{\mathfrak{p}}$ is factorial. Consider the injective scheme morphisms

$$\mathrm{Spek}(R_{\mathfrak{p}}) \longrightarrow W \longrightarrow X.$$

The open set U has nonempty intersection with W and with $\mathrm{Spek}(R_{\mathfrak{p}})$; write the latter intersection as

$$V \subseteq \mathrm{Spek}(R_{\mathfrak{p}}),$$

since the generic point of X corresponds to the zero ideal of $R_{\mathfrak{p}}$. The pullback of $\mathcal{L}$ to V is trivial by Lemma 20.13. Choose a trivialisation there. Since invertible sheaves and their isomorphisms are finitely presented data, after shrinking around P this trivialisation descends from the localisation to an isomorphism over the overlap with an open neighbourhood

$$P \in D(f) = W' \subseteq W.$$

Thus $\mathcal{L}$ on U can be glued to the trivial invertible sheaf on W' along $U \cap W'$, giving an extension to $U \cup W'$. We can therefore successively replace the open set by a strictly larger open set on which an extension exists. By Noetherianity, this process ends at the whole space.

> **Editorial note — descent and the set operation in the source proof.** The source introduces modules over $R_{\mathfrak{p}}$ and R without supplying the required descent comparison, then says that an extension has been found on $U \cap W'$ and immediately enlarges the open set. The edition makes the finite-presentation descent and gluing step explicit and uses the required union $U \cup W'$; the overlap on which the sheaves are identified remains $U \cap W'$.

Under the hypotheses above, the natural restriction homomorphism

$$\mathrm{Pic}(X) \longrightarrow \mathrm{Pic}(U)$$

is therefore surjective.

Example 20.15: the Picard group of a punctured A_n singularity

Consider the commutative ring

$$R = K[X, Y, Z]/(XY - Z^n)$$

over a field K, with maximal ideal

$$\mathfrak{m} = (X, Y, Z),$$

and open set

$$U = D(X, Y) = D(X) \cup D(Y) = \mathrm{Spek}(R) \setminus \{\mathfrak{m}\} \subseteq \mathrm{Spek}(R).$$

We have

$$R_X \cong K[X, X^{-1}, Z]$$

via $Y = Z^n/X$. This ring is a unique factorisation domain; hence, by Theorem 20.12, all invertible sheaves on $D(X)$ are trivial, and likewise on $D(Y)$. Furthermore,

$$R_{XY} = R_Z \cong K[X, X^{-1}, Z, Z^{-1}].$$

Thus an invertible sheaf on U is determined by an isomorphism

$$K[X, X^{-1}, Z, Z^{-1}] \cong \mathcal{O}_U|_{D(XY)} \longrightarrow K[X, X^{-1}, Z, Z^{-1}] \cong \mathcal{O}_U|_{D(XY)},$$

which in turn corresponds to a unit in $K[X, X^{-1}, Z, Z^{-1}]$. Let

$$cX^iZ^j$$

be such a unit. Units coming from R_X or R_Y, and multiplicative combinations of them, give a trivial invertible sheaf by Remark 20.2. The quotient group consists of

$$Z^j, \qquad j = 0, 1, \dots, n-1,$$

so the Picard group of U is $\mathbb{Z}/(n)$.

Example 20.16: the Picard group of the projective line

Consider the projective line $\mathbb{P}^1_K$ over a field K with its standard cover

$$\mathbb{P}^1_K = D_+(X) \cup D_+(Y),$$

consisting of the two affine lines

$$D_+(X) = \mathrm{Spek}\left(K\left[\frac{Y}{X}\right]\right) \cong \mathbb{A}^1_K$$

and

$$D_+(Y) = \mathrm{Spek}\left(K\left[\frac{X}{Y}\right]\right) \cong \mathbb{A}^1_K.$$

By Theorem 20.12 and Remark 20.2, the Picard group of the projective line can be calculated by taking the units in

$$\Gamma\Big(D_+(XY), \mathcal{O}_{\mathbb{P}^1_K}\Big) = K\left[\frac{X}{Y}, \frac{Y}{X}\right]$$

modulo the units on the two affine pieces. This gives the group

$$\left\{\left(\frac{X}{Y}\right)^k \mid k \in \mathbb{Z}\right\},$$

so the Picard group is isomorphic to $\mathbb{Z}$.

The last assertion holds more generally for projective space $\mathbb{P}^d_K$ with $d \geq 1$; see Theorem 22.12.

Worksheet 20: The Picard Group

None of the 13 exercises has a public solution at the frozen revision boundary. No solution stars are therefore used, and this edition does not invent new solutions.

Exercise 20.1

Let

$$\varphi : X \longrightarrow Y$$

be a scheme morphism. Prove that the assignment

$$\mathcal{L} \longmapsto \varphi^*\mathcal{L}$$

defines a group homomorphism

$$\mathrm{Pic}(Y) \longrightarrow \mathrm{Pic}(X).$$

For the next two exercises, take Remark 20.3 into account.

Exercise 20.2

Let

$$\varphi : X \longrightarrow Y$$

be a scheme morphism. Prove that the group homomorphism

$$\mathrm{Pic}(Y) \longrightarrow \mathrm{Pic}(X), \qquad \mathcal{L} \longmapsto \varphi^*\mathcal{L},$$

can be described using the cocycle description in Remark 20.2 as follows. The cocycle

$$r_{ij} \in \left(\Gamma(V_i \cap V_j, \mathcal{O}_Y)\right)^\times, \qquad i < j,$$

for an open cover

$$Y = \bigcup_{i \in I} V_i$$

is sent to the cocycle

$$\theta_{ij}(r_{ij}) \in \big(\Gamma\big(\varphi^{-1}(V_i) \cap \varphi^{-1}(V_j), \mathcal{O}_X\big)\big)^{\times}, \qquad i < j,$$

with respect to the cover

$$X = \bigcup_{i \in I} \varphi^{-1}(V_i),$$

where

$$\theta_{ij} : \Gamma(V_i \cap V_j, \mathcal{O}_Y) \longrightarrow \Gamma\big(\varphi^{-1}(V_i \cap V_j), \mathcal{O}_X\big)$$

denotes the associated ring homomorphism.

Exercise 20.3

Let

$$A = K[X_1, \dots, X_n]/\mathfrak{a}$$

be a standard-graded ring, with the open cover

$$U = \operatorname{Spek}(A) \setminus \{A_+\} = \bigcup_{i=1}^{n} D(X_i) \subseteq \operatorname{Spek}(A) = X$$

of the punctured spectrum and the open cover

$$Y = \operatorname{Proj}(A) = \bigcup_{i=1}^{n} D_+(X_i)$$

of the associated projective spectrum. Let $\ell \in \mathbb{Z}$. Prove the following statements.

1. The family of units

   $$X_i^{\ell} \in (\Gamma(D(X_i), \mathcal{O}_X))^{\times}, \qquad i = 1, \dots, n,$$

   defines the cocycle

   $$X_j^{\ell} X_i^{-\ell} \in \big(\Gamma(D(X_i X_j), \mathcal{O}_X)\big)^{\times}, \qquad i < j,$$

   representing the trivial invertible sheaf on U.
2. The cocycle from part (1) can be regarded as a cocycle on the projective spectrum Y.
3. The invertible sheaf on Y determined by the cocycle from part (2) is isomorphic to the twisted structure sheaf $\mathcal{O}_Y(\ell)$, or to $\mathcal{O}_Y(-\ell)$; there is a choice of sign here.
4. The pullback of a twisted structure sheaf under the cone map $U \to Y$ is trivial.

   Editorial note — the ring symbol in the source. The source defines $A = K[X_1, \dots, X_n]/\mathfrak{a}$ but prints $\operatorname{Spek}(R) = X$ in the cover of the punctured spectrum. This edition uses A, the ring defined in the exercise.

Exercise 20.4

Let R be a unique factorisation domain. Prove the following statements.

1. For $f \in R$, $f \neq 0$, we have

$$(f)R_{(p)} = (p^s)$$

   if and only if p occurs with exponent s in the prime factorisation of f.

2. Two principal ideals (f) and (g) agree if and only if, for every prime element p, the ideals

$$(f)R_{(p)} \quad \text{and} \quad (g)R_{(p)}$$

   agree in the localisation $R_{(p)}$.

The following exercise prepares for Lemma 20.13.

Exercise 20.5

Let R be a unique factorisation domain and let $f, g \in R$. Using Remark 20.2, prove that the Picard group of

$$D(f,g) \subseteq \mathrm{Spek}(R)$$

is trivial.

Exercise 20.6

Let R be a Noetherian integral domain such that all localisations $R_{\mathfrak{p}}$ are factorial. Let $U \subseteq \mathrm{Spek}(R)$ be an open subset and let $\mathfrak{p} \notin U$ be a point of codimension ≥ 2, so the prime ideal $\mathfrak{p}$ has height ≥ 2. Prove that an invertible sheaf $\mathcal{L}$ on U has a unique extension to an open set U' containing U and $\mathfrak{p}$.

Exercise 20.7

Consider the quadratic number ring

$$R = \mathbb{Z}[\sqrt{-5}]$$

with open subset $D(2) \subseteq \mathrm{Spek}(R)$. Prove that the structure sheaf on $D(2)$ can be extended in more than one way to an invertible sheaf on $\mathrm{Spek}(R)$.

Exercise 20.8

Prove that the Picard group of the punctured spectrum

$$U = D(X,Y,Z) \subseteq \mathrm{Spek}\Big((K[X,Y,Z]/(XY - Z^n))_{(X,Y,Z)} \Big)$$

of the local ring

$$(K[X,Y,Z]/(XY - Z^n))_{(X,Y,Z)}$$

is $\mathbb{Z}/(n)$; compare Example 20.15.

> **Editorial note — the scope of localisation in the source formula.** In both displays the source places the subscript (X,Y,Z) directly on $(XY - Z^n)$ within the quotient. Since the text calls this a local ring and asks for its punctured spectrum, this edition places the localisation on the quotient ring itself.

Exercise 20.9

Prove that on the punctured spectrum

$$U = D(X, Y, Z) \subseteq \text{Spek}(K[X, Y, Z]/(XY - Z^n)),$$

for $n \geq 2$, there are invertible sheaves that cannot be extended to invertible sheaves on

$$\text{Spek}(K[X, Y, Z]/(XY - Z^n)).$$

Exercise 20.10

Prove that the invertible sheaves on the punctured spectrum

$$U = D(X, Y, Z) \subseteq \text{Spek}(K[X, Y, Z]/(XY - Z^n))$$

are restrictions of the coherent ideal sheaves associated with the ideals

$$(X, Z^i), \qquad i = 0, \dots, n-1,$$

in $K[X, Y, Z]/(XY - Z^n)$.

Exercise 20.11

Let

$$R = K[X, Y, Z]/(XY - Z^n).$$

For $i = 0, 1, \dots, n-1$, consider the R-algebras

$$A_i = R[S, T]/(SX + TZ^i)$$

and the associated spectrum maps

$$\pi_i : \text{Spek}(A_i) \longrightarrow \text{Spek}(R).$$

Prove that over

$$U = D(X, Y, Z) \subseteq \text{Spek}(K[X, Y, Z]/(XY - Z^n)),$$

the maps π_i are line bundles that are nontrivial for $i \neq 0$. Prove also that for $i \neq 0$, the scheme $\text{Spek}(A_i)$ is not a line bundle over $\text{Spek}(R)$.

Exercise 20.12

Let

$$\mathbb{P}^d_K = \text{Proj}(K[X_0, X_1, \dots, X_d])$$

be projective space over a field K. Prove that the Picard group of the open subset

$$D_+(X_i, X_j) \subseteq \mathbb{P}^d_K, \qquad i \neq j,$$

is isomorphic to $\mathbb{Z}$.

Exercise 20.13

Let

$$\mathbb{P}^d_K = \mathrm{Proj}(K[X_0, X_1, \dots, X_d])$$

be projective space over a field K. Prove that for $d \geq 1$, the Picard group of $\mathbb{P}^d_K$ is isomorphic to $\mathbb{Z}$.

Public Solution Coverage for Worksheet 20

At the frozen revision boundary, all thirteen candidate solution pages, Exercises 20.1–20.13, have *missing* status in the official query evidence. There are therefore no public solutions to translate for this unit, and this edition neither invents nor implies new solutions.

The ordered exercise map and candidate evidence remain the record of source coverage, including the negative result for every exercise number.

Lecture 21: Normal Schemes

Normal rings

Definition 21.1: total quotient ring

Let R be a commutative ring and $S \subseteq R$ the set of all non-zero-divisors in R. The localisation R_S is called the *total quotient ring* of R and is denoted by

$$Q(R).$$

Definition 21.2: normal ring

A commutative ring is called *normal* if it is integrally closed in its total quotient ring.

Definition 21.3: normalisation

Let R be a commutative ring and $Q(R)$ its total quotient ring. The integral closure of R in $Q(R)$ is called the *normalisation* of R.

Example 21.4: normalisation of the coordinate cross

We determine the normalisation of the ring

$$K[X,Y]/(XY)$$

over a field K. The element $X+Y$ is a non-zero-divisor. For the element

$$\frac{X-Y}{X+Y} \in Q(R)$$

we have

$$\left(\frac{X-Y}{X+Y}\right)^2 = \frac{(X-Y)^2}{(X+Y)^2} = \frac{X^2+Y^2}{X^2+Y^2} = 1.$$

Thus this element satisfies an equation of integral dependence and hence belongs to the normalisation.

> **Editorial note - scope of the source example.** The source announces a determination of the normalisation, but the available passage only proves that $(X-Y)/(X+Y)$ is integral over R. This edition does not complete a calculation that the source does not supply. The source also switches to the notation $Q(R)$ without first assigning the symbol R to the displayed ring.

Discrete valuation rings

Definition 21.5: discrete valuation ring

A *discrete valuation ring* R is a principal ideal domain with exactly one prime element up to association.

Definition 21.6: order

Let $f \in R$, $f \neq 0$, be an element of a discrete valuation ring R with prime element p. The number $n \in \mathbb{N}$ satisfying

$$f = up^n,$$

where u is a unit, is called the *order* of f and is denoted by

$$\operatorname{ord}(f).$$

Lemma 21.7: properties of the order

Let R be a discrete valuation ring with maximal ideal $\mathfrak{m} = (p)$. The order map

$$R \setminus \{0\} \longrightarrow \mathbb{N}, \qquad f \longmapsto \operatorname{ord}(f),$$

has the following properties.

1. We have

$$\operatorname{ord}(fg) = \operatorname{ord}(f) + \operatorname{ord}(g).$$

2. We have

$$\operatorname{ord}(f + g) \geq \min\{\operatorname{ord}(f), \operatorname{ord}(g)\}.$$

3. We have $f \in \mathfrak{m}$ if and only if $\operatorname{ord}(f) \geq 1$.
4. We have $f \in R^\times$ if and only if $\operatorname{ord}(f) = 0$.

Proof See Exercise 21.7.

> **Editorial note - domain of the order function.** The source defines ord on $R \setminus \{0\}$, but part 2 does not exclude $f + g = 0$. The source statement is preserved; if $\operatorname{ord}(0)$ is not defined, the inequality is meaningful only for $f + g \neq 0$.

We quote the following characterisation theorem. In particular, it says that one-dimensional normal local integral domains are discrete valuation rings, and hence unique factorisation domains and regular. Consequently, for a normal Noetherian integral domain R, all localisations at prime ideals of height 1 are discrete valuation rings.

> **Editorial note - Noetherian hypothesis.** The first sentence in the source omits the Noetherian hypothesis. Its conclusion is intended under that hypothesis, as stated explicitly in Theorem 21.8; normality and dimension one alone do not imply that a local domain is a discrete valuation ring.

Theorem 21.8: characterisation of discrete valuation rings

Let R be a Noetherian local integral domain with exactly two prime ideals

$$0 \subset \mathfrak{m}.$$

The following statements are equivalent.

1. R is a discrete valuation ring.
2. R is a principal ideal domain.
3. R is a unique factorisation domain.
4. R is normal.
5. $\mathfrak{m}$ is a principal ideal.

Lemma 21.9: order as a vector-space dimension

Let K be a field and B a discrete valuation ring over K whose residue field is K. For every $f \in B$, $f \neq 0$, the order of f equals the dimension of $B/(f)$ as a vector space over K:

$$\operatorname{ord}(f) = \dim_K(B/(f)).$$

Proof This follows from Exercise 21.2 by induction on the order of f.

Normal schemes

Definition 21.10: normal scheme

A scheme X is called *normal* if every local ring $\mathcal{O}_x$, for $x \in X$, is a normal ring.

Lemma 21.11: affine criteria for normality

For a scheme $(X, \mathcal{O}_X)$, the following properties are equivalent.

1. X is normal.
2. For every affine open subset $U = \operatorname{Spek}(R)$ of X, the ring R is normal.
3. There is an affine open cover

$$X = \bigcup_{i \in I} U_i, \qquad U_i = \operatorname{Spek}(R_i),$$

with every R_i a normal ring.

Proof The implication $(2) \Rightarrow (3)$ is immediate. Suppose (3) holds. For every point $x \in X$, there is therefore an affine open neighbourhood

$$x \in U = \operatorname{Spek}(R) \subseteq X$$

with R normal. Here

$$\mathcal{O}_x = R_{\mathfrak{p}}$$

for a prime ideal $\mathfrak{p}$ of R. By Theorem 42.3 in Commutative Algebra, $R_{\mathfrak{p}}$ is also normal.

> **Editorial note - directions of implication in the source.** The source states three equivalent properties and gives $(2) \Rightarrow (3)$ and $(3) \Rightarrow (1)$, but does not supply an argument for $(1) \Rightarrow (2)$. This edition does not invent an unavailable proof.

Theorem 21.12: intersection of height-one localisations

Let R be a normal Noetherian integral domain. Then

$$R = \bigcap_{\mathfrak{p}} R_{\mathfrak{p}},$$

where $\mathfrak{p}$ ranges over all prime ideals of height 1 in R.

Proof Let $f = g/h \in Q(R)$ and suppose that $f \notin R$. By Lemma 44.12 in Commutative Algebra, there is a prime ideal $\mathfrak{p}$ associated to a quotient ring by a principal ideal such that

$$f \notin R_{\mathfrak{p}}.$$

Thus $\mathfrak{p}$ is the annihilator ideal of an element x modulo a principal ideal (y). By localising, we may assume that $\mathfrak{p}$ is the maximal ideal of R. Consider the R-submodule

$$N = \{q \in Q(R) \mid q\mathfrak{p} \subseteq R\} \subseteq Q(R).$$

We have

$$\mathfrak{p} \subseteq \mathfrak{p}N \subseteq R.$$

Since $\mathfrak{p}$ is maximal, either

$$\mathfrak{p} = \mathfrak{p}N$$

or

$$\mathfrak{p}N = R.$$

In the first case, Lemma 41.7 in Commutative Algebra says that the elements of N are integral over R. Normality of R then gives $N = R$. Since

$$x\mathfrak{p} \subseteq (y),$$

we also have

$$\frac{x}{y}\mathfrak{p} \subseteq R,$$

and hence

$$\frac{x}{y} \in N = R,$$

a contradiction. Thus the second case holds: $\mathfrak{p}N = R$. There must therefore be elements $a \in \mathfrak{p}$ and $q \in N$ satisfying

$$aq = 1.$$

For $b \in \mathfrak{p}$, we then have

$$bq = \frac{b}{a} \in R.$$

Thus $b \in (a)$, so $\mathfrak{p} = (a)$ is a principal ideal. By Theorem 21.8, R is a discrete valuation ring and $\mathfrak{p}$ has height 1.

> **Editorial note - details implicit in the source proof.** The fraction x/y need not equal g/h: the associated-prime description supplies a non-zero class of x modulo (y), which gives the required contradiction when $x/y \in R$. At the final step, the non-zero principal maximal ideal has height 1 by the principal ideal theorem; this supplies the dimension hypothesis needed to apply Theorem 21.8.

Worksheet 21: Discrete Valuation Rings

The stars mark exactly the three exercises with public solutions at the frozen revision boundary: Exercises 21.3, 21.9, and 21.10. The other ten exercises have negative candidate results; this edition does not invent new solutions.

Exercise 21.1

Describe the spectrum of a discrete valuation ring.

Exercise 21.2

Let R be a discrete valuation ring and $\mathfrak{m} = (\pi)$. Let

$$K = R/(\pi)$$

be the residue field of R. Show that for every $n \in \mathbb{N}$ there is an isomorphism of R-modules

$$(\pi^n)/(\pi^{n+1}) \longrightarrow K.$$

Exercise 21.3 ★

Let R be a discrete valuation ring with field of fractions Q. Show that there is no proper intermediate ring between R and Q.

Exercise 21.4

Let R be a discrete valuation ring with field of fractions Q. Characterise the finitely generated R-submodules of Q. To what form can a system of generators be reduced?

Exercise 21.5

Let K be a field of characteristic 0, let $f \in K[X]$ with $f \neq 0$, and let $a \in K$. Show that the following three "orders" of f at a coincide.

1. The order of vanishing of f at a, namely the least order of a formal derivative satisfying

$$f^{(k)}(a) \neq 0.$$

2. The exponent of the linear factor $X - a$ in the factorisation of f.
3. The order of f in the localisation

$$K[X]_{(X-a)}$$

of $K[X]$ at the maximal ideal $(X - a)$.

Editorial note - difference from the course PDF witness. The frozen semantic Web entity states the "least order" of a non-zero formal derivative, as in part 1 above. The course PDF witness on page 189 has the older wording "greatest order" of a derivative with $f^{(k)}(a) = 0$. This edition follows the frozen Web entity and records this substantive difference without silently combining the two formulations.

Exercise 21.6

Let K be a field and $K(T)$ the field of rational functions over K. Find a discrete valuation ring

$$R \subseteq K(T)$$

satisfying

$$Q(R) = K(T) \qquad \text{and} \qquad R \cap K[T] = K.$$

Exercise 21.7

Let R be a discrete valuation ring with maximal ideal $\mathfrak{m} = (p)$. Show that the order map

$$R \setminus \{0\} \longrightarrow \mathbb{N}, \qquad f \longmapsto \mathrm{ord}(f),$$

has the following properties.

1. We have

$$\mathrm{ord}(fg) = \mathrm{ord}(f) + \mathrm{ord}(g).$$

2. We have

$$\mathrm{ord}(f + g) \geq \min\{\mathrm{ord}(f), \mathrm{ord}(g)\}.$$

3. We have $f \in \mathfrak{m}$ if and only if $\mathrm{ord}(f) \geq 1$.
4. We have $f \in R^\times$ if and only if $\mathrm{ord}(f) = 0$.

   Editorial note - domain of the order function. The source defines ord on $R \setminus \{0\}$, but part 2 does not exclude $f + g = 0$. The source statement is preserved; if $\mathrm{ord}(0)$ is not defined, the inequality is meaningful only for $f + g \neq 0$.

Exercise 21.8

Fix a prime number p. For each integer $n \neq 0$, let $\nu_p(n)$ denote the exponent with which p occurs in the prime factorisation of n.

1. Show that the map

$$\nu_p : \mathbb{Z} \setminus \{0\} \longrightarrow \mathbb{N}$$

   is surjective.
2. Show that

$$\nu_p(nm) = \nu_p(n) + \nu_p(m).$$

3. Find an extension

$$\nu_p : \mathbb{Q} \setminus \{0\} \longrightarrow \mathbb{Z}$$

   of the given map that is a group homomorphism; here $\mathbb{Q}^\times = \mathbb{Q} \setminus \{0\}$ is equipped with multiplication and $\mathbb{Z}$ with addition.
4. Describe the kernel of the group homomorphism in part 3.

Exercise 21.9 ★

Let K be a field and let

$$\nu : (K^\times, \cdot, 1) \longrightarrow (\mathbb{Z}, +, 0)$$

be a surjective group homomorphism satisfying

$$\nu(f+g) \geq \min\{\nu(f), \nu(g)\}$$

for all $f, g \in K^\times$. Show that

$$R = \{f \in K^\times \mid \nu(f) \geq 0\} \cup \{0\}$$

is a discrete valuation ring.

> **Editorial note - domain of the valuation.** The source defines ν only on $K^\times$, but writes $\nu(f+g)$ without excluding $f+g=0$. The source statement is preserved; the inequality is meaningful only when $f+g \neq 0$, unless ν is extended to zero.

Exercise 21.10 ★

Let

$$P \in C = V(F) \subset \mathbb{A}^2_K$$

be a smooth point of an irreducible plane curve. Show that the corresponding local ring is a discrete valuation ring.

Exercise 21.11

Let

$$V = V(x^2 + y^2 - 1) \subseteq \mathbb{A}^2_K$$

be the unit circle over a field K, and let $P = (a, b) \in V$ be a point.

1. Show that the local ring R of V at P is a discrete valuation ring.
2. Deduce that the coordinate ring

   $$K[X,Y]/(X^2+Y^2-1)$$

   is normal. The source states that K may be assumed algebraically closed.
3. Show that

   $$K[X,Y]/(X^2+Y^2-1)$$

   is not a unique factorisation domain.
4. Determine the orders of X and $Y-1$ in the local ring at $(0,1)$.

   Editorial note - field hypotheses in the source. The source does not restrict the characteristic of K and also allows K to be assumed algebraically closed in part 2, whereas the assertion about unique factorisation in part 3 depends on the ground field; in characteristic 2, the displayed equation is even a square. This edition preserves the exercise as available and does not introduce new hypotheses.

Exercise 21.12

Let K be a field. A *power series in one variable* over K is a formal expression of the form

$$a_0 + a_1T + a_2T^2 + a_3T^3 + \cdots, \qquad a_i \in K.$$

Thus infinitely many coefficients a_i may be non-zero. Define a ring structure on the set of all power series that extends the ring structure on the polynomial ring in one variable. Show that this ring is a discrete valuation ring.

A module with no torsion elements other than 0 is called *torsion-free*.

Exercise 21.13

Show that every finitely generated torsion-free module M over a discrete valuation ring is free.

Public Solutions and Coverage of Worksheet 21

At the frozen revision boundary, the source provides exactly three public solutions among the 13 exercises: those for Exercises 21.3, 21.9, and 21.10. The exercise map and candidate evidence record negative results for Exercises 21.1–21.2, 21.4–21.8, and 21.11–21.13. The absence of a public solution page is not replaced by an invented solution.

Solution to Exercise 21.3

Let the maximal ideal of R be

$$\mathfrak{m} = (\pi).$$

The field of fractions of R is

$$Q(R) = R_\pi,$$

and every non-zero element of this field has the form

$$u\pi^n, \qquad u \in R^\times, \quad n \in \mathbb{Z}.$$

Let

$$R \subset T \subseteq Q(R).$$

Then there is an element

$$u\pi^n \in T$$

with $n < 0$. But we then also have $\pi^n \in T$ and

$$\pi^{-1} = \pi^{-n-1}\pi^n \in T.$$

Thus $T = Q(R)$.

Back to Exercise 21.3.

Solution to Exercise 21.9

We first show that R is a subring of the field K. We have

$$0 \in R.$$

Since ν is a group homomorphism, we must have

$$\nu(1) = 0.$$

For elements $f, g \in R$, we have

$$\nu(f), \nu(g) \geq 0,$$

and, since ν is a group homomorphism,

$$\nu(f \cdot g) = \nu(f) + \nu(g) \geq 0,$$

as well as, by hypothesis,

$$\nu(f+g) \geq \min\{\nu(f), \nu(g)\} \geq 0.$$

Thus R is closed under multiplication and addition. Furthermore,

$$\nu(-1) + \nu(-1) = \nu((-1)^2) = \nu(1) = 0,$$

so $\nu(-1) = 0$ and $-1 \in R$. Hence R is also closed under taking negatives and is a commutative ring.

Next, R must be a local ring. We claim that

$$\mathfrak{m} := \{f \in K^\times \mid \nu(f) \geq 1\} \cup \{0\} \subseteq R$$

is its only maximal ideal. This set contains 0, and since

$$\nu(f+g) \geq \min\{\nu(f), \nu(g)\} \geq 1,$$

it is closed under addition. For $f \in \mathfrak{m}$ and $g \in R$, we have

$$\nu(f) \geq 1 \qquad \text{and} \qquad \nu(g) \geq 0,$$

and hence

$$\nu(gf) = \nu(g) + \nu(f) \geq 1,$$

so the set is closed under multiplication by elements of R. Thus $\mathfrak{m}$ is an ideal.

The complement $R \setminus \mathfrak{m}$ consists exactly of the elements $h \in K$ with

$$\nu(h) = 0.$$

For such an element,

$$\nu(h^{-1}) = -\nu(h) = 0,$$

so $h^{-1} \in R$. All elements of $R \setminus \mathfrak{m}$ are therefore units. Consequently, $\mathfrak{m}$ is a maximal ideal.

It remains to show that R is a discrete valuation ring. Take $p \in K$ with

$$\nu(p) = 1.$$

Such an element exists because ν is assumed surjective. We show that p is a prime element. In general, for $x, y \in R$, the element y is a multiple of x precisely when

$$\nu(y) \geq \nu(x),$$

since this condition is equivalent to $y/x \in R$. Now suppose that $p \mid xy$ for $x, y \in R$. Then

$$1 = \nu(p) \leq \nu(xy) = \nu(x) + \nu(y).$$

Thus $\nu(x) \geq 1$ or $\nu(y) \geq 1$, so one of x and y is a multiple of p. Hence p is a prime element.

By the same argument, every non-zero element $x \in R$ with

$$n = \nu(x)$$

is associated to p^n. Thus R is a principal ideal domain with exactly the ideals

$$0 \qquad \text{and} \qquad (p^n), \quad n \in \mathbb{N}.$$

> **Editorial note - domain of the valuation.** The source defines ν only on $K^\times$, but in the solution it evaluates $\nu(f)$, $\nu(f+g)$, $\nu(gf)$, and $\nu(xy)$ for elements stated only to belong to R, although $0 \in R$. This edition preserves the source proof without adding the convention $\nu(0) = \infty$ or separating the zero cases.

Back to Exercise 21.9.

Solution to Exercise 21.10

First, R is a Noetherian local ring which, by the source fact about components at a smooth point, is an integral domain. Its only prime ideals are therefore the zero ideal and the maximal ideal $\mathfrak{m}_P$. We shall show that this maximal ideal is principal.

We may assume that P is the origin and write F as

$$F = F_d + \cdots + F_1$$

with $F_1 \neq 0$. Such a form exists because P is smooth. By a change of variables, we can arrange that

$$F_1 = Y.$$

In F, we can collect the isolated powers of X, namely the monomials not involving Y, and factor Y out of the remaining terms. The equation $F = 0$ can then be written as

$$Y(1+G) = XH(X),$$

where

$$G \in (X, Y).$$

The element $1 + G$ is a unit in $K[X,Y]_{(X,Y)}$, and therefore also in the local ring of the curve at the origin,

$$R = K[X,Y]_{(X,Y)}/(F).$$

Thus in R we have the relation

$$Y = \frac{H}{1+G} X.$$

The maximal ideal in the local ring R is therefore generated by X alone. By Theorem 21.8, R is a discrete valuation ring.

Back to Exercise 21.10.

Frozen negative results

There is no public solution page at the frozen revision for Exercises 21.1, 21.2, 21.4, 21.5, 21.6, 21.7, 21.8, 21.11, 21.12, or 21.13. This statement records the candidate checks; it does not assert that mathematical solutions do not exist.

Lecture 22: The Divisor Class Group

Weil divisors

We call an irreducible closed subset $Y \subset X$ of codimension 1 in an integral scheme X a *prime divisor.* If X is normal and Noetherian, the local ring $\mathcal{O}_{X,\eta}$ at the generic point η of Y is a discrete valuation ring. Thus every element

$$f \neq 0$$

of the function field $K(X)$ has a well-defined order along Y, which we denote by $\mathrm{ord}_Y(f)$. If π denotes a uniformiser, that is, a generator of the maximal ideal, in the discrete valuation ring $\mathcal{O}_{X,\eta}$, we can write

$$f = u\pi^n$$

with a unit u of that ring and $n \in \mathbb{Z}$. This exponent n is called the order of f along Y. Positive order means a zero, whereas negative order means a pole. If

$$U = \mathrm{Spek}(R) \subseteq X$$

is an affine open subset with $U \cap Y \neq \emptyset$, then Y corresponds to a prime ideal $\mathfrak{p}$ of height 1 in R, and the local ring satisfies

$$\mathcal{O}_{X,\eta} = R_{\mathfrak{p}}.$$

Definition 22.1: principal divisor

Let X be a normal Noetherian integral scheme with function field K, and let $f \in K$, $f \neq 0$. The formal sum

$$\mathrm{div}(f) = \sum_{Y \text{ prime divisor}} \mathrm{ord}_Y(f) \cdot Y,$$

where $\mathrm{ord}_Y(f)$ denotes the order of f in the local ring at Y, is called the *principal divisor* defined by f.

The principal divisor thus describes the zeros and poles of the function f. We first show that a principal divisor is a finite sum.

Lemma 22.2: a principal divisor has finite support

Let X be a normal Noetherian integral scheme with function field K, and let $f \in K$, $f \neq 0$. There are only finitely many prime divisors Y with

$$\mathrm{ord}_Y(f) \neq 0.$$

Proof Let $U \subseteq X$ be a non-empty affine open subset with

$$f \in R = \Gamma(U, \mathcal{O}_X).$$

Since the generic point of X belongs to U, the prime divisors not meeting U are irreducible components of $X \setminus U$. The set $X \setminus U$ is closed in X and hence Noetherian, so it has only finitely many components. We therefore need only consider prime divisors meeting U. Their generic points correspond to prime ideals of height 1 in R. We have

$$\mathrm{ord}_Y(f) = \mathrm{ord}_{\mathfrak{p}}(f) \geq 0,$$

and this is positive only if $f \in \mathfrak{p}$. The prime ideals $\mathfrak{p}$ of height 1 containing f are the minimal prime ideals of $R/(f)$; since the ring is Noetherian, there are only finitely many of them.

Definition 22.3: Weil divisor

Let X be a normal Noetherian integral scheme. A formal sum

$$\sum_Y n_Y \cdot Y,$$

where Y ranges over the prime divisors of X and only finitely many n_Y are non-zero, is called a *Weil divisor* on X.

A Weil divisor is an arbitrary prescription for the "theoretically possible" zeros and poles of a rational function. Such a prescription need not, however, be realised by a function. A divisor whose coefficients all satisfy $a_Y \geq 0$ is called *effective*. On an irreducible normal (hence smooth) curve X, a prime divisor is simply a closed point. In this case, a Weil divisor is a finite sum

$$\sum_{P \in X} n_P \cdot P.$$

Editorial note - coefficients and smoothness. The coefficients of a Weil divisor are integers. The source leaves this implicit and calls a normal curve smooth: a normal Noetherian curve is regular, but smoothness over its ground field additionally holds, for example, when that field is perfect.

Definition 22.4: Weil divisor group

Let X be a normal Noetherian integral scheme. The group of all Weil divisors, with componentwise addition, is called the *Weil divisor group* of X. It is denoted by

$$\mathrm{Div}(X).$$

Lemma 22.5: principal divisors define a group homomorphism

Let X be a normal Noetherian integral scheme with function field K. The map

$$K^\times \longrightarrow \mathrm{Div}(X), \qquad f \longmapsto \mathrm{div}(f),$$

is a group homomorphism.

Proof By Lemma 22.2, the principal divisor of f is indeed a Weil divisor. For a fixed prime divisor Y with its associated discrete valuation ring $\mathcal{O}_Y$, the homomorphism property follows from Lemma 21.7 (1).

The divisor class group

Definition 22.6: divisor class group

Let X be a normal Noetherian integral scheme with function field K. The quotient group

$$\mathrm{DKG}(X) = \mathrm{Div}(X)/\,\mathrm{HDiv}(X)$$

is called the *divisor class group* of X.

For a normal Noetherian integral domain R, similarly,

$$\mathrm{DKG}(R) = \mathrm{DKG}(\mathrm{Spek}(R))$$

is called the divisor class group of the ring R. In number theory, when R is the ring of integers in a finite extension of $\mathbb{Q}$, this group is also called the *ideal class group*. Divisors defining the same divisor class are called *linearly equivalent.*

Theorem 22.7: a divisor criterion for unique factorisation

Let R be a normal Noetherian integral domain, and let $\mathrm{DKG}(R)$ denote the divisor class group of R. The following statements are equivalent.

1. R is a unique factorisation domain.
2. Every prime ideal of height 1 is principal.
3. Every divisor is principal.
4. $\mathrm{DKG}(R) = 0$.

Proof Suppose (1) holds and $\mathfrak{p}$ is a prime ideal of height 1. There is an element $f \in \mathfrak{p}$, $f \neq 0$. It has a prime factorisation

$$f = p_1 \cdots p_n.$$

Since $\mathfrak{p}$ is prime, we must have $p_i \in \mathfrak{p}$ for some i. The height condition then gives

$$(p_i) = \mathfrak{p}.$$

Now suppose every prime ideal of height 1 is principal. Writing

$$\mathfrak{p} = (p),$$

we have the divisor relation

$$\mathrm{div}(p) = 1 \cdot \mathfrak{p},$$

since p belongs to no other prime ideal of height 1, and in $R_{\mathfrak{p}}$ the element p also generates $\mathfrak{p}R_{\mathfrak{p}}$. Thus the generators of the divisor class group are principal divisors, so all divisors are principal. The equivalence of (3) and (4) is clear.

> **Editorial note - group name in the source.** At this step, the source prints *Divisorenklassengruppe* (divisor class group), although the argument uses prime divisors as generators. This edition preserves the printed group name and does not silently replace it with the divisor group.

Next suppose every divisor is principal. For every prime ideal $\mathfrak{p}$ of height 1, there is an element $f \in Q(R)$, $f \neq 0$, with

$$\mathrm{div}(f) = 1 \cdot \mathfrak{p}.$$

Since this principal divisor is non-negative, Theorem 21.12 gives $f \in R$. Thus among prime ideals of height 1, f belongs only to $\mathfrak{p}$. Let

$$\mathfrak{p} = (g_1, \dots, g_n).$$

Then

$$\mathrm{div}(g_i) \geq \mathrm{div}(f),$$

so $g_i/f \in R$, that is, $g_i \in (f)$, and hence $\mathfrak{p} = (f)$.

Finally, suppose (2) holds and take $f \in R$, $f \neq 0$. Let $\mathfrak{p}_1, \dots, \mathfrak{p}_s$ be the minimal prime ideals containing f. By Krull's principal ideal theorem, they all have height 1. Let $\mathfrak{p}_i = (p_i)$ with prime element p_1. We have

$$\mathrm{div}(f) = \sum_{i=1}^{s} n_i \mathfrak{p}_i.$$

The element $\prod_{i=1}^s p_i^{n_i}$ has the same principal divisor. Therefore the quotient

$$f \Big/ \prod_{i=1}^{s} p_i^{n_i}$$

is a unit, and

$$f = u \prod_{i=1}^{s} p_i^{n_i}$$

for a unit u. Thus R is a unique factorisation domain.

> **Editorial note - subscript on the prime element.** In the final step, the source writes $\mathfrak{p}_i = (p_i)$ but then refers to the prime element p_1. This edition preserves the printed subscript; the subsequent product again uses p_i.

Example 22.8: the divisor class group of projective space

We shall describe the Weil divisors and the divisor class group of projective space $\mathbb{P}_K^d$ over a field K, with $d \geq 1$. Consider the disjoint decomposition

$$\mathbb{P}_K^d = D_+(X_0) \cup V_+(X_0) = \mathbb{A}_K^d \cup \mathbb{P}_K^{d-1}.$$

In other words, we fix the hyperplane

$$H = V_+(X_0) \cong \mathbb{P}_K^{d-1}$$

"at infinity". A prime divisor of projective space either equals the hyperplane on the right, or meets the affine space on the left non-trivially and can be viewed as a prime ideal of height 1 in the polynomial ring

$$K\left[\frac{X_1}{X_0}, \dots, \frac{X_d}{X_0}\right].$$

Every function f in the function field can be written uniquely, up to scaling and cancellation of common factors, as

$$f = \frac{P}{Q},$$

with

$$P, Q \in K\left[\frac{X_1}{X_0}, \dots, \frac{X_d}{X_0}\right].$$

Using prime factorisations of P and Q, we can write directly

$$f = \prod_{i=1}^{n} c P_i^{\nu_i},$$

with a constant $c \neq 0$ and $\nu_i \in \mathbb{Z}$, and read off the principal divisor of f as far as components in affine space are concerned. The order of f "at infinity", at $V_+(X_0)$, is obtained as follows. The local ring at this prime divisor is

$$\begin{aligned} K[X_0, X_1, \dots, X_n]_{((X_0))} &= \left(K[X_0, X_1, \dots, X_n]_{(K[X_0,X_1,\dots,X_n] \setminus (X_0)) \cap \{\text{homogeneous elements}\}}\right)_0 \\ &= K\left(\frac{X_2}{X_1}, \dots, \frac{X_d}{X_1}\right)\left[\frac{X_0}{X_1}\right]_{\left(\frac{X_0}{X_1}\right)}. \end{aligned}$$

We rewrite P (and similarly Q or f) by replacing each X_i/X_0 with

$$\frac{X_i}{X_1}\cdot\frac{X_1}{X_0}.$$

We view this expression as a rational function in the single variable X_0/X_1 over the field

$$K\left(\frac{X_2}{X_1},\dots,\frac{X_d}{X_1}\right).$$

Its degree in X_0/X_1, which is typically negative, is its order. For example,

$$\begin{aligned}
P &= \frac{X_1}{X_0}+\left(\frac{X_2}{X_0}\right)^3\\
&= \frac{X_1}{X_0}+\left(\frac{X_2}{X_1}\right)^3\left(\frac{X_1}{X_0}\right)^3\\
&= \left(\left(\frac{X_0}{X_1}\right)^2+\left(\frac{X_2}{X_1}\right)^3\right)\left(\frac{X_0}{X_1}\right)^{-3},
\end{aligned}$$

so the order is -3.

> **Editorial note - factorisation and order.** In the source product, the constant belongs outside the product: $f=c\prod_i P_i^{\nu_i}$, for $f\neq 0$. The source's "degree" at infinity means the valuation in X_0/X_1, namely the least Laurent exponent for a polynomial expressed in that variable, not the usual degree of a rational function. For $f=P/Q$, the order is $\deg Q-\deg P$.

Since the polynomial ring is a unique factorisation domain, on affine space every Weil divisor equals a principal divisor. Thus every Weil divisor is linearly equivalent to a divisor of the form

$$nV_+(X_0),\qquad n\in\mathbb{Z}.$$

The class of $V_+(X_0)$ is also called the *hyperplane class*. For $n\neq 0$, such a divisor is not principal: such a principal divisor would be trivial on affine space and would therefore have to come from a constant, whereas a constant also has order 0 at infinity. Thus the divisor class group of projective space is $\mathbb{Z}$, and any hyperplane can be chosen as a generator.

> **Editorial note - indices in the local-ring display.** The source uses d for the dimension of projective space but writes $K[X_0,X_1,\dots,X_n]$ in the local-ring display. This edition preserves both indices as printed.

The divisor class group and the Picard group

We now discuss the relationship between divisors and invertible subsheaves of the function-field sheaf $\mathcal{K}$, and between the divisor class group and the Picard group. An invertible subsheaf

$$\mathcal{L}\subseteq\mathcal{K}$$

defines, for every point $x\in X$, a free $\mathcal{O}_{X,x}$-submodule of rank 1,

$$\mathcal{L}_x\subseteq\mathcal{K}_x=K.$$

If $\mathcal{O}_{X,x}$ is a discrete valuation ring with uniformiser π, as happens on a normal scheme at the generic point of every prime divisor, then

$$\mathcal{L}_x=\pi^n\mathcal{O}_{X,x}$$

for a unique $n\in\mathbb{Z}$. We denote this number by $\operatorname{ord}_Y(\mathcal{L})$ when Y is the prime divisor in question.

Theorem 22.9: invertible subsheaves and Weil divisors

Let X be a locally factorial Noetherian integral scheme. The invertible $\mathcal{O}_X$-submodules of the constant function-field sheaf $\mathcal{K}$ and the Weil divisors correspond to one another via

$$\mathcal{L} \longmapsto \sum_Y \operatorname{ord}_Y(\mathcal{L})$$

and

$$D = \sum_Y a_Y Y \longmapsto \mathcal{L}_D,$$

where

$$\mathcal{L}_D(U) = \{f \in K \mid \operatorname{ord}_Y(f) \geq D \text{ for all } Y \in U\}$$

for an open subset $U \subseteq X$. These correspondences are compatible with the group structures; trivial subsheaves correspond to principal divisors. Invertible ideals

$$\mathcal{L} \subseteq \mathcal{O}_X \subseteq \mathcal{K}$$

correspond to effective divisors.

> **Editorial note - two source displays.** In the first map, the source omits the factor Y after $\operatorname{ord}_Y(\mathcal{L})$, although the proof later writes $D = \sum_Y \operatorname{ord}_Y(\mathcal{L})Y$. In the definition of $\mathcal{L}_D(U)$, the source compares $\operatorname{ord}_Y(f)$ directly with the divisor D, without naming its coefficient. Both displays are preserved as they stand. The intended first sum is $\sum_Y \operatorname{ord}_Y(\mathcal{L})Y$. For non-empty U, the intended section set is $\{0\} \cup \{f \in K^\times \mid \operatorname{ord}_Y(f) \geq a_Y \text{ for all } Y \text{ meeting } U\}$; the zero section must be included, since the order was defined only for non-zero functions. On the empty open set there is the unique zero section.

Proof There is a finite affine open cover

$$X = \bigcup_{i \in I} U_i$$

with

$$\mathcal{L} = (f_i)\mathcal{O}_X|_{U_i},$$

where $f_i \in K$ and $f_i \neq 0$. By Lemma 22.2, for each i there are only finitely many irreducible Weil divisors in U_i satisfying

$$\operatorname{ord}_Y(f_i) \neq 0.$$

Consequently,

$$D = \sum_Y \operatorname{ord}_Y(\mathcal{L})Y$$

is indeed a Weil divisor.

Conversely, let D be a Weil divisor and $\mathcal{L}$ the associated subsheaf of the constant sheaf of the function field. We must show that $\mathcal{L}$ is invertible. Take a point $x \in X$ and an affine open neighbourhood

$$x \in U \subseteq X.$$

By hypothesis, the local ring $\mathcal{O}_{X,x}$ is a unique factorisation domain. By Theorem 22.7, the divisor D_x consisting of all irreducible components of D passing through x is principal. By removing the components of D not passing through x, we can replace U with a smaller affine neighbourhood V of x on which the divisor is principal. There we have

$$D|_V = \mathrm{div}(f)|_V$$

for some $f \in K$, and then

$$\mathcal{L}_D|_V = (f)\mathcal{O}_X|_V.$$

We must now show that these correspondences are inverse to one another. Start with an invertible subsheaf and use the notation above. On U_i we have

$$D|_{U_i} = \mathrm{div}(f_i)|_{U_i}.$$

Hence for $g \in K$, membership

$$g \in \{f \in K \mid \mathrm{ord}_Y(f) \geq D \text{ for all } Y \in U\}$$

holds precisely when the relation between principal divisors

$$\mathrm{div}(g) \geq \mathrm{div}(f_i)$$

holds on U. By Theorem 21.12, this is equivalent to

$$g \in f \cdot \Gamma(X, \mathcal{O}_X).$$

If we start with a Weil divisor, it locally equals a principal divisor. An element of the function field with that principal divisor then locally generates the associated invertible sheaf, and the same element is used to recover the associated divisor.

> **Editorial note - notation at the end of the source proof.** After using the generator f_i on U_i, the source prints $g \in f \cdot \Gamma(X, \mathcal{O}_X)$. This edition does not silently change the index or the open set. On U_i, the intended conclusion is $g \in f_i\Gamma(U_i, \mathcal{O}_X)$. In the preceding shrinking argument, one removes the support of $D - \mathrm{div}(f)$, whose components avoid x, not just components of D avoiding x; this also removes any extra zeros or poles of the chosen f.

In the correspondence above, ideals correspond to effective divisors, and the principal ideal (f) corresponds to the principal divisor $\mathrm{div}(f)$. There are also good reasons to modify this correspondence by inserting a minus sign. With that convention, an effective divisor corresponds to a global section of the associated invertible sheaf.

Theorem 22.10: the divisor class group and the Picard group

Let X be a locally factorial Noetherian integral scheme. Then the divisor class group of X agrees with the Picard group of X.

Proof This follows from Lemma 20.6 and Theorem 22.9.

Corollary 22.11: the case of smooth schemes

Let X be a smooth scheme over an algebraically closed field K. Then the divisor class group of X agrees with the Picard group of X.

Proof On a smooth scheme, the local rings are regular by Theorem 18.16, and they are unique factorisation domains by Theorem 25.12 of *Singularity Theory (Osnabrück 2019).* Thus the statement follows from Theorem 22.10.

Theorem 22.12: the Picard group of projective space

The Picard group of projective space $\mathbb{P}^d_K$, with $d \geq 1$, over a field K is $\mathbb{Z}$. The invertible sheaves on projective space are represented by the twisted structure sheaves

$$\mathcal{O}_{\mathbb{P}^d_K}(\ell), \qquad \ell \in \mathbb{Z}.$$

Proof This follows from Theorem 22.10 and Example 22.8. Under the explicit correspondence in Theorem 22.9, the negative of the hyperplane class corresponds to the tautological bundle $\mathcal{O}_{\mathbb{P}^d_K}(1)$.

> **Editorial note - bundle name.** The negative hyperplane class does correspond to $\mathcal{O}(1)$ under the sign convention of Theorem 22.9. With the usual twisting notation also used in Exercise 22.13, however, the tautological line subbundle is $\mathcal{O}(-1)$; $\mathcal{O}(1)$ is its dual. The source's bundle name should be read with this distinction.

Worksheet 22: The Divisor Class Group

The star marks Exercise 22.19. The candidate-title checks recorded by QA found that this is the only exercise with a public solution page. The other nineteen candidate titles do not exist; this edition does not invent new solutions.

Exercise 22.1

Determine the principal divisor of

$$\frac{1000}{333}$$

on Spek($\mathbb{Z}$).

Exercise 22.2

Determine the principal divisor of

$$f = (t-3)^2(t-1)^{-5}t^2(t+2)^{-1}$$

on the projective line

$$\mathbb{P}^1_K = \text{Proj}(K[X,Y]).$$

Exercise 22.3

Determine the principal divisor of

$$f = \frac{t}{t^2+1}$$

on the projective line

$$\mathbb{P}^1_K = \text{Proj}(K[X,Y]), \qquad t = \frac{Y}{X},$$

for the fields $K = \mathbb{R}$ and $K = \mathbb{C}$.

Exercise 22.4

Consider the projective line

$$\mathbb{P}^1_K = \text{Proj}(K[X, Y])$$

over a field K, together with the affine line

$$\mathbb{A}^1_K \subseteq \mathbb{P}^1_K = D_+(X) \cup \{\infty\}$$

whose ring of global sections is

$$K\left[\frac{Y}{X}\right] = K[t].$$

Prove the following statements.

1. The principal divisor of a polynomial $P \in K[t]$ has no negative order (no pole) in $\mathbb{A}^1_K$.
2. The order of a polynomial $P \in K[t]$ at ∞ is the negative of the degree of P.
3. Let
$$D = \sum_P n_P \cdot P$$
and let K be algebraically closed. Then D is a principal divisor if and only if
$$\sum_P n_P = 0.$$

Editorial note - non-zero polynomial. Parts 1 and 2 require $P \neq 0$: the source defines principal divisors and finite orders only for non-zero functions.

Exercise 22.5

Let

$$\mathbb{P}^n_K = \text{Proj}(K[X_0, X_1, \dots, X_n])$$

be projective space over a field K. Show that effective Weil divisors on $\mathbb{P}^n_K$ correspond to normalised homogeneous polynomials

$$P \in K[X_0, X_1, \dots, X_n].$$

Exercise 22.6

Show that on projective space $\mathbb{P}^d_K$ over a field, any two hyperplanes

$$H_1 = V_+(a_0X_0 + a_1X_0 + \dots + a_dX_d)$$

and

$$H_2 = V_+(b_0X_0 + b_1X_0 + \dots + b_dX_d)$$

are linearly equivalent.

Editorial note - repeated variable in the source. Both source equations print X_0 in their first two terms. This edition preserves these formulas and does not silently change the second term.

Exercise 22.7

Let

$$V = V_+(F) \subseteq \mathbb{P}^d_K$$

be an irreducible hypersurface of degree d in projective space over a field, viewed as an element of the divisor class group. Show that V is linearly equivalent to dH, where H denotes the class of a hyperplane.

Exercise 22.8

Show that the set of all hyperplanes in a projective space itself forms a projective space of the same dimension.

Exercise 22.9

Let $P_0, P_1, \dots, P_N$ be (closed) points of projective space $\mathbb{P}^N_K$ not all contained in any one hyperplane. Let $0 \leq r \leq N$.

1. Show that $P_0, P_1, \dots, P_r$ are not contained in any projective subspace of dimension $< r$.
2. Show that the set of all hyperplanes containing the points $P_0, P_1, \dots, P_r$ forms a projective subspace of dimension $N - 1 - r$ in the space of all hyperplanes.

   Editorial note - rational points. The dimension assertion requires the P_i to be K-rational points, as is automatic over an algebraically closed field. Arbitrary closed points over a general field may impose more than one linear condition. For $r = N$, the set is empty, interpreted as projective dimension -1.

Exercise 22.10

Let X be a normal Noetherian integral scheme, and let $Z \subset X$ be a closed subset of codimension ≥ 2. Show that the divisor class groups of X and $X \setminus Z$ agree.

Exercise 22.11

Let X be a normal Noetherian integral scheme, and let $U \subset X$ be an open subset. Show that omitting the prime divisors not meeting U gives a surjective group homomorphism

$$\mathrm{Div}(X) \longrightarrow \mathrm{Div}(U).$$

Show that principal divisors map to principal divisors, so that there is a surjective group homomorphism

$$\mathrm{DKG}(X) \longrightarrow \mathrm{DKG}(U).$$

Exercise 22.12

Let X be a normal Noetherian integral scheme, and let $x \in X$ be a point. Show that omitting the prime divisors not passing through x gives a surjective group homomorphism

$$\mathrm{Div}(X) \longrightarrow \mathrm{Div}(\mathcal{O}_{X,x}).$$

Show that principal divisors map to principal divisors, so that there is a surjective group homomorphism

$$\mathrm{DKG}(X) \longrightarrow \mathrm{DKG}(\mathcal{O}_{X,x}).$$

Exercise 22.13

Let

$$D = \sum_Y a_Y \cdot Y = \sum_j a_j \cdot Y_j = \sum_j a_j \cdot V_+(G_j)$$

be a Weil divisor of projective space $\mathbb{P}^d_K$, where the prime divisors involved are described by homogeneous prime elements

$$G_j \in K[X_0, X_1, \dots, X_d].$$

Consider the polynomial

$$G := \prod_j G_j^{a_j}$$

of degree

$$\delta = \sum_j a_j \operatorname{grad}(G_j).$$

Show that its associated invertible subsheaf $\mathcal{L}_D$ of the function-field sheaf, in the sense of Theorem 22.9, agrees with the realisation of the twisted structure sheaf

$$\mathcal{O}_{\mathbb{P}^d_K}(-\delta)$$

by multiplication by G from Example 20.9.

> **Editorial note - the word "polynomial" in the source.** The source calls $G := \prod_j G_j^{a_j}$ a polynomial, although the coefficients a_j of a general Weil divisor may be negative. This edition preserves the source terminology and formula without assuming the divisor is effective.

In the following exercises, for a divisor D we work with

$$\mathcal{O}_X(D) = -\mathcal{L}_D.$$

Thus, for an open subset $U \subseteq X$,

$$\mathcal{O}_X(D)(U) = \{f \in K \mid \operatorname{ord}_Y(f) \geq -D \text{ for all } Y \in U\}.$$

> **Editorial note - the convention printed in the source.** The source writes $\mathcal{O}_X(D) = -\mathcal{L}_D$ and compares $\operatorname{ord}_Y(f)$ directly with $-D$, without naming the divisor coefficient. This edition preserves that notation. Here the minus sign means the inverse invertible sheaf, $\mathcal{O}_X(D) = \mathcal{L}_{-D} \cong \mathcal{L}_D^\vee$, not pointwise negation of its sections. The inequality means $\operatorname{ord}_Y(f) \geq -a_Y$ for prime divisors meeting U; the zero section is included separately, as in the note to Theorem 22.9.

Exercise 22.14

Let X be a locally factorial projective integral scheme over an algebraically closed field K, and let D be a Weil divisor on X with associated sheaf $\mathcal{O}_X(D)$. Show that there is a natural correspondence between effective Weil divisors linearly equivalent to D and non-trivial global sections of $\mathcal{O}_X(D)$, where sections are identified if they differ only by scaling.

Exercise 22.15

Let X be a locally factorial Noetherian integral scheme and $\mathcal{L}$ an invertible subsheaf of the constant function-field sheaf. Let

$$s \in \Gamma(X, \mathcal{L})$$

be a non-trivial section. Show that the zero locus of s, namely

$$Z(s) = X \smallsetminus X_s,$$

is naturally an effective Weil divisor D on X such that

$$\mathcal{L} \cong \mathcal{O}_X(D).$$

Exercise 22.16

Let

$$F = F_1^{a_1} \cdots F_r^{a_r}$$

be the prime factorisation of a homogeneous polynomial

$$F \in K[X_0, X_1, \dots, X_d]$$

of degree e over an algebraically closed field K, with homogeneous prime polynomials F_j, and let

$$D = \sum_{j=1}^{n} a_j V_+(F_j)$$

be the associated Weil divisor on projective space $\mathbb{P}_K^d$. Prove the following statements.

1. Every effective Weil divisor on projective space can be represented in this form, uniquely up to scaling.
2. Set-theoretically,
$$V_+(F) = \bigcup_{j=1}^{n} V_+(F_i).$$
3. Let $C \subseteq \mathbb{P}_K^d$ be a smooth projective curve not contained in $V_+(F)$. Then D induces a Weil divisor $D|_C$ on the curve C by taking, at each point $P \in C$, the order of F in $\mathcal{O}_{C,P}$.
4. The restricted invertible sheaf $\mathcal{O}_{\mathbb{P}_K^d}(e)|_C$ is isomorphic to the invertible sheaf on C associated to $D|_C$. Thus
$$\mathcal{O}_{\mathbb{P}_K^d}(D)|_C = \mathcal{O}_C(D|_C).$$
5. Linearly equivalent divisors on projective space induce linearly equivalent divisors on the curve.

   Editorial note - indices in the source. The source factorisation uses r factors, whereas the sum and union use the bound n; in the union, the running index is j but the term is printed as F_i. All these indices are preserved as in the source. In part 3, "the order of F" means the order of the local equation F/X_i^e on a chart with $X_i(P) \neq 0$. The homogeneous polynomial itself is a section of $\mathcal{O}(e)$, not a function in $\mathcal{O}_{C,P}$. Changing such a chart multiplies the local equation by a unit, so its order is well defined.

Exercise 22.17

Let D be an effective Weil divisor in projective space $\mathbb{P}^d_K$ with at least one positive component. Show that D has non-empty intersection with every projective curve $C \subseteq \mathbb{P}^d_K$.

Use the fact that $D_+(f)$ is affine and a projective curve is not affine.

In particular, two curves in the projective plane always have non-empty intersection. For smooth curves, one can also use Exercise 22.16. This property by no means holds on all projective surfaces, as the following examples show.

Exercise 22.18

Show that the projective surface

$$V_+(XY - ZW) \subseteq \mathbb{P}^3_K$$

contains disjoint lines, viewed as objects in projective space.

Exercise 22.19 ★

Show that the projective surface

$$V_+(X^3 + Y^3 - Z^3 - W^3) \subseteq \mathbb{P}^3_K$$

of degree 3 over an algebraically closed field K of characteristic $\neq 3$ contains disjoint lines, viewed as objects in projective space.

Exercise 22.20

Let C_1 and C_2 be two concentric circles in $\mathbb{R}^2$ centred at $(0, 0)$. Determine their intersection points, viewing the circles as projective curves, in the projective plane $\mathbb{P}^2_{\mathbb{C}}$.

Public Solutions and Coverage of Worksheet 22

The candidate-title checks recorded by QA, based on the sequence of 20 exercises in the frozen worksheet revision, found exactly one public solution page: the solution to Exercise 22.19. The results were negative for Exercises 22.1–22.18 and 22.20. The absence of a public solution page is not replaced by an invented solution.

Solution to Exercise 22.19

We write the defining equation as

$$\begin{aligned} F &= X^3 - Z^3 + Y^3 - W^3 \\ &= (X-Z)(X-\zeta Z)(X-\zeta^2 Z) + (Y-W)(Y-\zeta W)(Y-\zeta^2 W), \end{aligned}$$

where ζ is a third root of unity. Thus, for example,

$$F \in (X-Z, Y-W)$$

and

$$F \in (X-\zeta Z, Y-\zeta W).$$

Hence

$$L_1 = V_+(X-Z, Y-W), \qquad L_2 = V_+(X-\zeta Z, Y-\zeta W) \subseteq V_+(F).$$

As intersections of two distinct projective planes, L_1 and L_2 are lines. Their intersection is

$$\begin{aligned} L_1 \cap L_2 &= V_+(X-Z, Y-W),\ L_2 \cap V_+(X-\zeta Z, Y-\zeta W) \\ &= V_+(X-Z, Y-W, X-\zeta Z, Y-\zeta W) \\ &= V_+(X-Z, X-\zeta Z, Y-W, Y-\zeta W) \\ &= V_+(X, Z, Y, W) \\ &= \emptyset. \end{aligned}$$

Editorial note - first line of the intersection chain. The first line of the source display prints a comma, repeats L_2, and then writes a second intersection. This edition preserves that line as in the source; the subsequent lines give the combined ideal yielding the empty intersection. The root ζ must be primitive, so $\zeta \neq 1$; otherwise the two lines coincide and the factorisation is invalid in characteristic different from 3. Such a root exists under the exercise's hypotheses.

Negative candidate-check results

Those candidate checks found no public solution page for Exercises 22.1, 22.2, 22.3, 22.4, 22.5, 22.6, 22.7, 22.8, 22.9, 22.10, 22.11, 22.12, 22.13, 22.14, 22.15, 22.16, 22.17, 22.18, or 22.20. This statement records the candidate checks; it does not assert that mathematical solutions do not exist.

Lecture 23: Injective Modules

Injective modules

Definition 23.1: injective module

Let R be a commutative ring. An R-module I is called *injective* if, for every R-module M, every submodule $N \subseteq M$, and every R-module homomorphism

$$\varphi : N \longrightarrow I$$

there is an extension

$$\widetilde{\varphi} : M \longrightarrow I.$$

Over a field, every vector space is injective. Indeed, every vector subspace of a vector space has a direct complement, and the linear map can be extended arbitrarily on that complement. For $R = \mathbb{Z}$, the situation is already more complicated.

Definition 23.2: divisible group

An abelian group G is called *divisible* if, for every $n \in \mathbb{N}_+$ and every $g \in G$, there is an $h \in G$ with

$$g = nh.$$

The group $\mathbb{Z}$ itself is not divisible. In contrast, $\mathbb{Q}$ is divisible as an abelian group, since for every $n \in \mathbb{N}_+$ the multiplication map

$$\mathbb{Q} \longrightarrow \mathbb{Q}, \qquad x \longmapsto nx,$$

is surjective: we can divide by n, which explains the name *divisible*.

Lemma 23.3: quotient groups of divisible groups

If D is a divisible group, every quotient group D/H is also divisible.

Proof Let $n \in \mathbb{N}_+$. For every $d \in D$ there is an $e \in D$ with $d = ne$. Then in D/H we also have

$$[d] = n[e].$$

Lemma 23.4: embedding in a divisible group

For every abelian group G, there is a divisible group D with $G \subseteq D$.

Proof We write

$$G = \mathbb{Z}^{(J)}/H$$

for a suitable index set J indexing a system of generators of G. The free abelian group $\mathbb{Z}^{(J)}$ embeds in the divisible group $\mathbb{Q}^{(J)}$. There is therefore an embedding

$$G \subseteq \mathbb{Q}^{(J)}/H,$$

and the group on the right is divisible by Lemma 23.3.

We state the following result without proof.

Lemma 23.5: divisible if and only if injective

An abelian group G is divisible if and only if G is injective.

Lemma 23.6: short exact sequences containing an injective module

Let I be an injective module over a commutative ring R. Every short exact sequence of R-modules

$$0 \longrightarrow I \longrightarrow B \longrightarrow C \longrightarrow 0$$

splits.

Proof The identity $\mathrm{Id}_I : I \to I$ has an extension $\varphi : B \to I$. This map gives the splitting.

Lemma 23.7: change of rings for injective modules

Let R be a commutative ring, S a commutative R-algebra, and I an injective R-module. Then the S-module

$$\mathrm{Hom}_R(S, I)$$

is also injective.

Proof Let $A \subseteq B$ be S-modules and let

$$\varphi : A \longrightarrow \mathrm{Hom}_R(S, I), \qquad a \longmapsto \varphi_a,$$

be an S-module homomorphism. Explicitly, this means that

$$\varphi_a(s) = \varphi_{as}(1).$$

View A and B as R-modules and consider the composite R-module homomorphism

$$A \xrightarrow{\varphi} \mathrm{Hom}_R(S, I) \xrightarrow{\theta \mapsto \theta(1)} I.$$

Since I is injective as an R-module, this composite has an R-linear extension

$$\widetilde{\varphi} : B \longrightarrow I.$$

We claim that the map

$$B \longrightarrow \mathrm{Hom}_R(S, I), \qquad b \longmapsto (s \longmapsto \widetilde{\varphi}(sb)),$$

is an S-module homomorphism. First, the composite map

$$S \xrightarrow{1 \mapsto b} B \xrightarrow{\widetilde{\varphi}} I$$

clearly belongs to $\mathrm{Hom}_R(S, I)$. The overall assignment is S-linear by the S-module structure on $\mathrm{Hom}_R(S, I)$. For $a \in A$ we have

$$\widetilde{\varphi}(sa) = \varphi_{as}(1) = \varphi_a(s),$$

so the map is indeed an extension.

Injective resolutions

Corollary 23.8: embedding a module in an injective module

For an R-module M over a commutative ring R, there is an injective module I with $M \subseteq I$.

Proof By Lemma 23.4, for the abelian group M there is a divisible group D and an embedding $M \subseteq D$. By Lemma 23.5, D is an injective $\mathbb{Z}$-module. Lemma 23.7 then says that the R-module

$$\operatorname{Hom}_{\mathbb{Z}}(R, D)$$

is also injective. The source displays the commutative diagram

$$\begin{array}{ccc} M & \longrightarrow & D \\ \downarrow & & \downarrow \\ \operatorname{Hom}_{\mathbb{Z}}(R, M) & \longrightarrow & \operatorname{Hom}_{\mathbb{Z}}(R, D). \end{array}$$

The left vertical map is given by

$$v \longmapsto (r \longmapsto rv),$$

and the bottom horizontal map is induced by the embedding $M \hookrightarrow D$. Both are injective R-module homomorphisms. Their composite gives the R-submodule

$$M \subseteq \operatorname{Hom}_{\mathbb{Z}}(R, D).$$

> **Editorial note - right vertical arrow in the source diagram.** The source also states that the right vertical arrow is given by $v \mapsto (r \mapsto rv)$. However, the stated data make D only a divisible abelian group, not an R-module, so that arrow is not defined without additional structure. The conclusion needs only the well-defined left vertical and bottom horizontal arrows above. This edition preserves the source diagram but uses the composite of those two arrows for the resulting embedding.

Definition 23.9: injective resolution

An *injective resolution* of an R-module M over a commutative ring R is an exact complex of R-modules

$$0 \longrightarrow M \longrightarrow I_0 \longrightarrow I_1 \longrightarrow I_2 \longrightarrow \cdots,$$

where I_n is injective for every $n \geq 0$.

Lemma 23.10: existence of injective resolutions

Every R-module M over a commutative ring R has an injective resolution.

Proof By Corollary 23.8, there is an injective module I_0 with $M \subseteq I_0$. Similarly, for the quotient module I_0/M there is an injective module I_1 with $I_0/M \subseteq I_1$, and so on.

Lemma 23.11: extending an initial homomorphism to complexes

Let L and M be R-modules over a commutative ring R. Let

$$0 \longrightarrow L \longrightarrow L_0 \longrightarrow L_1 \longrightarrow \cdots$$

be an exact complex,

$$0 \longrightarrow M \longrightarrow I_0 \longrightarrow I_1 \longrightarrow \cdots$$

an injective resolution, and

$$\varphi : L \longrightarrow M$$

an R-module homomorphism. Then there are R-module homomorphisms

$$\varphi_n : L_n \longrightarrow I_n$$

commuting with the homomorphisms in the two complexes.

Proof We prove the existence of the commuting homomorphisms by induction on n. Since $L \subseteq L_0$ and I_0 is injective, the homomorphism $L \to M \subseteq I_0$ has a commuting extension

$$\varphi_0 : L_0 \longrightarrow I_0.$$

This establishes the base case. Now suppose the homomorphisms through φ_n already exist. Consider the commutative diagram

$$\begin{array}{ccccc} L_{n-1} & \longrightarrow & L_n & \longrightarrow & L_{n+1} \\ \downarrow\varphi_{n-1} & & \downarrow\varphi_n & & \downarrow \\ I_{n-1} & \longrightarrow & I_n & \longrightarrow & I_{n+1}, \end{array}$$

where the right vertical arrow remains to be constructed. There is an injection

$$L_n / \operatorname{bild} L_{n-1} \longrightarrow L_{n+1}.$$

By commutativity, all of L_{n-1} maps to zero in I_{n+1}. Thus there is a homomorphism

$$L_n / \operatorname{bild} L_{n-1} \longrightarrow I_{n+1},$$

and this homomorphism has an extension to L_{n+1}.

In general, there are several homomorphisms of chain complexes in the situation above. Nevertheless, they are homotopic to one another.

Lemma 23.12: uniqueness up to homotopy

Let M be an R-module over a commutative ring R. Let

$$0 \longrightarrow M \longrightarrow L_0 \longrightarrow L_1 \longrightarrow \cdots$$

be an exact complex and let

$$0 \longrightarrow M \longrightarrow I_0 \longrightarrow I_1 \longrightarrow \cdots$$

be a complex in which all the modules I_n are injective. If

$$\varphi, \psi : L_\bullet \longrightarrow I_\bullet$$

are homomorphisms of chain complexes, then φ and ψ are homotopic.

> **Editorial note - common initial map.** The source leaves implicit that φ and ψ extend the same map on the initial module M (in particular, the identity when comparing resolutions of M). This condition is needed for the induction at $n = 0$; arbitrary chain maps need not be homotopic.

Proof We define the homotopy maps inductively,

$$\Theta_n : L_{n+1} \longrightarrow I_n$$

and set

$$\Theta_{-1} : L_0 \longrightarrow M = I_{-1}$$

to be the zero map. Note that M is not injective in general. Suppose the homotopy maps through Θ_{n-1} have already been constructed. We have the diagram

$$\begin{array}{ccccc} L_{n-1} & \xrightarrow{d_n} & L_n & \xrightarrow{d_{n+1}} & L_{n+1} \\ \downarrow & & \downarrow & & \downarrow \\ I_{n-1} & \xrightarrow{e_n} & I_n & \xrightarrow{e_{n+1}} & I_{n+1}, \end{array}$$

together with the diagonal map $\Theta_{n-1} : L_n \to I_{n-1}$, and

$$\varphi_{n-1} - \psi_{n-1} = e_{n-1} \circ \Theta_{n-2} + \Theta_{n-1} \circ d_n.$$

Consider the homomorphism

$$\varphi_n - \psi_n - e_n \circ \Theta_{n-1}$$

from L_n to I_n. For $x \in L_{n-1}$ we have

$$\begin{aligned}
&(\varphi_n - \psi_n - e_n \circ \Theta_{n-1})(d_n(x)) \\
&\quad = (\varphi_n - \psi_n)(d_n(x)) - (e_n \circ \Theta_{n-1})(d_n(x)) \\
&\quad = (\varphi_n - \psi_n)(d_n(x)) - e_n(\Theta_{n-1}(d_n(x))) \\
&\quad = (\varphi_n - \psi_n)(d_n(x)) - e_n(-e_{n-1}(\Theta_{n-2}(x)) + \varphi_{n-1}(x) - \psi_{n-1}(x)) \\
&\quad = \varphi_n(d_n(x)) - \psi_n(d_n(x)) - e_n(\varphi_{n-1}(x)) + e_n(\psi_{n-1}(x)) \\
&\quad = \varphi_n(d_n(x)) - e_n(\varphi_{n-1}(x)) - \psi_n(d_n(x)) + e_n(\psi_{n-1}(x)) \\
&\quad = 0,
\end{aligned}$$

since φ and ψ commute with the differentials. Thus $\varphi_n - \psi_n - e_n \circ \Theta_{n-1}$ maps the image of d_n to zero. We obtain an induced homomorphism

$$L_n / \operatorname{bild} d_n \longrightarrow I_n.$$

Since the complex $L_\bullet$ is exact, there is an injective map

$$L_n / \operatorname{bild} d_n \longrightarrow L_{n+1}.$$

Since I_n is injective, we obtain an extension

$$-\Theta_n : L_{n+1} \longrightarrow I_n.$$

We therefore have

$$\varphi_n - \psi_n - e_n \circ \Theta_{n-1} = -\Theta_n \circ d_{n+1}.$$

Editorial note - sign in the source proof. The final two displays introduce $-\Theta_n$, whereas the induction hypothesis uses a plus sign. Choose the extension as Θ_n instead. The resulting identity is $\varphi_n - \psi_n = e_n \circ \Theta_{n-1} + \Theta_n \circ d_{n+1}$, consistent with the preceding induction.

Injective and flasque sheaves

By definition, an injective module is characterised by the existence of homomorphisms in certain situations. There is therefore a corresponding notion of an *injective object* in any category in which one can speak of injective homomorphisms. The usual setting is that of additive or abelian categories; see the appendices. The category of sheaves of abelian groups on a topological space, and the category of sheaves of modules on a ringed space, are such abelian categories. We essentially proved this in Lectures 5 and 6. We now show that in this setting too, a sheaf can be embedded in an injective sheaf.

Lemma 23.13: embedding a sheaf of modules in an injective sheaf

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{M}$ an $\mathcal{O}_X$-module. There is an injective sheaf of modules $\mathcal{I}$ on X with $\mathcal{M} \subseteq \mathcal{I}$.

Proof For every sheaf of modules $\mathcal{M}$, the map

$$\mathcal{M} \longrightarrow \prod_{x \in X} i_* \mathcal{M}_x$$

is an injective $\mathcal{O}_X$-module homomorphism. Here, for $x \in X$, $i_* \mathcal{M}_x$ is the pushforward of the $\mathcal{O}_{X,x}$-module $\mathcal{M}_x$, viewed as a sheaf on $\{x\}$, along the embedding

$$i : \{x\} \longrightarrow X.$$

By Corollary 23.8, for $\mathcal{M}_x$ there is an injective $\mathcal{O}_{X,x}$-module I_x at x. Set

$$\mathcal{I} := \prod_{x \in X} i_* I_x.$$

We thus obtain inclusions of $\mathcal{O}_X$-modules

$$\mathcal{M} \longrightarrow \prod_{x \in X} i_* \mathcal{M}_x \longrightarrow \prod_{x \in X} i_* I_x.$$

We must show that $\mathcal{I}$ is injective. Let $\mathcal{F} \subseteq \mathcal{G}$ be $\mathcal{O}_X$-modules and suppose we are given an $\mathcal{O}_X$-module homomorphism

$$\varphi : \mathcal{F} \longrightarrow \mathcal{I}.$$

By Exercise 3.18 and Lemma 4.3 of the Appendix, this corresponds to an element

$$(\varphi_x) \in \prod_{x \in X} \mathrm{Hom}_{\mathcal{O}_X}(\mathcal{F}, i_* I_x) = \prod_{x \in X} \mathrm{Hom}_{\mathcal{O}_{X,x}}(\mathcal{F}_x, I_x).$$

Each φ_x has an extension $\widetilde{\varphi}_x : \mathcal{G}_x \to I_x$, and these combine to give an extension

$$\widetilde{\varphi} : \mathcal{G} \longrightarrow \mathcal{I}.$$

Injective sheaves are closely related to flasque sheaves. The latter are often easier to work with computationally.

Definition 23.14: flasque sheaf

A sheaf $\mathcal{G}$ on a topological space is called *flasque* if for every pair of open subsets $U \subseteq V$, the restriction map

$$\mathcal{G}(V) \longrightarrow \mathcal{G}(U)$$

is surjective.

For a flasque sheaf, the restriction map

$$\mathcal{G}(V) \longrightarrow \mathcal{G}(U)$$

is thus surjective for arbitrary open subsets $U \subseteq V$.

Lemma 23.15: basic properties of flasque sheaves

Let X be a topological space and

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H} \longrightarrow 0$$

a short exact sequence of sheaves of abelian groups. The following properties hold.

1. If $\mathcal{F}$ is flasque, then the map on global sections

$$\Gamma(X, \mathcal{G}) \longrightarrow \Gamma(X, \mathcal{H})$$

   is surjective.
2. If $\mathcal{F}$ and $\mathcal{G}$ are flasque, then $\mathcal{H}$ is also flasque.

Proof For (1), let $t \in \Gamma(X, \mathcal{H})$ be given. We use Zorn's lemma and consider the set

$$\mathcal{M} = \{(U, s) \mid U \subseteq X \text{ open},\ s \in \Gamma(U, \mathcal{G}),\ s \longmapsto t|_U\}.$$

We order $\mathcal{M}$ by setting

$$(U, s) \preccurlyeq (U', s')$$

if $U \subseteq U'$ and s' extends s. By the sheaf property, every chain has an upper bound. Zorn's lemma therefore gives a maximal element (U, s) of $\mathcal{M}$. We must show that $U = X$.

Suppose $U \neq X$ and take $x \notin U$. Since the sheaf morphism $\mathcal{G} \to \mathcal{H}$ is surjective, there is an open neighbourhood $x \in V$ and a section $r \in \Gamma(V, \mathcal{G})$ mapping to $t|_V$. Consequently,

$$s|_{U \cap V} - r|_{U \cap V}$$

maps to zero and belongs to $\Gamma(U \cap V, \mathcal{F})$. Since $\mathcal{F}$ is flasque, there is a section

$$z \in \Gamma(X, \mathcal{F})$$

whose restriction to $U \cap V$ is $s|_{U \cap V} - r|_{U \cap V}$. Replace r with

$$r' = r + z|_V.$$

This element still maps to $t|_V$, and

$$s|_{U\cap V} - r'|_{U\cap V} = z|_V - z|_V = 0.$$

Thus s and r', as sections of $\mathcal{G}$ over U and V respectively, agree on the overlap and determine a section

$$s' \in \Gamma(U \cup V, \mathcal{G})$$

mapping to t. This contradicts the maximality of U.

> **Editorial note - the sheaf containing the glued section.** The frozen source prints $s' \in \Gamma(U \cup V, \mathcal{F})$. However, s and r' have just been specified as sections of $\mathcal{G}$, and the glued section must map to $t \in \Gamma(X, \mathcal{H})$. The argument therefore requires $s' \in \Gamma(U \cup V, \mathcal{G})$, as used above. This edition explicitly records that change of symbol and leaves the rest of the proof unchanged. In the preceding cancellation display, both copies of $z|_V$ must also be restricted to $U \cap V$, where the left-hand side is defined.

Statement (2) follows from (1).

Lemma 23.16: injective sheaves are flasque

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{I}$ an injective $\mathcal{O}_X$-module. Then $\mathcal{I}$ is flasque.

Proof Let $U \subseteq X$ be an open subset. Consider the presheaf

$$\mathcal{P}(V) := \begin{cases} \mathcal{O}_X(V), & \text{if } V \subseteq U, \\ 0, & \text{otherwise,} \end{cases}$$

and denote its sheafification by $\mathcal{O}_U$. By Lemma 5.2 (4), the natural presheaf homomorphism $\mathcal{P} \to \mathcal{O}_X$ induces a sheaf homomorphism

$$\mathcal{O}_U \longrightarrow \mathcal{O}_X.$$

This homomorphism is injective. Moreover,

$$\operatorname{Hom}(\mathcal{O}_U, \mathcal{I}) = \Gamma(U, \mathcal{I}).$$

Since $\mathcal{I}$ is injective, each element here extends to an element of

$$\operatorname{Hom}(\mathcal{O}_X, \mathcal{I}) = \Gamma(X, \mathcal{I}).$$

This means that the restriction map

$$\Gamma(X, \mathcal{I}) \longrightarrow \Gamma(U, \mathcal{I})$$

is surjective.

Worksheet 23: Injective Modules

There are no stars because the frozen source closure found no public solution page for any of the 21 exercises. All Exercises 23.1–23.21 have negative candidate results; this edition does not invent new solutions.

Exercise 23.1

Let $H \subseteq G$ be a subgroup of an abelian group G, and let

$$\varphi : H \longrightarrow \mathbb{Z}$$

be a group homomorphism. Show that there is a group homomorphism

$$G \longrightarrow \mathbb{Q}$$

extending φ, where φ is viewed as a map to $\mathbb{Q}$.

Exercise 23.2

Show that the group $(\mathbb{Q}_+, \cdot)$ is not divisible.

Exercise 23.3

Show that the groups $(\mathbb{R}, +)$ and $(\mathbb{R}_+, \cdot)$ are divisible. Is $(\mathbb{R}^\times, \cdot)$ also divisible?

Exercise 23.4

Show that the groups $\mathbb{Z}/(n)$ with $n \in \mathbb{N}_+$ are not injective.

> **Editorial note - exceptional value.** The source must exclude $n = 1$: $\mathbb{Z}/(1)$ is the zero group and is injective. The intended range is $n \geq 2$.

Exercise 23.5

Let K be an algebraically closed field. Show that the group of units $K^\times$ is divisible.

Exercise 23.6

For the cyclic group $\mathbb{Z}/(n)$, describe a divisible group D with

$$\mathbb{Z}/(n) \subseteq D.$$

Exercise 23.7

Let D be a divisible group and S a set. Show that the group

$$\text{Abb}(S, D)$$

is also divisible.

Exercise 23.8

Let D be a divisible group and S an abelian group. Show that the group

$$\text{Hom}(S, D)$$

is also divisible.

> **Editorial note - false assertion in the source.** This is false for general S: with $S = \mathbb{Z}/(2)$ and $D = \mathbb{Q}/\mathbb{Z}$, the group $\text{Hom}(S, D)$ is $\mathbb{Z}/(2)$ and is not divisible. A sufficient additional hypothesis is that S is free abelian, reducing the claim to Exercise 23.7. This is an editorial correction, not an available source solution.

Exercise 23.9

Discuss the differences and similarities between injective and projective modules over a commutative ring R.

Exercise 23.10

Give an example of an injective R-module M over a commutative ring R such that M is not divisible as an abelian group.

Exercise 23.11

Give an example of a non-injective R-module M over a commutative ring R such that M is divisible as an abelian group.

Hint. Consider

$$R = M = \mathbb{Q}[X].$$

Exercise 23.12

Let I be an injective R-module over a commutative ring R, and let $r \in R$ be a non-zero-divisor. Show that multiplication

$$\mu_r : I \longrightarrow I, \qquad v \longmapsto rv,$$

is surjective.

> **Editorial note - module variable in the source.** The frozen source begins the exercise by calling R an injective R-module, but the formula then uses $\mu_r : I \to I$. To make the object in the sentence agree with the formula, this edition uses I as the module variable without changing any other mathematical hypothesis or conclusion.

Exercise 23.13

Show that every abelian group G has an injective resolution of the form

$$0 \longrightarrow G \longrightarrow I_0 \longrightarrow I_1 \longrightarrow 0.$$

Exercise 23.14

Let $\mathcal{I}_j$, $j \in J$, be a family of injective sheaves of abelian groups on a topological space X. Show that the direct product

$$\prod_{j \in J} \mathcal{I}_j$$

is also injective.

Exercise 23.15

Show that the sheaf of real-valued continuous functions on $\mathbb{R}$ is not flasque.

Exercise 23.16

Show that every sheaf on a discrete topological space X is flasque.

> **Editorial note - sheaf category.** Read this in the context of sheaves of abelian groups (or modules). For arbitrary sheaves of sets, a stalk may be empty, preventing extension even on a discrete space.

Exercise 23.17

Show that a locally constant sheaf $\mathcal{G}$ on an irreducible topological space is flasque.

Exercise 23.18

Show that a locally constant sheaf $\mathcal{G}$ on a topological space need not be flasque.

Exercise 23.19

Let G be an abelian group and X a topological space. Show that the sheaf

$$U \longmapsto \mathrm{Abb}(U, G)$$

on X is flasque.

Exercise 23.20

Let $\mathcal{G}_j$, $j \in J$, be a family of flasque sheaves on a topological space X. Show that the direct product

$$\prod_{j \in J} \mathcal{G}_j$$

is also flasque.

Exercise 23.21

Let X be a topological space with a decomposition

$$X = Y \uplus Z$$

into disjoint non-empty open subsets. Let $\mathcal{G}$ be a sheaf on Y and $\mathcal{H}$ a sheaf on Z. Show that the sheaf $\mathcal{F}$ defined by

$$\mathcal{F}(U) = \mathcal{G}(U \cap Y) \times \mathcal{H}(U \cap Z)$$

is flasque if and only if $\mathcal{G}$ and $\mathcal{H}$ are flasque.

Editorial note - missing grammatical subject. The frozen source sentence literally says "show that the by $\mathcal{F}(U) = \cdots$ is flasque if and only if", without naming the object being defined. The formula and the rest of the sentence identify that object as the sheaf $\mathcal{F}$. This edition makes it explicit and preserves the source formula, hypotheses, and equivalence.

Public Solutions and Coverage of Worksheet 23

At the frozen revision boundary, the source provides no public solution page for any of the 21 exercises. The exercise map and candidate evidence record negative results for all Exercises 23.1–23.21. The absence of public solution pages is not replaced by invented solutions.

Frozen negative results

There is no public solution page at the frozen revision for Exercises 23.1–23.21. This statement records the candidate checks; it does not assert that mathematical solutions do not exist.

Lecture 24: Right Derived Functors

Abelian categories

The notion of an abelian category is described abstractly by a list of axioms, which we have collected in an appendix. The following four main examples are important for us.

1. The category of abelian groups with group homomorphisms.
2. The category of R-modules over a commutative ring, with R-module homomorphisms.
3. The category of sheaves of abelian groups on a topological space X, with sheaf homomorphisms.
4. The category of $\mathcal{O}_X$-modules on a ringed space $(X, \mathcal{O}_X)$, with $\mathcal{O}_X$-module homomorphisms.

In each of these categories, the meaning of a short exact sequence or an exact complex is clear. Moreover, in these categories every object can be embedded in an injective object of the category, so injective resolutions can also be constructed; see Corollary 23.8 and Lemma 23.13. This property even deserves a name of its own.

Definition 24.1: enough injective objects

An abelian category $\mathcal{A}$ is said to *have enough injective objects* if, for every object $M \in \mathcal{A}$, there is an injective object I and a monomorphism

$$M \longrightarrow I.$$

Left exact additive functors

Definition 24.2: additive functor

Let $\mathcal{A}$ and $\mathcal{B}$ be additive categories. A covariant functor

$$F : \mathcal{A} \longrightarrow \mathcal{B}$$

is called *additive* if, for objects $G, H \in \mathcal{A}$, the map

$$\mathrm{Mor}(G, H) \longrightarrow \mathrm{Mor}(F(G), F(H)), \qquad \varphi \longmapsto F(\varphi),$$

is a group homomorphism.

Definition 24.3: left exact functor

Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories. A covariant functor

$$F : \mathcal{A} \longrightarrow \mathcal{B}$$

is called *left exact* if it is additive and, for every short exact sequence

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

in $\mathcal{A}$, the sequence

$$0 \longrightarrow F(A) \longrightarrow F(B) \longrightarrow F(C)$$

is exact in $\mathcal{B}$.

Two functors with both of these properties will be important for us.

Example 24.4: the Hom functor

Let R be a commutative ring and A a fixed R-module. The assignment taking each R-module M to the module of homomorphisms

$$\mathrm{Hom}_R(A, M)$$

is left exact; see Exercise 24.1.

Example 24.5: the global sections functor

Let X be a topological space and $\mathcal{A}$ the category of sheaves of abelian groups on X, with the assignment

$$\mathcal{G} \longmapsto \Gamma(X, \mathcal{G}).$$

Let

$$\mathcal{B} = \text{ABEL}$$

be the category of abelian groups, and let F be evaluation on all of X. Then $\mathcal{A}$ has enough injective objects, and F is a covariant, additive, left exact functor. Left exactness follows from Lemma 6.8, and the existence of enough injective sheaves follows from Lemma 23.13.

The assignments in the examples above are not right exact; see Exercise 24.2 and Example 6.6. Among other things, the cohomology theories we shall study will provide a theoretical account of this failure of right exactness.

> **Edition note — scope of the right-exactness claim.** The source's statement is a general warning, not an assertion about every possible parameter or space. For example, `Hom(A,-)` is exact when `A` is projective, and global sections can be exact in special situations.

Derived functors

Definition 24.6: right derived functor

Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories, with $\mathcal{A}$ having enough injective objects. Let

$$F : \mathcal{A} \longrightarrow \mathcal{B}$$

be a covariant, additive, left exact functor. The *nth right derived functor*

$$R^n F : \mathcal{A} \longrightarrow \mathcal{B}, \qquad n \in \mathbb{N},$$

is defined as follows. For an object $M \in \mathcal{A}$, take an injective resolution $I^\bullet$ of M and set

$$R^n F(M) := H^n(F(I^\bullet)).$$

For a homomorphism $\varphi : M \longrightarrow N$ in $\mathcal{A}$, take an extension

$$\widetilde{\varphi} : I^\bullet \longrightarrow J^\bullet,$$

where $J^\bullet$ is an injective resolution of N, and set

$$R^nF(\varphi) := \big(H^n(\widetilde{\varphi}) : H^n(F(I^\bullet)) \longrightarrow H^n(F(J^\bullet))\big),$$

using the induced homomorphism on homology in the sense of Lemma 8.5 of the Appendix.

> **Edition note — induced-map notation.** The printed source labels this map $H^n(\widetilde{\varphi})$, although the displayed source and target are the cohomology of the complexes obtained after applying F; the intended map is therefore the one induced by $F(\widetilde{\varphi})$. The proof of Theorem 24.7 later switches from superscript H^n to subscript H_n for the same construction. This edition preserves both printed conventions while making their relationship explicit.

Theorem 24.7: delta properties of right derived functors

Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories, with $\mathcal{A}$ having enough injective objects. Let $F : \mathcal{A} \longrightarrow \mathcal{B}$ be a covariant, additive, left exact functor, and let R^nF denote its right derived functors. The following properties hold.

1. R^nF is a well-defined additive functor from $\mathcal{A}$ to $\mathcal{B}$.
2. There is a natural isomorphism

$$R^0F \cong F.$$

3. For every short exact sequence

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

in $\mathcal{A}$ and every $n \in \mathbb{N}$, there is a natural connecting homomorphism

$$\delta^n : R^nF(C) \longrightarrow R^{n+1}F(A)$$

such that there is an exact complex in $\mathcal{B}$,

$$\begin{aligned} \ldots \longrightarrow R^{n-1}F(C) \xrightarrow{\delta^{n-1}} R^nF(A) \longrightarrow R^nF(B) \longrightarrow R^nF(C) \\ \xrightarrow{\delta^n} R^{n+1}F(A) \longrightarrow R^{n+1}F(B) \longrightarrow \ldots . \end{aligned}$$

4. For a homomorphism of exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & 0, \end{array}$$

the diagram

$$\begin{array}{ccc} R^nF(C) & \xrightarrow{\delta^n} & R^{n+1}F(A) \\ \downarrow & & \downarrow \\ R^nF(C') & \xrightarrow{\delta^n} & R^{n+1}F(A') \end{array}$$

commutes.

Proof

1. To establish well-definedness, that is, independence of the chosen injective resolution, we give the proof when $\mathcal{A}$ is the category of R-modules; formulating the general case takes a little more work. Let

$$0 \longrightarrow N \longrightarrow L_0 \longrightarrow L_1 \longrightarrow \ldots$$

and

$$0 \longrightarrow N \longrightarrow I_0 \longrightarrow I_1 \longrightarrow \ldots$$

be injective resolutions of a module N. By Lemma 23.11, there are homomorphisms of chain complexes

$$\varphi : L_\bullet \longrightarrow I_\bullet$$

and

$$\psi : I_\bullet \longrightarrow L_\bullet.$$

By Lemma 23.12, the composites $\psi \circ \varphi$ and $\varphi \circ \psi$ are homotopic to the identity on $L_\bullet$ and $I_\bullet$, respectively. By Lemma 8.9 of the Appendix, the same holds for the associated homomorphisms on the complexes $F(L_\bullet)$ and $F(I_\bullet)$. Thus for the induced homomorphisms on homology, the composite

$$H_n(F(L_\bullet)) \overset{H_n(\varphi)}{\longrightarrow} H_n(F(I_\bullet)) \overset{H_n(\psi)}{\longrightarrow} H_n(F(L_\bullet))$$

is the identity. Hence $H(\varphi)$ is a canonical isomorphism. Additivity always holds on homology by Lemma 8.5 of the Appendix.

2. Let $I^\bullet$ be an injective resolution of the object M. The homology in degree 0 of the complex

$$0 \longrightarrow F(I^0) \longrightarrow F(I^1) \longrightarrow F(I^2) \longrightarrow \ldots$$

is simply the kernel of the homomorphism

$$F(I^0) \longrightarrow F(I^1).$$

Since F is left exact, this kernel equals $F(M)$.

3. By Lemma 9.9 of the Appendix, there is a commutative diagram

$$\begin{array}{ccccccccc}
 & & 0 & & 0 & & 0 & & \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & I^0 & \longrightarrow & J^0 & \longrightarrow & K^0 & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & I^1 & \longrightarrow & J^1 & \longrightarrow & K^1 & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
 & & \vdots & & \vdots & & \vdots & &
\end{array}$$

with exact rows and columns. Since every row except the initial sequence splits, for every $n \geq 0$ we obtain a short exact sequence

$$0 \longrightarrow F(I^n) \longrightarrow F(J^n) \longrightarrow F(K^n) \longrightarrow 0.$$

Thus there is a commutative diagram

$$\begin{array}{ccccccccc}
0 & \longrightarrow & F(I^{n-1}) & \longrightarrow & F(J^{n-1}) & \longrightarrow & F(K^{n-1}) & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & F(I^{n}) & \longrightarrow & F(J^{n}) & \longrightarrow & F(K^{n}) & \longrightarrow & 0 \\
 & & \downarrow & & \downarrow & & \downarrow & & \\
0 & \longrightarrow & F(I^{n+1}) & \longrightarrow & F(J^{n+1}) & \longrightarrow & F(K^{n+1}) & \longrightarrow & 0
\end{array}$$

with exact rows. In such a situation, by Lemma 8.6 of the Appendix, there is a homomorphism from the kernel of

$$F(K^{n-1}) \longrightarrow F(K^n)$$

to the kernel of

$$F(I^n) \longrightarrow F(I^{n+1}),$$

and hence also to $R^n F(A)$. The image of $F(K^{n-2})$ maps to 0, so this induces a homomorphism

$$R^{n-1}F(C) \longrightarrow R^n F(A).$$

Edition note — target of the connecting construction. The source compresses the diagram chase when it says that there is a homomorphism from the first kernel to the second kernel. Canonically, a cycle in `F(K)` lifts to `F(J)`, its differential lies in the cycle kernel of `F(I)`, and only its class modulo boundaries is independent of the chosen lift. Thus the canonical target is the cohomology quotient $R^n F(A)$, not in general the kernel itself; the fact that boundaries map to zero is what makes the displayed connecting homomorphism well defined.

4. See Exercise 24.5.

The map δ is also called the *connecting homomorphism*.

Theorem 24.8: injective objects are acyclic

Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories, with $\mathcal{A}$ having enough injective objects. Let $F : \mathcal{A} \longrightarrow \mathcal{B}$ be a covariant, additive, left exact functor. For every injective object I of $\mathcal{A}$ and every $n \geq 1$, the right derived functors satisfy

$$R^n F(I) = 0.$$

Proof This is immediate, since we can use the injective resolution

$$I_0 = I \longrightarrow 0 \longrightarrow 0 \longrightarrow \dots .$$

Definition 24.9: acyclic object

Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories, with $\mathcal{A}$ having enough injective objects. Let $F : \mathcal{A} \longrightarrow \mathcal{B}$ be a covariant, additive, left exact functor. An object Z of $\mathcal{A}$ is called *acyclic* (with respect to F) if for every $n \geq 1$ the right derived functors satisfy

$$R^n F(Z) = 0.$$

By Theorem 24.8, every injective object is acyclic.

Corollary 24.10: dimension shifting through an acyclic object

Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories, with $\mathcal{A}$ having enough injective objects. Let $F : \mathcal{A} \longrightarrow \mathcal{B}$ be a covariant, additive, left exact functor. Let A be an object of $\mathcal{A}$ and suppose

$$0 \longrightarrow A \longrightarrow Z$$

is exact, with Z an acyclic object. Then

$$R^1F(A) = F(Z/A)/\operatorname{bild}(F(Z))$$

and

$$R^nF(A) = R^{n-1}F(Z/A)$$

for $n \geq 2$.

Proof Consider the short exact sequence

$$0 \longrightarrow A \longrightarrow Z \longrightarrow Z/A \longrightarrow 0.$$

The statements follow from the long exact sequence, since by hypothesis the middle terms satisfy $R^nF(Z) = 0$.

Definition 24.11: the Ext functor

Let R be a commutative ring and M an R-module. The right derived functors of

$$N \longmapsto \operatorname{Hom}_R(M, N)$$

(from the category of R-modules to itself) are called the *Ext functors* and are denoted by

$$\operatorname{Ext}^n(M, N).$$

By definition, to compute the Ext modules we must take an injective resolution of the second module,

$$0 \longrightarrow N \longrightarrow I_0 \longrightarrow I_1 \longrightarrow I_2 \longrightarrow \dots,$$

and then determine the homology of the complex

$$\operatorname{Hom}(M, I_{n-1}) \longrightarrow \operatorname{Hom}(M, I_n) \longrightarrow \operatorname{Hom}(M, I_{n+1}).$$

Worksheet 24: Right Derived Functors

At the frozen revision boundary, all five exercises have negative candidate results: there are no public solution pages. This worksheet therefore uses no stars, and this edition does not invent new solutions.

Exercise 24.1

Let R be a commutative ring and A an R-module. Let

$$0 \longrightarrow L \longrightarrow M \longrightarrow N \longrightarrow 0$$

be a short exact sequence of R-modules. Show that

$$0 \longrightarrow \operatorname{Hom}(A, L) \longrightarrow \operatorname{Hom}(A, M) \longrightarrow \operatorname{Hom}(A, N)$$

is exact.

Exercise 24.2

Let R be a commutative ring and A an R-module. Let $M \longrightarrow N$ be a surjective R-module homomorphism. Show that the induced map

$$\operatorname{Hom}(A, M) \longrightarrow \operatorname{Hom}(A, N)$$

need not be surjective.

Consider $A = N = \mathbb{Z}/(k)$.

Exercise 24.3

Let R be a commutative ring, P a projective R-module, and M another R-module. Show that

$$\operatorname{Ext}^n(P, M) = 0$$

for $n \geq 1$.

Exercise 24.4

Using the short exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot k} \mathbb{Z} \longrightarrow \mathbb{Z}/(k) \longrightarrow 0,$$

show that

$$\operatorname{Ext}^1(\mathbb{Z}/(k), \mathbb{Z})$$

is not the zero module for $k \geq 2$.

Exercise 24.5

Let $\mathcal{A}$ and $\mathcal{B}$ be abelian categories, with $\mathcal{A}$ having enough injective objects. Let $F : \mathcal{A} \longrightarrow \mathcal{B}$ be a covariant, additive, left exact functor, and let $R^n F$ denote its right derived functors. Show that for a homomorphism of exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \downarrow \\ 0 & \longrightarrow & A' & \longrightarrow & B' & \longrightarrow & C' & \longrightarrow & 0, \end{array}$$

the diagram

$$\begin{array}{ccc} R^nF(C) & \xrightarrow{\delta^n} & R^{n+1}F(A) \\ \downarrow & & \downarrow \\ R^nF(C') & \xrightarrow{\delta^n} & R^{n+1}F(A') \end{array}$$

commutes.

Public Solutions and Coverage of Worksheet 24

At the frozen revision boundary, the source provides no public solution page for any of the five exercises. The exercise map and candidate evidence record negative results for Exercises 24.1–24.5. The absence of public solution pages is not replaced by invented solutions.

Frozen negative results

There is no public solution page at the frozen revision for Exercises 24.1, 24.2, 24.3, 24.4, or 24.5. This statement records the candidate checks; it does not assert that mathematical solutions do not exist.

Lecture 25: Sheaf Cohomology

Sheaf cohomology

Definition 25.1: sheaf cohomology

Let $\mathcal{G}$ be a sheaf of abelian groups on a topological space X. The nth right derived functor of the global sections functor $\Gamma(X, -)$ is called the *nth sheaf cohomology* of $\mathcal{G}$ on X. It is denoted by

$$H^n(X, \mathcal{G}) := R^n\Gamma(X, \mathcal{G}).$$

Corollary 25.2: basic properties of sheaf cohomology

Let X be a topological space. Sheaf cohomology has the following properties.

1. For every $n \in \mathbb{N}$, $H^n(X, -)$ is an additive functor from the category of sheaves of abelian groups on X to the category of abelian groups.
2. There is a natural isomorphism $H^0(X, \mathcal{G}) \cong \Gamma(X, \mathcal{G})$.
3. For a short exact sequence of sheaves

   $$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H} \longrightarrow 0,$$

   there is a long exact cohomology sequence

   $$\begin{gathered} 0 \longrightarrow \Gamma(X, \mathcal{F}) \longrightarrow \Gamma(X, \mathcal{G}) \longrightarrow \Gamma(X, \mathcal{H}) \longrightarrow H^1(X, \mathcal{F}) \\ \longrightarrow H^1(X, \mathcal{G}) \longrightarrow H^1(X, \mathcal{H}) \longrightarrow H^2(X, \mathcal{F}) \longrightarrow \cdots . \end{gathered}$$

Proof This is a special case of Theorem 24.7.

Cohomology groups are generally difficult to compute. Here are some basic approaches to calculation.

1. *Vanishing theorems:* one proves that the cohomology groups are 0 for certain spaces, sheaves, and indices. If a term in a long exact cohomology sequence is 0, the preceding map is surjective and the following map is injective.
2. Instead of injective sheaves, one can use other acyclic sheaves, for example flasque sheaves.
3. H^1 can be interpreted as a group classifying certain geometric objects, for example the Picard group.
4. If the sheaves are modules on a ringed space, their cohomology groups also have a module structure over the ring of global sections $\Gamma(X, \mathcal{O}_X)$; see Lemma 25.5. If this ring is a field, as is the case in particular for connected projective varieties, the cohomology groups are even vector spaces. When finite, their dimensions are important invariants.
5. Cohomology on X can be compared with cohomology on an open subset.
6. Sheaf cohomology can be compared with other cohomology theories: Čech cohomology, singular cohomology, and simplicial cohomology.

   Edition note — scope of the projective-variety parenthesis. The parenthetical statement uses "variety" in the usual reduced finite-type sense. It does not extend unchanged to arbitrary connected projective schemes: a connected non-reduced projective scheme can have a ring of global sections that is not a field.

Lemma 25.3: flasque sheaves are acyclic

A flasque sheaf $\mathcal{G}$ on a topological space X is acyclic; that is,

$$H^n(X, \mathcal{G}) = 0$$

for $n \geq 1$.

Proof By Lemma 23.13, there is an embedding of $\mathcal{G}$ in an injective sheaf $\mathcal{I}$. Consider the associated short exact sequence of sheaves,

$$0 \longrightarrow \mathcal{G} \longrightarrow \mathcal{I} \longrightarrow \mathcal{I}/\mathcal{G} \longrightarrow 0.$$

By Lemma 23.16, $\mathcal{I}$ is flasque. Then by Lemma 23.15 (2), the quotient sheaf $\mathcal{I}/\mathcal{G}$ is also flasque. Using Theorem 24.8, the long exact cohomology sequence gives, on the one hand,

$$0 \longrightarrow \Gamma(X, \mathcal{G}) \longrightarrow \Gamma(X, \mathcal{I}) \longrightarrow \Gamma(X, \mathcal{I}/\mathcal{G}) \longrightarrow H^1(X, \mathcal{G}) \longrightarrow 0,$$

and, on the other hand,

$$0 \longrightarrow H^n(X, \mathcal{I}/\mathcal{G}) \xrightarrow{\delta^n} H^{n+1}(X, \mathcal{G}) \longrightarrow 0$$

for $n \geq 1$. Since the map

$$\Gamma(X, \mathcal{I}) \longrightarrow \Gamma(X, \mathcal{I}/\mathcal{G})$$

is surjective by Lemma 23.15 (1), the first segment shows that $H^1(X, \mathcal{G}) = 0$. This holds for every flasque sheaf. Applying the second segment, which the source says is applied for $n = 2$, we obtain $H^2(X, \mathcal{G}) = 0$, and so on.

> **Editorial note - index in the induction step.** The displayed formula relates $H^n(X, \mathcal{I}/\mathcal{G})$ to $H^{n+1}(X, \mathcal{G})$, so the conclusion about H^2 formally uses $n = 1$. The printed source says $n = 2$; this edition preserves the source's explanation and explicitly records the discrepancy.

Remark 25.4: first cohomology classes of the sheaf of continuous functions

Let G be a topological abelian group and X a topological space. Consider the sheaf of continuous maps to G, namely $C^0(-, G)$, with

$$C^0(U, G) = \{f : U \longrightarrow G \text{ a map} \mid f \text{ continuous}\}.$$

> **Edition note — group hypothesis.** The German source says only "topological group". The displayed subtraction, quotient sheaf of groups, long exact sequence, and ordinary group-valued sheaf cohomology require `G` to be abelian. The discussion is read with that necessary hypothesis; it is not a claim about non-abelian first cohomology.

There is a natural inclusion of sheaves

$$C^0(-, G) \subseteq \text{Abb}(-, G),$$

and therefore a short exact sequence of sheaves

$$0 \longrightarrow C^0(-, G) \longrightarrow \text{Abb}(-, G) \longrightarrow \text{Abb}(-, G)/C^0(-, G) \longrightarrow 0.$$

The sheaf of maps in the middle is flasque, since every map extends to a larger set. Thus by Lemma 25.3,

$$H^1(X, \text{Abb}(-, G)) = 0,$$

so the long exact cohomology sequence begins

$$\begin{aligned}0 &\longrightarrow C^0(X,G) \longrightarrow \mathrm{Abb}(X,G) \longrightarrow \Gamma(X,\mathrm{Abb}(-,G)/C^0(-,G)) \\ &\longrightarrow H^1(X,C^0(-,G)) \longrightarrow 0.\end{aligned}$$

Every first cohomology class of $C^0(-,G)$ is therefore represented by a global element of the quotient sheaf $\mathrm{Abb}(-,G)/C^0(-,G)$. Two such representatives define the same class precisely when their difference comes from a map $X \to G$.

As for any quotient sheaf, by Lemma 5.9 (1), a global element is represented by an open cover

$$X = \bigcup_{i\in I} U_i$$

and sections $f_i \in \Gamma(U_i, \mathrm{Abb}(-,G))$, that is, maps $f_i : U_i \to G$, such that the differences

$$f_i - f_j\big|_{U_i\cap U_j}$$

come from the subsheaf, that is, are continuous functions on $U_i \cap U_j$. By Lemma 5.9 (2), such an element comes from the left, and thus maps to the trivial cohomology class, precisely when there is a function $f : X \to G$ such that

$$g_i := f_i - f\big|_{U_i}$$

is continuous for every i. In that case, on $U_i \cap U_j$ we have

$$g_i - g_j = f_i - f - (f_j - f) = f_i - f_j.$$

Conversely, if there is a family of continuous functions g_i on U_i whose differences agree with the prescribed differences, then on U_i we can define

$$f := f_i - g_i.$$

Since these definitions agree on overlaps, they give a global function on X. Thus the first cohomology group of the sheaf of continuous functions is trivial precisely when for every family (U_i, f_i) with continuous differences $f_i - f_j$, there is a family (U_i, g_i) of continuous functions g_i with the same differences.

Lemma 25.5: the module structure on sheaf cohomology

Let $(X, \mathcal{O}_X)$ be a ringed space and $\mathcal{M}$ an $\mathcal{O}_X$-module. Then the sheaf cohomology groups $H^i(X,\mathcal{M})$ are naturally $\Gamma(X,\mathcal{O}_X)$-modules.

Proof Every element $f \in \Gamma(X,\mathcal{O}_X)$ defines an $\mathcal{O}_X$-module homomorphism

$$f : \mathcal{M} \longrightarrow \mathcal{M}.$$

On each open set $U \subseteq X$, the restriction of f to $\Gamma(U,\mathcal{O}_X)$ acts by scalar multiplication, namely

$$\Gamma(U,\mathcal{M}) \longrightarrow \Gamma(U,\mathcal{M}), \qquad s \longmapsto fs.$$

Multiplication by f is, in particular, a homomorphism of sheaves of abelian groups. By functoriality of sheaf cohomology in Corollary 25.2, it induces a group homomorphism

$$H^n(f) : H^n(X,\mathcal{M}) \longrightarrow H^n(X,\mathcal{M}).$$

We must show that the map

$$\Gamma(X, \mathcal{O}_X) \times H^n(X, \mathcal{M}) \longrightarrow H^n(X, \mathcal{M}), \qquad (f, c) \longmapsto H^n(f)(c),$$

defines a module structure on $H^n(X, \mathcal{M})$. Since $H^n(f)$ is a group homomorphism, additivity in the module variable is assured. By functoriality, 1 maps to the identity, first as a sheaf homomorphism and then in cohomology. Compatibility with composition gives

$$H^n(fg) = H^n(f) \circ H^n(g).$$

For global ring elements f, g, scalar multiplication by $f + g$ at the level of sheaves of modules is the sum of scalar multiplication by f and by g. Since H^n is an additive functor, we also have

$$H^n(f + g) = H^n(f) + H^n(g).$$

Cohomology on schemes

We now consider the cohomology of sheaves on schemes.

Lemma 25.6: first cohomology via the function field

Let $(X, \mathcal{O}_X)$ be an integral scheme with function field K, viewed as the constant sheaf $\mathcal{K}$ on X. Then

$$H^1(X, \mathcal{O}_X) = \Gamma(X, \mathcal{K}/\mathcal{O}_X)/\operatorname{im}(K \longrightarrow \Gamma(X, \mathcal{K}/\mathcal{O}_X)).$$

Proof Since X is in particular irreducible, the constant presheaf with value K is a sheaf. By Lemma 11.16, there is an injective sheaf homomorphism

$$\mathcal{O}_X \longrightarrow \mathcal{K},$$

and hence a short exact sequence of sheaves

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{K} \longrightarrow \mathcal{K}/\mathcal{O}_X \longrightarrow 0.$$

The associated long exact cohomology sequence is

$$\begin{gathered} 0 \longrightarrow \Gamma(X, \mathcal{O}_X) \longrightarrow \Gamma(X, \mathcal{K}) \longrightarrow \Gamma(X, \mathcal{K}/\mathcal{O}_X) \\ \longrightarrow H^1(X, \mathcal{O}_X) \longrightarrow H^1(X, \mathcal{K}) \longrightarrow \cdots. \end{gathered}$$

As a constant sheaf, $\mathcal{K}$ is flasque and hence acyclic by Lemma 25.3. In particular,

$$H^1(X, \mathcal{K}) = 0.$$

Thus $H^1(X, \mathcal{O}_X)$ is the cokernel of the preceding map.

Lemma 25.7: vanishing on integral affine schemes

Let R be an integral domain and

$$\text{Spek}(R) = (X, \mathcal{O}_X)$$

the associated integral affine scheme. Then

$$H^1(X, \mathcal{O}_X) = 0.$$

Proof Consider the short exact sequence of R-modules

$$0 \longrightarrow R \longrightarrow K \longrightarrow K/R \longrightarrow 0$$

and the associated sequence of quasi-coherent sheaves, exact by Lemma 14.9,

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \widetilde{K} \longrightarrow \widetilde{K/R} \longrightarrow 0.$$

In particular,

$$\widetilde{K/R} = \widetilde{K}/\mathcal{O}_X,$$

and $\widetilde{K}$ is the constant sheaf of the function field. Evaluating this sheaf sequence globally recovers the original sequence. The result now follows from Lemma 25.6.

Example 25.8: the punctured affine plane

Consider the punctured affine plane

$$U = \mathbb{A}^2_K \setminus \{(0,0)\}$$

over a field K. We want to understand $H^1(U, \mathcal{O}_U)$ using Lemma 25.7. Its function field is $K(X,Y)$; denote the associated constant sheaf by $\mathcal{Q}$. The long exact cohomology sequence begins

$$0 \longrightarrow K[X,Y] \longrightarrow K(X,Y) \longrightarrow \Gamma(U, \mathcal{Q}/\mathcal{O}_U) \longrightarrow H^1(X, \mathcal{O}_U) \longrightarrow 0.$$

> **Editorial note - space symbol in the source.** The printed source writes $H^1(X, \mathcal{O}_U)$ in the sequence above, although the example, evaluation, and other terms concern U. This edition preserves the source symbol and does not silently replace it.

We have $U = D(X) \cup D(Y)$. Consider sections of $\Gamma(U, \mathcal{Q}/\mathcal{O}_U)$ of the form

$$(D(X), X^\alpha Y^\beta; D(Y), 0), \qquad \alpha, \beta \in \mathbb{Z}.$$

Such a section is specified on $D(X)$ by the rational function $X^\alpha Y^\beta$ and on $D(Y)$ by the rational function 0. Their difference is simply $X^\alpha Y^\beta$, which belongs to the structure sheaf on the intersection $D(X) \cap D(Y) = D(XY)$. Thus we do obtain a section of the quotient sheaf; compare Lemma 5.9.

Depending on α and β, we determine whether this section lies in the image. Equivalently, does it define the trivial element in first cohomology? Coming from the left means that there is a rational function $q \in K(X,Y)$ corresponding to the section. This means that the differences on $D(X)$ and $D(Y)$ come from the structure sheaf, so simultaneously

$$q - X^\alpha Y^\beta \in \Gamma(D(X), \mathcal{O}_U)$$

and

$$q - 0 \in \Gamma(D(Y), \mathcal{O}_U).$$

The second condition means

$$q = \frac{h}{Y^n},$$

while the first means

$$\frac{h}{Y^n} - X^\alpha Y^\beta = \frac{g}{X^m}.$$

The question is therefore whether the equation

$$X^\alpha Y^\beta = \frac{h}{Y^n} - \frac{g}{X^m}$$

has a solution with $g, h \in K[X, Y]$ and $m, n \in \mathbb{N}$. If $\alpha \geq 0$ or $\beta \geq 0$, such a solution exists. If $\alpha, \beta < 0$, no solution is possible, since the right-hand side equals

$$\frac{hX^m - gY^n}{X^m Y^n}.$$

Multiplication by $X^m Y^n$ shows the impossibility: the ideal (X^m, Y^n) contains only monomials divisible by one of its generators.

Example 25.9: cohomology of a syzygy sheaf

We continue Example 16.9. Over the polynomial ring

$$R = K[X_1, \dots, X_n], \qquad n \geq 2,$$

consider the short exact sequence

$$0 \longrightarrow \mathrm{Syz}(X_1, \dots, X_n) \longrightarrow R^n \longrightarrow (X_1, \dots, X_n) \longrightarrow 0.$$

Restricting the associated sheaf sequence to the punctured space

$$U = \mathbb{A}^n_K \setminus \{(0, \dots, 0)\}$$

gives

$$0 \longrightarrow \mathcal{S} = \mathrm{Syz}(\widetilde{X_1, \dots, X_n}) \longrightarrow \mathcal{O}_U^n \longrightarrow \mathcal{O}_U \longrightarrow 0.$$

Evaluating this sheaf sequence on U gives

$$\begin{aligned} 0 \longrightarrow \mathrm{Syz}(X_1, \dots, X_n) \longrightarrow R^n \longrightarrow R \longrightarrow H^1(U, \mathcal{S}) \\ \longrightarrow H^1(U, \mathcal{O}_U^n) \longrightarrow . \end{aligned}$$

> **Editorial note - end of the exact sequence.** The printed source ends the sequence above with an arrow without displaying the next term. This edition preserves that printed ending and adds neither an ellipsis nor a term absent from the source.

Since the image of $R^n \to R$ is still the maximal ideal, this map is not surjective. Therefore

$$H^1(U, \mathcal{S}) \neq 0.$$

Lemma 25.10: first cohomology of the sheaf of units

Let $(X, \mathcal{O}_X)$ be an integral scheme with function field K. Let $\mathcal{O}_X^\times$ be the sheaf of units on X, and let $\mathcal{U}$ be the constant sheaf with value $K^\times$. Then

$$H^1(X, \mathcal{O}_X^\times) = \Gamma(X, \mathcal{U}/\mathcal{O}_X^\times) / \operatorname{im}(K^\times \longrightarrow \Gamma(X, \mathcal{U}/\mathcal{O}_X^\times)).$$

Proof See Exercise 25.10.

We state the following important theorems without proof.

Theorem 25.11: a cohomological criterion for affineness

Let X be a Noetherian scheme. The following properties are equivalent.

1. X is an affine scheme.
2. For every quasi-coherent sheaf $\mathcal{F}$ on X, we have $H^i(X, \mathcal{F}) = 0$.
3. For every coherent ideal sheaf $\mathcal{I}$ on X, we have $H^1(X, \mathcal{I}) = 0$.

   Edition note — omitted degree range. In condition 2 the printed source leaves `i` unquantified. The standard criterion, and the only reading compatible with condition 1, is vanishing for every positive degree `i >= 1`; including degree zero would make the assertion false even for affine schemes.

Theorem 25.12: vanishing above the dimension of the space

Let X be a Noetherian topological space of dimension d. Then

$$H^i(X, \mathcal{G}) = 0$$

for $i > d$ and every sheaf of abelian groups $\mathcal{G}$.

Worksheet 25: Sheaf Cohomology

The star marks exactly one exercise with a frozen public solution: Exercise 25.1. The other twelve exercises have negative candidate results; this edition does not invent new solutions.

Exercise 25.1 ★

Let $I \subseteq \mathbb{R}$ be a real interval and $I = U \cup V$ a cover by intervals open in I. Show that a continuous function

$$f : U \cap V \longrightarrow \mathbb{R}$$

can be written as

$$f = g\big|_{U \cap V} - h\big|_{U \cap V}$$

with continuous functions $g : U \to \mathbb{R}$ and $g : V \to \mathbb{R}$.

> **Editorial note - function names in the source.** The last line of the printed source names both functions g, whereas the formula uses g and h. This edition preserves the printed statement and does not silently change the name of the second function.

Exercise 25.2

Let $I \subseteq \mathbb{R}$ be a real interval. On I, consider the short exact sequence of sheaves

$$0 \longrightarrow C^0(-, \mathbb{R}) \longrightarrow \mathrm{Abb}(-, \mathbb{R}) \longrightarrow \mathrm{Abb}(-, \mathbb{R})/C^0(-, \mathbb{R}) \longrightarrow 0.$$

Let $I = U \cup V$ be a cover by intervals open in I. Suppose we are given a global section of the quotient sheaf $\mathrm{Abb}(-, \mathbb{R})/C^0(-, \mathbb{R})$ represented by sections $s \in \mathrm{Abb}(U, \mathbb{R})$ and $t \in \mathrm{Abb}(V, \mathbb{R})$. Show that this section is represented by a map $r \in \mathrm{Abb}(I, \mathbb{R})$.

Exercise 25.3

Let $I \subseteq \mathbb{R}$ be a closed real interval. On I, consider the short exact sequence of sheaves

$$0 \longrightarrow C^0(-,\mathbb{R}) \longrightarrow \mathrm{Abb}(-,\mathbb{R}) \longrightarrow \mathrm{Abb}(-,\mathbb{R})/C^0(-,\mathbb{R}) \longrightarrow 0.$$

Show that

$$\mathrm{Abb}(-,\mathbb{R}) \longrightarrow \mathrm{Abb}(-,\mathbb{R})/C^0(-,\mathbb{R})$$

is surjective.

Exercise 25.4

Let $I \subseteq \mathbb{R}$ be a closed real interval. Show that

$$H^1(I, C^0(-,\mathbb{R})) = 0.$$

Exercise 25.5

Let G be a discrete topological abelian group with at least two elements $a \neq b$. On S^1, consider the exact sequence of sheaves

$$0 \longrightarrow G \longrightarrow \mathrm{Abb}(-,G) \longrightarrow \mathrm{Abb}(-,G)/G \longrightarrow 0,$$

where G here denotes the sheaf of locally constant functions with values in G, namely $C^0(-,G)$. Let $S^1 = U \cup V$ be an open cover of the unit circle by two overlapping arcs such that $U \cap V$ consists of two disjoint arcs, A and B.

> **Edition note — group hypothesis.** The source says only "group", but its quotient sheaf, exact sequence, and group H^1 use the abelian category of sheaves of abelian groups. Accordingly `G` is taken to be abelian here; no non-abelian cohomology assertion is intended.

Let

$$h \in \Gamma(S^1, \mathrm{Abb}(-,G)/G)$$

be a section represented on V by the zero map $0 \in \mathrm{Abb}(V,G)$ and on U by a map $g \in \mathrm{Abb}(U,G)$ that has the constant value a on A and the constant value b on B. Show that this section cannot be represented by an element of $\mathrm{Abb}(S^1,G)$, and consequently

$$H^1(S^1,G) \neq 0.$$

Exercise 25.6

Using Example 6.6, show that

$$H^1(\mathbb{C}^\times, \mathbb{Z}) \neq 0.$$

Here $\mathbb{Z}$ denotes the sheaf of continuous functions with values in the discrete topological group $\mathbb{Z}$.

Exercise 25.7

Let X be a topological space with a point $x \in X$ whose only open neighbourhood is the entire space.

1. Show that the spectrum of a local ring has this property.
2. Show that every sheaf of abelian groups $\mathcal{G}$ on X has no non-trivial cohomology.
3. Show that not every sheaf on $\mathrm{Spek}(R)$ for a local ring R is flasque.

Exercise 25.8

Let R be an integral domain and I an ideal of R, with associated quasi-coherent ideal sheaf $\tilde{I}$ on Spek(R). Using Lemma 25.7, show that

$$H^1(\text{Spek}(R), \tilde{I}) = 0.$$

Exercise 25.9

Let $R = K[X, Y]$ be the polynomial ring over a field K, with maximal ideal $\mathfrak{m} = (X, Y)$. Consider the short exact sequence of R-modules

$$0 \longrightarrow R \xrightarrow{e \mapsto (Y, -X)} R^2 \xrightarrow{e_1 \mapsto X,\ e_2 \mapsto Y} \mathfrak{m} \longrightarrow 0.$$

1. Write down the short exact sequence of sheaves of the associated quasi-coherent modules on $\mathbb{A}^2_K = \text{Spek}(R)$.
2. Show that evaluating the sheaf sequence from (1) on $U = D(X, Y) \subset \mathbb{A}^2_K$ does not give an exact sequence.
3. What is the image of

$$1 \in R = \Gamma(U, \mathcal{O}_X) = \Gamma(U, \widetilde{\mathfrak{m}})$$

in $H^1(U, \mathcal{O}_X)$ under the connecting homomorphism?

Exercise 25.10

Let $(X, \mathcal{O}_X)$ be an integral scheme with function field K. Let $\mathcal{O}_X^\times$ be the sheaf of units on X, and let $\mathcal{U}$ be the constant sheaf with value $K^\times$. Show that

$$H^1(X, \mathcal{O}_X^\times) = \Gamma(X, \mathcal{U}/\mathcal{O}_X^\times)/\operatorname{im}(K^\times \longrightarrow \Gamma(X, \mathcal{U}/\mathcal{O}_X^\times)).$$

Exercise 25.11

Let R be a unique factorisation domain. Show that

$$H^1(\text{Spek}(R), \mathcal{O}_X^\times) = \{1\}.$$

Exercise 25.12

Consider the quadratic number ring

$$R = A_{-5} = \mathbb{Z}[\sqrt{-5}] \cong \mathbb{Z}[T]/(T^2 + 5).$$

Using Example 14.6, show that

$$H^1(\text{Spek}(R), \mathcal{O}^\times_{\text{Spek}(R)}) \neq \{1\}.$$

Exercise 25.13

Let $f : X \to Y$ be a continuous map between topological spaces. Show that pushforward

$$\mathcal{G} \longmapsto f_*\mathcal{G}$$

is a left exact covariant functor from the category of sheaves of abelian groups on X to the category of sheaves of abelian groups on Y.

Remark. The associated right derived functors are called *higher direct image sheaves.*

Public Solutions and Coverage of Worksheet 25

At the frozen revision boundary, the source provides exactly one public solution among the 13 exercises: the solution to Exercise 25.1. The exercise map and candidate evidence record negative results for Exercises 25.2–25.13. The absence of public solution pages is not replaced by invented solutions.

Solution to Exercise 25.1

Let

$$U =]a, b[$$

and

$$V =]c, d[$$

with

$$a < c < b < d.$$

If the intersection is empty, the statement is trivial. The outer endpoints may also belong to the intervals.

> **Edition note — scope of the published solution.** The source explicitly treats the nonempty, finite, strictly interlacing endpoint configuration displayed above. It does not spell out the nested, coincident-endpoint, or unbounded configurations allowed by the exercise; those cases require their own elementary reductions or cutoff choices.

Choose $\epsilon > 0$ such that

$$2\epsilon < b - c.$$

Define

$$g(x) := \begin{cases} f(x), & x \geq b - \epsilon, \\ f(x)\dfrac{x - c - \epsilon}{b - c - 2\epsilon}, & x \in]c + \epsilon, b - \epsilon[, \\ 0, & x \leq c + \epsilon, \end{cases}$$

and

$$-h(x) := \begin{cases} 0, & x \geq b - \epsilon, \\ f(x)\dfrac{-x + b - \epsilon}{b - c - 2\epsilon}, & x \in]c + \epsilon, b - \epsilon[, \\ f(x), & x \leq c + \epsilon. \end{cases}$$

These functions are continuous because their values agree at the transition points. For

$$x \in]c + \epsilon, b - \epsilon[$$

we have

$$\begin{aligned} g(x) - h(x) &= f(x)\frac{x - c - \epsilon}{b - c - 2\epsilon} + f(x)\frac{-x + b - \epsilon}{b - c - 2\epsilon} \\ &= f(x)\frac{x - c - \epsilon - x + b - \epsilon}{b - c - 2\epsilon} \\ &= f(x). \end{aligned}$$

The same equation holds outside this interval.

The source solution credits “substantially Tarek Emmrich”.

Frozen negative results

There is no public solution page at the frozen revision for Exercises 25.2, 25.3, 25.4, 25.5, 25.6, 25.7, 25.8, 25.9, 25.10, 25.11, 25.12, or 25.13. This statement records the candidate checks; it does not assert that mathematical solutions do not exist.

Lecture 26: Čech cohomology

We ask whether a finite topological space, that is, a space with only finitely many points, can have nontrivial cohomology. If the space is discrete, so that every point is both open and closed, this is impossible because every sheaf on it is flasque. Nor can there be nontrivial cohomology on the spectrum of a discrete valuation ring—or, more generally, on a local space such as the spectrum of a local ring. Nevertheless, cohomology already occurs on a three-element space, as the following example shows.

Example 26.1: cohomology on a three-point space

We consider the topological space

$$X = \{a, b, c\}$$

with open sets

$$\emptyset, \quad X, \quad U = \{a, c\}, \quad V = \{b, c\}, \quad U \cap V = \{c\}.$$

This space has two closed points a and b, is irreducible, and has c as its generic point. Apart from the empty set, its open sets form the inclusion diagram

$$\begin{array}{ccc} X & \longleftarrow & U \\ \uparrow & & \uparrow \\ V & \longleftarrow & U \cap V. \end{array}$$

A sheaf of commutative groups on X is specified by assigning groups and restriction homomorphisms to these subsets and checking the compatibility condition. We consider the sheaf $\mathcal{F}$ given by

$$\begin{array}{ccc} 0 & \longrightarrow & 0 \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & \mathbb{Z}. \end{array}$$

This sheaf embeds in the constant sheaf $\mathcal{G}$ (with identity maps)

$$\begin{array}{ccc} \mathbb{Z} & \longrightarrow & \mathbb{Z} \\ \downarrow & & \downarrow \\ \mathbb{Z} & \longrightarrow & \mathbb{Z}. \end{array}$$

Edition note: the source calls the constant sheaf in this sentence `\mathcal F`, but immediately afterwards uses `\mathcal G/\mathcal F` and `\Gamma(X,\mathcal G)`. This edition uses `\mathcal G` for the constant sheaf to keep the two sheaves’ roles distinct.

The quotient sheaf $\mathcal{G}/\mathcal{F}$ is given by

$$\begin{array}{ccc} \mathbb{Z} \times \mathbb{Z} & \xrightarrow{p_2} & \mathbb{Z} \\ p_1 \downarrow & & \downarrow \\ \mathbb{Z} & \longrightarrow & 0. \end{array}$$

The values on $U \cap V$, U, and V are obtained directly by taking quotients; sheafification has no effect. On X we obtain the product $\mathbb{Z} \times \mathbb{Z}$, since sections on U and V are automatically compatible. Thus the global map

$$\Gamma(X, \mathcal{G}) = \mathbb{Z} \longrightarrow \Gamma(X, \mathcal{G}/\mathcal{F}) = \mathbb{Z} \times \mathbb{Z}$$

is not surjective. The long exact cohomology sequence instead has the form

$$0 \longrightarrow 0 \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Z} \times \mathbb{Z} \xrightarrow{\delta} H^1(X, \mathcal{F}) = \mathbb{Z} \longrightarrow 0.$$

The map at the front is $n \mapsto (n, n)$, and the one at the back is $(r, s) \mapsto (r - s)$; the latter follows from exactness.

An important question in the opposite direction is whether the cohomology of a complicated topological space can be captured and computed using finite data. In many situations this is indeed possible by means of Čech cohomology, which refers to a finite open cover together with all its intersections.

Example 26.2: gluing data as a Čech complex

We continue Remark 20.2. Let $(X, \mathcal{O}_X)$ be a ringed space, and suppose we are interested in invertible sheaves on X, specifically those admitting trivialisations with respect to a fixed open cover

$$X = \bigcup_{i \in I} U_i.$$

These invertible sheaves correspond to collections of data

$$\left(U_i, r_{ij} \in \Gamma(U_i \cap U_j, \mathcal{O}_X^\times) \text{ with } r_{kj}\, r_{ki}^{-1}\, r_{ji} = 1 \text{ in } \Gamma(U_i \cap U_j \cap U_k, \mathcal{O}_X^\times)\right).$$

Such a collection of data must, however, be regarded as trivial if there are elements

$$s_i \in \Gamma(U_i, \mathcal{O}_X^\times)$$

with

$$s_i\, s_j^{-1} = r_{ij}$$

for all i, j. This entire situation can be expressed by the complex

$$\prod_{i \in I} \Gamma(U_i, \mathcal{O}_X^\times) \longrightarrow \prod_{i<j} \Gamma(U_i \cap U_j, \mathcal{O}_X^\times) \longrightarrow \prod_{i<j<k} \Gamma(U_i \cap U_j \cap U_k, \mathcal{O}_X^\times),$$

after fixing a total order on I. The first map is given by

$$(s_i) \longmapsto (s_j s_i^{-1})_{i<j},$$

and the second by

$$(r_{ij}) \longmapsto (r_{jk} r_{ik}^{-1} r_{ij}).$$

An element in the middle belongs to the kernel of the second map precisely when it satisfies the cocycle condition, and belongs to the image of the first precisely when it represents the trivial invertible sheaf.

Čech cohomology

Let

$$X = \bigcup_{i \in I} U_i$$

be an open cover of a topological space X. For a subset $J \subseteq I$, we set

$$U_J := \bigcap_{i \in J} U_i.$$

If $J \subseteq L \subseteq I$, then $U_L \subseteq U_J$. For a sheaf $\mathcal{G}$ of commutative groups on X, we consider the values $\mathcal{G}(U_J)$ for the various J; to $J \subseteq L$ there corresponds the restriction map

$$\mathcal{G}(U_J) \longrightarrow \mathcal{G}(U_L).$$

For $s \in \mathcal{G}(U_J)$, we use the abbreviation

$$s|_L = s|_{U_L},$$

and often write simply s. We fix a well-ordering on I (the case of finite I is the one mainly needed). We can now define the Čech complex and Čech cohomology, an important tool for computing sheaf cohomology.

Definition 26.3: the Čech complex

Let

$$X = \bigcup_{i \in I} U_i$$

be an open cover of a topological space X, and let $\mathcal{G}$ be a sheaf of commutative groups on X. For $k \in \mathbb{N}$, set

$$\check{C}^k(\mathcal{U}, \mathcal{G}) = \prod_{\{J \mid \#(J) = k+1\}} \mathcal{G}(U_J),$$

and define group homomorphisms

$$\delta_k : \check{C}^k(\mathcal{U}, \mathcal{G}) \longrightarrow \check{C}^{k+1}(\mathcal{U}, \mathcal{G}), \qquad s = (s_J)_J \longmapsto \delta_k(s) = (\delta_k(s)_L)_L,$$

by

$$(\delta_k(s))_L = \sum_{\ell=0}^{k+1} (-1)^\ell s_{L \setminus \{i_\ell\}}|_{U_L},$$

where $L = \{i_0, i_1, \ldots, i_{k+1}\}$ is written in the order induced from I. The complex

$$\check{C}^\bullet(\mathcal{U}, \mathcal{G}) = (\check{C}^k(\mathcal{U}, \mathcal{G}),\ k \geq 0,\ \delta_k)$$

is called the *Čech complex* of the sheaf $\mathcal{G}$ with respect to this cover.

For $k = 0$, we have

$$\check{C}^0(\mathcal{U}, \mathcal{G}) = \prod_{i \in I} \mathcal{G}(U_i),$$

and, if I is finite and nonempty, for $k = \#(I) - 1$ we have

$$\check{C}^k(\mathcal{U},\mathcal{G}) = \mathcal{G}\left(\bigcap_{i\in I} U_i\right).$$

Edition note: the source writes $k = \#(I)$ for the last term and $k > \#(I)$ for vanishing. The convention $\#(J) = k+1$ instead gives last degree $\#(I)-1$ and vanishing for $k \geq \#(I)$. In the differential above, the source's shorthand has also been made explicit: take the indicated component and restrict it to U_L.

If I is finite and $k \geq \#(I)$, the index set for $\check{C}^k(\mathcal{U},\mathcal{G})$ is empty, and this term is 0. For negative k, the complex is likewise defined to be 0. For a cover consisting of two open sets U and V, the complex is

$$0 \longrightarrow \Gamma(U,\mathcal{G}) \times \Gamma(V,\mathcal{G}) \longrightarrow \Gamma(U\cap V,\mathcal{G}) \longrightarrow 0.$$

For a cover consisting of three open sets U, V, W, the complex is

$$\begin{aligned}
0 &\longrightarrow \Gamma(U,\mathcal{G}) \times \Gamma(V,\mathcal{G}) \times \Gamma(W,\mathcal{G}) \\
&\longrightarrow \Gamma(V\cap W,\mathcal{G}) \times \Gamma(U\cap W,\mathcal{G}) \times \Gamma(U\cap V,\mathcal{G}) \\
&\longrightarrow \Gamma(U\cap V\cap W,\mathcal{G}) \longrightarrow 0.
\end{aligned}$$

To understand the homomorphisms, it is useful even in these cases to use the numbered names U_1, U_2, U_3.

Lemma 26.4: the Čech complex is indeed a complex

The Čech complex is indeed a complex.

Proof Let $(s_J)_J \in \check{C}^k(\mathcal{U},\mathcal{G})$ be a tuple. For a fixed index set

$$L = \{i_0, i_1, \dots, i_k, i_{k+1}, i_{k+2}\},$$

we obtain

$$\begin{aligned}
(\delta(\delta s))_L &= \sum_{p=0}^{k+2} (-1)^p (\delta s)|_{L\setminus\{i_p\}} \\
&= \sum_{p=0}^{k+2} (-1)^p \left(\sum_{q=0}^{p-1} (-1)^q s|_{(L\setminus\{i_p\})\setminus\{i_q\}} + \sum_{q=p+1}^{k+2} (-1)^{q+1} s|_{(L\setminus\{i_p\})\setminus\{i_q\}} \right) \\
&= \sum_{0\leq p<q\leq k+2} (-1)^{p+q} \left(s|_{L\setminus\{i_p,i_q\}} - s|_{L\setminus\{i_p,i_q\}} \right) \\
&= 0.
\end{aligned}$$

Note that the sign inside the parentheses depends on the position of i_q in $L \setminus \{i_p\}$. Here each displayed $s|_{L\setminus\{i_p,i_q\}}$ means the component $s_{L\setminus\{i_p,i_q\}}$ restricted to U_L; the same convention applies to δs.

Edition note: the source ends the second inner sum at $k+1$ and indexes the final sum by an undefined J. Since L has $k+3$ elements, the corrected bound is $k+2$, with one cancelling pair for each $0 \leq p < q \leq k+2$.

Definition 26.5: Čech cohomology

Let

$$X = \bigcup_{i\in I} U_i$$

be an open cover of a topological space X, and let $\mathcal{G}$ be a sheaf of commutative groups on X. For $k \in \mathbb{N}$, the *kth Čech cohomology*

$$\check{H}^k(\mathcal{U}, \mathcal{G})$$

is defined to be the kth homology of the Čech complex $\check{C}^\bullet(\mathcal{U}, \mathcal{G})$.

As with the homology of any complex, at each position we form the quotient group of the kernel modulo the image. Elements of the kth kernel are also called *Čech cocycles*, and elements of the kth image are also called *Čech coboundaries*. The element of the kth Čech cohomology associated with a Čech cocycle is also called a *Čech cohomology class*. The zeroth Čech cohomology group is simply $\mathcal{G}(X)$, as follows directly from the sheaf property; see Exercise 26.2.

Example 26.6: cocycles on the circle

On the circle, we consider the cover by two open circular arcs, each homeomorphic to a real interval,

$$S^1 = U \cup V,$$

whose intersection

$$U \cap V = S \cup T$$

is a union of two intervals. We consider several sheaves of commutative groups, written multiplicatively. Let h be the function on $U \cap V$ with constant value 1 on S and value -1 on T. This is a nontrivial Čech cocycle for the sheaf of locally constant functions with values in the group of units $K^\times$ of a field K of characteristic different from 2. The same holds for the sheaf of continuous functions with values in $\mathbb{K}^\times$, where $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$.

This cocycle defines a trivial Čech cohomology class in

$$\check{H}^1(\{U, V\}, \mathcal{G})$$

if and only if there are functions—locally constant or continuous, as appropriate—

$$f : U \longrightarrow K^\times, \qquad g : V \longrightarrow K^\times$$

with $h = fg^{-1}$. In the locally constant case this is impossible, because locally constant functions on the connected arcs U and V are constant; consequently fg^{-1} is also constant and hence differs from h. It is likewise impossible for the sheaf of nowhere-zero continuous real-valued functions. In this case, f and g have constant signs, so the sign of fg^{-1} agrees with that of h on exactly one interval of the intersection. The corresponding nontrivial first Čech cohomology class,

$$[h] \in \check{H}^1(\{U, V\}, C^0(-, \mathbb{R}^\times)),$$

represents the Möbius strip over the unit circle.

In the complex case, by contrast, h can be written as the quotient of two nowhere-zero continuous complex-valued functions. We can take $g = 1$ and choose f to have constant value 1 on S, constant value -1 on T, and, in between—that is, on $U \setminus (S \cup T)$—to take values continuously along the complex unit circle.

Edition note: the source omits the exclusion of characteristic 2, where $h = 1$ is trivial, and reverses "trivial" and "nontrivial" in the coboundary criterion. Both points are corrected above. In the real case it is the signs, not necessarily the function values, that agree on exactly one component. The source's set-difference braces have been replaced by parentheses.

Example 26.7: the Čech complex for a module on a quasi-affine scheme

For a commutative ring R and elements $f_1, \dots, f_n \in R$ generating an ideal $\mathfrak{a}$, there is an open cover

$$D(\mathfrak{a}) = \bigcup_{i=1}^{n} D(f_i)$$

of the quasi-affine scheme $D(\mathfrak{a}) \subseteq \operatorname{Spek}(R)$. For an R-module M, the Čech complex of the module sheaf $\widetilde{M}$ on $D(\mathfrak{a})$ can be written down directly, without considering the sheafification process. By Lemma 14.5, on the relevant open sets we have

$$\Gamma\left(D\left(\prod_{i\in J} f_i\right), \widetilde{M}\right) = M_{\prod_{i\in J} f_i}.$$

The Čech complex is therefore

$$0 \longrightarrow \prod_{1\le i\le n} M_{f_i} \longrightarrow \prod_{1\le i<j\le n} M_{f_i f_j} \longrightarrow \prod_{1\le i<j<k\le n} M_{f_i f_j f_k} \longrightarrow \cdots.$$

Computing the homology of this complex is generally still difficult, but it is now purely a problem in commutative algebra.

Čech cohomology and sheaf cohomology

We now discuss situations in which Čech cohomology for certain covers agrees with the "actual" sheaf cohomology, defined using injective resolutions.

Lemma 26.8: first Čech cohomology and sheaf cohomology

Let $\mathcal{F}$ be a sheaf of commutative groups on a topological space X, and let

$$X = \bigcup_{i\in I} U_i$$

be an open cover with

$$H^1(U_i, \mathcal{F}) = 0 \qquad \text{and} \qquad H^1(U_i \cap U_j, \mathcal{F}) = 0$$

for all i, j. Then

$$\check{H}^1(U_i, \mathcal{F}) = H^1(X, \mathcal{F}).$$

Proof Let $\mathcal{F} \subseteq \mathcal{I}$ be an embedding in an injective sheaf, and let

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I} \longrightarrow \mathcal{H} \longrightarrow 0$$

be the associated short exact sequence. By the long exact cohomology sequence—see Corollary 25.2 (3)—and Theorem 24.8, we have

$$H^1(X, \mathcal{F}) = \Gamma(X, \mathcal{H}) / \operatorname{bild}(\Gamma(X, \mathcal{I}) \longrightarrow \Gamma(X, \mathcal{H})).$$

We first define a homomorphism

$$\Gamma(X, \mathcal{H}) \longrightarrow \check{H}^1(U_i, \mathcal{F}).$$

A section $t \in \Gamma(X, \mathcal{H})$ determines restrictions $t_i = t|_{U_i}$. Since $H^1(U_i, \mathcal{F}) = 0$, there are

$$s_i \in \Gamma(U_i, \mathcal{I})$$

mapping to t_i. For $i < j$, the elements

$$r_{ij} := s_i - s_j$$

map to 0 in $\Gamma(U_i \cap U_j, \mathcal{H})$, so

$$r_{ij} \in \Gamma(U_i \cap U_j, \mathcal{F}).$$

For indices $i < j < k$, we have

$$r_{ij} - r_{ik} + r_{jk} = s_i - s_j - (s_i - s_k) + s_j - s_k = 0.$$

Thus the cocycle condition holds. The family $(r_{ij})_{i<j}$ is a Čech cocycle and defines an element of $\check{H}^1(U_i, \mathcal{F})$. This assignment is independent of the choice of s_i and is a group homomorphism; see Exercise 26.5.

Now let $t \in \Gamma(X, \mathcal{H})$ be the image of a global element $s \in \Gamma(X, \mathcal{I})$. We can take $s_i = s|_{U_i}$, so all the r_{ij} constructed from t are 0. Such an element therefore maps to 0. By Theorem 47.4 (Linear Algebra (Osnabrück 2024-2025)), we obtain a factorisation

$$H^1(X, \mathcal{F}) = \Gamma(X, \mathcal{H}) / \operatorname{bild}(\Gamma(X, \mathcal{I}) \longrightarrow \Gamma(X, \mathcal{H})) \longrightarrow \check{H}^1(U_i, \mathcal{F}).$$

Conversely, suppose we are given a first Čech cocycle of $\mathcal{F}$, represented by

$$(r_{ij})_{i<j} \in \prod_{i<j} \Gamma(U_i \cap U_j, \mathcal{F})$$

with $r_{ij} - r_{ik} + r_{jk} = 0$ on triple intersections.

Edition note: the source extends each r_{ij} to a global section of the flasque sheaf $\mathcal{I}$ and sets $s_i = r_{i1}$. Arbitrary extensions need not preserve the cocycle relations outside the original intersections. The following compatible construction repairs that step. The preceding quotient also corrects the source's target $\Gamma(X, \mathcal{I})$ to $\Gamma(X, \mathcal{H})$, and the tuple's ambient group is written as a product.

Set $r_{ji} = -r_{ij}$ and $r_{ii} = 0$. Using the fixed well-ordering of I, construct $s_i \in \Gamma(U_i, \mathcal{I})$ successively, with $s_i - s_j = r_{ij}$ on overlaps. At stage i, the sections $s_j + r_{ij}$ on $U_i \cap U_j$, for $j < i$, agree on their overlaps by the cocycle relation. They therefore glue on $U_i \cap \bigcup_{j<i} U_j$. Flasqueness extends this section to U_i; call the extension s_i. At the first stage take $s_i = 0$. This construction, including limit stages, gives

$$s_i - s_j = r_{ij} \quad \text{on } U_i \cap U_j.$$

The elements s_i determine elements

$$t_i \in \Gamma(U_i, \mathcal{H}).$$

Since their differences come from $\mathcal{F}$, these are compatible and determine a global element

$$t \in \Gamma(X, \mathcal{H}).$$

Via the connecting homomorphism δ, this determines a cohomology class

$$\delta(t) \in H^1(X, \mathcal{F}).$$

If the Čech cocycle r_{ij} is represented by other elements $s'_i \in \Gamma(U_i, \mathcal{J})$, then the elements $s_i - s'_i$, $i \in I$, are compatible because

$$(s_i - s'_i) - (s_j - s'_j) = s_i - s_j - (s'_i - s'_j) = r_{ij} - r_{ij} = 0,$$

and determine a global element of $\Gamma(X, \mathcal{J})$. Hence the difference between the two representations maps to 0 in $H^1(X, \mathcal{F})$. Altogether we obtain a well-defined map

$$\check{Z}^1(U_i, \mathcal{F}) \longrightarrow H^1(X, \mathcal{F}) = \Gamma(X, \mathcal{H}) / \operatorname{bild}(\Gamma(X, \mathcal{J}) \longrightarrow \Gamma(X, \mathcal{H})).$$

Now suppose the Čech cocycle determines the zero class in first Čech cohomology. By definition, there are elements

$$r_i \in \Gamma(U_i, \mathcal{F})$$

with

$$r_i - r_j = r_{ij}.$$

Regard these as local sections of $\mathcal{J}$ on U_i; the r_i can directly play the role of the s_i above. (Edition note: no global extensions are needed here, or for the alternative representatives s'_i.) Then all t_i are 0, so their image in $H^1(X, \mathcal{F})$ is also 0. Thus there is a map

$$\check{H}^1(U_i, \mathcal{F}) \longrightarrow H^1(X, \mathcal{F}).$$

This is a group homomorphism and is inverse to the map constructed previously.

Lemma 26.9: comparison using an acyclic resolution

Let $\mathcal{F} = \mathcal{F}_0$ be a sheaf of commutative groups on a topological space X, and suppose an acyclic resolution $\mathcal{Z}^\bullet$ of $\mathcal{F}$ is given, with associated short exact sequences

$$0 \longrightarrow \mathcal{F}_n \longrightarrow \mathcal{Z}_n \longrightarrow \mathcal{F}_{n+1} \longrightarrow 0.$$

Let

$$X = \bigcup_{i \in I} U_i$$

be an open cover with

$$H^k\left(\bigcap_{i \in J} U_i, \mathcal{F}_n\right) = 0$$

for all nonempty subsets $J \subseteq I$, all $n \in \mathbb{N}$, and all $k \in \mathbb{N}_+$. Then

$$\check{H}^k(U_i, \mathcal{F}) = H^k(X, \mathcal{F}).$$

Proof We use induction on k, simultaneously for all n. The case $k = 1$ was treated in Lemma 26.8, and the following argument follows that lemma. We consider the short exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{Z}_0 \longrightarrow \mathcal{F}_1 = \mathcal{H} \longrightarrow 0$$

together with the isomorphisms

$$H^k(X, \mathcal{F}) = H^{k-1}(X, \mathcal{H}) = \check{H}^{k-1}(U_i, \mathcal{H}).$$

The left-hand isomorphism comes from the connecting homomorphism and the acyclicity of $\mathcal{Z}_0$, and the right-hand one from the induction hypothesis applied to $\mathcal{H}$. It therefore remains to show that there is an isomorphism

$$\check{H}^{k-1}(U_i, \mathcal{H}) = \check{H}^k(U_i, \mathcal{F}).$$

A class on the left is represented by a tuple

$$(t_J) \in \prod_{\#(J)=k} \Gamma(U_J, \mathcal{H})$$

subject to the condition

$$\sum_{\ell=0}^{k} (-1)^{\ell} t_{L \setminus \{i_\ell\}} = 0$$

for all $(k+1)$-element subsets $L \subseteq I$. Since the cover is acyclic for $\mathcal{F}$, there is a tuple

$$(s_J) \in \prod_{\#(J)=k} \Gamma(U_J, \mathcal{Z}_0)$$

mapping to (t_J). This in turn determines a tuple of differences (r_L), where L ranges over the $(k+1)$-element subsets of I, by

$$r_L := \sum_{\ell=0}^{k} (-1)^{\ell} s_{L \setminus \{i_\ell\}}.$$

Since s_J maps to t_J, the condition above implies that all r_L map to 0. Thus

$$(r_L)_L \in \prod_{\#(L)=k+1} \Gamma(U_L, \mathcal{F}).$$

Further considerations show that this tuple is a cocycle, that the map is well-defined, and that it is a bijective group homomorphism.

Edition note: the induction step is for $k > 1$; degree 0 is the sheaf axiom. In the source's two sums over a $(k+1)$-element set, $k+1$ has been corrected to k, the first exponent i to ℓ, and the lifting product's $\#(J) = n, \mathcal{Z}$ to $\#(J) = k, \mathcal{Z}_0$. Restrictions to U_L are understood. To justify the abbreviated bijectivity step without an extra assumption on the chosen acyclic resolution, one may compute with an injective resolution instead. Its terms are flasque and have exact augmented Čech complexes. The successive cokernels are acyclic on every finite nonempty cover intersection, by the long exact sequence and the assumed acyclicity of $\mathcal{F}$. The resulting degreewise exact short sequences of Čech complexes give the asserted connecting isomorphism. This also supplies the induction argument for the stated comparison.

Theorem 26.10: cohomology on projective schemes

Let $(X, \mathcal{O}_X)$ be a projective scheme over a commutative ring R, and let $\mathcal{F}$ be a quasi-coherent module on X. Then the sheaf cohomology of $\mathcal{F}$ agrees with Čech cohomology for the affine cover by the sets $D_+(X_i)$.

Proof Let $X_0, X_1, \dots, X_n$ be the variables of the homogeneous coordinate ring of the projective scheme X, so that

$$X = \mathrm{Proj}(R[X_0, X_1, \dots, X_n]/\mathfrak{a}).$$

All intersections

$$\bigcap_{i \in J} D_+(X_i) = D_+\left(\prod_{i \in J} X_i\right)$$

are affine by Lemma 12.9. For $\mathcal{F}$, there is a flasque quasi-coherent sheaf $\mathcal{Z}$ with a short exact sequence

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{Z} \longrightarrow \mathcal{H} \longrightarrow 0.$$

By Exercise 14.21, the quotient sheaf is again quasi-coherent. By Theorem 25.11, quasi-coherent sheaves on affine schemes have no cohomology. We can therefore apply Lemma 26.9.

Edition note: for the arbitrary base ring stated here, the comparison need not rely on the source's unproved existence of a flasque quasi-coherent embedding. Take an injective resolution in sheaves of abelian groups. Its terms are flasque, and its successive cokernels are acyclic on every finite nonempty affine intersection by Theorem 25.11 and the long exact cohomology sequence. Lemma 26.9 then applies, without requiring the injective terms or the cokernels to be quasi-coherent.

The same assertion holds for quasi-affine schemes. The decisive property is that intersections of affine subsets are again affine. This often holds, but not for every scheme.

Worksheet 26: Čech cohomology

None of the eight exercises has a public solution at the frozen revision boundary. This edition does not create new solutions.

Exercise 26.1

Define a sheaf on $\mathrm{Spek}(\mathbb{Z})$ with nontrivial first cohomology.

Exercise 26.2

Let

$$X = \bigcup_{i \in I} U_i$$

be an open cover of a topological space X, and let $\mathcal{G}$ be a sheaf of commutative groups on X. Prove that

$$\check{H}^0(U_i,\ i \in I,\ \mathcal{G}) = \Gamma(X, \mathcal{G}).$$

Exercise 26.3

Let

$$X = \bigcup_{i \in I} U_i$$

be an open cover of a topological space X, and let $\mathcal{G}$ be a sheaf of commutative groups on X. Let

$$s \in \check{C}^k(U_i, \mathcal{G})$$

be a Čech cocycle which, for a particular

$$J = \{i_0, i_1, \dots, i_k\} \subseteq I$$

has value

$$a \in \Gamma(U_J, \mathcal{G}),$$

and has value 0 on all other $(k+1)$-element subsets $J' \neq J$. Determine

$$\delta(s) \in \check{C}^{k+1}(U_i, \mathcal{G}).$$

Edition note: the frozen source calls s a cocycle, not an arbitrary cochain. That hypothesis is retained; it restricts which single-component tuples are admissible. This note does not supply a source solution.

Exercise 26.4

Let K be a field and let

$$\mathbb{P}^1_K = \mathrm{Proj}(K[X, Y])$$

be the projective line over K. Determine the first Čech cohomology

$$\check{H}^1\left(D_+(X), D_+(Y), \mathcal{O}^\times_{\mathbb{P}^1_K}\right).$$

How does this relate to Example 26.1?

Exercise 26.5

Prove that the assignment in the proof of Lemma 26.8, which associates to a section

$$t \in \Gamma(X, \mathcal{H}),$$

a Čech cohomology class for $\mathcal{F}$, is independent of the chosen local representatives in $\mathcal{I}$ and is a group homomorphism.

Exercise 26.6

Let X be an irreducible topological space and let $\mathcal{G}$ be the constant sheaf associated with a commutative group G. Determine the Čech complex and Čech cohomology of $\mathcal{G}$ for a finite open cover

$$X = \bigcup_{i \in I} U_i.$$

Exercise 26.7

Let $(X, \mathcal{O}_X)$ be a ringed space, let

$$X = \bigcup_{i \in I} U_i$$

be an open cover, and let $\mathcal{F}$ be an $\mathcal{O}_X$-module. Prove that the Čech complex of $\mathcal{F}$ for this cover is a complex of $\Gamma(X, \mathcal{O}_X)$-modules, and hence that the corresponding Čech cohomology groups are also $\Gamma(X, \mathcal{O}_X)$-modules.

Exercise 26.8

Let R be a commutative ring and let

$$X = \mathrm{Spek}(R) = D(f) \cup D(g)$$

with $f, g \in R$. Prove that

$$\check{H}^1(\{D(f), D(g)\}, \mathcal{O}_X) = 0.$$

Public solutions and coverage of Worksheet 26

At the frozen revision boundary, the source provides no public solutions for any of the eight exercises on Worksheet 26. The exercise map and candidate evidence record negative results for Exercises 26.1–26.8. The absence of a public solution page is not replaced by a fabricated solution.

Frozen negative results

There is no public solution page at the frozen revision for Exercise 26.1, 26.2, 26.3, 26.4, 26.5, 26.6, 26.7, or 26.8. This statement records the result of checking the candidates, not a claim that mathematical solutions do not exist.

Lecture 27: Cohomology on projective schemes

Čech cohomology on the polynomial ring

Let R be a commutative ring and let

$$A = R[X_1, \dots, X_n]$$

be the polynomial ring in n variables over R. In particular, one may keep in mind the case where R is a field. We consider the open set

$$U = \mathbb{A}^n_R \setminus V(X_1, \dots, X_n) = D(X_1, \dots, X_n) = \bigcup_{i=1}^{n} D(X_i),$$

and will use the displayed affine cover, with $D(X_i) = \mathrm{Spek}(A_{X_i})$. For an A-module M, the Čech complex of the sheaf $\widetilde{M}$ on U with respect to this cover has the form

$$\prod_{1 \le i \le n} M_{X_i} \longrightarrow \prod_{1 \le i < j \le n} M_{X_i X_j} \longrightarrow \prod_{1 \le i < j < k \le n} M_{X_i X_j X_k} \longrightarrow \cdots \longrightarrow M_{X_1 \cdots X_n} \longrightarrow 0.$$

Thus

$$\check{C}^p(D(X_i), \widetilde{M}) = \prod_{1 \le i_0 < i_1 < \cdots < i_p \le n} M_{X_{i_0} \cdots X_{i_p}},$$

> **Edition note—missing relation sign.** The frozen source displays `i_1<\cdots i_p`, with no further `<` before `i_p`. The missing relation sign has been restored above, since the product is indexed by strictly increasing tuples, as in the preceding two displayed terms of the Čech complex.

and the pth Čech cohomology is the homology of the complex written above. Componentwise, the maps are simply the canonical maps into the localisations: at each step one additional variable is admitted as a denominator. These maps also carry the signs specified in the definition of the Čech complex.

We will describe this complex more precisely for the structure sheaf, that is, for $M = A$. It is useful to split it into simpler complexes using the fine monomial grading by the group $\mathbb{Z}^n$. We begin with small dimensions.

Example 27.1: two variables

Let $A = R[X, Y]$. The Čech complex of the structure sheaf $\mathcal{O}_{\mathbb{A}^2_R}$ on $U = D(X) \cup D(Y) \subset \mathbb{A}^2_R$ is

$$0 \longrightarrow A_X \times A_Y \longrightarrow A_{XY} \longrightarrow 0.$$

This complex is compatible with the fine monomial grading. The component corresponding to $(\alpha, \beta) \in \mathbb{Z}^2$ depends essentially on whether the two exponents are negative or nonnegative.

If α and β are both nonnegative, the entire complex in that component is

$$(A_X \times A_Y)_{(\alpha,\beta)} = R \cdot X^\alpha Y^\beta \oplus R \cdot X^\alpha Y^\beta \longrightarrow R \cdot X^\alpha Y^\beta \longrightarrow 0.$$

This complex is exact at the right-hand term, and the kernel at the left-hand term is isomorphic to $R \cdot X^\alpha Y^\beta$. If α is negative and β is nonnegative, or vice versa, the entire complex is

$$(A_X \times A_Y)_{(\alpha,\beta)} = R \cdot X^\alpha Y^\beta \oplus 0 \longrightarrow R \cdot X^\alpha Y^\beta \longrightarrow 0,$$

and this complex is exact. If α and β are both negative, the entire complex is

$$(A_X \times A_Y)_{(\alpha,\beta)} = 0 \oplus 0 \longrightarrow R \cdot X^\alpha Y^\beta \longrightarrow 0,$$

and the homology at the right-hand term is $R \cdot X^\alpha Y^\beta$. Consequently,

$$\check{H}^0(D(X), D(Y), \mathcal{O}_{\mathbb{A}^2_R}) = \bigoplus_{(\alpha,\beta)\in\mathbb{N}^2} R \cdot X^\alpha Y^\beta = A$$

and

$$\check{H}^1(D(X), D(Y), \mathcal{O}_{\mathbb{A}^2_R}) = \bigoplus_{(\alpha,\beta)\in\mathbb{Z}^2_-} R \cdot X^\alpha Y^\beta.$$

Example 27.2: three variables

Let $A = R[X, Y, Z]$. The Čech complex of the structure sheaf is

$$0 \longrightarrow A_X \times A_Y \times A_Z \longrightarrow A_{XY} \times A_{XZ} \times A_{YZ} \longrightarrow A_{XYZ} \longrightarrow 0.$$

This complex is compatible with the fine monomial grading. Here $\check{H}^0 = A$, while $\check{H}^1 = 0$ (see Theorem 27.3), and $\check{H}^2$ is the free R-module with basis

$$X^i Y^j Z^k, \qquad (i, j, k) \in \mathbb{Z}^3_-.$$

Theorem 27.3: cohomology of the punctured cover

Let R be a commutative ring and let $A = R[X_1, \dots, X_n]$ be the polynomial ring in $n \geq 2$ variables over R. Then the Čech cohomology of the structure sheaf on the open set $U = D(X_1, \dots, X_n)$ with respect to the cover $D(X_i)$, $i = 1, \dots, n$, is

$$\check{H}^p(D(X_i), \mathcal{O}_{\mathbb{A}^n_R}) = \begin{cases} A, & p = 0, \\ 0, & 1 \leq p \leq n-2, \\ \bigoplus_{\alpha\in\mathbb{Z}^n_-} R \cdot X^\alpha, & p = n-1. \end{cases}$$

Proof We consider the Čech complex with the fine $\mathbb{Z}^n$-grading given by the monomials. For a fixed tuple

$$\alpha = (\alpha_1, \dots, \alpha_n) \in \mathbb{Z}^n,$$

let $N = N_\alpha \subseteq \{1, \dots, n\}$ be the set of indices with negative entries. For this α,

$$
\begin{aligned}
\left(\check{C}^p(D(X_i), \mathcal{O}_{\mathbb{A}^n})\right)_\alpha &= \left(\prod_{1\le i_0<i_1<\dots<i_p\le n} A_{X_{i_0}X_{i_1}\cdots X_{i_p}}\right)_\alpha \\
&= \prod_{\substack{N\subseteq L\\ \#(L)=p+1}} R\cdot e_L \\
&\cong \prod_{\substack{J\subseteq\{1,\dots,n\}\setminus N\\ \#(J)=p+1-\#(N)}} R\cdot e_J.
\end{aligned}
$$

The middle identification rests on the fact that the component of $A_{\prod_{i\in L} X_i}$ is 0 when $N \nsubseteq L$ and is $R\cdot X^\alpha$ when $N\subseteq L$. The monomial X^α in this localisation corresponds to e_L. In the right-hand identification, e_J corresponds to the basis element $e_L = e_{N\cup J}$.

Thus, after shifting degrees by $\#(N)-1$ and changing basis signs in the usual way, the complex at index α corresponds to an ascending binomial complex for the index set $\{1,\dots,n\}\setminus N$ over the ring R (rather than $\mathbb{Z}$). If $N\neq\emptyset$, its empty-set summand occurs in Čech degree $\#(N)-1$; only when $N=\emptyset$ would that summand occur in degree -1, and it is then absent from the Čech complex.

> **Edition note—empty-set summand.** The frozen source says without qualification that the corresponding binomial complex lacks the free summand indexed by the empty set. The component formula shows that this is true only for $N=\emptyset$; for nonempty N the summand occurs in degree $\#(N)-1$. The degree shift and harmless basis-sign changes needed to identify the differentials have also been made explicit.

For $p=0$ and at least one negative exponent, there is at most one isolated $R\cdot X^\alpha$ on the right. Since $n\ge 2$, however, this term does not map to 0, and so contributes nothing to H^0. If all exponents are nonnegative, by contrast, the elements have the form

$$(c_1X^\alpha,\dots,c_nX^\alpha),$$

and such an element maps to 0 precisely when all coefficients $c_i\in R$ agree. The zeroth Čech cohomology is therefore the polynomial ring

$$A=\bigoplus_{\alpha\in\mathbb{N}^n} R\cdot X^\alpha.$$

Now let $p\ge 1$. If $N\neq\{1,\dots,n\}$, the situation is isomorphic to an ascending binomial complex on a nonempty index set. Its homology is therefore trivial by Appendix Lemma 8.11. Hence the homology is trivial for every p between 1 and $n-2$.

It remains to consider $p=n-1$ and $N=\{1,\dots,n\}$. These are precisely the α with all exponents negative. The complex, corresponding to the ascending binomial complex on the empty set, is

$$0\longrightarrow R\cdot X^\alpha\longrightarrow 0.$$

Therefore,

$$H^{n-1}=\bigoplus_{\alpha\in\mathbb{Z}_-^n} R\cdot X^\alpha.$$

Cohomology on projective schemes

Theorem 27.4: cohomology of twisted structure sheaves

Let R be a commutative ring, let $A=R[X_0,X_1,\dots,X_d]$ be the polynomial ring in $d+1\ge 2$ variables over R, and let

$$\mathbb{P}^d_R=\mathrm{Proj}(A)$$

be the associated projective space. Then the cohomology of the twisted structure sheaf $\mathcal{O}_{\mathbb{P}^d_R}(n)$ is

$$\check{H}^p(\mathbb{P}^d_R, \mathcal{O}_{\mathbb{P}^d_R}(n)) = \begin{cases} A_n, & p = 0, \\ 0, & 1 \le p \le d-1, \\ \bigoplus_{\substack{\alpha \in \mathbb{Z}_-^{d+1} \\ \sum_{j=1}^{d+1} \alpha_j = n}} R \cdot X^\alpha, & p = d. \end{cases}$$

Proof This follows from Theorem 27.3.

In particular, for the canonical sheaf $\mathcal{O}_{\mathbb{P}^d_R}(-d-1)$ (compare Corollary 19.10),

$$H^d(\mathbb{P}^d_R, \mathcal{O}_{\mathbb{P}^d_R}(-d-1)) = RX_0^{-1}X_1^{-1} \cdots X_d^{-1} \cong R,$$

and

$$H^d(\mathbb{P}^d_R, \mathcal{O}_{\mathbb{P}^d_R}(n)) = 0$$

for $n > -d-1$.

> **Edition note—index in the canonical generator.** The frozen source specifies the variables $X_0, \dots, X_d$, but writes X_n^{-1} as the final factor of the generator. The display above corrects that final index to d; its product then uses each of the $d+1$ variables exactly once and has degree $-d-1$.

Theorem 27.5: finiteness of cohomology on projective space

Let $\mathbb{P}^n_R$ be projective space over a Noetherian ring R, and let $\mathcal{F}$ be a coherent sheaf on $\mathbb{P}^n_R$. Then $H^i(\mathbb{P}^n_R, \mathcal{F})$ is a finitely generated R-module.

Proof For the twisted structure sheaves $\mathcal{O}_{\mathbb{P}^n_R}(\ell)$, the assertion follows from Theorem 27.4. It therefore also holds for finite direct sums of such sheaves.

We prove the general case by descending induction on the cohomological index i. If this index exceeds n, there is only trivial cohomology by Theorem 26.10. If R has finite dimension, the same also follows from Theorem 25.12. This establishes the base case.

Suppose, then, that the assertion has been proved for some i and every coherent sheaf. Let $\mathcal{F}$ be a coherent sheaf. By Theorem 15.13, there are a finite direct sum

$$\bigoplus_{j \in J} \mathcal{O}_{\mathbb{P}^n_R}(\ell_j)$$

and a surjective $\mathcal{O}_{\mathbb{P}^n_R}$-module homomorphism

$$\bigoplus_{j \in J} \mathcal{O}_{\mathbb{P}^n_R}(\ell_j) \longrightarrow \mathcal{F}.$$

Let $\mathcal{G}$ be the kernel of this map; by Exercise 14.20, $\mathcal{G}$ is also coherent. The long exact cohomology sequence associated with the short exact sequence of sheaves

$$0 \longrightarrow \mathcal{G} \longrightarrow \bigoplus_{j \in J} \mathcal{O}_{\mathbb{P}^n_R}(\ell_j) \longrightarrow \mathcal{F} \longrightarrow 0$$

contains the portion

$$\cdots \longrightarrow H^{i-1}\left(\mathbb{P}^n_R, \bigoplus_{j \in J} \mathcal{O}_{\mathbb{P}^n_R}(\ell_j)\right) \xrightarrow{\epsilon} H^{i-1}(\mathbb{P}^n_R, \mathcal{F}) \xrightarrow{\delta} H^i(\mathbb{P}^n_R, \mathcal{G}) \longrightarrow \cdots.$$

This gives a short exact sequence of R-modules

$$0 \longrightarrow \operatorname{im} \epsilon = \ker \delta \longrightarrow H^{i-1}(\mathbb{P}^n_R, \mathcal{F}) \longrightarrow \operatorname{im} \delta \longrightarrow 0.$$

By the preceding observation and the induction hypothesis,

$$H^{i-1}\left(\mathbb{P}^n_R, \bigoplus_{j \in J} \mathcal{O}_{\mathbb{P}^n_R}(\ell_j)\right) \quad \text{and} \quad H^i(\mathbb{P}^n_R, \mathcal{G})$$

are finitely generated R-modules. Hence $\operatorname{im} \epsilon$ is finitely generated. Moreover, since R is Noetherian, $\operatorname{im} \delta$, as a submodule of $H^i(\mathbb{P}^n_R, \mathcal{G})$, is finitely generated by Theorem 10.4 in *Algebraic Curves (Osnabrück 2025-2026)*. By Lemma 23.2 in *Commutative Algebra*, $H^{i-1}(\mathbb{P}^n_R, \mathcal{F})$ is likewise finitely generated.

> **Edition note—two inconsistencies in the source proof.** The source calls the displayed homomorphism an $\mathcal{O}_{\mathbb{P}^d_R}$-module homomorphism, although its source and target lie on $\mathbb{P}^n_R$; the index has been corrected to n. It also repeats finite generation of $\ker \delta = \operatorname{im} \epsilon$, whereas the second end term needed in the displayed short exact sequence is $\operatorname{im} \delta$. The proof above corrects that object and uses the stated Noetherian hypothesis to pass to the submodule $\operatorname{im} \delta \subseteq H^i(\mathbb{P}^n_R, \mathcal{G})$.

Theorem 27.6: cohomology under a closed embedding

Let X be a projective scheme over a Noetherian ring R, with a closed embedding $X \subseteq \mathbb{P}^n_R$ in a projective space. Let $\mathcal{G}$ be a quasi-coherent sheaf on X, and let $j_*\mathcal{G}$ be its direct image sheaf. Then

$$H^i(X, \mathcal{G}) = H^i(\mathbb{P}^n_R, j_*\mathcal{G})$$

for every i.

Proof The direct image sheaf is again quasi-coherent. By Theorem 26.10, both sides can be computed using Čech cohomology for the standard affine cover $D_+(X_s)$ of projective space and the cover $X \cap D_+(X_s)$ of X. The resulting Čech complexes agree in their entirety, and hence so do their Čech cohomology groups.

Theorem 27.7: finiteness of cohomology on projective schemes

Let X be a projective scheme over a Noetherian ring R and let $\mathcal{G}$ be a coherent sheaf on X. Then $H^i(X, \mathcal{G})$ is a finitely generated R-module.

Proof This follows from Theorem 27.6 and Theorem 27.5.

Note that these are R-modules, not modules over the coordinate ring of X. In the most important case, where $R = K$ is a field, the cohomology groups are finite-dimensional vector spaces over K. Their dimensions are natural numbers associated with coherent sheaves on X and, in a certain sense, characteristic of those sheaves. Taking the structure sheaf or the tangent sheaf on X gives numbers, or invariants, characteristic of X itself. In this context one uses the abbreviation

$$h^i(\mathcal{F}) = \dim_K\big(H^i(X, \mathcal{F})\big).$$

For example, for a smooth projective curve X over an algebraically closed field K, the vector-space dimension of $H^1(X, \mathcal{O}_X)$ is called the *genus* of the curve. This is its most important invariant. In the complex case, there is a direct connection with the topological shape of the curve as a one-dimensional complex manifold and a two-dimensional real manifold.

> **Edition note—source typo.** In the sentence about modules over the coordinate ring, the German source writes *Koordinatenring*. The translation renders the intended term as “coordinate ring” without changing its mathematical content.

The Euler characteristic

Definition 27.8: the Euler characteristic

Let X be a projective scheme over a field K. For a coherent sheaf $\mathcal{G}$, the number

$$\chi(\mathcal{G}) := \sum_{i=0}^{\dim(X)} (-1)^i h^i(X, \mathcal{G})$$

is called the *Euler characteristic* of $\mathcal{G}$.

By Theorem 27.7, this expression is a well-defined integer. Since cohomology is 0 above the dimension, the alternating sum could also be continued to infinity.

Lemma 27.9: additivity of the Euler characteristic

Let X be a projective scheme over a field K. The Euler characteristic of coherent sheaves on X is additive in short exact sequences. In other words, for a short exact sequence of coherent sheaves

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H} \longrightarrow 0,$$

we have

$$\chi(\mathcal{G}) = \chi(\mathcal{F}) + \chi(\mathcal{H}).$$

Proof This follows from the associated long exact cohomology sequence, Theorem 25.12, and the dimension formula.

Worksheet 27: Cohomology on projective schemes

At the frozen revision boundary, none of the fourteen exercises has a public solution. The candidate map records negative results for Exercises 27.1–27.14; this edition does not create new solutions.

Exercise 27.1

Let $A = R[X]$ over a commutative ring R. Determine the Čech complex of the structure sheaf for the one-element cover consisting of $D(X)$ of the punctured line. What homology results?

Exercise 27.2

Let $A = R[X, Y]$ over a commutative ring R. Determine the Čech complex of the structure sheaf for the standard cover of the punctured plane at the monomials

1. X^2Y^3,
2. X^5Y^{-4},
3. $X^{-3}Y^{-6}$.

What homology results in each case?

Exercise 27.3

Let $A = R[X, Y, Z]$ over a commutative ring R. Determine the Čech complex of the structure sheaf for the standard cover of punctured space at the monomial $X^3Y^{-2}Z^7$. What homology results?

Exercise 27.4

Let $A = R[X, Y, Z]$ over a commutative ring R. Determine the Čech complex of the structure sheaf for the standard cover of punctured space at the monomial $X^{-5}YZ^{-4}$. What homology results?

Exercise 27.5

Let $A = R[X, Y, Z, W]$ over a commutative ring R. Determine the Čech complex of the structure sheaf for the standard cover of punctured space at the monomial $X^{-3}YZ^3W^{-2}$. What homology results?

Exercise 27.6

Let $K[X_1, \ldots, X_d]$ be the polynomial ring over a field K, and let H be the K-vector space spanned by all monomials X^ν in the variables $X_1, \ldots, X_d$ with $\nu_j \leq -1$ for every j, that is,

$$H = K \left\langle X_1^{\nu_1} \cdots X_d^{\nu_d},\ \nu_j \leq -1 \right\rangle .$$

Define a natural $K[X_1, \ldots, X_d]$-module structure on H.

Exercise 27.7

Let $K[Y_1, \ldots, Y_d]$ be the polynomial ring over a field K, and let H be the K-vector space spanned by all monomials X^ν in the variables $X_1, \ldots, X_d$ with $\nu_j \leq -1$ for every j, that is,

$$H = K \left\langle X_1^{\nu_1} \cdots X_d^{\nu_d},\ \nu_j \leq -1 \right\rangle .$$

Prove that the map

$$K[Y_1, \ldots, Y_d] \longrightarrow H, \qquad Y^\mu \longmapsto X^{-\mu-(1,1,\ldots,1)},$$

is a K-linear isomorphism of K-vector spaces.

Exercise 27.8

Let $K[Y_1, \ldots, Y_d]$ be the polynomial ring over a field K of characteristic 0, and let H be the K-vector space spanned by all monomials X^ν in the variables $X_1, \ldots, X_d$ with $\nu_j \leq -1$ for every j, that is,

$$H = K \left\langle X_1^{\nu_1} \cdots X_d^{\nu_d},\ \nu_j \leq -1 \right\rangle .$$

Prove that the map

$$\theta : K[Y_1, \ldots, Y_d] \longrightarrow H, \qquad Y^\mu \longmapsto \mu! X^{-\mu-(1,1,\ldots,1)},$$

is a K-linear isomorphism of K-vector spaces.

Exercise 27.9

Let K be a field of characteristic 0, and let $K[X_1, \ldots, X_d]$, $K[Y_1, \ldots, Y_d]$, and $K[D_1, \ldots, D_d]$ be polynomial rings in d variables. Let H be the K-vector space described in Exercise 27.6, with its natural $K[X_1, \ldots, X_d]$-module structure.

The polynomial ring $K[D_1, \ldots, D_d]$ acts on $K[Y_1, \ldots, Y_d]$ by letting D_i act as the ith partial derivative, namely

$$D_i = \partial_i = \frac{\partial}{\partial Y_i}.$$

Prove that the correspondence $D_i \mapsto X_i$, together with the map

$$\theta : K[Y_1, \ldots, Y_d] \longrightarrow H$$

from Exercise 27.8, gives an isomorphism of modules.

For the next exercise, note that in the smooth case, by Corollary 19.12, one is computing the dimension of the space of global differential forms.

Exercise 27.10

Let

$$C = V_+(f) \subset \mathbb{P}^2_K$$

be a projective plane curve of degree d over a field K. Using the long exact cohomology sequence arising from the short exact sequence of sheaves (compare Exercise 13.23)

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^2_K}(-3) \xrightarrow{f} \mathcal{O}_{\mathbb{P}^2_K}(d-3) \longrightarrow \mathcal{O}_C(d-3) \longrightarrow 0$$

on the projective plane and Theorem 27.4, prove that the dimension of

$$H^0(C, \mathcal{O}_C(d-3))$$

is

$$\frac{(d-1)(d-2)}{2}.$$

For the next two exercises, compare Theorem 22.12.

Exercise 27.11

Compute the Čech complex of the sheaf of units $\mathcal{O}^\times_{\mathbb{P}^1_K}$ for the standard affine cover of the projective line $\mathbb{P}^1_K$ over a field K, and its first Čech cohomology

$$\check{H}^1\left(D_+(X), D_+(Y), \mathcal{O}^\times_{\mathbb{P}^1_K}\right).$$

Exercise 27.12

Compute the Čech complex of the sheaf of units $\mathcal{O}^\times_{\mathbb{P}^2_K}$ for the standard affine cover of the projective plane $\mathbb{P}^2_K$ over a field K, and its first Čech cohomology

$$\check{H}^1\left(D_+(X), D_+(Y), D_+(Z), \mathcal{O}^\times_{\mathbb{P}^2_K}\right).$$

Exercise 27.13

Let R be a commutative ring and let M_i, $i \in \mathbb{N}$, be R-modules with fixed R-module homomorphisms

$$\varphi_i : M_i \longrightarrow M_{i+1}.$$

The sequence

$$\cdots \longrightarrow M_i \longrightarrow M_{i+1} \longrightarrow M_{i+2} \longrightarrow M_{i+3} \longrightarrow \cdots$$

is called exact if, for every $i \geq 1$,

$$\ker(\varphi_i) = \operatorname{im}(\varphi_{i-1}).$$

1. Prove that for a short exact sequence, this definition agrees with the definition of a short exact sequence referenced in the source.

2. Now let $R = K$ be a field, assume that the displayed sequence is exact, suppose all M_i are finitely generated, $M_0 = 0$, and $M_i = 0$ for every $i \geq n$, for some n. Prove that

$$\sum_{i=0}^{n} (-1)^i \dim_K M_i = 0.$$

Edition note—incomplete reference label. In the comparison sentence above, the frozen source displays the link text "Definition ." without a number, but the link target is available. This edition preserves the target and supplies a descriptive English label without inventing a definition number.

Edition note—missing exactness hypothesis. Part 2 of the frozen source does not explicitly say that the displayed sequence is exact. Without that hypothesis the asserted alternating-dimension formula is false in general, so the necessary hypothesis has been supplied above.

Edition note—index range. The frozen source requires the equality above "for every i", although the modules and maps are indexed by $\mathbb{N}$ and φ_{-1} is therefore not defined at $i = 0$. The condition has been stated for $i \geq 1$, the indices for which both sides are defined.

Exercise 27.14

Compute the Euler characteristic of the twisted structure sheaves $\mathcal{O}_{\mathbb{P}^d_K}(n)$ on projective space $\mathbb{P}^d_K$ over an algebraically closed field K.

Public solutions and coverage of Worksheet 27

At the frozen worksheet revision, the source contains exactly 14 exercises and provides no public solution pages. Each candidate was checked using the exact exercise-page title with the suffix `/Lösung` appended. The official candidate evidence establishes that all fourteen pages are absent. This result is not inferred from stars or from the presence or absence of solution links on the worksheet.

Official candidates checked

1. Exercise 27.1
    - Exercise page: Polynomring/1/NenneraufnahmeanX/Cech-Kohomologie/Aufgabe
    - Solution candidate: Polynomring/1/NenneraufnahmeanX/Cech-Kohomologie/Aufgabe/Lösung
    - Official result: page absent.
2. Exercise 27.2
    - Exercise page: Polynomring/2/Cech-Komplex/Monom/Aufgabe
    - Solution candidate: Polynomring/2/Cech-Komplex/Monom/Aufgabe/Lösung
    - Official result: page absent.
3. Exercise 27.3
    - Exercise page: Polynomring/3/Cech-Komplex/Monom/1/Aufgabe
    - Solution candidate: Polynomring/3/Cech-Komplex/Monom/1/Aufgabe/Lösung
    - Official result: page absent.
4. Exercise 27.4
    - Exercise page: Polynomring/3/Cech-Komplex/Monom/2/Aufgabe
    - Solution candidate: Polynomring/3/Cech-Komplex/Monom/2/Aufgabe/Lösung
    - Official result: page absent.
5. Exercise 27.5
    - Exercise page: Polynomring/4/Cech-Komplex/Monom/Aufgabe
    - Solution candidate: Polynomring/4/Cech-Komplex/Monom/Aufgabe/Lösung
    - Official result: page absent.
6. Exercise 27.6
    - Exercise page: Polynomring/HöchstelokaleKohomologie/Modulstruktur/Direkt/Aufgabe
    - Solution candidate: Polynomring/HöchstelokaleKohomologie/Modulstruktur/Direkt/Aufgabe/Lösung
    - Official result: page absent.
7. Exercise 27.7
    - Exercise page: Polynomring/HöchstelokaleKohomologie/K-IsomorphiezuPolynomring/Aufgabe
    - Solution candidate: Polynomring/HöchstelokaleKohomologie/K-IsomorphiezuPolynomring/Aufgabe/Lösung
    - Official result: page absent.
8. Exercise 27.8
    - Exercise page: Polynomring/HöchstelokaleKohomologie/Fakultäten/K-IsomorphiezuPolynomring/Aufgabe
    - Solution candidate: Polynomring/HöchstelokaleKohomologie/Fakultäten/K-IsomorphiezuPolynomring/Aufgabe/Lösung
    - Official result: page absent.
9. Exercise 27.9
    - Exercise page: Polynomring/HöchstelokaleKohomologie/Differentialoperatoren/IsomorpheSituation/Aufgabe
    - Solution candidate: Polynomring/HöchstelokaleKohomologie/Differentialoperatoren/IsomorpheSituation/Aufgabe/Lösung
    - Official result: page absent.

10. Exercise 27.10
    - Exercise page: ProjektiveEbene/Kurve/GetwisteteStrukturgabezud-3/GlobaleSchnitte/Aufgabe
    - Solution candidate: ProjektiveEbene/Kurve/GetwisteteStrukturgabezud-3/GlobaleSchnitte/Aufgabe/Lösung
    - Official result: page absent.
11. Exercise 27.11
    - Exercise page: ProjektiveGerade/Einheitengarbe/Cech-Komplex/ErsteKohomologie/Aufgabe
    - Solution candidate: ProjektiveGerade/Einheitengarbe/Cech-Komplex/ErsteKohomologie/Aufgabe/Lösung
    - Official result: page absent.
12. Exercise 27.12
    - Exercise page: ProjektiveEbene/Einheitengarbe/Cech-Komplex/ErsteKohomologie/Aufgabe
    - Solution candidate: ProjektiveEbene/Einheitengarbe/Cech-Komplex/ErsteKohomologie/Aufgabe/Lösung
    - Official result: page absent.
13. Exercise 27.13
    - Exercise page: Modultheorie/ExakteKomplexe/KurzeexakteSequenzen/Aufgabe
    - Solution candidate: Modultheorie/ExakteKomplexe/KurzeexakteSequenzen/Aufgabe/Lösung
    - Official result: page absent.
14. Exercise 27.14
    - Exercise page: ProjektiverRaum/GetwisteteStrukturgarben/Euler-Charakteristik/Aufgabe
    - Solution candidate: ProjektiverRaum/GetwisteteStrukturgarben/Euler-Charakteristik/Aufgabe/Lösung
    - Official result: page absent.

Frozen negative results

There are no public solution pages at the frozen revision for Exercises 27.1 through 27.14. There is therefore no source-solution text to translate. This statement records the check of each official candidate, not a claim that mathematical solutions do not exist.

Lecture 28: Morphisms to projective space

By definition, a projective variety X over a field K can be realised as a closed subvariety $X \subseteq \mathbb{P}^n_K$. Two competing viewpoints arise here.

On the one hand, realising X as part of a projective space lets us use concepts, structures, and properties of the ambient space by restricting them to X. We can investigate how X intersects other subvarieties Y, look for relationships with the open complement $\mathbb{P}^n_K \setminus X$, and visualise X inside an ambient space. This is the *extrinsic* viewpoint.

On the other hand, we can ask which properties belong to the variety X itself, independently of a particular realisation. This is the *intrinsic* viewpoint. Typically, X is isomorphic to an “other” variety X' given as a closed subset $X' \subseteq \mathbb{P}^{n'}_K$. Which properties of X and X' are independent of their respective embeddings?

The two viewpoints meet in the following questions. How many embeddings does a given X have? Can one understand all embeddings of X into projective space at once? Is there a best embedding, for example into an ambient space of small dimension or with a particularly transparent relationship to it? Is there a natural embedding related to characteristic objects on X?

For example, consider the closed projective curve

$$C = V_+(Y^2 - XZ) \subseteq \mathbb{P}^2_K.$$

This curve has degree 2, and its intersection with any line consists of two points, counted with multiplicities. The map

$$\mathbb{P}^1_K \longrightarrow \mathbb{P}^2_K, \qquad (s,t) \longmapsto (s^2, st, t^2),$$

induces an isomorphism $\mathbb{P}^1_K \to C \subseteq \mathbb{P}^2_K$. Thus C is isomorphic to the projective line and can be regarded as an "unnecessarily curved" version of it. Curves of degree two—quadrics or conic sections—are nevertheless natural objects in the plane. From the projective line's viewpoint, the elements s^2, st, t^2 form a basis of the second homogeneous component $K[s,t]_2$ of the homogeneous coordinate ring $K[s,t]$. These elements also occur as global sections of the invertible sheaf $\mathcal{O}_{\mathbb{P}^1_K}(2)$. We will see that different embeddings of X into projective space are related to global sections of invertible sheaves on X.

Invertible sheaves and morphisms to projective space

Lemma 28.1

Let X be a scheme over a commutative ring R, let $\mathcal{L}$ be an invertible sheaf on X, and let

$$s_0, s_1, \dots, s_n \in \Gamma(X, \mathcal{L}).$$

If $U \subseteq X$ is the union of the open sets X_{s_i}, then

$$U \longrightarrow \mathbb{P}^n_R, \qquad x \longmapsto (s_0(x), s_1(x), \dots, s_n(x))$$

is a morphism.

Proof First consider the situation on $X_i = X_{s_i}$. By Lemma 13.22,

$$\mathcal{O}_X|_{X_i} \longrightarrow \mathcal{L}|_{X_i}, \qquad 1 \longmapsto s_i,$$

is an isomorphism of $\mathcal{O}_X$-modules. Under this isomorphism, the section s_k corresponds to a function

$$f_{ki} \in \Gamma(X_i, \mathcal{O}_X), \qquad f_{ki} = \frac{s_k}{s_i}.$$

This quotient is well-defined. By Corollary 10.13, the functions f_{ki}, for $k \neq i$, define a morphism

$$\varphi_i : X_i \longrightarrow D_+(x_i) \cong \mathbb{A}^n_R \subseteq \mathbb{P}^n_R.$$

Altogether we have a commutative diagram

$$\begin{array}{ccc} X_i & \xrightarrow{\varphi_i} & D_+(x_i) \cong \mathbb{A}^n_R \subseteq \mathbb{P}^n_R \\ \uparrow & & \uparrow \\ X_i \cap X_j & \to & D_+(x_i) \cap D_+(x_j) \\ \downarrow & & \downarrow \\ X_j & \xrightarrow{\varphi_j} & D_+(x_j) \cong \mathbb{A}^n_R \subseteq \mathbb{P}^n_R. \end{array}$$

On $X_i \cap X_j$, these two morphisms correspond to the same gluing as on the intersection $D_+(x_i) \cap D_+(x_j)$ in projective space. They therefore glue to a single morphism on the union of all X_i. □

> **Edition note (source).** The source restricts the trivialisation and the functions f_{ki} to U in the middle of an argument taking place on X_i, and then writes $\mathbb{P}^n_K$ at a corner of the diagram although the base is Spec(R). This edition consistently displays X_i and $\mathbb{P}^n_R$; the mathematical content of the argument is unchanged.

Definition 28.2: the morphism defined by sections

Let X be a scheme over a commutative ring R, let $\mathcal{L}$ be an invertible sheaf on X, and let $s_0, s_1, \dots, s_n \in \Gamma(X, \mathcal{L})$ be global sections. The morphism of Lemma 28.1 defined on

$$U = \bigcup_{i=0}^{n} X_{s_i},$$

namely

$$U \longrightarrow \mathbb{P}^n_R, \qquad x \longmapsto (s_0(x), s_1(x), \dots, s_n(x)),$$

is called the *morphism defined by the sections* $s_0, \dots, s_n$, or the *morphism defined by the linear system* $s_0, \dots, s_n$. It is denoted by $\varphi_{s_0,\dots,s_n}$ or $\varphi_{\mathcal{L};s_0,\dots,s_n}$.

Definition 28.3: a linear system

Let X be a scheme over a commutative ring R and let $\mathcal{L}$ be an invertible sheaf on X. An R-submodule

$$T \subseteq \Gamma(X, \mathcal{L})$$

is called a *linear system* on X.

By Exercise 28.9, the morphism defined by a family of sections depends primarily on the submodule they generate. This is particularly clear when the sections are linearly independent, as is often required. If $T = \Gamma(X, \mathcal{L})$, we speak of a *complete linear system.*

A linear system has a geometric meaning. Each section $s \in T$ determines its invertibility locus X_s and its zero locus

$$Z(s) := X \setminus X_s.$$

If X is integral, a nonzero section s defines an effective Cartier divisor. When its support is nonempty and X is locally Noetherian, $Z(s)$ has pure codimension 1 and is therefore a hypersurface in X; a nowhere-vanishing section instead has $Z(s) = \emptyset$. Thus the sets $Z(s)$ for $s \in T$, $s \neq 0$, form a family of zero loci—often hypersurfaces—associated with the linear system; this family itself is also often called the linear system. If X is normal, the nonempty $Z(s)$ can be viewed as a family of linearly equivalent divisors.

> **Edition note (source).** The source calls $Z(s)$ a codimension-one hypersurface for every nonzero section, while only informally suggesting that X be integral. A nonzero section can be nowhere vanishing, in which case $Z(s)$ is empty; the codimension-one assertion above therefore includes the necessary nonemptiness and local Noetherian hypotheses.

Example 28.4

On the projective line

$$\mathbb{P}^1_R = \mathrm{Proj}(R[X, Y])$$

over a commutative ring R, the morphism associated with the complete linear system

$$(X, Y) \subseteq \Gamma(\mathbb{P}^1_R, \mathcal{O}_{\mathbb{P}^1_R}(1))$$

is the identity.

Example 28.5

On $\mathbb{P}^1_R = \mathrm{Proj}(R[X,Y])$ over a commutative ring R, the complete linear system

$$(X^2, XY, Y^2) \subseteq \Gamma(\mathbb{P}^1_R, \mathcal{O}_{\mathbb{P}^1_R}(2))$$

gives the morphism

$$\mathbb{P}^1_R \longrightarrow \mathbb{P}^2_R, \qquad (x,y) \longmapsto (x^2, xy, y^2).$$

The point with homogeneous coordinates (x,y) maps to the point with homogeneous coordinates $(x^2, xy, y^2) = (u,v,w)$. Its image satisfies $uw = v^2$ and thus lies on the plane curve

$$V_+(uw - v^2) \subseteq \mathbb{P}^2_R.$$

In fact, there is an isomorphism $\mathbb{P}^1_R \cong V_+(uw - v^2)$.

Definition 28.6: base-point-free

Let X be a scheme over a commutative ring R and let $\mathcal{L}$ be an invertible sheaf on X. A linear system $T \subseteq \Gamma(X, \mathcal{L})$ is called *base-point-free* if for every $x \in X$ there is an $s \in T$ such that $x \in X_s$.

This term is mainly used for schemes over a field. We also say that sections $s_0, s_1, \dots, s_n$ are base-point-free if the linear system they generate is base-point-free.

Lemma 28.7

Let X be a scheme over a commutative ring R, let $\mathcal{L}$ be an invertible sheaf on X, and let $s_0, s_1, \dots, s_n \in \Gamma(X, \mathcal{L})$ be global sections. The following statements are equivalent.

1. $X = \bigcup_{i=0}^n X_{s_i}$.
2. The morphism to $\mathbb{P}^n_R$ defined by the linear system $(s_0, s_1, \dots, s_n)$ is defined on all of X.
3. The linear system $(s_0, s_1, \dots, s_n)$ is base-point-free.

Proof See Exercise 28.14. □

Theorem 28.8

Let X be a scheme over a commutative ring R. The following concepts correspond to one another.

1. An invertible sheaf $\mathcal{L}$ on X together with base-point-free sections
$$s_0, s_1, \dots, s_n \in \Gamma(X, \mathcal{L}).$$
2. A morphism
$$\varphi : X \longrightarrow \mathbb{P}^n_R$$
over $\mathrm{Spec}(R)$.

The sections in (1) are assigned the morphism $\varphi_{s_0, s_1, \dots, s_n}$. Conversely, the morphism φ in (2) is assigned the invertible sheaf $\varphi^*(\mathcal{O}_{\mathbb{P}^n_R}(1))$ together with the sections $\varphi^*(x_i)$, $i = 0, 1, \dots, n$.

Proof First suppose that $\mathcal{L}$ and the sections $s_0, \dots, s_n$ are given. We must show

$$\varphi^* \mathcal{O}_{\mathbb{P}^n_R}(1) \cong \mathcal{L}.$$

On projective space there are $\mathcal{O}_{\mathbb{P}^n_R}$-module homomorphisms

$$\Psi_i : \mathcal{O}_{\mathbb{P}^n_R} \longrightarrow \mathcal{O}_{\mathbb{P}^n_R}(1), \qquad 1 \longmapsto x_i,$$

which become isomorphisms when restricted to $D_+(x_i)$. Pullback gives homomorphisms

$$\mathcal{O}_X \longrightarrow \varphi^*\mathcal{O}_{\mathbb{P}^n_R}(1), \qquad 1 \longmapsto \varphi^*(x_i),$$

and isomorphisms on X_{s_i}. Combined with the isomorphism

$$\mathcal{O}_X|_{X_{s_i}} \longrightarrow \mathcal{L}|_{X_{s_i}}, \qquad 1 \longmapsto s_i,$$

we obtain local isomorphisms

$$\varphi^*\mathcal{O}_{\mathbb{P}^n_R}(1)|_{X_{s_i}} \longrightarrow \mathcal{L}|_{X_{s_i}}$$

which identify $\varphi^*(x_i)$ with s_i. Their restrictions to $X_{s_i s_j}$ agree. By Corollary 4.10, they therefore glue to a global isomorphism

$$\varphi^*\mathcal{O}_{\mathbb{P}^n_R}(1) \longrightarrow \mathcal{L}.$$

Conversely, suppose $\varphi : X \to \mathbb{P}^n_R$ is given. This morphism determines sections $s_i = \varphi^*(x_i)$, which in turn determine a morphism φ'. Since a morphism is determined locally, it suffices to compare them on

$$\varphi^{-1}(D_+(x_i)) \longrightarrow D_+(x_i) \cong \mathbb{A}^n_R.$$

Here the variables x_k/x_i pull back to s_k/s_i, exactly as in the definition of φ'. Thus $\varphi' = \varphi$. □

> **Edition note (source).** Although Theorem 28.8 is stated over a commutative ring R with target $\mathbb{P}^n_R$, the converse direction of the source proof switches to $\mathbb{P}^n_K$ and $\mathbb{A}^n_K$ without introducing K. Both occurrences have been corrected to the stated base R above; this is a notation repair, not a new generalisation or restriction.

Lemma 28.9

Let X be a scheme over a commutative ring R, let $\mathcal{L}$ be an invertible sheaf on X, let $s_0, s_1, \dots, s_n \in \Gamma(X, \mathcal{L})$ be global sections, and let $\varphi : X \to \mathbb{P}^n_R$ be the associated morphism. The inverse image of the zero locus

$$V_+(a_0X_0 + a_1X_1 + \cdots + a_nX_n) \subseteq \mathbb{P}^n_R,$$

with $a_i \in R$ is the zero locus of the pulled-back section,

$$Z(a_0s_0 + a_1s_1 + \cdots + a_ns_n) = X \setminus X_{a_0s_0+a_1s_1+\cdots+a_ns_n}.$$

Proof This follows from Appendix Lemma 4.3. If $(a_0, \dots, a_n) = R$, the displayed zero locus is a relative hyperplane. Over a field, this condition is equivalent to the coefficients not all being zero. □

The twisted structure sheaf $\mathcal{O}_{\mathbb{P}^n_R}(1)$, through the global sections whose coefficient tuples generate the unit ideal, determines the family of relative hyperplanes in projective space. Over a field, these are precisely its nonzero global sections. Similarly, an invertible sheaf on a scheme determines the family of zero loci of its nonzero global sections. Under the correspondence of Theorem 28.8, inverse images of relative hyperplanes agree with the corresponding zero loci. If φ factors through a closed subvariety, that is, if there is a map

$$X \longrightarrow Y \subseteq \mathbb{P}^n_R,$$

then these corresponding zero loci are also inverse images of intersections $Y \cap H$ with a relative hyperplane H. In Example 28.5, for instance, the zero loci arising from unimodular linear combinations in the complete linear system on $\mathcal{O}_{\mathbb{P}^1_R}(2)$ agree with the intersections $V_+(uw - v^2) \cap H$, where H is a relative line.

Edition note (source). Over an arbitrary commutative ring, the source calls the zero locus of every nonzero linear form a hyperplane. Such a form defines a relative hyperplane only when its coefficients generate the unit ideal; the distinction disappears over a field. The inverse-image identity itself remains valid for every coefficient tuple.

Very ample sheaves

Definition 28.10: very ample

Let X be a scheme over a commutative ring R and let $\mathcal{L}$ be an invertible sheaf on X. The sheaf $\mathcal{L}$ is called *very ample* if there is an embedding $\varphi : X \to \mathbb{P}^n_R$, for some n, such that

$$\varphi^*(\mathcal{O}_{\mathbb{P}^n_R}(1)) \cong \mathcal{L}.$$

Lemma 28.11

Let X be a scheme over a commutative ring R and let $\mathcal{L}$ be an invertible sheaf on X. The sheaf $\mathcal{L}$ is very ample precisely when there are base-point-free global sections

$$s_0, s_1, \dots, s_n \in \Gamma(X, \mathcal{L})$$

such that the associated morphism $\varphi_{s_0, s_1, \dots, s_n}$ is an embedding.

Proof This follows from Theorem 28.8. □

Example 28.12

On projective space $\mathbb{P}^n_R$ over a commutative ring R, the invertible sheaf $\mathcal{O}_{\mathbb{P}^n_R}(k)$ is very ample for every $k \geq 1$. We have

$$\Gamma(\mathbb{P}^n_R, \mathcal{O}_{\mathbb{P}^n_R}(k)) = R[X_0, X_1, \dots, X_n]_k.$$

Consider the linear system generated by all monomials of degree k in the $n+1$ variables of $R[X_0, X_1, \dots, X_n]$, and its associated morphism

$$\varphi : \mathbb{P}^n_R \longrightarrow \mathbb{P}^m_R,$$

where m is one less than the number of these monomials. On $D_+(X_0) = (\mathbb{P}^n_R)_{X_0^k}$, this map is given by

$$\mathbb{A}^n_R \longrightarrow D_+(Y_\nu) \subseteq \mathbb{P}^m_R, \qquad \left(\frac{X_1}{X_0}, \dots, \frac{X_n}{X_0}\right) \longmapsto \left(\frac{X^\mu}{X_0^k} : \mu \text{ monomial of degree } k\right),$$

and analogously on each $D_+(X_i)$. At the level of polynomial rings, this is the substitution homomorphism

$$R[S_\mu : \mu \in I_k] \longrightarrow R[T_1, \dots, T_n], \qquad S_\mu \longmapsto T^\mu,$$

where I_k is the index set of all monomials in n variables of degree at most k (note the inequality). This map is surjective, so the morphism above is a closed embedding.

For $k \leq 0$ and $n \geq 1$, the sheaf $\mathcal{O}_{\mathbb{P}^n_R}(k)$ is not very ample.

Definition 28.13: ample

Let X be a scheme over a commutative ring R and let $\mathcal{L}$ be an invertible sheaf on X. The sheaf $\mathcal{L}$ is called *ample* if $\mathcal{L}^n$ is very ample for some $n \geq 1$.

Worksheet 28: Morphisms to projective space

Exercises

Exercise 28.1

Describe the cone map

$$\mathbb{A}_K^{d+1} \setminus \{(0, 0, \dots, 0)\} \longrightarrow \mathbb{P}_K^d$$

using a linear system in an invertible sheaf.

Exercise 28.2

Describe projection away from a point using a linear system in an invertible sheaf.

Exercise 28.3

Let K be a field and let $\mathbb{P}_K^n$ be the associated projective space. Let $\varphi : K^{n+1} \to K^{n+1}$ be a bijective linear map.

1. Prove that φ induces an automorphism

$$\varphi : \mathbb{P}_K^n \longrightarrow \mathbb{P}_K^n, \qquad (x_0, x_1, \dots, x_n) \longmapsto \varphi(x_0, x_1, \dots, x_n).$$

2. Determine the inverse image of $D_+(X_i)$ in the situation of part (1). What does the morphism look like on these affine sets?
3. Prove that φ_1 and φ_2 induce the same automorphism of projective space precisely when one is a nonzero scalar multiple of the other.
4. Does every linear map $\varphi : K^{n+1} \to K^{n+1}$ induce a morphism $\varphi : \mathbb{P}_K^n \to \mathbb{P}_K^n$?

In the situation above, we speak of a *projective linear automorphism*.

Exercise 28.4

Describe a projective linear automorphism

$$\mathbb{P}_K^d \longrightarrow \mathbb{P}_K^d$$

using a linear system in an invertible sheaf.

Exercise 28.5

Let $P, Q \in \mathbb{P}_K^n$ be points of projective space over a field K. Prove that there is an automorphism $\varphi : \mathbb{P}_K^n \to \mathbb{P}_K^n$ with $\varphi(P) = Q$.

Exercise 28.6 ★

Let V be a two-dimensional vector space over a field K. Let v_1, v_2, v_3 and w_1, w_2, w_3 be vectors in V, with each pair of vectors in each family linearly independent. Prove that there is a bijective linear map $\varphi : V \to V$ such that

$$\varphi(v_i) \in K w_i$$

for $i = 1, 2, 3$.

Exercise 28.7

Let $P_1, P_2, P_3 \in \mathbb{P}_K^1$ and $Q_1, Q_2, Q_3 \in \mathbb{P}_K^1$ each be three distinct points on the projective line over a field K. Prove that there is a K-automorphism $\varphi : \mathbb{P}_K^1 \to \mathbb{P}_K^1$ with $\varphi(P_i) = Q_i$ for $i = 1, 2, 3$.

Exercise 28.8

Let K be a field. Prove that every K-automorphism of projective space $\mathbb{P}^n_K$ is projective linear.

Use the fact that the pullback of $\mathcal{O}_{\mathbb{P}^n_K}(1)$ must again be $\mathcal{O}_{\mathbb{P}^n_K}(1)$.

Exercise 28.9

Let X be a scheme over a field K, let $\mathcal{L}$ be an invertible sheaf on X, and let $s_0, s_1, \dots, s_n \in \Gamma(X, \mathcal{L})$ be global sections determining the linear system

$$\langle s_0, s_1, \dots, s_n \rangle \subseteq \Gamma(X, \mathcal{L}).$$

Let $t_0, t_1, \dots, t_n$ be another generating system for the same linear system. Prove the following assertions.

1. $\bigcup_{i=0}^n X_{s_i} = \bigcup_{i=0}^n X_{t_i}$.
2. For the morphisms defined by the two generating systems, there is a projective linear automorphism

$$\theta : \mathbb{P}^n_K \longrightarrow \mathbb{P}^n_K$$

such that

$$\theta \circ \varphi_{s_0, s_1, \dots, s_n} = \varphi_{t_0, t_1, \dots, t_n}.$$

Exercise 28.10

Consider the projective line $\mathbb{P}^1_K$ and the complete linear system

$$L := \langle s, t \rangle = \Gamma(\mathbb{P}^1_K, \mathcal{O}_{\mathbb{P}^1_K}(1)).$$

Prove that choosing a generating system of three elements for L, up to scalar multiplication, corresponds to an embedding of the projective line into the projective plane as a line. How can the image line be described?

Exercise 28.11

Consider the projective line $\mathbb{P}^1_K$ and the complete linear system

$$L := \langle s^2, st, t^2 \rangle = \Gamma(\mathbb{P}^1_K, \mathcal{O}_{\mathbb{P}^1_K}(2)).$$

Prove that choosing a basis of L, up to scalar multiplication, corresponds to an embedding of the projective line into the projective plane. How can the image curve be described?

Exercise 28.12

Consider the projective line $\mathbb{P}^1_K$ and the complete linear system

$$L := \langle s^3, s^2t, st^2, t^3 \rangle = \Gamma(\mathbb{P}^1_K, \mathcal{O}_{\mathbb{P}^1_K}(3)).$$

Prove that the associated map

$$\mathbb{P}^1_K \longrightarrow \mathbb{P}^3_K$$

gives an embedding of the projective line into projective space. Give as many equations as possible satisfied by the image curve.

> **Edition note (source).** The title of the transcluded exercise page contains the segment `O(4)`, but the exercise itself uses four cubic sections and $\mathcal{O}(3)$. This edition preserves the formula displayed in the exercise; the exact source title remains recorded in the entity marker above.

Exercise 28.13

Let X be a scheme over a commutative ring R, let $\mathcal{L}$ be an invertible sheaf on X, and let $s_0, s_1, \ldots, s_n \in \Gamma(X, \mathcal{L})$ be global sections. Let $L \to X$ be the line bundle in the sense of Theorem 17.10 associated with the dual invertible sheaf $\mathcal{L}^*$, so that the s_i can be regarded as morphisms

$$L \longrightarrow X \times \mathbb{A}^1_R \longrightarrow \mathbb{A}^1_R.$$

Put $U = \bigcup_{i=0}^n X_{s_i}$, and let $L^\times|_U$ be the restriction of L to U with its zero section removed. Prove that there is a commutative diagram

$$\begin{tikzcd} L^\times|_U \arrow[r, "(s_0,s_1,\ldots,s_n)"] \arrow[d] & \mathbb{A}^{n+1}_R \setminus \{0\} = \bigcup_{i=0}^n D(x_i) \arrow[d, "\pi"] \\ U \arrow[r, "\varphi_{s_0,s_1,\ldots,s_n,\mathcal{L}}"] & \mathbb{P}^n_R, \end{tikzcd}$$

with the cone map on the right.

> **Edition note (source).** In both unions in the source diagram, the printed index starts at $i = 1$, although the family of sections is $s_0, \ldots, s_n$ and the cone map uses all coordinates. Moreover, the source places all of L in the upper left, but the zero section maps to the origin, where the cone map π is undefined; over a base point outside U, every fibre point also maps to the origin. The diagram has therefore been restricted to $L^\times|_U$, and both unions use the consistent bounds $i = 0, \ldots, n$.

Exercise 28.14

Let X be a scheme over a commutative ring R, let $\mathcal{L}$ be an invertible sheaf on X, and let $s_0, s_1, \ldots, s_n \in \Gamma(X, \mathcal{L})$ be global sections. Prove that the following statements are equivalent.

1. $X = \bigcup_{i=0}^n X_{s_i}$.
2. The morphism to $\mathbb{P}^n_R$ defined by the linear system $(s_0, s_1, \ldots, s_n)$ is defined on all of X.
3. The linear system $(s_0, s_1, \ldots, s_n)$ is base-point-free.

Public solutions and coverage of Worksheet 28

For each of the 14 official exercise-page titles, the literal candidate page `<exercise>/Lösung` was checked directly. This check found exactly one public solution page, namely the solution to Exercise 28.6. The other thirteen candidates led to the official page-creation view stating that the page did not yet exist. The absence of a visible link on the worksheet page was not used as negative evidence.

Solution to Exercise 28.6

Since v_1, v_2 and w_1, w_2 are bases, the theorem on specifying a linear map on a basis gives a bijective linear map

$$\psi : V \longrightarrow V$$

with

$$\psi(v_1) = w_1 \qquad \text{and} \qquad \psi(v_2) = w_2.$$

Under ψ, the assumptions of pairwise linear independence remain valid.

Replacing the first family by its image under ψ and renaming the common basis, we therefore only need to consider two families of vectors of the form v_1, v_2, y and v_1, v_2, z. Let

$$y = av_1 + bv_2$$

and

$$z = cv_1 + dv_2.$$

Here

$$a, b, c, d \neq 0,$$

since otherwise y, respectively z, would be linearly dependent with one of the v_i. We now consider the linear map ϕ given by

$$v_1 \longmapsto \frac{c}{a} v_1 \qquad \text{and} \qquad v_2 \longmapsto \frac{d}{b} v_2.$$

Then

$$\begin{aligned} \phi(y) &= \phi(av_1 + bv_2) \\ &= a\phi(v_1) + b\phi(v_2) \\ &= a\frac{c}{a} v_1 + b\frac{d}{b} v_2 \\ &= cv_1 + dv_2 \\ &= z. \end{aligned}$$

Thus ϕ satisfies the required condition in the reduced situation, and $\phi \circ \psi$ satisfies it for the original two families.

> **Edition note (source).** The source solution introduces a map ψ but writes its two basis values with the symbol ϕ, and it does not name the final composition. The notation and the composition have been made explicit above. The source also writes $cv_1 + bv_2 = z$ in the last line, although it previously specified $z = cv_1 + dv_2$; the second coefficient has therefore been corrected to d.

Negative results established by direct candidate checks

The literal candidate pages `<exercise>/Lösung` do not exist at the checked boundary for Exercises 28.1, 28.2, 28.3, 28.4, 28.5, 28.7, 28.8, 28.9, 28.10, 28.11, 28.12, 28.13, and 28.14. Each candidate URL led to the official page-creation view stating "Diese Seite existiert noch nicht" ("This page does not yet exist"). This statement records source-page coverage, not a claim that mathematical solutions do not exist.

Lecture 29: The genus of a curve

Smooth projective curves and their genus

Definition 29.1: genus

For a smooth projective curve C over an algebraically closed field K, the number

$$g := \dim_K H^1(C, \mathcal{O}_C)$$

is called the *genus* of the curve.

By Theorem 27.7, the dimension of $H^1(C, \mathcal{O}_C)$ is finite. Thus the genus of a curve is a natural number.

> **Source illustrations.** The following three surfaces illustrate genus one, two, and three through their number of handles. They occur in this order in the official lecture and retain their component public-domain status.

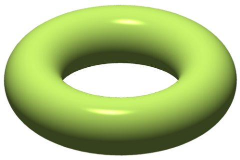

Figure 6: A green torus with one handle, a surface of genus one

Figure 7: A green double torus with two handles, a surface of genus two

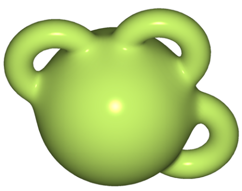

Figure 8: A green sphere-like surface with three handles, a surface of genus three

Example 29.2

By Theorem 27.4, the genus of the projective line

$$\mathbb{P}^1_K = \operatorname{Proj}(K[X, Y])$$

is 0.

Definition 29.3: an elliptic curve

A smooth projective curve C of genus 1 over an algebraically closed field K is called an *elliptic curve*.

Remark 29.4

If the complex numbers $\mathbb{C}$ are chosen as the ground field, the genus of a smooth projective curve has a simple topological interpretation. Such a curve can be regarded as a compact one-dimensional complex manifold—a Riemann surface—and also as a compact oriented two-dimensional real manifold. Manifolds of the latter kind have a simple topological classification: each is homeomorphic to the surface of a sphere with g handles attached. This number is called the topological genus of the real surface, and hence also of the curve.

It can be proved that the genus defined algebraically using the first cohomology of the structure sheaf agrees with this topological genus. The complex projective line is a two-dimensional sphere with no handles, so its topological genus is 0. A surface of genus 1 is a torus—like a tyre—homeomorphic to $S^1 \times S^1$. Projective curves of genus 1, namely elliptic curves, have this topological shape.

Theorem 29.5

Let

$$C = V_+(f) \subseteq \mathbb{P}^2_K$$

be a projective plane curve of degree d over an algebraically closed field K. Then

$$\dim_K H^1(C, \mathcal{O}_C) = \frac{(d-1)(d-2)}{2}.$$

Proof Consider the short exact sequence (compare Exercise 13.23)

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^2_K}(-d) \xrightarrow{f} \mathcal{O}_{\mathbb{P}^2_K} \longrightarrow \mathcal{O}_C \longrightarrow 0$$

of coherent sheaves on the projective plane. Here the structure sheaf $\mathcal{O}_C$ is regarded as a sheaf on the projective plane with support C. The relevant portion of the associated long exact cohomology sequence is

$$H^1(\mathbb{P}^2_K, \mathcal{O}_{\mathbb{P}^2_K}) = 0 \longrightarrow H^1(\mathbb{P}^2_K, \mathcal{O}_C) \longrightarrow H^2(\mathbb{P}^2_K, \mathcal{O}_{\mathbb{P}^2_K}(-d)) \longrightarrow H^2(\mathbb{P}^2_K, \mathcal{O}_{\mathbb{P}^2_K}) = 0.$$

Both vanishings follow from Theorem 27.4. By the same theorem, the space

$$H^2(\mathbb{P}^2_K, \mathcal{O}_{\mathbb{P}^2_K}(-d))$$

has a basis consisting of all monomials $x^i y^j z^k$ whose exponents are all negative and satisfy $i + j + k = -d$. Thus we must count tuples (α, β, γ) of degree $d - 3$. By Exercise 12.4, their number is

$$\binom{d-3+2}{2} = \binom{d-1}{2} = \frac{(d-1)(d-2)}{2}.$$

By Theorem 27.6, $H^1(\mathbb{P}^2_K, \mathcal{O}_C) = H^1(C, \mathcal{O}_C)$, which proves the assertion. □

Edition note (PDF witness). The official 2020 PDF witness still refers to Exercise 11.4; the frozen semantic revision controlling this edition has updated the reference to Exercise 12.4.

In the smooth case, this theorem gives a formula for computing the genus of a plane curve:

d	1	2	3	4	5
g	0	0	1	3	6

For $d = 1$, we obtain a projective line of genus 0. For $d = 2$, we obtain a projective plane quadric—a conic section—which also has genus 0 and is indeed isomorphic to the projective line. For $d = 3$, the genus is 1, so the curve is elliptic. It can be proved that every elliptic curve can be realised as a plane cubic curve. It is by no means obvious that there are smooth projective curves of every genus. By Theorem 29.5, not all can be realised as plane curves.

Remark 29.6

The cohomologically defined genus of a smooth projective curve agrees with the vector-space dimension of the global sections of its canonical sheaf. In dimension one, the canonical sheaf is the sheaf of Kähler differentials $\Omega_{C|K}$, that is, the cotangent sheaf, dual to the tangent sheaf. Thus

$$\dim_K H^1(C, \mathcal{O}_C) = \dim_K \Gamma(C, \omega_C).$$

For plane curves this is immediate. By Theorem 29.5, the genus is $(d-1)(d-2)/2$. Corollary 19.12 gives $\omega_C \cong \mathcal{O}_C(d-3)$, and by Exercise 27.10 the dimension of $\Gamma(C, \mathcal{O}_C(d-3))$ is also $(d-1)(d-2)/2$.

In the general case, Serre duality applies. Among other things, it states that for a locally free sheaf $\mathcal{F}$ on a smooth projective curve C, the cohomology group $H^1(C, \omega_C)$ is a one-dimensional K-vector space, and the natural map

$$\operatorname{Hom}(\mathcal{F}, \omega_C) \times H^1(C, \mathcal{F}) \longrightarrow H^1(C, \omega_C) \cong K$$

gives a perfect duality. In other words, $\operatorname{Hom}(\mathcal{F}, \omega_C)$ and $H^1(C, \mathcal{F})$ are dual to one another and in particular have the same dimension. For the structure sheaf $\mathcal{F} = \mathcal{O}_C$, the equality $\operatorname{Hom}(\mathcal{O}_C, \omega_C) = \Gamma(C, \omega_C)$ from Theorem 13.10 gives the duality between $H^1(C, \mathcal{O}_C)$ and $\Gamma(C, \omega_C)$.

Edition note (source). In the corresponding Hom display, the frozen source has the malformed nested expression `\mathcal{\mathcal O_C}`. The intended structure sheaf $\mathcal{O}_C$ is displayed above.

Divisors on curves

On a smooth projective curve C, as on any one-dimensional normal scheme, a Weil divisor is simply a formal sum

$$\sum_{P \in C} n_P \, P$$

over closed points P. In this case, those points are the prime divisors, that is, the irreducible closed subsets of codimension 1. The coefficients satisfy $n_P \in \mathbb{Z}$, and only finitely many are nonzero. By Corollary 22.11, the divisor class group agrees with the Picard group.

We will discuss how divisors on curves behave under morphisms. A morphism between two irreducible curves is either constant or has dense image. A nonconstant morphism $\varphi : C_1 \to C_2$ induces an extension of function fields

$$Q(C_2) \subseteq Q(C_1).$$

First we show that an element of the function field of a smooth curve can be regarded as a morphism to the projective line. In general, for a nonconstant element q of the function field of a normal scheme X, the principal divisor $\operatorname{div}(q)$ can be decomposed into the divisor of zeros and the divisor of poles, with positive coefficients. These two effective divisors are linearly equivalent and, in the locally factorial case by Exercise 22.15, correspond to sections of the associated invertible sheaf $\mathcal{O}_X(\text{divisor of zeros}(q))$. The results of the preceding lecture show that these two sections determine a morphism to the projective line on an open set $U \subseteq X$. The following result is stronger for smooth curves: the domain of definition is the entire curve.

Edition note (PDF witness). The official 2020 PDF witness still refers to Exercise 22.13; the frozen semantic revision controlling this edition has updated the reference to Exercise 22.15.

Lemma 29.7

Let C be a smooth irreducible curve over an algebraically closed field K, and let Q be its function field. Every rational function $q \in Q$ naturally determines a morphism

$$q : C \longrightarrow \mathbb{P}^1_K.$$

Proof Let

$$U := \{P \in C \mid q \in \mathcal{O}_{C,P}\}$$

be the domain of definition of q as a function to the affine line. If $q = 0$, the constant map with image $(1, 0)$ proves the assertion. Hence assume $q \neq 0$ and let

$$V := \{P \in C \mid q^{-1} \in \mathcal{O}_{C,P}\}$$

be the domain of definition of q^{-1}. We have $C = U \cup V$, since every $\mathcal{O}_{C,P}$ is a discrete valuation ring and there $q = \pi^n u$ for a unit $u \in \mathcal{O}_{C,P}$, a local parameter $\pi \in \mathcal{O}_{C,P}$, and $n \in \mathbb{Z}$. By Corollary 10.12, there are a morphism

$$q : U \longrightarrow \mathbb{A}^1_K = \operatorname{Spec} K\left[\frac{Y}{X}\right] \cong D_+(X) \subseteq \mathbb{P}^1_K$$

and a morphism

$$q^{-1} : V \longrightarrow \mathbb{A}^1_K = \operatorname{Spec} K\left[\frac{X}{Y}\right] \cong D_+(Y) \subseteq \mathbb{P}^1_K.$$

These correspond to the substitution homomorphisms $Y/X \mapsto q$ and $X/Y \mapsto q^{-1}$. On $U \cap V$ the two morphisms agree, so they glue to a morphism to the projective line. □

> **Edition note (source).** The source defines V only when $q \neq 0$ but then uses V as though it existed for every q. The trivial case $q = 0$ has been separated above; the gluing argument then applies under the stated assumption $q \neq 0$.

Definition 29.8: ramification index

For an injective ring homomorphism $R \subseteq S$ between discrete valuation rings, the order in S of a local uniformiser of R is called the *ramification index* of the extension.

The ramification order is denoted by $\operatorname{Ram}(S \mid R)$. If $\varphi : C_1 \to C_2$ is a nonconstant morphism between smooth curves over an algebraically closed field, then for each closed point $Q \in C_1$ with image point $\varphi(Q) \in C_2$ there is an extension of discrete valuation rings

$$\mathcal{O}_{C_2,\varphi(Q)} \subseteq \mathcal{O}_{C_1,Q}.$$

The corresponding ramification order is also called the ramification order of φ at Q and is denoted by $\operatorname{Ram}(Q \mid \varphi(Q))$.

> **Edition note (source and PDF witness).** The frozen semantic revision uses *Verzweigungsindex* (“ramification index”) in the definition, then *Verzweigungsordnung* (“ramification order”) in the following paragraph. The official 2020 PDF witness also uses *Verzweigungsordnung* in the definition. This edition explicitly preserves the terminology difference between the witnesses.

Definition 29.9: pullback of a Weil divisor

Let $\varphi : C_1 \to C_2$ be a nonconstant morphism between smooth curves over an algebraically closed field, and let

$$D = \sum_P a_P\, P$$

be a Weil divisor on C_2. The Weil divisor

$$\varphi^* D := \sum_{Q \in C_1} \mathrm{Ram}(Q \mid \varphi(Q)) a_{\varphi(Q)}\, Q$$

is called the *pullback Weil divisor.*

For a single point $P \in C_2$, the pullback divisor is

$$\sum_{Q \in \varphi^{-1}(P)} \mathrm{Ram}(Q \mid P)\, Q.$$

Thus it is essentially the fibre over P, but ramification points—points with ramification order at least 2—are counted with multiplicity according to their orders.

Lemma 29.10

Let $\varphi : C_1 \to C_2$ be a nonconstant morphism between smooth irreducible curves over an algebraically closed field, and let

$$D = \sum_P a_P\, P = \mathrm{div}(q)$$

be a principal divisor on C_2, with $q \in Q(C_2)$ and $q \neq 0$. Then $\varphi^*(D)$ agrees with the principal divisor of $q \in Q(C_1)$ on C_1.

Proof Since φ is nonconstant, there is a field extension $Q(C_2) \subseteq Q(C_1)$. For every $Q \in C_1$, there is a commutative diagram of injective homomorphisms

$$\begin{array}{ccc} \mathcal{O}_{C_2,\varphi(Q)} & \to & \mathcal{O}_{C_1,Q} \\ \downarrow & & \downarrow \\ Q(C_2) & \to & Q(C_1), \end{array}$$

with discrete valuation rings in the first row. If $q = u\pi_2^n$, where $u \in \mathcal{O}_{C_2,\varphi(Q)}$ is a unit and π_2 is a local uniformiser, the source writes

$$q = u\pi_2^n = u\big(u'\pi_1^{\mathrm{Ram}(Q|\varphi(Q))}\big)^n = u(u')^n \pi_1^{n\,\mathrm{Ram}(Q|\varphi(Q))},$$

where π_1 is a local uniformiser of $\mathcal{O}_{C_1,Q}$. This proves the assertion. □

Edition note (source and PDF witness). The frozen semantic revision uses the argument order $\mathrm{Ram}(Q \mid \varphi(Q))$ above, whereas the official 2020 PDF witness reverses it to $\mathrm{Ram}(\varphi(Q) \mid Q)$. This edition follows the frozen semantic revision. When expanding $(u'\pi_1^e)^n$, both witnesses write the unit factor u' instead of $(u')^n$; that exponent has been corrected above.

Corollary 29.11

Let C be a smooth irreducible curve over an algebraically closed field K, let Q be its function field, and let $q \in Q \setminus K$. For the associated morphism

$$q : C \longrightarrow \mathbb{P}^1_K$$

we have

$$q^*((0) - (\infty)) = \mathrm{div}(q).$$

Proof The function field of the projective line $\mathbb{P}^1_K = \mathrm{Proj}(K[X, Y])$ is $K(t)$, with $t = Y/X$. The extension of function fields is given by

$$K(t) \longrightarrow Q(C), \qquad t \longmapsto q.$$

The principal divisor of t is $(0) - (\infty) = (Y) - (X)$, using two descriptions of the points. The assertion therefore follows from Lemma 29.10. □

The degree of a divisor

Definition 29.12

Let C be a smooth projective curve over an algebraically closed field K. The degree of a Weil divisor $D = \sum_{P \in C} n_P P$ is defined as

$$\deg(D) := \sum_{P \in C} n_P.$$

Theorem 29.13

Let C be a smooth projective curve over an algebraically closed field K. Every principal divisor has degree 0.

This theorem is stated without proof. Consequently, the group homomorphism

$$\mathrm{Div}(C) \longrightarrow \mathbb{Z}, \qquad D \longmapsto \deg(D),$$

factors through the divisor class group of C. The following definition therefore makes sense.

Definition 29.14

Let C be a smooth projective curve over an algebraically closed field K. The degree of an invertible sheaf $\mathcal{L}$ on C is defined as the degree of an associated Weil divisor.

"Associated" means that a divisor D corresponds to the invertible sheaf $\mathcal{O}_C(D)$; in particular, effective divisors correspond to sections of this sheaf.

Worksheet 29: The genus of a curve

Exercise 29.1

Prove that the following data or constructions determine the same morphism $\mathbb{P}^1_K \to \mathbb{P}^1_K$, where $(a, b), (c, d) \in K^2$ are linearly independent.

1. The morphism induced, as in Theorem 12.11, by the homogeneous ring homomorphism $K[X, Y] \to K[S, T]$ with $X \mapsto aS + bT$ and $Y \mapsto cS + dT$.
2. The morphism of Lemma 28.1 defined by the two sections

$$as + bt,\ cs + dt \in \Gamma(\mathbb{P}^1_K, \mathcal{O}_{\mathbb{P}^1_K}(1)).$$

3. The morphism of Lemma 29.7 defined by the rational function
$$\frac{cs+dt}{as+bt} \in K\Big(\frac{s}{t}\Big).$$
Edition note (source). The source prints the reciprocal rational function. Under the convention $Y/X \mapsto q$ in Lemma 29.7, the first two constructions send X to $as+bt$ and Y to $cs+dt$, so the corresponding rational function is $(cs+dt)/(as+bt)$, as displayed above.

Exercise 29.2

Let K be a field. Prove that there is a morphism

$$V(X^2-Y^3) \supseteq U = V(X^2-Y^3) \setminus \{(0,0)\} \longrightarrow \mathbb{A}^1_K$$

which cannot be extended to all of $V(X^2-Y^3)$.

Edition note (source). The source leaves the ground field unstated and prints the target with an empty base subscript. The curve and affine line have been placed over an explicit field K; no further hypothesis on K is used.

Exercise 29.3

Let K be a field, let $C \subseteq \mathbb{P}^2_K$ be a projective plane curve, let $P \in C$, and let

$$\varphi : C \setminus \{P\} \longrightarrow \mathbb{P}^1_K$$

be the morphism defined by projection away from P. Thus a point $Q \in C$, $Q \neq P$, maps to the secant line through Q and P.

1. Let $K = \mathbb{C}$ and let (Q_n) be a sequence on C converging to P in the complex topology. Does $\varphi(Q_n)$ converge?
2. Does $\varphi(Q_n)$ have an accumulation point?
3. Let P be a smooth point. Prove that the morphism extends to all of C.

Exercise 29.4

Discuss the situation of Exercise 29.3 for the crossing of axes

$$V_+(YZ) \subseteq \mathbb{P}^2_K$$

and its crossing point $(1,0,0)$.

Exercise 29.5 ★

Let K be an algebraically closed field of characteristic 0, let $C \subseteq \mathbb{P}^2_K$ be an irreducible projective plane curve of degree d, and let

$$\varphi : C \longrightarrow \mathbb{P}^1_K$$

be the morphism given by projection away from a point $P \notin C$. Prove that, with at most finitely many exceptions, the fibre over each $t \in \mathbb{P}^1_K$ consists of exactly d points.

Exercise 29.6

Let K be an algebraically closed field and let $C \subseteq \mathbb{P}^2_K$ be a smooth curve of degree $d \geq 2$. Prove that there is a morphism $C \to \mathbb{P}^1_K$ such that every fibre consists of at most $d-1$ points.

Exercise 29.7

Let

$$C = V_+(X^3 + Y^3 + Z^3) \subseteq \mathbb{P}^2_K$$

be the Fermat cubic over an algebraically closed field of characteristic other than 3. Describe explicitly a morphism $C \to \mathbb{P}^1_K$ with at most two points over each point.

Exercise 29.8

Let C be a smooth irreducible projective curve over a field K, with function field $Q(C)$, and let $q \in Q(C)$ have associated morphism $q : C \to \mathbb{P}^1_K$. Let $a \in K$. Prove that there is an automorphism

$$\theta : \mathbb{P}^1_K \longrightarrow \mathbb{P}^1_K$$

such that the diagram

$$\begin{array}{ccc} C & \xrightarrow{q} & \mathbb{P}^1_K \\ {\scriptstyle q-a}\downarrow & & \downarrow{\scriptstyle \theta} \\ \mathbb{P}^1_K & = & \mathbb{P}^1_K \end{array}$$

commutes.

> **Edition note (source).** The frozen semantic revision and the official 2020 PDF witness display $Q(C) =$ with an empty right-hand side and use K without first naming it as the ground field. This edition repairs the visible defect to "with function field $Q(C)$" and makes the implicit ground field explicit; it does not impose algebraic closedness here.

Exercise 29.9

Let C be a smooth irreducible projective curve over an algebraically closed field K, and let $q \in Q(C)$ be nonconstant with associated morphism $q : C \to \mathbb{P}^1_K$. Prove that, as P ranges over $\mathbb{P}^1_K$, the pullback divisors $q^*(P)$ are linearly equivalent to one another. Use Exercise 29.8.

> **Edition note (source).** The frozen source says only "over an algebraically closed field" but then writes $\mathbb{P}^1_K$. The field has been named K above so that the base in the target is bound.

Exercise 29.10

Let $\mathbb{P}^1_K = \text{Proj}(K[X, Y])$ have function field $K(t)$, with $t = Y/X$. Describe the associated morphism of schemes

$$\mathbb{P}^1_K \longrightarrow \mathbb{P}^1_K, \qquad t \longmapsto t^n,$$

for $n \in \mathbb{N}_+$. What is the inverse image of zero? What is the inverse image of the point at infinity? What are the ramification orders?

Exercise 29.11

Let $\mathbb{P}^1_K = \text{Proj}(K[X, Y])$ have function field $K(t)$, with $t = Y/X$, and let $P \in K[t]$ be a polynomial of degree $e \geq 1$. Describe the ramification order at ∞ of the associated morphism of schemes

$$\mathbb{P}^1_K \longrightarrow \mathbb{P}^1_K, \qquad t \longmapsto P.$$

Exercise 29.12 ★

Let $\mathbb{P}^1_K = \text{Proj}(K[X,Y])$ and $\mathbb{P}^1_K = \text{Proj}(K[W,Z])$ be two projective lines with function fields $K(t)$, $t = Y/X$, and $K(u)$, $u = Z/W$, over an algebraically closed field K of characteristic other than 2. On the second projective line, consider the linear system given by

$$WZ,\ W^2 + Z^2 \in \Gamma(\mathbb{P}^1_K, \mathcal{O}_{\mathbb{P}^1_K}(2))$$

with associated map

$$\varphi : \mathbb{P}^1_K \longrightarrow \mathbb{P}^1_K, \qquad (w,z) \longmapsto (wz, w^2 + z^2).$$

1. Is this linear system complete?
2. Determine the inverse images of $D_+(X)$ and $D_+(Y)$, and describe the induced maps between the affine open subsets.
3. Is this linear system base-point-free?
4. Describe the associated extension of function fields $K(t) \subseteq K(u)$. What is its degree?
5. For each $(x,y) \in \mathbb{P}^1_K$, determine its inverse image under φ and the respective ramification orders.
6. Describe the pullback divisor $\varphi^*(0 - \infty) = \varphi^*((Y) - (X))$.

Exercise 29.13

Let X be a normal Noetherian integral scheme over a field K, and let $q \in Q(X)$. Prove that q defines a morphism

$$q : U \longrightarrow \mathbb{P}^1_K$$

on an open set $U \subseteq X$ such that $X \setminus U$ has codimension at least 2.

> **Edition note (source).** The frozen source writes the target as $\mathbb{P}^1$ although X is a scheme over the named field K. The base subscript has been supplied above.

Exercise 29.14

Let $\mathcal{L}$ be an invertible sheaf of negative degree on a smooth irreducible projective curve C over an algebraically closed field. Prove

$$\Gamma(C, \mathcal{L}) = 0.$$

Exercise 29.15

Prove that, for a smooth projective curve C over an algebraically closed field, the degree of invertible sheaves gives a surjective group homomorphism

$$\text{Pic}(C) \longrightarrow \mathbb{Z}, \qquad \mathcal{L} \longmapsto \deg(\mathcal{L}).$$

Public solutions and coverage of Worksheet 29

At the frozen official-page revisions, the source provides exactly two public solutions among the 15 exercises, namely those to Exercises 29.5 and 29.12. The other thirteen exercise pages only offer to create a new solution. The absence of a public solution page is not replaced by a fabricated solution.

Solution to Exercise 29.5

We consider a line $V_+(L)$ through the point P and its associated affine complement,

$$D_+(L) \cong \mathbb{A}_K^2.$$

Without loss of generality, take

$$P = (1, 0, 0)$$

and $L = Z$. We may therefore assume that we are considering the affine projection

$$\mathbb{A}_K^2 \cong D_+(Z) \longrightarrow \mathbb{A}_K^1 \cong D_+(Z), \qquad (x, y) \longmapsto y,$$

and an affine curve $V(F)$ of degree d. The term X^d occurs in F, since otherwise $P \in C$. We regard the polynomial F as

$$F = G_d X^d + G_{d-1} X^{d-1} + \cdots + G_1 X + G_0 \in K[Y][X] \subset K(Y)[X].$$

Here $G_i \in K[Y]$, while G_d is constant. Since the curve is irreducible, $G_0 \neq 0$ if $d \geq 2$ (for $d = 1$, the whole assertion is immediate). We must show that, for all but finitely many $y \in K$, the polynomial

$$F(y) = G_d(y) X^d + G_{d-1}(y) X^{d-1} + \cdots + G_1(y) X + G_0(y)$$

has d distinct roots. Since $F \in K(Y)[X]$ is irreducible and we are in characteristic 0, the polynomial F is separable. Thus F and F' are coprime, where F' denotes the formal derivative with respect to X. Hence there are $S, T \in K(Y)$ such that

$$SF + TF' = 1.$$

This means that there are polynomials $A, B, C \in K[Y]$ satisfying

$$AF + BF' = C$$

with $C \neq 0$. The polynomial C has only finitely many roots. For $y \in K$ with $C(y) \neq 0$, we have

$$A(y)F(y) + B(y)F(y)' = C(y),$$

which means that $F(y)$ and $F(y)' = F'(y)$ are coprime in $K[X]$. Hence $F(y)$ and its derivative $F(y)'$ have no common root, so no root of $F(y)$ is multiple.

Solution to Exercise 29.12

1. The linear system is not complete, because $\Gamma(\mathbb{P}^1_K, \mathcal{O}_{\mathbb{P}^1_K}(2))$ contains three linearly independent sections, namely

$$WZ, \qquad W^2, \qquad Z^2.$$

2. By definition, the morphism associated with a family of global sections is defined on the invertibility loci of those sections. These are $D_+(WZ)$ and $D_+(W^2 + Z^2)$. On the first locus, the map is given by

$$D_+(WZ) \cong \mathbb{A}^1_K \setminus \{(0)\} \longrightarrow D_+(X) \cong \mathbb{A}^1_K,$$

with

$$\frac{Y}{X} \longmapsto \frac{W^2 + Z^2}{WZ} = \frac{W}{Z} + \frac{Z}{W} = u^{-1} + u.$$

On the second locus, the map is given by (here i denotes a square root of -1)

$$D_+(W^2 + Z^2) \cong \mathbb{P}^1_K \setminus \{(1, \mathrm{i}), (1, -\mathrm{i})\} \longrightarrow D_+(Y) \cong \mathbb{A}^1_K,$$

with

$$\frac{X}{Y} \longmapsto \frac{WZ}{W^2 + Z^2} = \frac{1}{u + u^{-1}} = \frac{u}{u^2 + 1}.$$

3. The system is base-point-free, since the two sets $D_+(WZ)$ and $D_+(W^2 + Z^2)$ cover the projective line. Indeed, if

$$P \notin D_+(WZ) \cup D_+(W^2 + Z^2),$$

then initially one coordinate must be 0, but then the second coordinate is also 0.

4. The extension of function fields

$$K(t) \subseteq K(u)$$

is given by $t \mapsto u + u^{-1}$. Its degree is 2, since u satisfies the quadratic equation

$$u^2 - (u + u^{-1})u + 1 = 0$$

over $K(t)$. The extension cannot be the identity extension, since, for example, $u \mapsto u^{-1}$ gives a nontrivial automorphism of the field $K(u)$ over $K(t)$.

5. Take $(x, y) \in \mathbb{P}^1_K$, and first suppose that both coordinates are nonzero. We use the affine description above, namely the map

$$\mathbb{A}^1_K \setminus \{0\} \longrightarrow \mathbb{A}^1_K, \qquad u \longmapsto u + u^{-1}.$$

The inverse image of a point b consists of the solutions of $u + u^{-1} = b$, that is,

$$u^2 - bu + 1 = 0,$$

so

$$u = \pm\frac{1}{2}\sqrt{b^2-4} + \frac{b}{2}.$$

Let $b \neq 2, -2$, and let a be an inverse image of b. For the local ring homomorphism

$$K[t]_{(t-b)} \longrightarrow K[u]_{(u-a)}, \qquad t \longmapsto u + u^{-1},$$

we have

$$t - b = u + u^{-1} - b = \frac{1}{u}(u^2 - bu + 1) = \frac{1}{u}(u - a^{-1})(u - a).$$

The factor $\frac{1}{u}(u - a^{-1})$ is a unit, since $a \neq a^{-1}$ when $b \neq 2, -2$, while $u - a$ is a uniformiser in $K[u]_{(u-a)}$. The ramification order is therefore 1. For

$$b = 2, -2,$$

there is only the inverse image 1, respectively -1. For the local ring homomorphism

$$K[t]_{(t-2)} \longrightarrow K[u]_{(u-1)},$$

we have

$$t - 2 = u + u^{-1} - 2 = \frac{1}{u}(u^2 - 2u + 1) = \frac{1}{u}(u-1)^2,$$

and the ramification order is 2. The same holds for $b = -2$.

The inverse image of zero in $D_+(X)$, namely $(1, 0)$ (or (Y)), consists of $(W - \mathrm{i}Z)$ and $(W + \mathrm{i}Z)$. The local ring homomorphism is

$$K[t]_{(t)} \longrightarrow K[u]_{(u-\mathrm{i})},$$

and since

$$t = u + u^{-1} = \frac{1}{u}(u^2 + 1) = \frac{1}{u}(u + \mathrm{i})(u - \mathrm{i}),$$

the ramification order is 1. The inverse image of the point at infinity, namely $(0, 1)$ or (X), consists of $(1, 0)$ and $(0, 1)$. The local ring homomorphism is, on the one hand,

$$K[t^{-1}]_{(t^{-1})} \longrightarrow K[u]_{(u)},$$

with

$$t^{-1} = \frac{u}{u^2 + 1},$$

so its ramification index is 1, and, on the other hand,

$$K[t^{-1}]_{(t^{-1})} \longrightarrow K[u^{-1}]_{(u^{-1})},$$

with

$$t^{-1} = \frac{u}{u^2+1} = \frac{u \cdot u^{-2}}{(u^2+1) \cdot u^{-2}} = \frac{u^{-1}}{1 + u^{-2}},$$

and this also has ramification order 1.

Edition note (source). For a root a of $u^2 - bu + 1$, the other root is a^{-1}, so the source's factorisation $(u+a)(u-a)$ has been corrected to $(u-a^{-1})(u-a)$. The source also calls the point $(0,1)$ the ideal (Y) immediately after using the convention identifying (Y) with $(1,0)$; it has been corrected to (X).

6. We need to determine the principal divisor of $u + u^{-1}$ on the projective line. If

$$u = 0, \infty,$$

there is a pole, in each case of order 1. Elsewhere $u + u^{-1}$ is defined and has two simple roots, i and −i. Its principal divisor is therefore, in coordinate notation viewed from the affine line $D_+(W)$,

$$1 \cdot (\mathrm{i}) + 1 \cdot (-\mathrm{i}) - 1 \cdot (0) - 1 \cdot (\infty),$$

or, expressed in terms of homogeneous prime ideals of height 1,

$$1 \cdot (W - \mathrm{i}Z) + 1 \cdot (W + \mathrm{i}Z) - 1 \cdot (W) - 1 \cdot (Z).$$

Frozen negative results

There are no public solution pages at the checked exercise-page revisions for Exercises 29.1, 29.2, 29.3, 29.4, 29.6, 29.7, 29.8, 29.9, 29.10, 29.11, 29.13, 29.14, or 29.15. On each of these pages, the solution control reads "Eine Lösung erstellen" ("Create a solution"). This statement records the result of checking the official links, not a claim that mathematical solutions do not exist.

Lecture 30: The Riemann–Roch theorem

The degree of twisted structure sheaves on plane curves

Lemma 30.1

Let

$$C = V_+(F) \subseteq \mathbb{P}^2_K$$

be a smooth projective plane curve of degree $d = \deg(F)$ over an algebraically closed field K. The restriction of $\mathcal{O}_{\mathbb{P}^2_K}(e)$ to C has degree de.

Edition note (source). The source uses the symbol K throughout the statement but calls the base only "an algebraically closed field". The missing name has been supplied above.

Proof It suffices to take $e = 1$, since pullback of sheaves is compatible with tensor products and, by Exercise 29.15, degree is additive under tensor products of invertible sheaves. Let

$$G \in \Gamma(\mathbb{P}^2_K, \mathcal{O}_{\mathbb{P}^2_K}(1)) = K[X,Y,Z]_1$$

be a section which, as a polynomial in $K[X,Y,Z]$, is not a multiple of F. Then G can also be regarded as a nonzero section in

$$\Gamma(C, \mathcal{O}_{\mathbb{P}^2_K}(1)|_C) = \Gamma(C, \mathcal{O}_C(1)).$$

We must compute the degree of the divisor of zeros of G on $C = V_+(F)$. Let $P = (a,b,c) \in V_+(F)$. The order of vanishing of a section of an invertible sheaf can be computed in an affine neighbourhood of the point. Without loss of generality, take $c = 1$ and $P \in D_+(Z)$. The affine equation of the curve is the dehomogenisation F' with respect to Z, and under the identification

$$\mathcal{O}_{\mathbb{P}^2_K}|_{D_+(Z)} \longrightarrow \mathcal{O}_{\mathbb{P}^2_K}(1)|_{D_+(Z)}, \qquad 1 \longmapsto Z,$$

the section becomes the dehomogenisation G' of G. The local ring of the curve is

$$\mathcal{O}_{C,P} = \left(K\left[\frac{X}{Z}, \frac{Y}{Z}\right]/(F')\right)_{(X/Z-a,\, Y/Z-b)}.$$

By Lemma 21.9, the order of G' in this ring equals the K-dimension of

$$\mathcal{O}_{C,P}/(G') = \left(K\left[\frac{X}{Z}, \frac{Y}{Z}\right]/(F')\right)_{(X/Z-a,\, Y/Z-b)}/(G') = K\left[\frac{X}{Z}, \frac{Y}{Z}\right]_{(X/Z-a,\, Y/Z-b)}/(F', G').$$

This description is symmetric in F and G. Hence the degree of the divisor of zeros of G on $V_+(F)$ equals the degree of the divisor of zeros of F on $V_+(G) = \mathbb{P}^1_K$. For a homogeneous polynomial of degree d on the projective line, the sum of all orders of vanishing is d. $\square$

Riemann–Roch for invertible sheaves

Let $C = V_+(F) \subseteq \mathbb{P}^2_K$ be a smooth projective plane curve of degree d over an algebraically closed field K, and let

$$\mathcal{L} = \mathcal{O}_C(e) = \mathcal{O}_{\mathbb{P}^2_K}(e)|_C.$$

Edition note (source). The source omits smoothness here, although the subsequent appeal to the cohomological genus in Definition 29.1 and the transition to Riemann–Roch use a smooth projective curve. The missing hypothesis has been made explicit.

We want to compute the number of global sections of $\mathcal{O}_C(e)$. Consider the short exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^2_K}(e-d) \xrightarrow{F} \mathcal{O}_{\mathbb{P}^2_K}(e) \longrightarrow \mathcal{O}_C(e) \longrightarrow 0$$

on the projective plane and the beginning of the associated long exact cohomology sequence:

$$0 \longrightarrow H^0(\mathbb{P}^2_K, \mathcal{O}_{\mathbb{P}^2_K}(e-d)) \longrightarrow H^0(\mathbb{P}^2_K, \mathcal{O}_{\mathbb{P}^2_K}(e)) \longrightarrow H^0(\mathbb{P}^2_K, \mathcal{O}_C(e)) \longrightarrow H^1(\mathbb{P}^2_K, \mathcal{O}_{\mathbb{P}^2_K}(e-d)) = 0.$$

The equality on the right follows from Theorem 27.4. For $e \geq d$, the dimensions of the vector spaces involved can be computed directly using Exercise 12.4:

$$\begin{aligned}
\dim_K H^0(\mathbb{P}^2_K, \mathcal{O}_C(e)) &= \dim_K H^0(\mathbb{P}^2_K, \mathcal{O}_{\mathbb{P}^2_K}(e)) - \dim_K H^0(\mathbb{P}^2_K, \mathcal{O}_{\mathbb{P}^2_K}(e-d)) \\
&= \binom{e+2}{2} - \binom{e-d+2}{2} \\
&= \frac{(e+2)(e+1) - (e+2-d)(e+1-d)}{2} \\
&= \frac{2de + 3d - d^2}{2} \\
&= de - \frac{(d-1)(d-2)}{2} + 1.
\end{aligned}$$

By Theorem 27.6, $H^0(\mathbb{P}^2_K, \mathcal{O}_C(e)) = H^0(C, \mathcal{O}_C(e))$. By Lemma 30.1, de is the degree of $\mathcal{O}_C(e)$, and by Theorem 29.5, $(d-1)(d-2)/2$ is the cohomological genus g of the curve. Thus, for $e \geq d$,

$$\dim_K H^0(C, \mathcal{O}_C(e)) = \deg(\mathcal{O}_C(e)) - g + 1.$$

For $e < 0$, this formula cannot be correct: the left-hand side is zero, while the right-hand side can be arbitrarily negative. The Riemann–Roch theorem shows that an analogous formula holds for an invertible sheaf $\mathcal{L}$ on a smooth projective curve C, but the left-hand side must be replaced by

$$\dim_K H^0(C, \mathcal{L}) - \dim_K H^1(C, \mathcal{L}).$$

Thus first cohomology appears as a correction term.

Theorem 30.2: Riemann–Roch

Let C be a smooth irreducible projective curve of genus g over an algebraically closed field K, and let $\mathcal{L}$ be an invertible sheaf on C. Then

$$h^0(C, \mathcal{L}) - h^1(C, \mathcal{L}) = \deg(\mathcal{L}) + 1 - g.$$

Proof The assertion is true for the structure sheaf. For a closed point $P \in C$, consider the short exact sequence

$$0 \longrightarrow \mathcal{I}_P \longrightarrow \mathcal{O}_C \longrightarrow \kappa(P) \longrightarrow 0,$$

where $\mathcal{I}_P = \mathcal{O}_C(-P)$ is the reduced invertible ideal sheaf of P, and $\kappa(P)$ is the structure sheaf of the point, regarded as a skyscraper sheaf on C. Tensoring this sequence with the invertible sheaf $\mathcal{L}$ gives

$$0 \longrightarrow \mathcal{I}_P \otimes \mathcal{L} \longrightarrow \mathcal{L} \longrightarrow \kappa(P) \otimes \mathcal{L} = \kappa(P) \longrightarrow 0.$$

This sequence relates two invertible sheaves differing by the point P. The long exact cohomology sequence gives

$$h^0(C, \mathcal{L}) - h^1(C, \mathcal{L}) = h^0(C, \mathcal{I}_P \otimes \mathcal{L}) - h^1(C, \mathcal{I}_P \otimes \mathcal{L}) + 1,$$

since $h^0(C, \kappa(P)) = 1$ and $h^1(C, \kappa(P)) = 0$, as its support is zero-dimensional. Since $\deg(\mathcal{I}_P) = -1$, we also have

$$\deg(\mathcal{L}) = \deg(\mathcal{I}_P \otimes \mathcal{L}) + 1.$$

Thus the degree changes in exactly the same way as the difference between the dimensions of zeroth and first cohomology. The Riemann–Roch formula holds for $\mathcal{L}$ precisely when it holds for $\mathcal{I}_P \otimes \mathcal{L}$. By Corollary 22.11, every invertible sheaf on the curve has the form $\mathcal{O}_C(-D)$ for a Weil divisor D. Hence every invertible sheaf can be obtained from the structure sheaf by adding or removing finitely many points. The formula therefore holds for all invertible sheaves. □

Corollary 30.3

Let C be a smooth irreducible projective curve of genus g over an algebraically closed field K, and let $\mathcal{L}$ be an invertible sheaf on C. Then

$$h^0(C, \mathcal{L}) \geq \deg(\mathcal{L}) + 1 - g.$$

If the degree of $\mathcal{L}$ is at least the genus of the curve, then $\mathcal{L}$ has nontrivial global sections.

Proof This follows immediately from Theorem 30.2. □

Corollary 30.4

Let C be a smooth irreducible projective curve over an algebraically closed field K. For every closed point $P \in C$, there is a nonconstant rational function $f \in Q(C)$ defined outside P.

Proof By Corollary 30.3, for sufficiently large n the invertible sheaf $\mathcal{O}_C(nP)$ has nontrivial global sections, with arbitrarily many as n grows. These correspond to rational functions on C whose principal divisors are greater than or equal to $-nP$. Such a function can have a pole only at P, and is therefore defined on $C \setminus \{P\}$. Among these functions are nonconstant ones. □

Riemann–Roch for locally free sheaves

We will generalise the Riemann–Roch theorem to locally free sheaves. First we must define the degree of a locally free sheaf.

Definition 30.5

Let C be a smooth projective curve over an algebraically closed field K. The degree of a locally free sheaf $\mathcal{G}$ of rank r on C is defined as the degree of its determinant sheaf

$$\bigwedge^r \mathcal{G}.$$

Theorem 30.6

Let C be a smooth projective curve over an algebraically closed field K. The degree of locally free sheaves on C is additive in short exact sequences.

Proof This follows from Theorem 16.11. □

Thus we have three additive invariants for locally free sheaves on a smooth projective curve: rank, degree, and Euler characteristic.

Lemma 30.7

Let C be a smooth irreducible projective curve over an algebraically closed field K. Every nonzero coherent ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_C$ is invertible.

Proof Since invertibility can be checked locally at the stalks $\mathcal{O}_{C,x}$, the assertion follows from the fact that these local rings are discrete valuation rings and hence principal ideal domains. □

Theorem 30.8

Let C be a smooth irreducible projective curve over an algebraically closed field K, and let $\mathcal{F}$ be a locally free sheaf of rank r on C. Then there is a filtration

$$0 = \mathcal{F}_0 \subset \mathcal{F}_1 \subset \cdots \subset \mathcal{F}_{r-1} \subset \mathcal{F}_r = \mathcal{F}$$

by locally free sheaves $\mathcal{F}_i$ such that the quotient sheaves $\mathcal{F}_{i+1}/\mathcal{F}_i$ are invertible for $0 \leq i < r$.

> **Edition note (source).** The source statement omits the symbol r after "of rank" and prints the filtration starting with $0 = \mathcal{F}_1$, giving a rank-r sheaf only $r-1$ quotients. This edition displays the rank parameter r and the indexing $\mathcal{F}_0, \dots, \mathcal{F}_r$, consistently with the induction and the number of rank-one factors.

Proof For sufficiently large n, Theorem 15.12 gives a nontrivial global section $s \in \Gamma(C, \mathcal{F}^*(n))$. This section corresponds to a nontrivial module homomorphism

$$\mathcal{O}_C \longrightarrow \mathcal{F}^*(n).$$

Dualising gives a nontrivial module homomorphism

$$\mathcal{F}(-n) \longrightarrow \mathcal{O}_C.$$

Its image is an ideal sheaf $\mathcal{I} \neq 0$, which is invertible by Lemma 30.7 (cited as "Lemma 30.6" in the historical source PDF). Thus there is a surjective sheaf homomorphism

$$\mathcal{F}(-n) \longrightarrow \mathcal{I},$$

and therefore a surjective sheaf homomorphism

$$\mathcal{F} \longrightarrow \mathcal{I} \otimes \mathcal{O}_C(n) =: \mathcal{L}.$$

Since $\mathcal{L}$ is invertible, the kernel $\mathcal{G} \subset \mathcal{F}$ is locally free of smaller rank by Theorem 16.7. Applying this procedure inductively to $\mathcal{F}_{r-1} := \mathcal{G}$ yields the filtration. □

> **Edition note (source).** The historical source PDF proof refers to "Lemma 30.6" for invertibility of nonzero coherent ideal sheaves. That result is Lemma 30.7 in this lecture, as the frozen semantic proof now records; the historical printed number is preserved explicitly in the proof above.

Theorem 30.9

Let C be a smooth irreducible projective curve of genus g over an algebraically closed field K, and let $\mathcal{F}$ be a locally free sheaf of rank r on C. Then

$$h^0(C, \mathcal{F}) - h^1(C, \mathcal{F}) = \deg(\mathcal{F}) + r(1-g).$$

Proof We use induction on the rank r. The base case $r = 1$ is Theorem 30.2. For a locally free sheaf of rank r, use the filtration with invertible quotients from Theorem 30.8,

$$0 = \mathcal{F}_0 \subset \mathcal{F}_1 \subset \cdots \subset \mathcal{F}_{r-1} \subset \mathcal{F}_r.$$

In particular, there is a short exact sequence

$$0 \longrightarrow \mathcal{F}_{r-1} \longrightarrow \mathcal{F} \longrightarrow \mathcal{F}_r/\mathcal{F}_{r-1} \longrightarrow 0.$$

By the induction hypothesis, the Riemann–Roch formula holds for $\mathcal{F}_{r-1}$, and by Theorem 30.2 it holds for the invertible sheaf $\mathcal{F}_r/\mathcal{F}_{r-1}$. The Euler characteristic

$$\chi(\mathcal{G}) = h^0(C, \mathcal{G}) - h^1(C, \mathcal{G})$$

is additive in short exact sequences by Lemma 27.9, and the degree of locally free sheaves is likewise additive in short exact sequences by Theorem 30.6 (cited as "Theorem 30.7" in the historical source PDF). Hence the formula also holds for $\mathcal{F}$. □

> **Edition note (source).** The degree-additivity result used in the proof above is Theorem 30.6, not Theorem 30.7. The frozen semantic proof uses the correct reference. This edition preserves the historical source number explicitly in the proof and discloses the incorrect historical cross-reference here.

Worksheet 30: The Riemann–Roch theorem

Exercise 30.1

Let K be an algebraically closed field. Prove the Riemann–Roch theorem directly for the projective line $\mathbb{P}^1_K$.

> **Edition note (source).** The source leaves the base field implicit. The algebraically closed field K required by the version of Riemann–Roch stated in Lecture 30 has been made explicit.

Exercise 30.2

Let $f \in K[X, Y, Z]$ be a homogeneous polynomial of degree e over an algebraically closed field K such that

$$C = \text{Proj}(K[X, Y, Z]/(f)) \subseteq \mathbb{P}^2_K$$

is a smooth projective curve. Let $g_1, \dots, g_n \in K[X, Y, Z]$ be homogeneous elements of degrees $d_1, \dots, d_n$ such that the $D_+(g_i)$ cover the curve. Regard g_i as a sheaf homomorphism

$$\mathcal{O}_C(-d_i) \longrightarrow \mathcal{O}_C, \qquad h \longmapsto hg_i,$$

or, for $m \in \mathbb{Z}$,

$$\mathcal{O}_C(m - d_i) \longrightarrow \mathcal{O}_C(m), \qquad h \longmapsto hg_i.$$

1. Prove that the sheaf homomorphism

$$\bigoplus_{i=1}^{n} \mathcal{O}_C(m - d_i) \longrightarrow \mathcal{O}_C(m)$$

   is surjective.
2. Let $\text{Syz}(g_1, \dots, g_n)(m)$ be the kernel sheaf of the homomorphism in part (1). Prove that this sheaf is locally free.
3. Determine the rank of $\text{Syz}(g_1, \dots, g_n)(m)$.
4. Determine the degree of $\text{Syz}(g_1, \dots, g_n)(m)$.

Exercise 30.3

Let K be an algebraically closed field. Give examples on the projective line $\mathbb{P}^1_K$ of locally free sheaves of rank 2 and degree 0 whose spaces of global sections have arbitrarily large dimension.

> **Edition note (source).** The historical source exercise omits the value after "of degree", whereas the official entity title states `Grad 0`. This edition restores degree 0 so that the exercise agrees with its entity identity and the frozen semantic content.
>
> **Edition note (source).** The source uses K in $\mathbb{P}^1_K$ without declaring it. The algebraically closed base field assumed by the lecture's degree formalism has been stated explicitly.

Exercise 30.4

Let K be a field and, as in Theorem 19.8, let

$$\text{Syz}(x, y, z) \cong \Omega_{\mathbb{P}^2_K|K}$$

be the cotangent sheaf on the projective plane, and let $L \subseteq \mathbb{P}^2_K$ be a projective line. Prove

$$\text{Syz}(x, y, z)|_L \cong \mathcal{O}_L(-1) \oplus \mathcal{O}_L(-2).$$

> **Edition note (source).** The source uses K without declaring the base field. No algebraic-closure hypothesis is needed for this exercise, so K has been stated to be an arbitrary field.

Exercise 30.5

Let K be an algebraically closed field and, as in Theorem 19.8, let

$$\text{Syz}(x, y, z) \cong \Omega_{\mathbb{P}^2_K|K}$$

be the cotangent sheaf on the projective plane, and let $C = V_+(F) \subseteq \mathbb{P}^2_K$ be a smooth quadric. Prove that $\text{Syz}(x, y, z)|_C$ decomposes as a direct sum of two invertible sheaves. Use an isomorphism $\mathbb{P}^1_K \cong C$.

Edition note (source). The source leaves K undeclared while instructing the reader to use an isomorphism $\mathbb{P}^1_K \cong C$. A smooth conic need not have such an isomorphism over an arbitrary field; the algebraically closed hypothesis that guarantees it has therefore been supplied.

Public solutions and coverage of Worksheet 30

The official Worksheet 30 page contains exactly five exercises at the frozen boundary. For each exact exercise-page title, the official page with the suffix `/Lösung` was checked directly. All five candidate pages are absent, so there is no public solution text to translate. This conclusion is not inferred from stars or from the presence or absence of worksheet links.

Official candidates checked

1. Exercise 30.1
   - Exercise page: ProjektiveGerade/Riemann-Roch/Direkt/Aufgabe
   - Solution candidate: ProjektiveGerade/Riemann-Roch/Direkt/Aufgabe/Lösung
   - Official result: page absent.
2. Exercise 30.2
   - Exercise page: GlatteEbeneKurve/Syzygienbündel/Gradberechnung/Aufgabe
   - Solution candidate: GlatteEbeneKurve/Syzygienbündel/Gradberechnung/Aufgabe/Lösung
   - Official result: page absent.
3. Exercise 30.3
   - Exercise page: ProjektiveGerade/Rang2/Grad0/Schnitte/Aufgabe
   - Solution candidate: ProjektiveGerade/Rang2/Grad0/Schnitte/Aufgabe/Lösung
   - Official result: page absent.
4. Exercise 30.4
   - Exercise page: ProjektiveEbene/Kotangentialbündel/Einschränkung/Gerade/Aufgabe
   - Solution candidate: ProjektiveEbene/Kotangentialbündel/Einschränkung/Gerade/Aufgabe/Lösung
   - Official result: page absent.
5. Exercise 30.5
   - Exercise page: ProjektiveEbene/Kotangentialbündel/Einschränkung/Quadrik/Aufgabe
   - Solution candidate: ProjektiveEbene/Kotangentialbündel/Einschränkung/Quadrik/Aufgabe/Lösung
   - Official result: page absent.

Negative results established by the official pages

None of the five official candidate titles above has a public page. This statement records only the status of the source pages at this check; it is not a claim that mathematical solutions do not exist.

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2. **Inclusion–exclusion diagram for three sets** — File:Inclusion-exclusion.svg; creator/attribution: Burn~commonswiki (named in the course source; Commons metadata records the creator as 'Unknown Unknown' and the uploader as Chris-martin); licence/rights status: Public domain. Alternative text: Diagram of three mutually intersecting circular sets, with each intersection region distinguished by colour. Rights note: The inline Wikiversity label is CC-by-sa 3.0; the frozen Commons metadata offers Public domain. Reuse in this edition follows that Commons option.
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Retained source caption: **Visualisation of the spectrum of the integers: discrete prime points, a bold zero point as a dense point, and a connecting line indicating a one-dimensional object**. The edition's alternative text: A conceptual line with discrete prime points and a bold zero point illustrating the spectrum of the integers.

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